ANDHRA PRADESH GRANITE (MIDWEST)
PRIVATE LIMITED

CIN No. : U14102AP2007PTC054390
Regd. Office : 8-2-684/3/25&26, Road No. 12, Banjara Hills, Hyderabad - 500 034. INDIA
Tel : +91-40-23305193, 23305194, 23305195 Fax : +91-40-23305167
E-mail : info@midwestgranite.com

To
The Environmental Engineer,
Andhra Pradesh Pollution Control Board
Regional Office, Ongole
Prakasam District

Sir,

Sub: Quarry lease of Black Galaxy Granite over an extent of 19.425 Ha in Sy.no 55/5 of Rajupalem Lakshmipuram Village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh, held by Andhra Pradesh Granite (Midwest) Private Limited-**Submission of Draft EIA Report for Public Hearing –Reg.**

Ref: 1. TOR Application submitted on 15.05.2021.

2. Standard TOR Granted by SEIAA, Andhra Pradesh, File no SEIAA/AP/ PKM/ MIN/ VIO/ 08/ 2018/ 672 dated 01.07.2021.

Andhra Pradesh Granite (Midwest) Private Limited is holding a quarry lease for Black Galaxy Granite mining over an extent of 19.425 Ha in Sy.no. 55/5 of Rajupalem Lakshmipuram Village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh and Environmental Clearance for the production enhancement Granted by SEIAA, AP on 13.09.2019 for the production of 54,643 m3/ annum.

Andhra Pradesh Granite (Midwest) Private Limited proposed to enhance the production capacity from 54,643 m3 /annum to 79,401.0 m3 /annum.

As per the EIA Notification 2006, the project category comes under category B1 and the mine category comes under 1(a).

With the reference to the 1st cited the mining unit submitted Application of TOR and considered by the State Expert Appraisal committee(SEAC), Andhra Pradesh and the SEIAA issued the TOR and directed to prepare EIA Report as per the Standard Terms of Reference(TORs) issued by the MoEF&CC, GoI vide reference 2nd cited.

Now, we are here with submitting following documents for Public consultation along with EIA and Form I with a request of take necessary action for public notification and to initiate public hearing process.

- Draft EIA Report for Public Consultation Process
- Executive Summary (Telugu and English)
- Form I

We shall submit the necessary number of EIA copies along with English and Telugu versions of Executive summary to proceed further on receipt of communication from you.

Thanking you,

Yours faithfully,

For, Andhra Pradesh Granite (Midwest) Private Limited

(K.RAMACHANDRA)

DIRECTOR

ANDHRA PRADESH GRANITE (MIDWEST) PVT. LTD. Hyderabad

DRAFT ENVIRONMENTAL IMPACT ASSESSMENT REPORT

Of

M/s. Andhra Pradesh Granite (Midwest) Private Limited

Sy. No. 55/5 of Rajupalem- Lakshmi Puram village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh.

Proposal for enhance the production of Black Galaxy Granite form 54,643 m3/Annum to 79,401 m3/Annum
Category B Project under 1(a) Sector



Baseline Study period – January 2019 to March 2019

File No: SEIAA/AP/PKM/MIN/VIO/08/2018/672

Project Proponent

M/s. Andhra Pradesh Granite (Midwest) Private Limited
Sy. No. 55/5 of Rajupalem- Lakshmi Puram village, Chimakurthy Mandal,
Prakasam District,
Andhra Pradesh.
Ph.no:8790132121
Email:info@midwestgranite.com

Environmental Consultant

M/s Pridhvi Envirotech (P) Ltd.
Plot No: 184/C, Lawn house, 4th Floor,
Vengalrao Nagar,
Hyderabad-500038
Ph: 040-40179770
Email:info@pridhvienviro.com
pvr@pridhvienviro.com.

SUBMITTED TO
ANDHRA PRADESH POLLUTION CONTROL BOARD
ONGOLE

ANDHRA PRADESH GRANITE (MIDWEST)
PRIVATE LIMITED

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Regd. Office : 8-2-684/3/25&26, Road No. 12, Banjara Hills, Hyderabad - 500 034. INDIA
Tel : +91-40-23305193, 23305194, 23305195 Fax : +91-40-23305167
E-mail : info@midwestgranite.com

Undertaking by project Proponent

M/s Andhra Pradesh Granite (Midwest) Private Limited has carried out an Environmental Impact Assessment (EIA) Study to enhance the Production of Black Galaxy Granite from 54,643 m3 /annum to 79,401.0 m3 /annum at Sy.no. 55/5 of Rajupalem Lakshmipuram Village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh.

As per MoEF office memorandum No J-11013/41/2016-IA.II dated October 05th 2011, Andhra Pradesh Granite (Midwest) Private Limited herewith declares the ownership of contents (Information and data) of this EIA Report pertaining to Andhra Pradesh Granite (Midwest) Private Limited at Sy.no. 55/5 of Rajupalem Lakshmipuram Village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh.

Yours Sincerely,

For, Andhra Pradesh Granite (Midwest) Private Limited

(K.RAMACHANDRA)
DIRECTOR

ANDHRA PRADESH GRANITE (MIDWEST) PVT. LTD. Hyderabad

FORM – 1

(As per new Notification of MoEF dated 01-12-2009 vide S.O 3067 (E))

I) BASIC INFORMATION

<table>
<tr><th>S No.</th><th>Item</th><th>Details</th></tr>
<tr><td>1</td><td>Name of the project/s</td><td>M/s Andhra Pradesh Granite (Midwest) Pvt.Ltd</td></tr>
<tr><td>2</td><td>S. No. in the schedule</td><td>Item -1(a) Mining of Minerals</td></tr>
<tr><td>3</td><td>Proposed capacity /area/length/ tonnage to be handled/command area/lease area/ number of wells to be drilled.</td><td>Black Galaxy Granite
54,643m3/Annum
Quarry Lease area – 19.425 Hectares</td></tr>
<tr><td>4</td><td>New/Expansion/Modernization</td><td>Production Enhancement:
Enhancement of production capacity of Black Galaxy Granite from 54,643 m3/Annum to 79,401 m3/Annum</td></tr>
<tr><td>5</td><td>Existing Capacity/Area etc.</td><td>Existing mine
54,643m3/Annum
Mine Lease area- 19.425 Hectares</td></tr>
<tr><td>6</td><td>Category of Project i.e. ‘A’ or ‘B’</td><td>Category B (B1)</td></tr>
<tr><td>7</td><td>Does it attract the general condition? If yes, please specify.</td><td>No</td></tr>
<tr><td>8</td><td>Does it attract the specific condition? If yes, please specify.</td><td>No</td></tr>
<tr><td>9</td><td>Location

Plot / Survey/ Khasra No.
Village
Tehsil
District
State</td><td>Latitude and Longitude
<table>
<tr><th>S.no</th><th>Latitude</th><th>Longitude</th></tr>
<tr><td>1.</td><td>N 15^{0}35’13.21”</td><td>E 79^0 49’19.32”</td></tr>
<tr><td>2.</td><td>N 15^{0}35’06.03”</td><td>E 79^{0}49’20.63”</td></tr>
<tr><td>3.</td><td>N 15^{0}35’06.11”</td><td>E 79^{0}49’25.34”</td></tr>
<tr><td>4.</td><td>N 15^{0}35’07.22”</td><td>E 79^{0}49’25.61”</td></tr>
<tr><td>5.</td><td>N 15^{0}35’06.12”</td><td>E 79^{0}49’29.00”</td></tr>
<tr><td>6.</td><td>N 15^{0}35’05.00”</td><td>E 79^{0}49’29.23”</td></tr>
<tr><td>7.</td><td>N 15^{0}35’02.91”</td><td>E 79^{0}49’30.31”</td></tr>
<tr><td>8.</td><td>N 15^{0}35’04.63”</td><td>E 79^{0}49’40.46”</td></tr>
<tr><td>9.</td><td>N 15^{0}35’04.00”</td><td>E 79^{0}49’52.64”</td></tr>
<tr><td>10.</td><td>N 15^{0}35’09.52”</td><td>E 79^{0}49’51.61”</td></tr>
</table>
Sy.No: 55/5,
Rajupalem-Laxmipuram village,
Chimakurthy Mandal,
Prakasam District, Andhra Pradesh.</td></tr>
<tr><td>10</td><td>Nearest railway station/airport along</td><td>Ongole Railway Station – 27 Km due SE</td></tr>
</table>

	with distance in kms.	from lease area Gannavaram Airport – 148 Km due NE from lease area
11	Nearest Town, city, district Head - quarters with distance in Km.	Chimakurthy-4.5 Km
12	Village Panchayat, Zilla Parishad, Municipal Corporation, Local body (complete postal addresses with telephone nos. to be given)	Rajupalem Lakshmi Puram village Panchayat Chimakurthy Mandal Prakasam District
13	Name of the applicant	Mr.K. Rama Chandra
14	Registered Address	H. No. :8-2-684/3/25&26, Road No: 12;Banjara Hills, Hyderabad
15	Address for correspondence:	
	Name	Mr.K. Rama Chandra
	Designation(Owner/Partner/CEO)	Director
	Address	H. No. :8-2-684/3/25&26, Road No: 12;Banjara Hills, Hyderabad
	Pin code	500034
	E-mail	info@midwestgranite.com
	Telephone No.	8790132121
	Fax No	------
16	Details of Alternative Sites examined, if any. Location of these sites should be shown on a topo sheet.	Not applicable
17	Interlinked Projects	Not Applicable
18	Whether separate application of interlinked project has been submitted?	Not Applicable
19	If yes, date of submission	Not Applicable
20	If no, reason	Not Applicable
21	Whether the proposal involves approval/clearance under: if yes, details of the same and their status to be given. i. The Forest (Conservation) Act, 1980? ii. The Wildlife (Protection) Act, 1972? iii. The C.R.Z. Notification, 1991?	None
22	Whether there is any Government Order/Policy relevant /relating to the site?	Yes, Director of Mines and Geology, A.P vide Proceedings.No.545/Q/2005 dated: 03.02.2005. Mining scheme approval vide Lr. No. 7141/MMS/BGG/OGL/2020 Dated: 22.01.2021
23	Forest land involved (hectares)	No Forest Land is involved
24	Whether there is any litigation pending against the project and/or land in which the project is propose to be set up? a) Name of the Court b) Case No. c) Orders/directions of the Court, if	No Court cases

	any and its relevance with the proposed project.	

II) ACTIVITY

1. **Construction, operation or decommissioning of the Project involving actions, which will cause physical changes in the locality (topography, land use, changes in water bodies, etc.)**

S No.	Information/Checklist confirmation	Yes/ No	Details thereof (with approximate quantities/ rates, wherever possible) with source of information data
1.1	Permanent or temporary change in land use, land cover or topography including increase in intensity of land use (with respect to local land use plan)	Yes	Entire mining lease area of 19.425 Ha is Government land. The quarry lease area is located on plain with some worked out pits. There is no change in land use as the mining already under Progress.
1.2	Clearance of existing land, vegetation and buildings?	No	Area is barren with boulders of black galaxy Granite. There is no buildings in the lease area. Hence no clearance of existing vegetation and building is envisaged.
1.3	Creation of new land uses?	Yes	Mined out area will be utilized for water harvesting, green belt will be developed all along the dump.
1.4	Pre-construction investigations e.g. bore houses, soil testing?	Yes	Preliminary investigations have been carried out at the quarry lease area to estimate the reserves
1.5	Construction works?	No	No construction works proposed in the quarry lease area
1.6	Demolition works?	No	No demolition works involved.
1.7	Temporary sites used for construction works or housing of construction workers?	No	There is no need for construction of housing for workers, as the labor come from nearby villages and go back at the end of shift.
1.8	Above ground buildings, structures or earthworks including linear structures, cut and fill or excavation.	No	Not Applicable
1.9	Underground works including mining or tunneling?	No	No underground mining or tunneling is involved
1.10	Reclamation works?	Yes	The quarry pits would be developed as water reservoirs with proper fencing. The water can be used for

			local farming, developing in to fish pond and/or as tourist attraction.
1.11	Dredging?	No	Not applicable
1.12	Offshore structures?	No	Not applicable
1.13	Production and manufacturing process	Yes	Proposed average annual production of Black Galaxy Granite is 79,401 M3. Mining will be carried out by opencast semi mechanized method by using an excavator, tippers, compressors and jackhammers.
1.14	Facilities for storage of goods or materials?	Yes	Stock yard facilities are identified for storage of goods and materials in ML area.
1.15	Facilities for treatment or disposal of solid waste or liquid effluents?	Yes	The firm proposes to obtain permission from concerned authority under Rule 12 (5) (h) (xvi) of APMMCR 1966 to utilize the waste rock by supplying to local crusher units as Crusher Feed. The local crushers will crush the waste rock and produce rock aggregate of different sizes viz: 40mm, 20mm, 12mm, 6mm to use in concrete and other Road works. The final product in this process is fine aggregate / Rock dust which can also be used as filling material in place of ordinary sand in cement mix and also as top layer in black topped roads. The firm also propose to supply the waste for agencies to prepare Kerbs and cubes for pavements. The firm will try to dispose off the rock waste to the maximum possible extent for the purpose of production of road metal as well as other building materials. The generated waste will be dumped in Sy. Nos. 25/3B2 of R.L. Puram Village, Chimakurthy Mandal & 938/1 to 7 of Chimakurthy Village & Mandal during the present scheme period
1.16	Facilities for long term housing of operational workers?	No	Not envisaged
1.17	New road, rail or sea traffic during	Yes	Additional traffic will be created due

	construction or operation?		to the movement of mined granite blocks. The approximate increase of trucks / tippers will be 12 dumpers / day during operations.
1.18	New road, rail, air waterborne or other transport infrastructure including new or altered routes and stations, ports, airports etc.?	No	Not required
1.19	Closure or diversion of existing transport routes or infrastructure leading to changes in traffic movements?	No	Closure or diversion of existing routes or infrastructure is not required.
1.20	New or diverted transmission lines or pipelines?	No	Not envisaged
1.21	Impoundment, damming, culverting, realignment or other changes to the hydrology of water courses or aquifers?	No	Not required
1.22	Stream crossings?	No	There is no stream crossing in the mining lease area.
1.23	Abstraction or transfers of water from ground or surface waters?	Yes	Total water requirement will be 80 KLD. Water will be sourced from tankers in the nearby village. During monsoon season rain water accumulated in the quarry pit will be used for other purposes except human consumption.
1.24	Changes in water bodies or the land surface affecting drainage or run-off?	No	There are no perennial/seasonal water courses exist in this area
1.25	Transport of personnel or materials for construction, operation or decommissioning?	No	Not envisaged
1.26	Long-term dismantling or decommissioning or restoration works?	Yes	Not envisaged
1.27	Ongoing activity during decommissioning which could have an impact on the environment?	No	Not envisaged
1.28	Influx of people to an area in either temporarily or permanently?	No	No influx of other people is been envisaged.
1.29	Introduction of alien species?	No	No alien species will be introduced. Only local species will be used for green belt.
1.30	Loss of native species or genetic diversity?	No	There will be no loss of native species or genetic diversity.
1.31	Any other actions?	No	Not Applicable

2 .Use of Natural resources for construction or operation of the Project (such as land, water, materials or energy, especially any resources which are non-renewable or in short supply):

S No.	Information/checklist confirmation	Yes/ No	Details thereof (with approx. -mate quantities / rates, wherever possible) with source of

			information data
2.1	Land especially undeveloped or agricultural land (Ha)	Yes	The quarry lease area is 19.425 Ha. The area is covered by rocky out crops and bushy thorny vegetation.
2.2	Water (expected source & competing users) unit: KLD	Yes	Water requirement - 80 KLD Source – Tankers nearby village
2.3	Minerals (MT)	No	No minerals are required for quarrying operations
2.4	Construction material–stone, aggregates, sand / soil (expected source – MT)	No	As no construction is proposed in the lease area, construction material is not used
2.5	Forests and timber (source – MT)	No	No Forest and timber products are involved.
2.6	Energy including electricity and fuels (source, competing users) Unit: fuel (MT), energy (MW)	Yes	About 5000 lit/annum of Diesel will be used in transportation vehicles, excavators and D G set.
2.7	Any other natural resources (use appropriate standard units)	No	Not applicable

3. Use, storage, transport, handling or production of substances or materials, which could be harmful to human health or the environment or raise concerns about actual or perceived risks to human health.

S No	Information/Checklist confirmation	Yes/ No	Details thereof (with approximate quantities / rates, wherever possible) with source of information data
3.1	Use of substances or materials, which are hazardous (as per MSIHC rules) to human health or the environment (flora, fauna, and water supplies)	No	No Hazardous substance or materials will be used for quarrying.
3.2	Changes in occurrence of disease or affect disease vectors (e.g. insect or water borne diseases)	No	Not applicable
3.3	Affect the welfare of people e.g. by changing living conditions?	Yes	The proposed activity will provided employment to the nearby village thereby increasing their economical status.
3.4	Vulnerable groups of people who could be affected by the project e.g. hospital patients, children, the elderly etc.	No	No Vulnerable groups of people will be affected, as no hazardous, toxic or noxious substances are involved in the project activity.
3.5	Any other causes	No	Not applicable

4. Production of solid wastes during construction or operation or Decommissioning (MT/ month)

S No.	Information/Checklist confirmation	Yes/	Details thereof (with approximate

		No	**quantities/ rates, wherever possible) with source of information data**
4.1	Spoil, overburden or mine wastes	Yes	The firm proposes to obtain permission from concerned authority under Rule 12 (5) (h) (xvi) of APMMCR 1966 to utilize the waste rock by supplying to local crusher units as Crusher Feed. The local crushers will crush the waste rock and produce rock aggregate of different sizes viz: 40mm, 20mm, 12mm, 6mm to use in concrete and other Road works. The final product in this process is fine aggregate / Rock dust which can also be used as filling material in place of ordinary sand in cement mix and also as top layer in black topped roads.The firm also propose to supply the waste for agencies to prepare Kerbs and cubes for pavements. The firm will try to dispose off the rock waste to the maximum possible extent for the purpose of production of road metal as well as other building materials. The remaining overburden and rock waste will be dumped in the dump yards located over an extent of 6.17 acres in sy.no- 938/1 to 7 in chimakurthy village and mandal, Sy.No. 941/1 to 4 & 944/1 to 7 over an extent of 11.19 acrs in Chimakurthy (V&M), Prakasam District, Sy.No.937/1 of Chimakurthy village & Mandal, Prakasam District over 4.95 Ac. and 12.98 Ac. in Sy.No. 25/3B2 of R.L.Puram (V), Chimakurthy Mandal, Prakasam District.
4.2	Municipal waste (domestic and or commercial wastes)	No	Insignificant quantity of domestic waste water generated will be treated in septic tank followed by soak pit
4.3	Hazardous wastes (as per Hazardous Waste Management Rules)	No	No Hazardous waste is generated.
4.4	Other industrial process wastes	No	Not applicable
4.5	Surplus product	No	Not envisaged
4.6	Sewage sludge or other sludge from effluent treatment	No	Not envisaged
4.7	Construction or demolition wastes	No	Not envisaged

4.8	Redundant machinery or equipment	No	Not applicable
4.9	Contaminated soils or other materials	No	No contamination of soils or other material is involved.
4.10	Agricultural wastes	No	Not applicable
4.11	Other solid wastes	No	None

5. Release of pollutants or any hazardous, toxic or noxious substances to air (Kg/hr)

S.No.	Information/Checklist confirmation	Yes/ No	Details thereof (with approximate quantities/ rates, wherever possible) with source of information data
5.1	Emissions from combustion of fossil fuels from stationary or mobile sources	Yes	Emissions will be generated from the mining machinery, DG set and transportation vehicles.
5.2	Emissions from production processes	Yes	Dust will be generated from quarrying operations , which will be controlled by water sprinkling.
5.3	Emissions from materials handling including storage or transport	Yes	Emissions will be generated from transportation of quarried stone and dump waste Adequate control measures are proposed
5.4	Emissions from construction activities including plant and equipment	No	No construction activity is proposed at the quarry site. Hence no emissions from any such activities.
5.5	Dust or odour from handling of materials including construction materials, sewage and waste	Yes	Dust will be generated during mining and transportation of materials. However, there is no generation of any odor.
5.6	Emissions from incineration of waste	No	No such waste will be generated, which may be incinerated.
5.7	Emissions from burning of waste in open air (e.g. slash materials , construction debris)	No	No waste of any kind will be burnt in the open air.
5.8	Emissions from any other sources	No	There are no other sources

6. Generation of Noise and Vibration, and Emissions of Light and Heat:

S.No.	Information/Checklist confirmation	Yes/ No	Details thereof (with approximate quantities/rates, wherever possible) with source of information data with source of information data
6.1	From operation of equipment e.g. engines, ventilation plant, crushers	Yes	Noise up to 100 dB will be generated from operations of drilling machinery. Ear plugs are provided to all workers
6.2	From industrial or similar processes	No	Not applicable
6.3	From construction or demolition	No	Not applicable
6.4	From blasting or piling	Yes	Noise will be generated during

			quarrying operations involving blasting and will be instantaneous in nature and minimum. As there is no habitation nearby, there will be no impact.
6.5	From construction or operational traffic	Yes	The noise generated from operational traffic ie. Tippers will be about 75-85 dB(A)
6.6	From lighting or cooling systems	No	No lighting or cooling systems are involved
6.7	From any other sources	No	There are no other sources to generate Noise and Vibrations.

7.Risks of contamination of land or water from releases of pollutants into the ground or into sewers, surface waters, groundwater, coastal waters or the sea:

S.No.	Information/Checklist confirmation	Yes/ No	Details thereof (with approximate quantities/ rates, wherever possible) with source of information data
7.1	From handling, storage, use or spillage of hazardous materials	Yes	There may be a risk of contamination of land due to handling, storage, use or spillage of diesel. Proper handling and disposal methods will be followed.
7.2	From discharge of sewage or other effluents to water or the land (expected mode and place of discharge)	Yes	8.0 KLD domestic sewage discharged in to septic tank followed by soak pit.
7.3	By deposition of pollutants emitted to air into the land or into water	Yes	Dust emissions will be limited to the site itself which will be suppressed by regular water sprinkling.
7.4	From any other sources	No	Not applicable
7.5	Is there a risk of long term build up of pollutants in the environment from these sources?	No	Not applicable

8.Risk of accidents during construction or operation of the Project, which could affect human health or the environment

S No.	Information/Checklist confirmation	Yes /No	Details thereof (with approximate quantities/ rates, wherever possible) with source of information data
8.1	From explosions, spillages, fires etc from storage, handling, use or production of hazardous substances	No	Not applicable
8.2	From any other causes	No	Not applicable

8.3	Could the project be affected by natural disasters causing environmental damage (e.g. floods, earth -quakes, landslides, cloudburst etc)?	No	Not applicable

9. Factors which should be considered (such as consequential development) which could lead to environmental effects or the potential for cumulative impacts with other existing or planned activities in the locality.

S No.	Information/Checklist confirmation	Yes /No	Details thereof (with approx. -mate quantities / rates, wherever possible) with source of information data
9.1	Lead to development of supporting facilities, ancillary development or development stimulated by the project which could have impact on the environment e.g.: • Supporting infrastructure (roads, power supply, waste or waste water treatment, etc.) • housing development • extractive industries • supply industries • other	Yes	Generation of employment and improved standard of living Increased revenue to the state by way of Royalty, Taxes and Duties.
9.2	Lead to after-use of the site, which could have an impact on the environment	Yes	There would be minor impact on the environment as quarry pits will be converted in to water reservoirs, which recharge the ground water levels in the area.
9.3	Set a precedent for later developments	Yes	A better after use scenario, which is likely to increase aesthetic beauty of the area by converting the excavated region into water reservoirs and green cover
9.4	Have cumulative effects due to proximity to other existing or planned projects with similar effects	No	Not applicable

(III) ENVIRONMENTAL SENSITIVITY

S No.	Areas	Name/ Identity	Aerial distance (within 15 km) Proposed project location boundary
1	Areas protected under international conventions, national or local legislation for their ecological, landscape, cultural or other related	None	Not applicable

	value		
2	Areas which are important or sensitive for ecological reasons - Wetlands, watercourses or other water bodies, coastal zone, biospheres, mountains, forests	**Water Bodies Reserved Forest**	**Water Bodies:** Ramathirtham- 4.6(NW) Reservior-4.4Km(SW) Musi River-3.1 Km(E) Jawahar canal-5.6 Km(W) Gaddi Vagu-8.1 Km (NW) **Reserve Forests:** Chimakurthy RF-2.3Km (N) Devarapalem RF-6.0 Km(N)
3	Areas used by protected, important or sensitive species of flora or fauna for breeding, nesting, foraging, resting, over wintering, migration	No	Not applicable
4	Inland, coastal, marine or underground waters	No	Not applicable
5	State, National boundaries	No	Not applicable
6	Routes or facilities used by the public for access to recreation or other tourist, pilgrim areas.	No	Not applicable
7	Defense installations.	No	Not applicable
8	Densely populated or built-up area	Yes	RL Puram - 0.75 Km
9	Areas occupied by sensitive man-made land uses (hospitals, schools, places of worship, community facilities)		Primary Hospital RL Puram - 0.75 Km Major Hospital Ongole – 27 Km
10	Areas containing important, high quality or scarce resources(ground water resources, surface resources, forestry, agriculture, fisheries, tourism, minerals)	yes	**Water Bodies:** Ramathirtham- 4.6(NW) Reservior-4.4Km(SW) Musi River-3.1 Km(E) Jawahar canal-5.6 Km(W) Gaddi Vagu-8.1 Km (NW) **Reserve Forests:** Chimakurthy RF-2.3Km (N) Devarapalem RF-6.0 Km(N)
11	Areas already subjected to pollution or environmental damage. (Those where existing legal environmental standards are exceeded)	No	Not applicable
12	Areas susceptible to natural hazard which could cause the project to present environmental problems (earthquakes, subsidence, landslides, erosion, flooding or adverse climatic conditions)	No	Not applicable

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PRIVATE LIMITED

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E-mail : info@midwestgranite.com

Declaration

I hereby give an undertaking that the data and information given in the application and enclosures are true to the best of my knowledge and belief and I am aware that if any part of the data and information submitted is found to be false or misleading at any stage, the project will be rejected and clearance, if any given to the project, will be revoked at our risk and cost.



K.Ramachandra,
Director.
Signature of the applicant

LIST OF CONTENTS

S.NO	Description	Page No.
Executive Summary		
CHAPTER-1 INTRODUCTION		
1.0	Introduction	1-1
1.1	Purpose of report	1-1
1.2	Background of the project	1-1
1.3	Identification of project & Project Proponent	1-2
1.4	Project Location & Study area Description	1-3
1.5	Project Cost	1-8
1.6	Importance of the Project to the Country, Region	1-8
1.7	Project Justifications	1-8
1.8	Cluster of mines Details	1-8
1.9	Scope of the Study	1-15
1.10	Required Statutory Licenses at A Glance	1-16
1.11	Structure of the EIA Report	1-16
CHAPTER 2 PROJECT DESCRIPTION		
2.1	Details of the Mining Lease Area	2-1
2.2	Location of the Project	2-1
2.3	Road Connectivity to the site	2-3
2.4	Physiography of the Site	2-3
2.5	Site Description and Present land use	2-3
2.6	Mine Drainage	2-4
2.7	Geology	2-9
2.7.1	Regional Geology	2-9
2.7.2	Local Geology	2-9
2.8	Present Land Use Pattern	2-13
2.9.	Area for Dump	2-13
2.9.1	Area of Green belt	2-13
2.10	Method of Mining	2-14
2.10.1	Description of Existing method for excavation	2-14
2.11	Year wise Production and Development	2-16
2.12	Conceptual Plan	2-16
2.13	Anticipated life of mine	2-18
2.14	Top Soil &Waste Generation and Disposal	2-18
2.15	Requirements for the Project	2-19
2.15.1	Site Services	2-19
2.15.2	Water Requirement	2-19
2.15.3	Power Requirement	2-19
2.15.4	Man power requirements	2-19
CHAPTER-3 BASELINE ENVIRONMENTAL STATUS		
3.1	Preamble	3-1
3.1.1	Study Area	3-1
3.1.2	Data Collection Period & Components	3-1
3.2	Methodology of Base line data collection	3-1
3.2.1	Scoping	3-1
3.2.2	Reconnaissance Surveys	3-1
3.2.3	Assembly and Analysis of Data	3-2

3.2.4	Assessment of Alternatives	3-2
3.2.5	Assessment of Potential Impacts	3-2
3.2.6	Integration of Environmental Impacts in the Design Process	3-2
3.3	Land Environment	3-2
3.3.1	Physiography	3-2
3.3.2	Topography	3-2
3.4	Drainage	3-3
3.5	Land use Pattern	3-6
3.6	Geology	3-8
3.6.1	Climate and Rain Fall	3-9
3.7	Water Environment	3-10
3.7.1	Ground water conditions	3-11
3.7.2	Water-shed and Neighborhood Areas	3-11
3.7.3	Field study and discussion	3-12
3.7.4	Ground Water Resource Estimation	3-13
3.8	Ground Water Quality	3-14
3.9	Surface Water Quality	3-20
3.10	Soil Environment	3-24
3.11	Air Environment	3-26
3.11.1	Meteorology	3-26
3.11.2	Meteorological Station at Industry Site	3-29
3.11.3	Ambient Air Quality	3-32
3.11.4	Scope of Field Study	3-32
3.11.5	Description of Sampling Locations	3-32
3.11.6	Ambient Air Quality Status	3-33
3.11.7	Noise Environment	3-35
3.11.8	Observations of Noise Levels	3-38
3.12	Traffic Studies	3-39
3.13	Socio Economic Environment	3-42
3.13.1	Objectives of socio-economic study	3-42
3.13.2	Scope	3-42
3.13.3	Methodology	3-43
3.13.4	Project Site	3-44
3.13.5	Geographical Coverage	3-44
3.13.6	Impact Zones	3-46
3.13.7	Sample Households	3-46
3.13.8	Field Surveys	3-48
3.13.9	Data Analysis	3-49
3.13.10	Household Survey Results	3-49
3.13.10.1	Demographics	3-49
3.13.10.2	Occupational Distribution	3-50
3.13.11	Livelihoods	3-50
3.13.11.1	Agriculture	3-50
3.13.11.2	Industry	3-50
3.13.11.3	Mineral-Based Industries	3-51
3.13.11.4	Lifestyles	3-51
3.13.11.5	Assets Owned	3-51
3.13.11.6	Sources of Livelihood	3-52

3.13.11.7	Family Incomes	3-52
3.13.11.8	Family Expenditures	3-52
3.13.11.9	Educational Facilities	3-52
3.13.12	Health Status	3-53
3.13.12.1	Health and Family Welfare	3-53
3.13.12.2	Health Status of Surveyed Villages	3-54
3.13.13	Physical Infrastructure	3-54
3.13.14	Social Infrastructure	3-55
3.13.15	Archaeological Monuments	3-55
3.14	Aspirations and Apprehensions	3-55
3.15	Flora & Fauna	3-56
3.15.1	Scope of work	3-56
3.15.2	Methodology	3-57
3.15.3	Equipment / Instruments deployed	3-57
3.15.4	Floral analysis	3-57
3.15.5	Faunal analysis	3-58
3.15.6	Mine Lease Area	3-60
3.15.7	Dump	3-60
3.15.8	Reserve forest vegetation	3-60
3.15.9	Quantitative analysis	3-60
3.15.10	Vegetation Analysis	3-61
3.15.11	Terrestrial Fauna	3-62
3.15.12	Terrestrial fauna of the Core area and the Buffer zone	3-62
3.15.13	Flora and fauna of Aquatic ecosystem	3-63
CHAPTER -4 ANTICIPATED ENVIRONMENTAL IMPACTS & MITIGATIVE MEASURES		
4.1	Identification of Environmental Impacts	4-1
4.2.2	Impacts on Air Quality	4-1
4.2.2.1	Micro Meteorological Data	4-3
4.3.3	Air Quality Predictions	4-7
4.3.3.1	Post Project Scenario	4-8
4.3.3.2	Study Period	4-8
4.4	Noise Environment	4-10
4.4.1	Prediction of Impacts Due to Quarry Activity	4-10
4.4.2	Noise Due to vehicular Traffic	4-10
4.5	Mine Wastewater	4-12
4.5.1	Mine Pit Water	4-12
4.6	Land Environment	4-12
4.7	Vibration Levels	4-12
4.7.1	Disasters	4-12
4.7.2	Mitigation Measures	4-12
4.8	Air pollution mitigation measure	4-12
4.8.1	Water pollution mitigation measure	4-13
4.9	Impact on Traffic & Road Network	4-13
4.10	Land Reclamation	4-13
4.11	Predicted Impacts on Socio-Economic Environment	4-14
CHAPTER 5 ANALYSIS OF ALTERNATIVES (TECHNOLOGY & SITE)		
5.1	Analysis of Alternative Technology	5-1
5.2	Analysis of Alternative Sites	5-1

CHAPTER-6 ENVIRONMENTAL MONITORING PROGRAMME		
6.1	Environmental Monitoring	6-1
6.2	Environmental Management Cell	6-2
6.3	Environment Policy	6-2
6.4	Responsibility of Monitoring and Reporting System	6-2
6.5	Environmental Monitoring Budget	6-3
CHAPTER – 7 ADDITIONAL STUDIES		
7.1	Public Consultation	7-1
7.2	Risk Assessment of Black Galaxy Granite Mining	7-1
7.3	Disaster Management	7-5
7.4	Social Impact Assessment	7-6
CHAPTER – 8 PROJECT BENEFITS		
8.1	Employment Potential	8-1
8.2	Social Welfare Measures	8-1
8.3	Corporate Environmental Responsibility	8-1
8.4	Other Intangible benefits	8-1
CHAPETR – 9 ENVIRONMENTAL COST BENEFIT ANALYSIS		
CHAPTER – 10 ENVIRONMENAL MANAGEMENT PLAN		
10.1	Environnemental Management Plan	10-1
10.2	Air Pollution Control Measures	10-1
10.3	Noise Pollution Control Measures	10-1
10.3.1	Measures to minimise ground vibrations and noise	10[2
10.4	Water Pollution Control Measures	10-2
10.4.1	Water Conservation Measures	10-3
10.5	Socio-Economic Environment	10-3
10.6	Ecological Management	10-4
10.6.1	Greenbelt	10-4
10.6.2	Mitigation Measures proposed	10-6
10.7	Occupational Safety & Health Management	10-6
10.7.1	General Workplace Health and Safety	10-6
10.8	Use of explosives	10-7
10.8.1	Machine and Equipment Safety	10-8
10.9	Environment Management Cell	10-8
10.9.1	Environmental Policy and management reporting system	10-8
10.9.2	Environmental Policy	10-8
10.9.3	Reporting system to the Board of Directors	10-10
10.9.4	Reporting System	10-10
10.9.5	Records maintenance and reporting to statutory bodies	10-11
10.10	Reclamation	10-11
10.11	Environmental Management Budget	10-11
10.12	Environnemental Social responsibilities	10-12
CHAPTER – 11 SUMMARY & CONCLUSION		
11.1	Introduction	11-1
11.2	Project Description	11-1
11.3	Description of the Environment	11-1
11.3.1	Land Use Details	11-2
11.4	Project Cost	11-2
11.5	Anticipated Environmental Impacts & Mitigation	11-3

	Measures	
11.5.1	Environment	11-3
11.5.1.1	Air Environment	11-3
11.5.1.2	Noise Environment	11-4
11.5.1.3	Water Environment	11-4
11.5.2	Traffic & Road Network	11-5
11.5.3	Afforestation	11-5
11.6	Socio Economic Environment	11-5
11.7	Occupational Health and Safety Measures	11-6
11.8	Environmental Management Plan	11-6
11.9	Social Impact Assessment	11-6
11.10	Project Benefits	11-6
11.11	Environmental Social Responsibility (ESR)	11-7
11.12	Budget for Environmental Management Plan	11-7
11.13	Conclusion	11-7
CHAPTER-12 DISCLOSURE OF CONSULTANT		

LIST OF TABLES

Table	Description	Page No.
Table 1.0	Salient Features of the mine	1-3
Table 2.1	Details of the Mining Lease Area	2-1
Table 2.2	Land Use Pattern of the Mine	2-13
Table 2.3	Extent of Mechanization	2-15
Table 2.4	Year Wise Production from Initiation of Mine Starting Period to Till Date	2-16
Table 3.1	Land Use/Land Cover classes in Study area	3-7
Table 3.2	Climate Data of Ongole, District Head Quarters, Prakasam District, AP	3-10
Table 3.3	Ground Water Resource Estimations	3-14
Table 3.4	Summarized Chemical analysis of Ground Water	3-16
Table 3.5	Summarized Chemical Analysis of Surface Water	3-22
Table 3.6	Analytical data of Soil Samples	3-25
Table 3.7	Soil Test Results – Reference Tables	3-26
Table 3.8	Frequency Distribution of Wind Speeds and Wind Directions	3-30
Table 3.9	Locations of Ambient Air Quality Monitoring Stations	3-33
Table 3.10	Ambient Air Quality Status	3-33
Table 3.11	Noise Monitoring Locations	3-36
Table 3.12	Effects on Human Beings at Different Noise Levels	3-37
Table 3.13	Equivalent Noise levels in the Study Area	3-38
Table 3.14	Ambient Noise Standards	3-38
Table 3.15	Traffic Data of Ongole –Kurnool state highway	3-40
Table 3.16	The geographical coverage for SEIA of the Study area	3-44
Table 3.17	Sample Villages Selected for Socio-Economic Impact Assessment	3-47
Table 3.18	Category of Educational Intuitional Details in Mandal wise	3-52
Table 3.17	The status of health Facilities of Praksam	3-53

Table 3.18	Total Fertility and Under 5 Mortality rates from 2013 to 2015 of Prakasam	3-54
Table 3.19	Disease patterns observed in sampled villages	3-54
Table 3.20	Phytosociological data of study area	3-61
Table 3.21	Biodiversity indices values of the sampling sites in project site	3-61
Table 4.1	Frequency Distribution of Wind Speeds and Wind Directions	4-4
Table 4.2	Emission Details	4-6
Table 4.3	Emission Details	4-8
Table 4.4	Expected Noise from Mining Machinery	4-11
Table 7.1	Identified Risks & Control measures	7-2
Table 7.2	Sample Villages Selected for Socio-Economic Impact Assessment	7-6
Table 10.1	List of existing plantation	10-4
Table 10.2	List of Trees recommended for plantation under Greenbelt	10-5
Table 10.3	Proposed financial Budget for the Green belt development (Rs in Lakhs)	10-5
Table 10.4	Budget for EMP	10-12
Table 11.1	Salient features of the Project	11-1
Table 11.2	Land Use Pattern	11-2

LIST OF FIGURES

Figure	Description	Page No
Figure 1.1	Maps Showing the Details of Location	1-6
Figure 1.2	Google Map Showing 500 Mts radius of the Mine Area	1-7
Figure 2.1	Location Map showing project site	2-2
Figure 2.2	Topographical Map Sowing 10 Km Radius	2-5
Figure 2.3	Base Map of the Study Area	2-6
Figure 2.4	Google Map Showing Mine Lease Area	2-7
Figure 2.5	Photographs of the mine lease area	2-8
Figure 2.6	Surface Geological Plan	2-11
Figure 2.7	Geological cross Sections	2-12
Figure 2.8	Year wise Production Plan	2-17
Figure 3.1	Google map showing the project area and its surroundings	3-3
Figure 3.2	Regional Map of the Area showing the Paleru River	3-4
Figure 3.3	Drainage Distribution and Watershed Demarcation of the Study Area.(Watershed boundary is shown with Blue Outline, and Musi river is seen southwest of the watershed)	3-5
Figure 3.4	Satellite image of the study area	3-6

Figure 3.5	Land Use and Land Cover map of the study area	3-7
Figure 3.6	Pie Diagram Showing Land use & Land cover area	3-8
Figure 3.7	Topography and Drainage of the Project Site Neighborhood	3-12
Figure 3.8	Hydrogeology Map of Prakasam District (Source: CGWB, 2013)	3-13
Figure 3.9	Ground water sampling locations in Study area	3-15
Figure 3.10	Locations of surface water samples in the study area	3-21
Figure 3.11	Soil Sampling Locations	3-24
Figure 3.12	IMD Data of Meteorological Station ongole	3-27
Figure 3.13	Wind rose Diagram for the study January-2019 to March 2019	3-31
Figure 3.14	Location Map of Ambient Air Quality Monitoring Stations	3-35
Figure 3.15	Location Map for Noise Monitoring stations	3-36
Figure 3.16	Google Map of SEIA Sample Villages	3-48
Figure 3.17	Graph showing Important Value Index of dominant tree species within study area	3-61
Figure 4.1	Wind rose Diagram for the study period January 2019 to March 2019	4-5
Figure 4.2	Predicted 24 hourly of PM10 in µg/m3	4-9
Figure 10.1	AP Midwest Granites Pvt Limited Group Companies Organo Gram	10-10

LIST OF ANNEXURES

Annexure	Description
Annexure I	Copy of Current Environmental Clearance
Annexure II	Copy of combined CFE and CFO
Annexure III	Copy of certified compliance report
Annexure IV	Ambient Air Quality data
Annexure V	The list of Flora and Fauna of the study area
Annexure VI	Copy of the ToR and ToR Compliance Report

Chapter- 1

Introduction

CHAPTER-1

INTRODUCTION

1.0 Introduction

M/s. Andhra Pradesh Granite (Midwest) Private Limited (APGMPL) has been granted a Black Galaxy Granite mining lease over an area of 19.425 ha. in Sy. No. 55/5 of Rajupalem- Lakshmi Puram village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh.

1.1Purpose of the Report

The Ministry of Environment, Forest and Climate Change (MoEF&CC), Government of India (GOI) issued notifications vide SO 1533, dt. 14.9.2006, its amendments based on the directions issued by National Green Tribunal (NGT), New Delhi, vide O. A. No. 16 of 2016, dt.13.09.2018 and vide E.A.NO 55/2018 of O. A. No. 520/2016, dt. 11.12.2018 on MoEF&CC Notification S.O.2269 (E), dt.01.07.2016 and S.O.3977 (E), Dt.14.08.2018 and Office Memorandum Dt.12.12.2018 issued in compliance with the Hon'ble NGT orders, New Delhi, mandate prior environmental clearance for all mining leases across the country. The notification and its amendments based on NGT directions, and stipulate a three-stage process of scoping, public consultation, and appraisal by State Environmental Impact Assessment Authority (SEIAA) to issue prior environmental clearance. Accordingly scoping for preparation of environmental impact assessment (EIA) was done by the state expert for enhancement of production capacity of Black Galaxy Granite from 54,643 m3/Annum to 79,401 m3/Annum in an area of 19.425 Ha. The present study follows the prescribed TOR's to prepare Draft EIA report for public appraisal committee (SEAC), which issued terms of reference vide Letter No. SEIAA/AP/PKM/MIN/VIO/08/2018/672 Dt.01.07.2021 for the enhancement of production capacity consultation to facilitate informed view on the project by public/stakeholders.

1.2Background of the Project

- Andhra Pradesh Granite (Midwest) Private Limited is a Joint Venture Company constituted with APMDC and Midwest Granite Private Limited with its consortium partner under the Agreement dated 04.06.2007.

- The Environmental Clearance was given for 540 CBM of Primary Blocks and 486 CBM of Secondary Blocks of Black Galaxy Granite per annum vide order No. SEIAA/AP/PKM/119/2013-1593 dated 10.06.2013.
- Later applied for the Environmental Clearance under violation category in the year 2017 and obtained Environmental Clearance for the expansion from 1,026m3/annum to 54,643m3/annum vide order no. SEIAA/ AP/ PKM/ MIN/ VIO/ 08/ 2018/ 672-1021 Dated: 13.09.2019 with validity up to 02.02.2025. Copy of Current Environmental Clearance enclosed as **Annexure-I.**
- The unit obtained Combined CFE and CFO for expansion order No APPCB/VJA/NLR/188/HO/CFO/2017 dated21.10.2019 valid upto 31st July 2022. copy of combined CFE and CFO enclosed as **Annexure-II.**
- The modified mining plan approved by the Deputy Director of mines and Geology, Ongole for the period of 2020-2023 vide Lr.no.7141/MMS/BGG/OGL/2020 dated 22.01.2021.
- As per the Gazette Notification SO. 3977(E) dated 14th August, 2018 this comes under category B1 which is which is more than 5 Hectares hence cluster condition is applicable.
- The Certified compliance report obtained from MOEF &CC Integrated Regional Office Vijayawada vide File No. IRO/VIJ/EPA/MISC/111-01/2021. Copy of certified compliance report enclosed as **Annexure -III**
- SEIAA, Andhra Pradesh has granted terms of Reference (TORs) for enhance the production of Black Galaxy Granite form 54,643 m3/Annum to 79,401 m3/Annum.
- Draft EIA Report prepared based on the terms of Reference issued by SEIAA, Andhra Pradesh.

1.3Identification of Project & Project Proponent

Midwest Granite Private Limited., a Midwest Group company, was incorporated in the year 1981 by Mr. K. Ragahava Reddy with the core objective of developing competence and expertise in the field of natural stone, namely Granite, Marble, Limestone, and Travertine. In the ensuing years, Midwest has established itself as a very strong player in the domain of Exploration, Mining, Processing and export of natural Stone products. Midwest is among the top 3 natural stone players in India. Midwest mines over 3.5 million tons of granite per annum, which includes saleable product and waste granite.

Mr. K. Raghava Reddy is a visionary entrepreneur with a track record of building diversified and socially responsible mining enterprises. He is a natural stone specialist covering mineral

prospecting, mining and project management. He established the A.P. Granite Association and served as National Executive Member of All India Granite Stone Association for several years. Under his leadership, the company is continuously striving to establish a lead over its competitors by introducing new stone varieties in the market by acquiring new deposits, modernizing the existing quarries, adopting and deploying latest technology and equipment for quality and productivity enhancement, adhering to the stringent safety norms of operation and achieving higher level of efficiency

Midwest has presence in over 30 countries. Midwest has established market linkages with a number of direct customers in all important markets, through agency and distribution system. Midwest Company participates in most of the important International Trade Shows. Midwest manufactures environmental friendly diamond tools for mining industry with facilities in India and Srilanka. Midwest is a recipient of several awards at National and State level. Midwest is recognized as "Star Export House" by Government of India. Midwest has joint ventures and key relations with Central, State Governments and Large Corporations.

Midwest institutionalized its Corporate Social Responsibility (CSR) Program. Since Midwest's conscience dictates its CSR program rather than mere Government regulatory compliances, Midwest's contribution to communities constantly exceeds Government expectations in all the countries it operates. Through successful implementation of its CSR programs, Midwest gains local support and Government approvals faster than its competitors.

1.4 Project Location & study area Description

The Proposed activity is expansion of production from current 54,643 M3/annum to 79,401 M3/annum. The salient features of the project are given below.

Table 1.0: Salient Features of the mine

Name of the Project	M/s Andhra Pradesh Granite(Midwest) Private Limited
Nature	Quarrying of Black Galaxy Granite
Quarry Lease area	19.425 Ha
Location	Sy.No,55/5 Rajupalem-Lakshmipuram village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh
Name of the Proponent & Address	Sri. K. Rama Chandra, Director 8-2-684/3/25 and 26, Road No: 12, Banjara Hills, Hyderabad.
Type of Land	Government Land

<table>
<tr><td>Latitude & Longitude</td><td>
<table>
<tr><th>S.no</th><th>Latitude</th><th>Longitude</th></tr>
<tr><td>1.</td><td>N 15^{0}35’13.21”</td><td>E 79^0 49’19.32”</td></tr>
<tr><td>2.</td><td>N 15^{0}35’06.03”</td><td>E 79^{0}49’20.63”</td></tr>
<tr><td>3.</td><td>N 15^{0}35’06.11”</td><td>E 79^{0}49’25.34”</td></tr>
<tr><td>4.</td><td>N 15^{0}35’07.22”</td><td>E 79^{0}49’25.61”</td></tr>
<tr><td>5.</td><td>N 15^{0}35’06.12”</td><td>E 79^{0}49’29.00”</td></tr>
<tr><td>6.</td><td>N 15^{0}35’05.00”</td><td>E 79^{0}49’29.23”</td></tr>
<tr><td>7.</td><td>N 15^{0}35’02.91”</td><td>E 79^{0}49’30.31”</td></tr>
<tr><td>8.</td><td>N 15^{0}35’04.63”</td><td>E 79^{0}49’40.46”</td></tr>
<tr><td>9.</td><td>N 15^{0}35’04.00”</td><td>E 79^{0}49’52.64”</td></tr>
<tr><td>10.</td><td>N 15^{0}35’09.52”</td><td>E 79^{0}49’51.61”</td></tr>
</table>
</td></tr>
<tr><td>Toposheet No.</td><td>57M/14 (1:50,000 Scale)</td></tr>
<tr><td>Surroundings of the Mine</td><td>North East-M/s.Swathi Granite Mine
South- M/s.Golden Granites
East-Surya Granite Mine
West-Alliance Black Galaxy Granite Mine
Belongs to APMDC.
Southwest-M/s.Midwest Granite Mine</td></tr>
<tr><td>Nearest Habitation</td><td>R.L.Puram 0.75</td></tr>
<tr><td>Nearest Town</td><td>Podili-23 Km</td></tr>
<tr><td>Nearest Sea Port</td><td>Krishna Patnam Port -150 kms</td></tr>
<tr><td>Nearest Railway Station</td><td>Ongole-27 Km</td></tr>
<tr><td>Nearest Air Port</td><td>Vijayawada International Airport- 148 Kms (ESE)</td></tr>
<tr><td>Water Bodies</td><td>Musi River-3.1 Km E (Tributary of Paleru)
Ramathirtham Reservoir-4.6 (NW)
Jawahar canal (Right canal of Nagarjuna Sagar)- 5.6 Km (W)
Gaddi Vagu-8.1 km (NW)</td></tr>
<tr><td>Reserve Forests</td><td>Chimakuthy RF-2.3 Km (N)
Devarapalem RF – 6.0Km (N)</td></tr>
<tr><td>Historical/Important Places</td><td>None</td></tr>
<tr><td>Inter State Boundary</td><td>None</td></tr>
<tr><td>Production Capacity</td><td>Current - 54,643 M3/annum
After expansion– 79,401 M3/ annum</td></tr>
<tr><td>Method of Mining</td><td>Open cast Semi Mechanized method</td></tr>
<tr><td>Dump Site</td><td>The lessee is having waste dump in Sy.No 938/1 to 7 of Chimakurthy Village & Mandal over 6.17 Acres, Lease was granted vide DMG proceedings No 25711/R3-2/2007 dt. 08.11.2007 and executed vide ADMG proceedings No 5483/Q/2007 dt. 20.11.2007</td></tr>
</table>

	The Lessee also having waste dumps in Sy.No. 941/1 to 4, 944/1 & 944/4 to 7 of Chimakurthy Village & Mandal over 11.19 Acres and Sy.No 25/3B2 of R.L.Puram (V) over an extent of 11.00Acres, in Sy.No 938/1 to 7, over an extent of 6.17 acres of Chimakurthy (V) for dumping the waste. The Lessee also has entered in to lease agreement over an extent of 1.98 acres Survey.No.25/3B2 of R.L.Puram (V) with Midwest Granite Pvt Ltd for dumping the waste. The generated waste will be dumped in Sy. Nos. 25/3B2 of R.L. Puram Village, Chimakurthy Mandal & 938/1 to 7 of Chimakurthy Village & Mandal during the present scheme period.
Manpower Requirement	Current -440 After expansion-40
Water Requirement source of water	Existing – 60 KLD Proposed-80 KLD Trough Tankers
Total Cost	Current 30.51 Cr For expansion -5.0 Cr Total-35.31Cr
Total Budget for EMP	Rs. 56.0 lakhs (capital) Rs. 39.0lakhs (recurring)

Project location map is given at **Figure 1.1**

Google map of 500Mts radius of the site are presented at **Figure 1.2**

Figure 1.1: Maps Showing the Details of Location

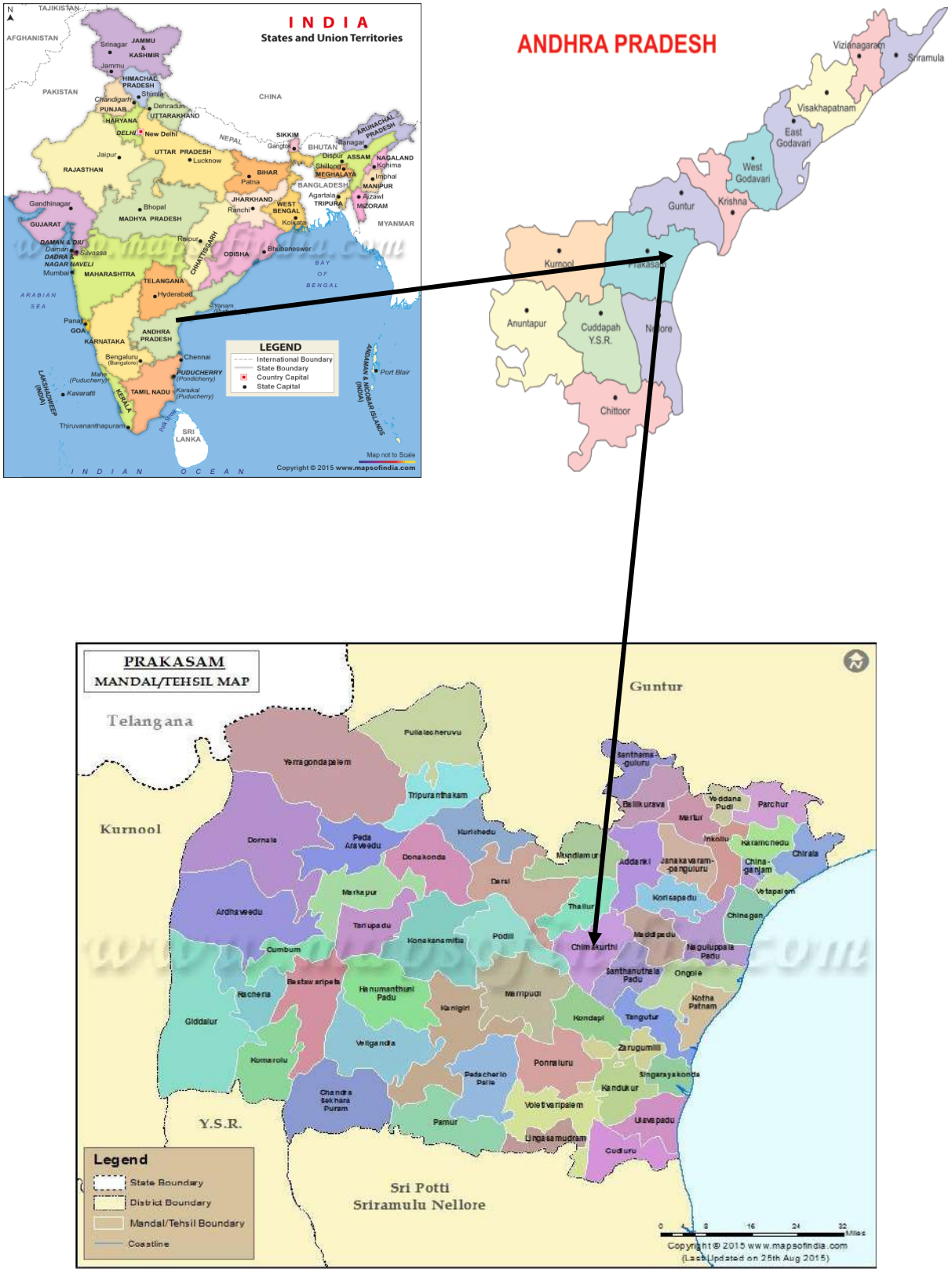


Figure 1.2: Google Map Showing 500 Mts radius of the Mine Area



1.5 Project Cost

Total cost of the project is Rs. 3500.51 Lakhs and the proponent proposed to incur about Rs. 56.0 Lakhs towards capital cost and Rs. 39.0 Lakhs as recurring cost for implementing Environment Management Plan.

1.6 Importance of the Project to the Country, Region

Granite is an important building material used from ages in our country. There is huge export potential for raw and polished granites. The Black Galaxy is Indian black granite, with a great market demand. This is a unique type of black granite due to its abundant number of silver flakes. These are from a specific mineral and give this granite a unique and beautiful look

The Black Galaxy is a black coloured granite, with a deep black colour, thin to medium grain and clear signs of silver shiny flakes. This is a well-known black granite from India, famous for its uniform and deep black colour, as well as its silver shiny flakes. Its main variations depend on the grain thickness and abundance of silver flakes.

Black Galaxy is actively traded as a commodity and owing to the large volume of quarrying, large blocks in First Choice material are becoming scarcer and much more expensive. Nowadays Indian Granite Star Galaxy aka Black Galaxy Granite has become most demanding natural building stone in USA, UK, Poland, Bulgaria, Australia, New Zealand, South Africa, Italy, Mexico and other countries.

Although granite is a minor mineral, it is a major contributor in foreign exchange earnings. India is the second largest exporter of raw granite after China and ahead of Brazil and South Africa.

1.7 Project Justifications

As Andhra Pradesh Granite (Midwest) Private Limited is operating the mine at the site, all necessary infrastructure is in place. Due to market presence in many countries, it will be able market the enhanced production quantity with adequate environmental safe guards, the project is justified.

1.8 Cluster of mines Details

Assistant Director of Mines and Geology, Ongole, provided a cluster certificate vide his letter 545/Q/2005dated 29.10.2021, on our request. A total of 27 quarry leases were existing out of which 26 quarry leases are having valid E.C and 1 quarry lease was non-working and 1 Quarry lease area issued LoI surrounded within 500 meters from the peripheral boundary of lease held by M/s. AP Granite (Midwest) Private Limited with a total extent of 138.78 Hectares. As

per the list of leases granted all of the leases were granted prior to 9th September, 2013. As per the Gazette Notification, SO. 2269 (E) dated 1st July 2016, the cluster mining will be applicable only for those mines whose mining leases were granted after 9th September, 2013. Thus though number of mines within 500meters distance are more, cluster rule does not applicable here as per the MOEF notification. Copy of proceedings from Assistant Director Mines, Ongole is given along with sketch.

Copy of Cluster Letter

GOVERNMENT OF ANDHRAPADESH
DEPARTMENT OF MINES & GEOLOGY

From
Sri N.Obula Reddy, M.sc.,
Asst. Director of Mines & Geology,
Ongole

To
M/s AP Granite(Midwest) Pvt.,Ltd.,
R.L.Puram Village,
Mg. Pa.Sri K. Rama Chandra,
D.No:8-2-684/3/25&26,
Road No. 12, Banjara Hills,
Hyderabad

Letter No:545/Q/2005 dt:28-10-2021

Sir,

Sub: Environment clearance -SEIAA,A.P.-Quarry Lease details in 500Mts cluster Surrounding of the mining projects of M/s AP Granite(Midwest) Pvt., Ltd over an extent of 19.425 Hects in Sy.No: 55/5P,of R.L. Puram Village, Chimakurthy Mandal, Prakasam District- As evidence to establish the status of cluster formation as per MOEF &CC Notification S.O.No.141(E) dt:15th January 2016 and its amendment Notification S.O.No:2269 (E) dt:1st July 2016 for Environmental Clearance enhanced quantity -Certain information- Requested-Furnished - Regarding.

Ref:- Letter Dated: 23.10.2021 of by M/s AP Granite(Midwest) Pvt., Ltd.,

***

The attention to the subject and reference cited, wherein by M/s AP Granite(Midwest) Pvt.,Ltd., has requested this office to furnish the information with regard to type of mining projects located within 500mts radius of their quarry lease held for Black Galaxy Granite, over an extent of 19.425 Hects in Sy.No: 55/5P,of R.L.Puram Village, Chimakurthy Mandal, Prakasam District as evidence to establish the status of cluster formation as per MOEF & CC Notification S.O.No.141 (E), dated 01.07.2016, Consideration for issue of Environmental Clearance.

In this connection it is informed that, there are 27 quarry leases existing (out of which 26 quarry leases are having valid E.C and 1 QL non working condition) and 1 Quarry Lease area issued LoI surrounded within 500 mts from the peripheral boundary of the leased area held by M/s AP Granite(Midwest) Pvt.,Ltd., for Black Galaxy Granite over an extent of 19.425 Hects in Sy.No: 55/5P,of R.L.Puram Village,ChimakurthyMandal, Prakasam District.

The map showing the above details are enclosed for information.

Sl. No.	Name of the Lease	Location	Extent in Hects	Details of Lease	Lease Period	EC	Remarks
1	M/s AP Granite (Midwest) Pvt.,Ltd	RL Puram	19.425	Granted vide Proc No: 20827/R3(2)/2007, dt: 08/06/2007 and executed vide this office Proc No; 545/Q/2005, dt:06.13.2007.	02-02-2025	SEIAA/AP/PKM /MIN/VIO/08/ 2018/672-1021, dt:13-09-2019.	Proposed for enhance ment of EC quantity
2	M/s Alleance APBGG (P) Ltd	RL Puram	21.887	Granted Vide Proc.No: 20826/R3(2)/2007-1, dt. 08-06-2007 and executed Vide Proc.No: 4241/Q/2003 dt:14-06-2027	19-06-2023	-	Non Working

1

3	M/s Pallava Red Granite Industries (INDIA) Pvt Ltd.,	RL Puram	14.170	Granted Vide Proc.No: 1800/R3-2/2008, dt: 26/04/08 and executed Vide Proc.No: 4361/Q2/08 dt: 30-04-08.	From 30-04-2008 to 19-06-2028	SEIAA/AP/PKM -115/2013-1531 dt:10-06-13	Working
4	M/s Bash Granites	RL Puram	2.125	Granted Vide Proc.No: 1225/R3-1/11, dt: 3/07/2013 And executed Vide Proc.No: 10164/Q/2010 dt:23-01-12	From 23-01-2012 to 22-01-2032	SEIAA/AP/PKM -140/2013-2359 dt: 21-01-12	Working
5	M/s Mid West Granite Pvt., Ltd.,	RL Puram	9.510	Granted Vide Proc.No: 5873/R3-2/ 2008, dt: 04/01/2008 And Executed VideProc.No:1219 /Q/2008 dt:04-04-09	From 04-04-2009 to 03-04-2029	SEIAA/AP/PKM -120/2013-1591 Dt: 10-06-2013	Working
6	Sri Sudarsana Granites	R.L Puram	1.141	Granted Vide Proceeding No: 34476/ R3(2)/2000, Dt:01.04.2002 and executed on 20-07-2002	From 20.07.2002 to 19.07.2022	SEIAA/AP/PKM -217/2014, Date: 20.11.2015	Working
7	M/s Hari Priya Granites	R.L Puram	2.429	The lease was granted Vide Proc.No: 32869/R3-1/1998 dt: 01-04-2002 by the DM&G and executed Vide This office Proc.No: 5121/Q/1998 dt: 20-07-2002	From 20.07.2002 to 19.07.2022	SEIAA/AP/PKM -137/2013-2373, Date: 10.07.2013	Working
8	M/s Ongole Galaxy	R.L Puram	2.720	Granted Vide Proc.No: 6386/Q2/1991, dt:08-04-1992 and executed on 18-08-1993. Subsequently, the lease was renewed under 1st renewal Vide Proc No: 25807/R3-1/ 2009 dt:16-07-2009 and executed on 20-07-2009.	From 20-08-2009 to 19-08-2028	DEIAA/AP/PK M/07/MIN/201 7-54 Dt: 03-08-2017	Working
9	M/s Ongole Galaxy	R.L Puram	1.234	Granted Vide Proc.No: 9673/ R3-2/2013, dated 05/01/2017 and executed Vide Proc.No: 7356/Q2/2012, dt.12/01/2018	From 12-01-2018 to 11-01-2018	DEIAA/AP/PK M/07/MIN/201 7-54 Dt: 03-08-2017	Working
10	M/s Gem Granites (P) Ltd.	R.L Puram	0.680	Granted Vide Proc.No:28683/R3(2)/2 003,dt: 30/12/2006 and executed Vide Proc.No:6217/Q/2002 dt: 14-02-2007.	From 14.12.2007 to 13.12.2027.	SEIAA/AP/PKM -32/2013-1989, Date: 04.04.2013	Working
11	M/s Gem Granites (P) Ltd.	R.L Puram	4.889	The lease was transferred from M/s BMR Industries Vide Proc.No: 15920/R3-2/ 97 dt:08-07-1997 and executed on 01-08-1997, Suseeqently, Granted under 1st renewal vide Procs No: 24817/R2-2/2010, Dt: 25.05.2015 and executed on 14-07-2015.	From 02.09.2011 to 01.09.2031	SEIAA/AP/PKM -32/2013-1989, Date: 04.04.2013	M/s Gem Granites (P) Ltd.
12	M/s Gem Granites(P) Ltd.	R.L Puram	0.818	Granted Vide Proc.No: 29925/R3-1/2009, 30/09/2009 and executed Vide Proc.No: 5769/Q/2006, Dt.28-10-2009	From 28-10-2009 To 12-11-2011	DEIAA/AP/PK M/36/MIN/201 7-133 dt:13-11-17	Renewal filed and Proposal Submitte d
13	M/s Anand Granite Exports Pvt Ltd.,	R.L Puram	1.773	Orginally Granted Vide Proc.No: 26507/R3(2)/1996 dt: 28-12-2001 and the lease deed was executed on 27-11-1996	From 27-11-1996 to 26-11-2016	SEIAA/AP/PKM -72/2013-492 dt:17-04-2013	Renewal filed and grant proposal Submitte d on 26-08-16

2

14	M/s Pearl Mineral & Mines Pvt. Ltd	R.L Puram	7.252	1.Renewed under 1st renewal vide Proc. No: 30194/R3-2/ 2006, dt: 29-06-2007 and Executed vide Proc No: 6701/Q/2006 dt:12-07-2007 2. Transferred Vide Proc. No: 50985/ R3-1/2010 dt: 07-01-2011 and executed vide Proc.No: 6701/Q/2006 dt:10-01-11	Upto 14.07.2027	SEIAA/AP/PKM -167/2013, dt: 08.10.2015,	Working
15	M/s Apple Granites	R.L Puram	7.272	Renewed Vide Proceeding No: 2341/R3-2/2008, Date: 30.08.2008 and executed Vide Proc No: 9666/Q/2007 dt: 15-09-2008	From 06.12.2008 to 05.12.2028	SEIAA/AP/PKM/MIN/06/2017/334 dt: 11-06-2018	Working
16	M/s Golden Granites	R.L Puram	2.745	Granted Proceeding No: 34354/R3(2)/2001, Date: 12.11.2001. and executed Vide Proc.No: 7400/Q1/2001 dt: 16-11-2001	From 16.11.2001 to 15.11.2021	SEIAA/AP/PKM -245/2015 dt: 08-10-2015	Working
17	M/s APMDC Ltd	RL Puram	6.382				LoI issued
18	M/s VIVA Granites,	Chimakurthy	1.611	Granted Vide Proc.No: 24476/R3-2/2008, dt: 12/02/2009 and executed Vide Proc.No: 5968/Q/2008 dt: 19-02-2009	Upto 06-07-2029	SEIAA/AP/PKM -89/2013-708	Working
19	M/s Manikanta Minerals	Chimakurthy	2.203	Granted Vide Proc.No: 11996/R3(2)/2006, dt: 01/08/2006 and executed vide Proc.No:2569/Q/2006 dt: 19-10-2007	From 19-10-2007 to 18-10-2027	SEIAA/AP/PKM-/2013-6331 dt:26-03-2013	Working
20	M/s Krishana Sai Granites	Chimakurthy	3.122	Granted vide Proc.No: 38192/R3(2)/2006 dt: 22/01/2007 and executed Vide Proc.No:8849/Q/2006, dt: 25/01/2007	From 29-12-2007 to 28-12-2027	SEIAA/AP/PKM/2013-4004 Dt:03-10-2013	Working
21	M/s Swathi Granites,	Chimakurthy	1.429	Granted Vidfe Proc No;13183/R3-1/2010,dt: 07/01/2011 and executed vide This office Proc No; 9649/Q/2009 dt: 27-06-2011	From 27-06-2011 to 26-06-2031	SEIAA/AP/PKM -201/2013-4911 Dt: 25-10-2013	Working
22	M/s Sri Vasavi Granites	Chimakurthy	0.583	The 2nd renewal of lease was granted vide Proceeding No: 45571/R3-1/2010, Date: 12.07.2012 in favour of M/s Sri Vasavi Granites and same was transferred vide Proceeding No: 8003/R2-2/2015, Date: 30.04.2015 in favour of M/s Sri Vasavi Granites.	Up to 06.12.2031	SEIAA/AP/PKM -87/2013-1245, Date: 21.05.2013	Working
23	Sri Surya Granites	Chimakurthy	3.846	Proceeding No: 30222/R3-1/2010, Date: 08.04.2011.	From 13.06.2011 to 12.06.2031	SEIAA/AP/PKM -134/2013-2494, Date: 10.07.2013	Working
4	M/s Sri Vasavi Granites	Chimakurthy	3.889	The 2nd renewal of lease was granted vide Proceeding No: 21274/R3-1/2011, Date: 29.09.2011 in favour of M/s Sri Vasavi Granites and same was transferred vide Proceeding No: 8004/R2-2/2015, Date: 30.04.2015 in favour of M/s Sri Vasavi Granites.	Up to 06.12.2031	SEIAA/AP/PKM -132/2013, Date: 10.07.2013	Working
25	M/s Krishna Sai Granites	Chimakurthy	1.174	Transferred Vide Proc.No; 28689/R3-1/2009 dt: 07/08/2009	From 31-08-2009 to 28-12-2027	SEIAA/AP/PKM -184/2013-4488	Working

				and Executed Vide Proc.No: 7314/Q/09 Dt: 31-08-2009		Dt:dt: 23-10-2013	
26	M/s Anand Granite Exports Pvt. Ltd.	Chimakurthy	5.890	Renewed under1st renewal Vide Memo No: 3551/R3-1/2007, dt: 08/07/2011 and executed Vide Proc.No: 7703/Q/2007 dt:03-09-2011	From 14.12.2008 to 13.12.2028	SEIAA/AP/PKM -168/2013-4297, Date: 08.10.2015	Working
27	M/s Sri Surya Granites	Chimakurthy	1.976	Originally granted into Two leases over an extents of 1.44Acs & 3.44Acs . And 1.44Acs is granted Vide Proc.No:4426/Q2/1991 dt: 28-11-1991 and Executed Vide Proc No: 1845/Q1/1991 dt: 07-12-1991 And 3.44Acs is Granted Vide Proc.No: 3228/Q2/1991 dt: 27-09-1991 and executed Vide Proc No: 2089/Q1/1991 dt: 07-12-1991 for a period upto 06-12-1996 And Clubbed into single lease Vide Proc.No:12805/K4/1995 dt:27.06.1996 and executed Vide this office Proc.No: 4709/Q/1996 dt:25.07.1996 for a period upto 06-12-1996 and renewed under 1st renewal Vide Proc.No: 28303/R3(2)/1996 dt: 17-12-1996 for a period w.e.f. 07-12-96 to 06-12-2011, Subsequently, renewed under 2nd renewal Proceeding No: 30221/ R3-1/2010, Dt:08.04.2011	From 07.12.2011 to 06.12.2031	SEIAA/AP/PKM -135/2013, Date: 10.07.2013.	Working
28	M/s Indian Mineral Granite Company	Chimakurthy	3.064	Renewed under 1st renewal Vide Proc.No: 21127/R3-/ Ogl/11/96 dt: 22-10-1996 and executed Vide This office Proc.No: 5675/Q/1996 dt:30-11-1996 and 2nd renewal Vide Proc No: 13813/R3-1/ 2011, Date: 20.09.2011 and executed vide T.O. Proc.No: 8558/Q/2010 dt: 3110/2011	From 12.12.2011 to 11.12.2031	SEIAA/AP/PKM -59/2013-316, Date: 10.04.2013.	Working
29	M/s Indian Mineral Granite Company	Chimakurthy	3.541	Granted Vide Proceeding No: 19423/R3(2)/2003, Date: 4.07.2003	From 14.07.2003 to 13.07.2023	SEIAA/AP/PKM -63/2013-477, Date: 17.04.2013	Working

Yours faithfully,

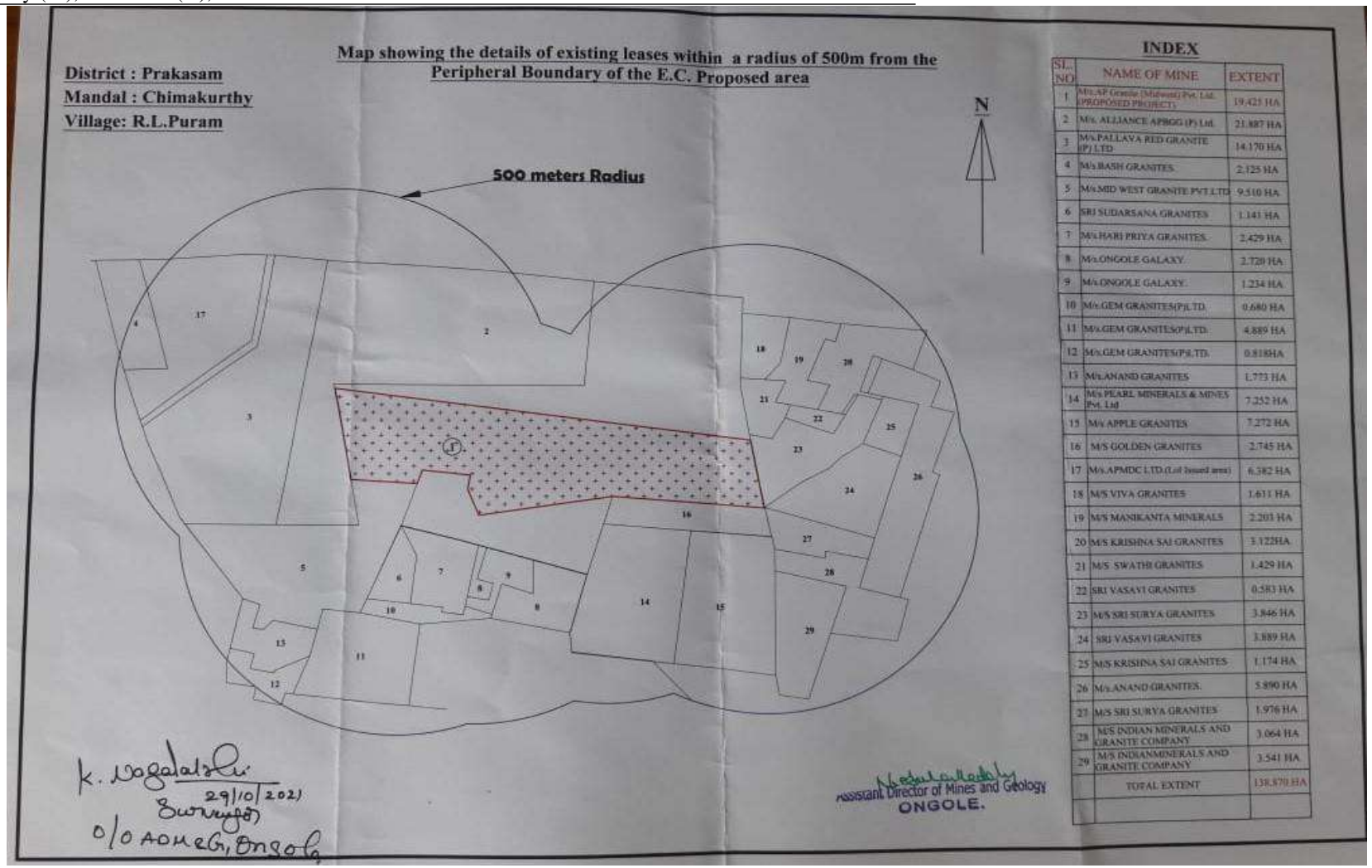
Encl: As above.

Asst. Director of Mines & Geology,
Ongole.

Copy submitted the Director of Mines and Geology, Ibrahimpatnam for information.

Copy submitted to the Deputy Director of Mines and Geology, Ongole for information.

4



SL. NO	NAME OF MINE	EXTENT
1	M/s.AP Granite (Midwest) Pvt. Ltd. (PROPOSED PROJECT)	19.425 HA
2	M/s. ALLIANCE APBGG (P) Ltd.	21.887 HA
3	M/s.PALLAVA RED GRANITE (P) LTD	14.170 HA
4	M/s.BASH GRANITES.	2.125 HA
5	M/s.MID WEST GRANITE PVT LTD.	9.510 HA
6	SRI SUDARSANA GRANITES	1.141 HA
7	M/s.HARI PRIYA GRANITES.	2.429 HA
8	M/s.ONGOLE GALAXY.	2.720 HA
9	M/s.ONGOLE GALAXY.	1.234 HA
10	M/s.GEM GRANITES(P)LTD.	0.680 HA
11	M/s.GEM GRANITES(P)LTD.	4.889 HA
12	M/s.GEM GRANITES(P)LTD.	0.818HA
13	M/s.ANAND GRANITES	1.773 HA
14	M/s PEARL MINERALS & MINES Pvt. Ltd	7.252 HA
15	M/s APPLE GRANITES	7.272 HA
16	M/S GOLDEN GRANITES	2.745 HA
17	M/s.APMDC LTD.(Lsd leased area)	6.382 HA
18	M/S VIVA GRANITES	1.611 HA
19	M/S MANIKANTA MINERALS	2.203 HA
20	M/S KRISHNA SAI GRANITES	3.122HA
21	M/S SWATHI GRANITES	1.429 HA
22	SRI VASAVI GRANITES	0.583 HA
23	M/S SRI SURYA GRANITES	3.846 HA
24	SRI VASAVI GRANITES	3.889 HA
25	M/S KRISHNA SAI GRANITES	1.174 HA
26	M/s.ANAND GRANITES.	5.890 HA
27	M/S SRI SURYA GRANITES	1.976 HA
28	M/S INDIAN MINERALS AND GRANITE COMPANY	3.064 HA
29	M/S INDIANMINERALS AND GRANITE COMPANY	3.541 HA
	TOTAL EXTENT	138.870 HA

1.9 Scope of the Study

Andhra Pradesh Granite (Midwest) Private Limited has obtained Terms of Reference form SEIAA, Andhra Pradesh Lr. No. SEIAA/AP/PKM/MIN/VIO/08/2018/672 Dt.01.07.2021.

The subject EIA Report is prepared incorporating prescribed Standard Terms of Reference for enhancement of Black Galaxy Granite of production from current 54,643 M3/annum to 79,401 M3/annum from mining lease area of 19.425 Ha.

Environmental Impact Assessment study has been carried out within an area of 10 km radius around the Mine area as per the Terms of Reference issued by SEIAA, Andhra Pradesh. The various steps involved in Environmental Impact Assessment study of the project site are divided into the following phases:

- Identification of significant environmental parameters and assessing the existing status within the impact zone with respect to air, water, noise, soil and socioeconomic components of environment.
- Study of various mining activities of the proposed mine to identify the areas leading to impact/ change in environmental quality.
- Quantification/prediction of impact for the identified activities and to study the level of impact on various environmental components.
- Evaluation of impacts after superimposing the predicted/ quantified scenario over the baseline scenario.
- Formulation of Environmental Management Plan for implementation in the project.

Baseline data collected for the project during January 2019 to March 2019, has been used to assess the existing environmental status of various environmental parameters within the 10 km radius study area of the project.

The report presents the baseline scenario, prediction of impacts due to enhancement production of Black Galaxy Granite production from 54,643 M3/annum to 79,401 M3/annum from mining lease area of 19.425 Ha,

Draft EIA report with the above scope incorporating the TORs issued by SEIAA, Andhra Pradesh is presented in subsequent chapters for public consultations.

1.10 Required Statutory Licenses at A Glance:

The table gives the requirement & status of various applicable clearances for the project and its current status.

S.NO	Nodal Agency	License/NOC /Approval requirements	Current Status
1	Ministry of Environment, forest and Climate Change	Environmental Clearance for proposed expansion	Obtained TOR and Draft EIA report prepared for expansion and waiting for Public Hearing
2	Andhra Pradesh State Pollution Control Board	Consent for Establishment (CFE) Consent for operation (CFO) Under Water & Air Acts hazardous Waste Management Rules	Obtained CFE and CFO for the existing production and Will be applied after getting expansion environmental Clearance
3	Department of Mines and Geology Andhra Pradesh	Mining Plan	Mining plan is approved by DDMG, Ongole District
4	Directorate General of Mine safety	NOC Mining Safety	Obtained NOC form Mining safety department
5	State Ground water Board	No Ground water proposed to be extracted by Proponent	Purchasing water through water tankers and source is ground water.

1.11 Structure of the EIA Report

The report is organized in 12 chapters as follows

Chapter 1 is Introduction chapter, which highlights briefly about the project, introduction to project proponents, nature & size of the project and its importance & Scope of study.

Chapter 2 Outlines project description provides the complete description of the proposed project of Andhra Pradesh Granite (Midwest) Private Limited. The details are presented from the environmental perspective along with the salient features such as land use, Process adopted, Water consumption and Wastewater discharges, Air emissions sources Resource requirement for the project and current compliance levels of the unit vis-à-vis previous clearances from MOEF&CC.

Chapter 3 Discuss the study area, period, components & Methodology adopted for base line data established and environmental components

Chapter 4 Discuss the details of investigated environmental impacts due to project at various stages of development and operations. Also the chapter outlines the measures for minimizing or off-setting adverse impacts identified and mitigation measures.

Chapter 5 Discuss the details of alternative technologies.

Chapter 6 Discuss the details of environmental monitoring program, frequencies & costs during implementation and operation.

Chapter 7 Discuss the potential hazards and risks associated with project & risk mitigation measures proposed.

Chapter 8 Discuss the benefits from the project to various stakeholders.

Chapter 9 Discuss the Environmental Cost Benefit analysis (Not specified in TOR in this case).

Chapter 10 Discuss Detailed Environmental Management Plan, which addresses the measures to be adopted for effective environmental conservation measures. It also gives estimated budget for implementation of EMP, both capital investment and recurring costs.

Chapter 11 gives the Summary & Conclusion of the EIA and justification of implementation of project.

Chapter 12 Gives the Credentials of M/s. Pridhvi Envirotech (P) Ltd, their accreditation details and team which conducted the EIA studies.

Chapter - 2

Project Description

CHAPTER-2
PROJECT DESCRIPTION

2.1 Details of the Mining Lease Area

The mining lease area of the proposed project is a Government wasteland covered in an area of 19.425 ha. Entire mining lease area is a hilly area and barren land. The proposed project activity is mining of Black Galaxy Granite of 79,401 m^3/ annum at Prakasam District of Andhra Pradesh. The details of the mining lease area are as below.

Table 2.1: Details of the Mining Lease Area

State /District	Mandal	Village	Survey No.s	Extent in Ha.	Ownership
Andhra Pradesh / Prakasam	Chimakurthy	Rajupalem Lakshmi puram	55/5	19.425	Govt. barren Land
Topo sheet : 57 M/14					

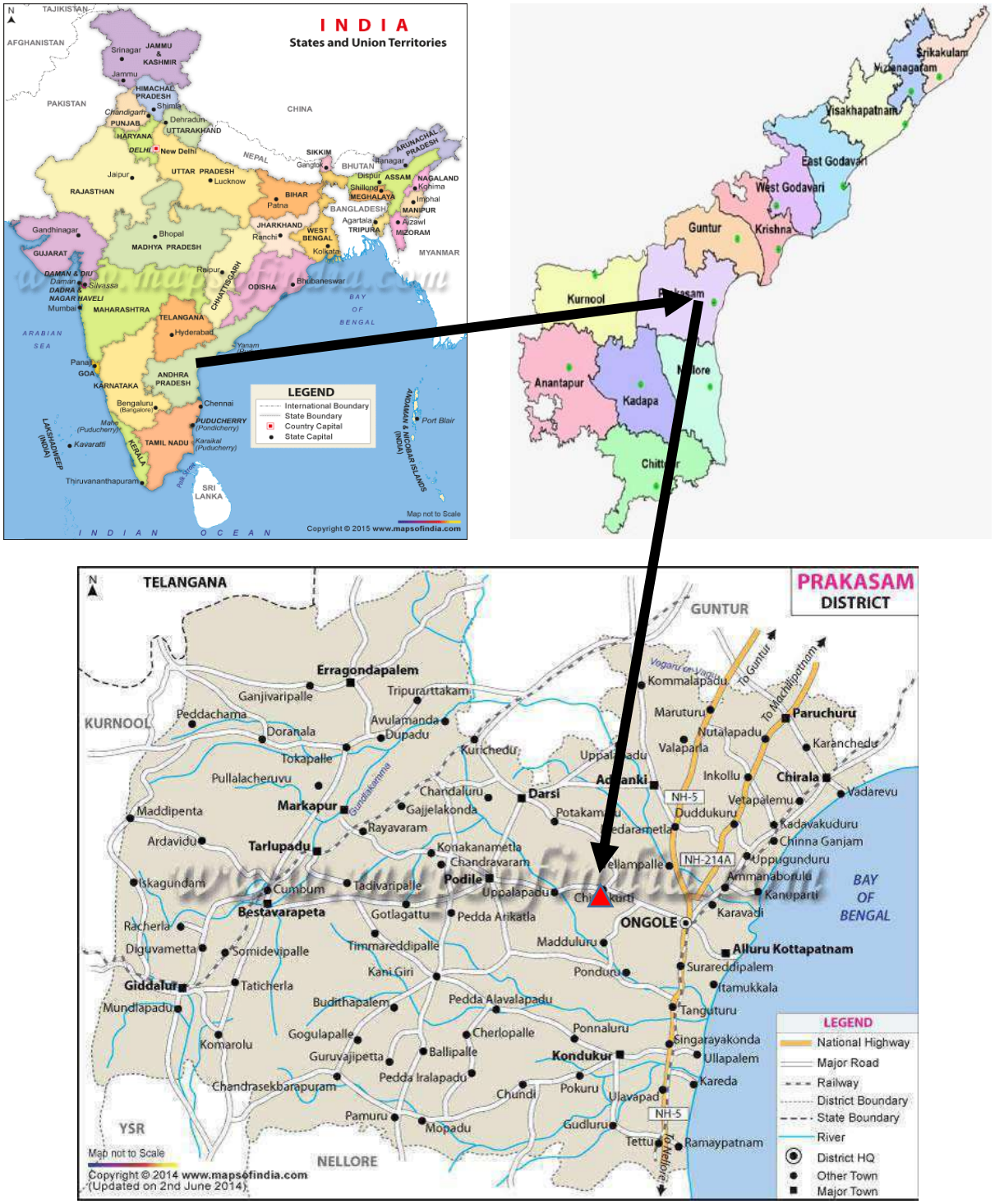
Boundaries of the lease area are given as below:

North – East	M/s. Swathi Granites
South	M/s. Golden Granites
East	M/s. Surya Granites
West	M/s. Alliance Black galaxy Granites/APMDC
SW	M/s. Midwest Granites

2.2 Location of the Project

The mine lease area is made up of plain ground and is falling in Survey of India Topo-sheet no 57 M/14. The area is bounded by Latitude 15°35'03.12" – 15°35' 13.10" & 79^0 49' 19.20" - 79^0 49' 53.06" E Longitude. Location Map showing project site location in Prakasam District is given in **Figure – 2.1.**

Figure 2.1: Location Map showing project site



The proposed mine site is situated at 0.75 Km South of RL Puram village. Major town and District head quarter, Ongole is at 22.2 km in ESE, the Mandal headquarter Chimakurthy is at 2.6 km in East direction from the mine site.

The project site is located at a distance of 0.3 Km from Ongole – Podili Road. National Highway NH-5 from Visakhapatnam to Chennai is 22.2 km in ENE direction from the mine site. The nearest railway station is Ongole Railway station at distance of 26.6 km. The nearest airport is Gannavaram Airport at a distance of 147.3 Km.

Topographical Map Showing 10 km radius around the project site is given as **Figure – 2.2**.

Base map of the study area giving features in 10 KM radius of study area is given as **Figure – 2.3 & Figure – 2.4** shows google image of the mine lease area with co-ordinates and **Figure – 2.5** shows the photographs of the mine lease area.

No National Parks, Wild life sanctuaries, critically polluted areas are located within 10 km radius from the mine lease area.

2.3 Road Connectivity to the site

The project site is located at a distance of 0.3 Km from Ongole – Podili Road. National Highway NH-5 from Visakhapatnam to Chennai is 22.2 km in ENE direction from the mine site.

2.4 Physiography of the Site

The quarry lease area is located on plain land with a bench mark with value of 66.398 M. The quarry lease area with Topo relief in south direction. The entire lease area excepting north central portion is under exploitation. The deepest pit floor level is with RL 12 .10 M i.e., 80 M depth in Pit 1 and 2, Pit 3-64 M, Pit 4-55 M Fig – 2.5 shows the surface plan of the mine lease area.

2.5 Site Description and Present land use

The quarry lease area is plain land with sparse vegetation with small bushes from the vegetal cover. The quarry area is surrounded by other quarries.

There are two reserve forests with in 10 kms of the mine which are Chimakuthy RF-2.3 Km (N) Devarapalem RF – 6.0Km (N).

There are no National Parks, Sanctuaries, Bio Sphere Reserves, Bird nesting places, Animal Crossing corridors, and avian routes of migratory birds within 10 KM radius of the site.

There are no historical places, archaeological sites and important piligrim centres within 10 KM radius of the site. No Recreation centres within 10 kms radius of the site.

2.6 Mine Drainage

The area regionally forms part of the Paleru River Catchment System. The Musi river located 5.50 km west of the quarry site forms part of its stream. The Musi River originates in the Darsi Donakonda hills and upland areas of Prakasam district, and flows 33 km southwardly to reach the study area, and further flows 23 km southeasterly and joins the Paleru River near Kamepalli (6 km south of Kondapi), which then flows 30 km eastwards to join the Bay of Bengal.

Figure 2.2: Topographical Map Sowing 10 Km Radius

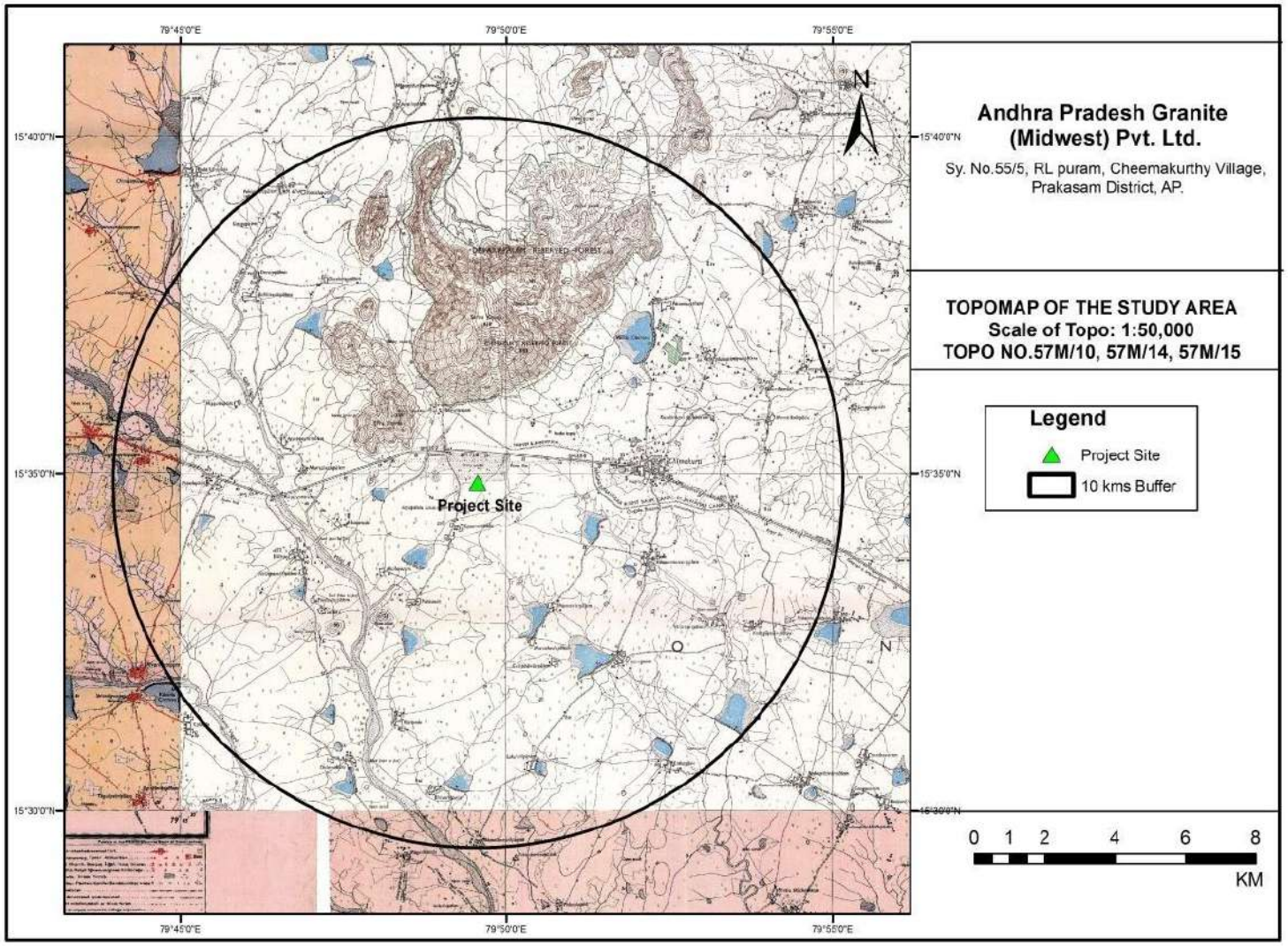


Figure-2.3: Base Map of the Study Area

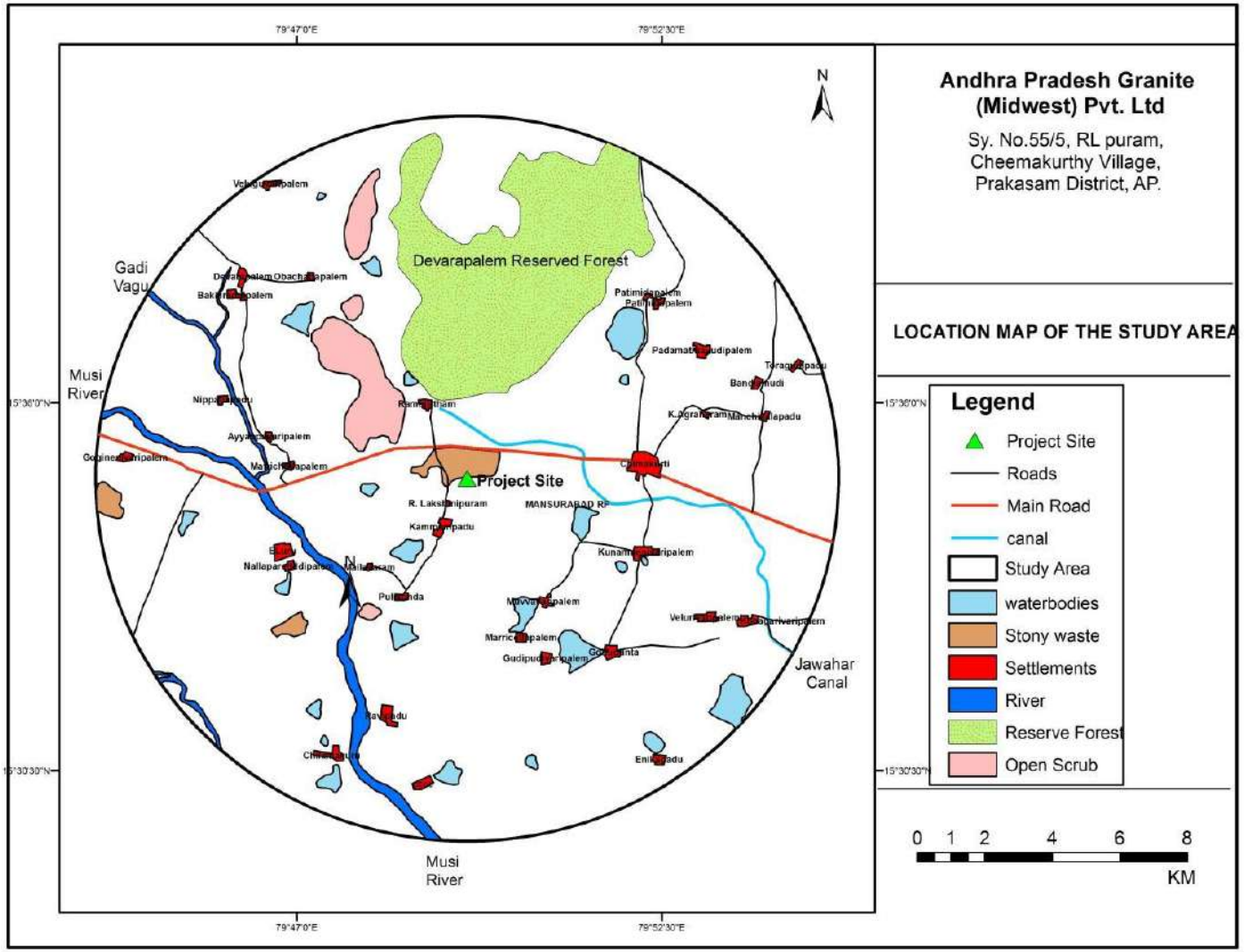


Figure -2.4: Google Map Showing Mine Lease Area



Figure: 2.5: Photographs of the mine lease area





2.7 GEOLOGY

2.7.1 Regional Geology

The Chimakurthy igneous complex is an oval shaped igneous body, comparable to Bushveld Igneous Complex of South Africa and skeeargard Intrusive of Greenland. Similar to these occurrences, the Chimakurthy igneous Complex is considered as "Massive Concentric Type" intrusive magmatic body. As the source of magma is from deeper levels, the depth continuity of the bronzite gabbro may be few kilometers below the surface.

The geological field investigations indicate that the pre-existing older migmatisebiotite hornblende schist and quartzo-Felspathic formation is introduces by the emplacement of huge igneous body formed by in-situ fraction differentiation and crystallization process. Single mafic igneous body has resulted into the formation of different rock types occurring in the forms of oval shaped or Elliptical shaped body with 15 km long (NS), 4-7 km wide (EW) and occupies 75 Km area. Hence it is named as Cheemakurthy Igneous Complex. Geomorphological expression shows that the outer bronzite gabbro rim and central core of anorthosite and proximate forms plains, whereas the semicircular, crescentic shaped clino-pyroxene gabbro occurs as a very high ridge with steep flanks.

The other rim of bronzite gabbro assumes economic importance by virtue of its rare occurrence of Bronzite bearing Gabbro and demand in international as well as domestic markets. The deposit is formed by intrusion of a major igneous episode, where the bronzite gabbro is formed by formed as an outer rim of theigneos body accomplished by in situ fractional crystallization, manifested by the process of differentiation from single magmatic body. In fact, it is the process which has contributed to the conversion of enstatite ortho pyroxene (OPX), normally present in gabbro into bronzite due to excess availability of iron under Oxygen deficient conditions.

Hence the concentration bronzite enriched in the periphery of igneous complex i.e. the peripheral 'gabbroic rim'. Even during the process of the fractional crystallization of the magma with excess presence of iron in the magmatic liquid fail to form magnetite (iron oxide), and instead contributed to the formation of the brozite under "reducing environment", resulting into the formation of then iron-rich enstatite, i.e.bronzite. In the lease area, the concentration of bronzite gives an aesthetic look for commercial marketing.

2.7.2 Local Geology

The geomorphology plan in this area reflects the presentence of sub-surface bronzite bearing gabbro (galaxy granite); located at the central swollen portion of bronzite gabbro rim. The quarry gabbro formation and its exploration in the quarries around the quarry lease area. From

the working quarries i.e. towards the northern and southern, south-eastern parts of the quarry lease area amply tesify the sub-surface continuity of formation in the quarry lease area. All these evidences undoubtedly confirm the lateral continuity and presence of black galaxy (bronzite gabbro) at sub-surface levels in this lease area. However the premise for occurrence of workable deposits at shallower level is positively indicated by the excavation carried in the quarry lease area.

The detailed geological investigations suggest that the lineation of the gabbro in this area is generally trending in N 60º E direction, which continues across the boundary. In view of the preferred orientation of the minerals, it is inferred that the area is subjected to some syn-emplacement/crystallization deformation.

Geology of the lease area

The black Galaxy Granite also known as bronzite bearing gabbro.norite is being quarried extensively in the area between RL Puram in south and Ramathirtham in north.30% of the lease area is under soil cover of 1 m-3m with scanty exposures and rest of the area quarried and there are four worked out pits in the area. From the geological data from the existing workings in the quary and adjacent quarries and also the borehole data, oit is observed that the bornzite bearing gabbrao/norite occurs under an overburden thickness of about 8-12m. The overburden comprises 1-3 m thick soil on surface followed by weathered bronzite-bearing gabbro/norite.

Gabbro/norite is melanocratic, light dark gery in colour, medium coarse grained, massive with scattered and randomly oriented coarse, euhedral, sub hedral, anhedral grains of bronzite. It mainly consists of plagioclase, pyroxene and traces of olivine, biotite and opaques

Soil Cover: The area represented by bouldery zone whose thickness varying between 1 to 3 M and the weathered zone about 8 M.

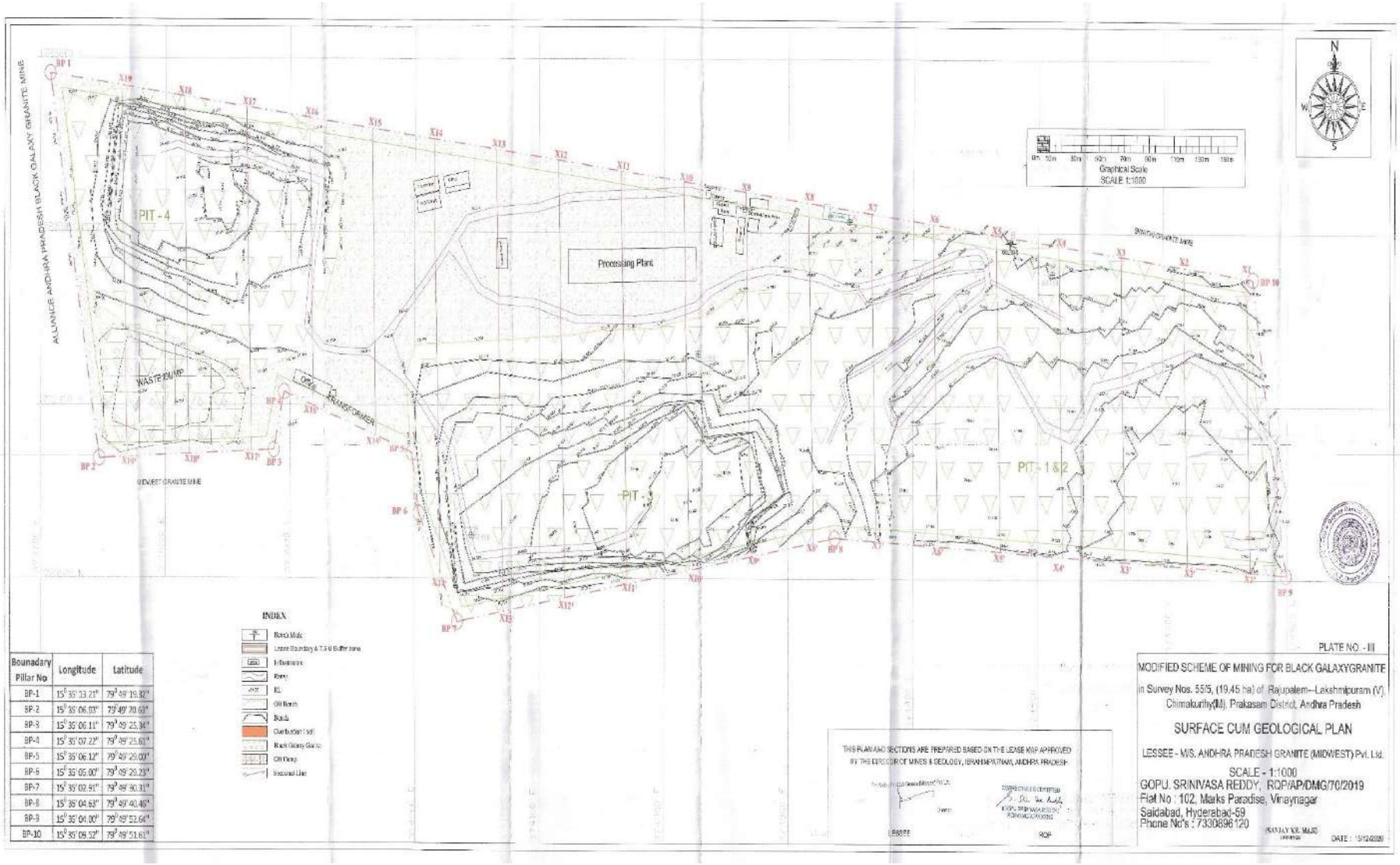
Weathered zone: Underlying the bouldery zone of 9M thick, brownish colour weathered mantle exists in the area; it is mixed with fresh rock boulders; the degree of weathering decreases with depth.

Gabbro: Black coloured, medium grained, equigranular gabbro with evenly distributed brown coloured bronzite specks occurring within the pit. It is sheet type exposures. The Massive type with underlying surface. Hence, the deposit can be classified as solid type.

Petrographically the gabbro essentially consists of plagioclase and pyroxenes, with accessory mineral of bronzite, ilmenite, Garnet and the specks of bronzite impart galaxy nature for the gabbro.

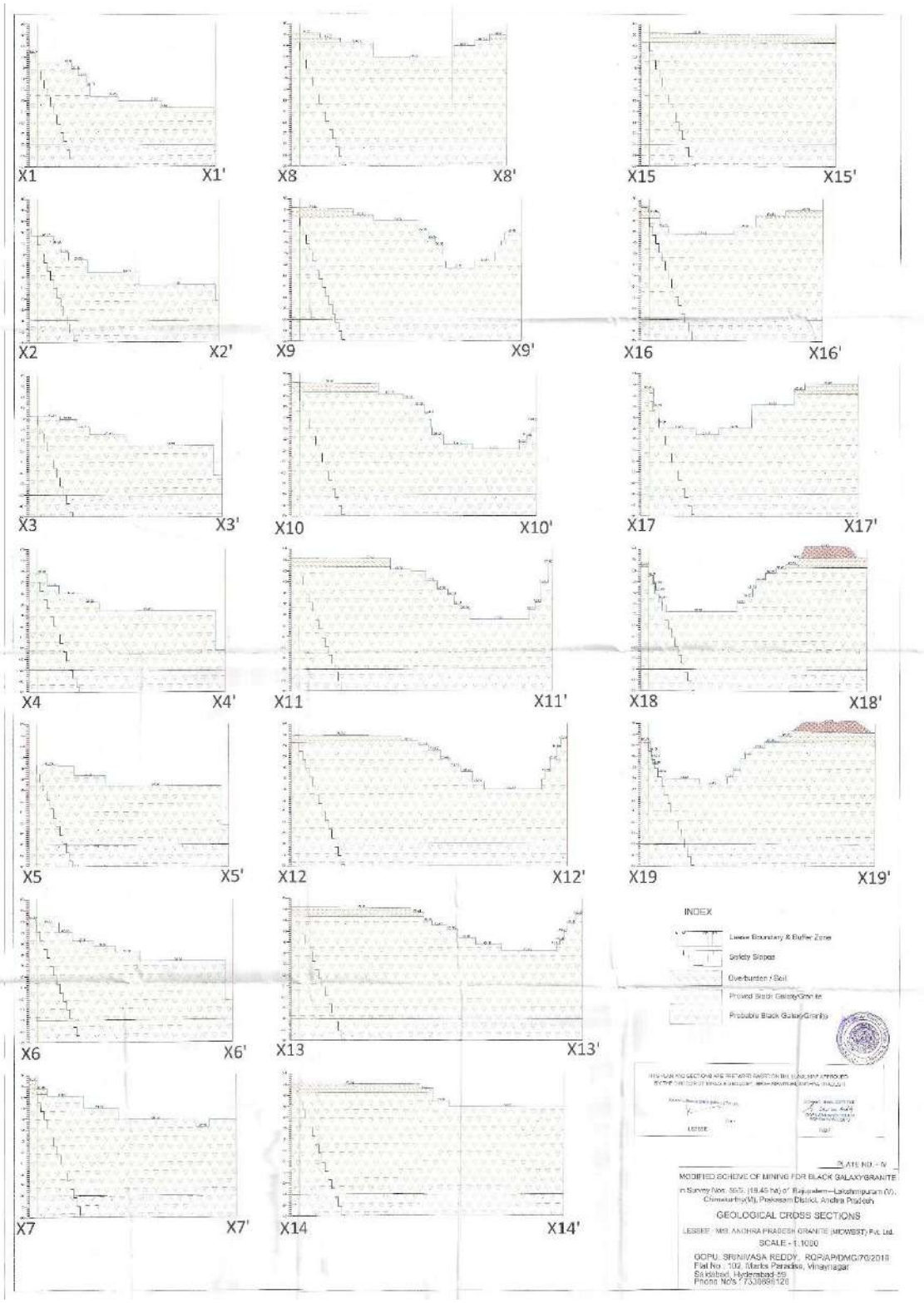
Figure – 2.6 shows the geological plan and **Figure – 2.7**shows geological cross sections of the mine lease area.

Figure 2.6: Surface Geological Plan



Bounadary Pillar No	Longitude	Latitude
BP-1	15° 35' 13.21"	79° 49' 19.32"
BP-2	15° 35' 06.53"	79° 49' 20.63"
BP-3	15° 35' 06.11"	79° 49' 25.34"
BP-4	15° 35' 07.22"	79° 49' 25.61"
BP-5	15° 35' 06.12"	79° 49' 29.00"
BP-6	15° 35' 05.00"	79° 49' 29.23"
BP-7	15° 35' 02.91"	79° 49' 30.31"
BP-8	15° 35' 04.63"	79° 49' 40.45"
BP-9	15° 35' 04.00"	79° 49' 52.64"
BP-10	15° 35' 09.52"	79° 49' 51.61"

Figure-2.7: Geological cross Sections



2.8 Present Land Use Pattern

The total mining lease area of 19.425 ha. About15.0 ha will be used for mining during the life of the mine. Details of the land use pattern of the mining lease area given in table

Table 2.2 - Land Use Pattern of the Mine

S No	Purpose	Area in Ha.
1	Area under Quarrying	15.00
2	Hauling Road	1.00
3	Finished Granite storage area	2.50
4	Other Infrastructure	0.925
Total		**19.425**

2.9. Area for Dump

The lessee is having waste dump in Sy.No 938/1 to 7 of Chimakurthy Village & Mandal over 6.17 Acres, Lease was granted vide DMG proceedings No 25711/R3-2/2007 dt. 08.11.2007 and executed vide ADMG proceedings No 5483/Q/2007 dt. 20.11.2007 and The Lessee also having another waste dumps in Sy.No. 941/1 to 4, 944/1 & 944/4 to 7 of Chimakurthy Village & Mandal over 11.19 Acres and Sy.No 25/3B2 of R.L.Puram (V) over an extent of 11.00Acres, in Sy.No 938/1 to 7, over an extent of 6.17 acres of Chimakurthy (V) for dumping the waste The lease agreement over an extent of 1.98 acres Survey.No.25/3B2 of R.L.Puram (V) with Midwest Granite Pvt Ltd for dumping the waste.

The generated waste will be dumped in Sy. Nos. 25/3B2 of R.L. Puram Village, Chimakurthy Mandal & 938/1 to 7 of Chimakurthy Village & Mandal during the present scheme period.

2.9.1 Area of Green belt

As mine site is hard sheet rock, a separate area of 2.74acres in sy.no 954/1 of chimakurthy village developed for green belt. Now developed 1184 trees in this area and proposed to develop additional 900 plants. The proposed green belt site approximately at a distance of 3.0 km from the mine site in Chimakurthy Village

2.10 Method of Mining

2.10.1 Description of Existing method for excavation

The method of the mining is opencast semi mechanized mining, employing deep drilling and blasting for breaking the overburden and black galaxy granite using heavy earth moving machinery viz. hydraulic excavators to excavate the broken material and load the same into tippers and dumpers. The overburden will be hauled to the dumps and stacking yards respectively.

Extraction of blocks: The natural joints in the sheet rock is taken for splitting them or a line of drill holes are drilled and split it with the help of feathers and wedges. The separated pieces will be examined for defects and lines, then the block or blocks are marked in clear area and holes are drilled along the line of marking with the help of feathers and wedges and the waste portions are separated forming a rectangular blocks.

Separation primary blocks from the mother rock: Diamond wire saw use for separation of primary block from the mother rock. Working faces are developed parallel to the direction of lineation. For effective use of wire saw, two sides free face is required. The bench height depends on the reach of the excavator. The main cut can be planned at 8m height x 9m x 8 to 9m lengths.

The Diamond wire is passed through two horizontal holes drilled at 90° and the bottom is cut first then the wire is passed through the vertical hole and one of the horizontal holes and vertical face is cut. Thus the entire block of 6m x 9m x 9m size is separated from the mother rock. The block is divided into number of slices, with each slice 1.8 m to 2m thickness. Slicing can be done by either with a wire saw or with a line of holes by using lack hammers/ Line Drill or Tam Rock and wedging.

The slice is brought down with the help of excavator and washed with water to check for defects and cracks. The blocks are marked and separated by drilling and wedging.

Secondary Splitting

After separation from the mother rock, it is properly checked by washing with water to identify defects and cracks. Based on this observation the size is decided and the blocks are marked and separated by drilling and wedging.

The separated pieces are examined for defects and lines, then the block or blocks are marked in clear area and holes are drilled along the line of marking, with the help of feathers and wedges and the waste portions are separated forming a rectangular blocks. Any bulges are

removed by drilling and wedging making it perfect blocks. A perfect block is that all the sides shall make with each other 90º.

Dressing

After secondary separation, the rock mass will be carefully examined to avoid hair-line cracks, mineral segregations and veins etc. The dressing of the rough blocks will be made by chipping the edges and geometrically equating the edges of the block at the dressing yard. The rough blocks obtained after secondary cut, it will be dressed for obtaining good geometric shape of galaxy granite. Adequate care is required to dress the block depending on the surface to be polished (lineation planes), as golden spots are visible only in one direction in lineation plane. However, bronzite is softer than other minerals, it is foliaceous and tabular and often flaky, oriented in the direction of general lineation. The dressing and sawing of the blocks should be oriented parallel to the direction of foliae / lineation to expose the plane parallel to 001-Crystallographic plane of the metallic luster, so as to expose more bronzite flakes on polished surface.

Drilling

Drill holes of 6 M depth will be drilled in a single row with spacing of less than 0.3 M and burden of 1.6 M or more shall be maintained uniformly. This arrangement will yield rock size of 3 M x 2 M cross section. A bench height of 6.0 M will be tackled in two steps.

Explosive type

The lessee proposes to use low intensity blasting using light charge of Gelatin (explosive) for controller blast OB and waste boulders

Loading & Transport

The mined blocks are loaded into trucks by cranes and transported to the processing plant.

Extent of mechanization

The following table describes the machinery used for mining

Table 2.3: Extent of Mechanization

S.no	Machinery	No's
1	Hydraulic Excavators	14
2	Dumpers	18
3	Compressors	16
4	Electrical Compressors	02
5	Hydraulic Drills	08

6	Wheel Loader	01
7	Slotters	13
8	Jack hammers	40
9	Wire Saw	12
10	Derrick k Crane	03
11	DG Sets(125,200, 500 KVA)	04
12	Dewatering Pump(60 HP, 20HP)	2+2
13	Escort Cranes and loaders	02
	Total	**137**

2.11 Year wise Production and Development

The year wise production and solid waste details are given **Table2.4** as per the modified mining plan approved by Deputy Director Mines & Geology, Ongole dated 22.01.2021.

Table 2.4: Year Wise Production from Initiation of Mine Starting Period to Till Date

Year	Production(M3)
2007-2008	3228.282
2008-2009	7370.467
2009-2010	11289.617
2010-2011	27709.123
2011-2012	35538.639
2012-2013	36095.831
2013-2014	26052.528
2014-2015	28073.993
2015-2016	26988.071
2016-2017	30937.487
2017-2018	23448.769
2018-2019	1506.272
2019-2020	15196.225
2020-2021	54408.740
April-2021 to September-2021	31834.672
TOTAL	**359678.716**

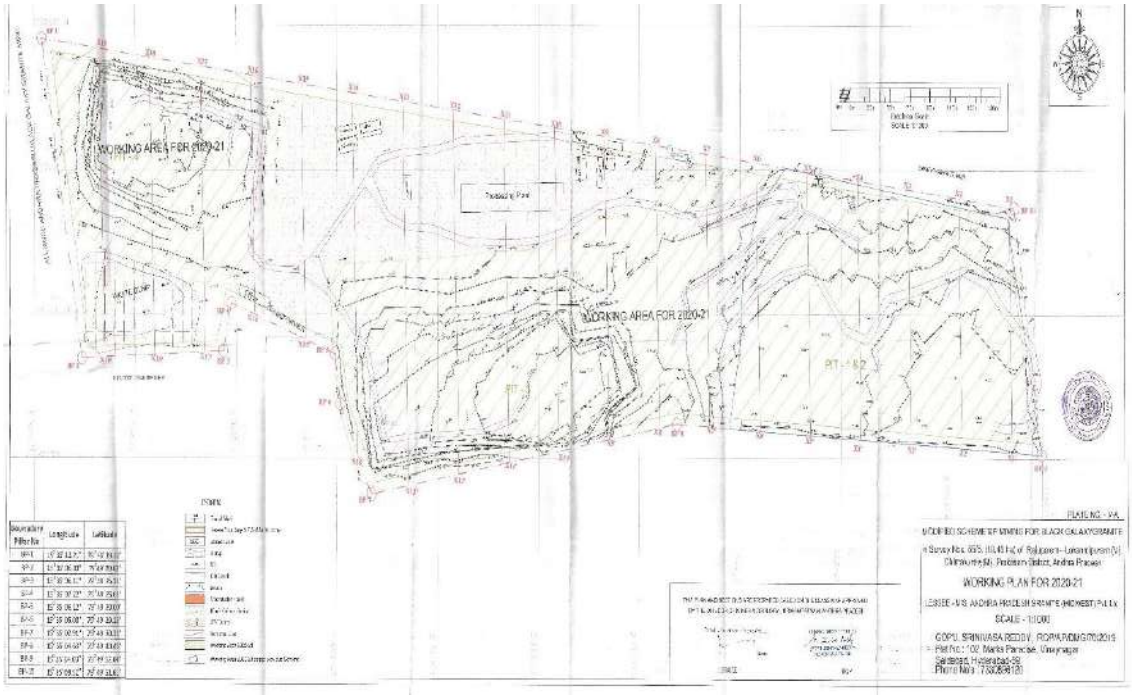
2.12 Conceptual Plan

Of the total mining lease area of 19.425 ha, about 15.00 ha.will be used for mining activities during the life of the mine. The waste generated during the life of the mine is stored in dump

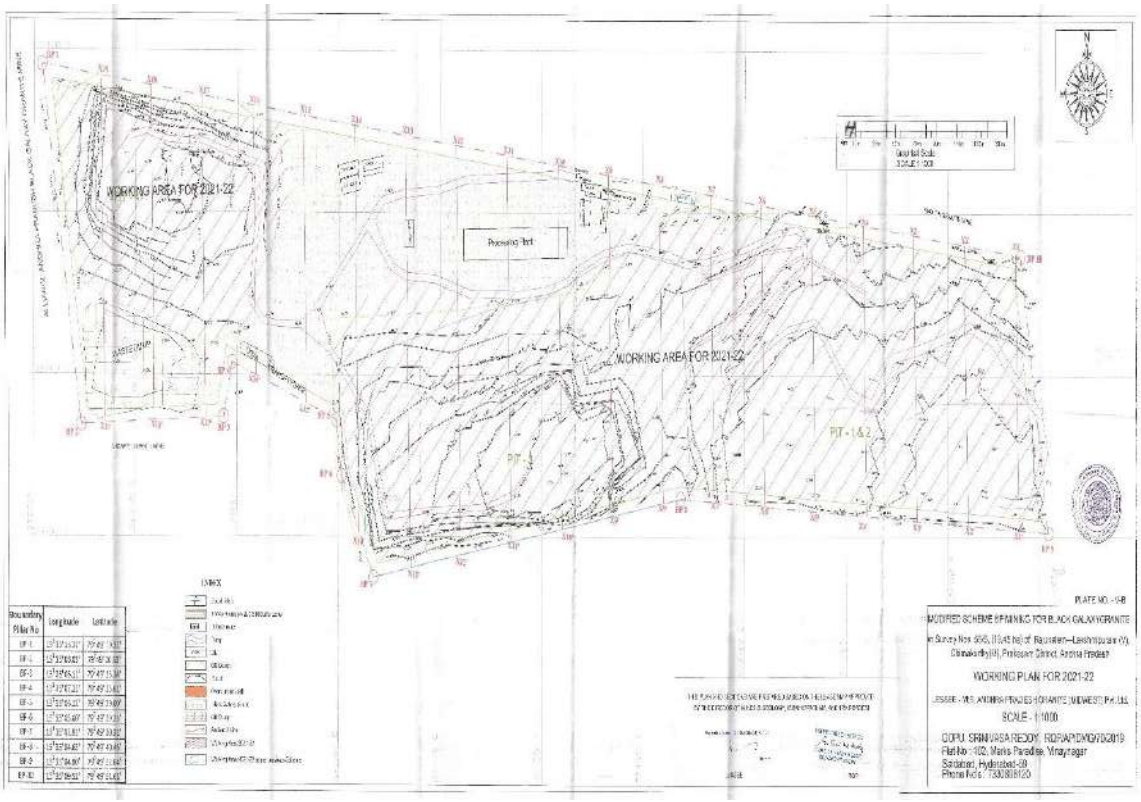
site and the matured pits will be back filled by re-handling of waste as a reclamation measure and balance the pits will be utilized as water harvesting structure

Figure 2.8: Year wise Production Plan

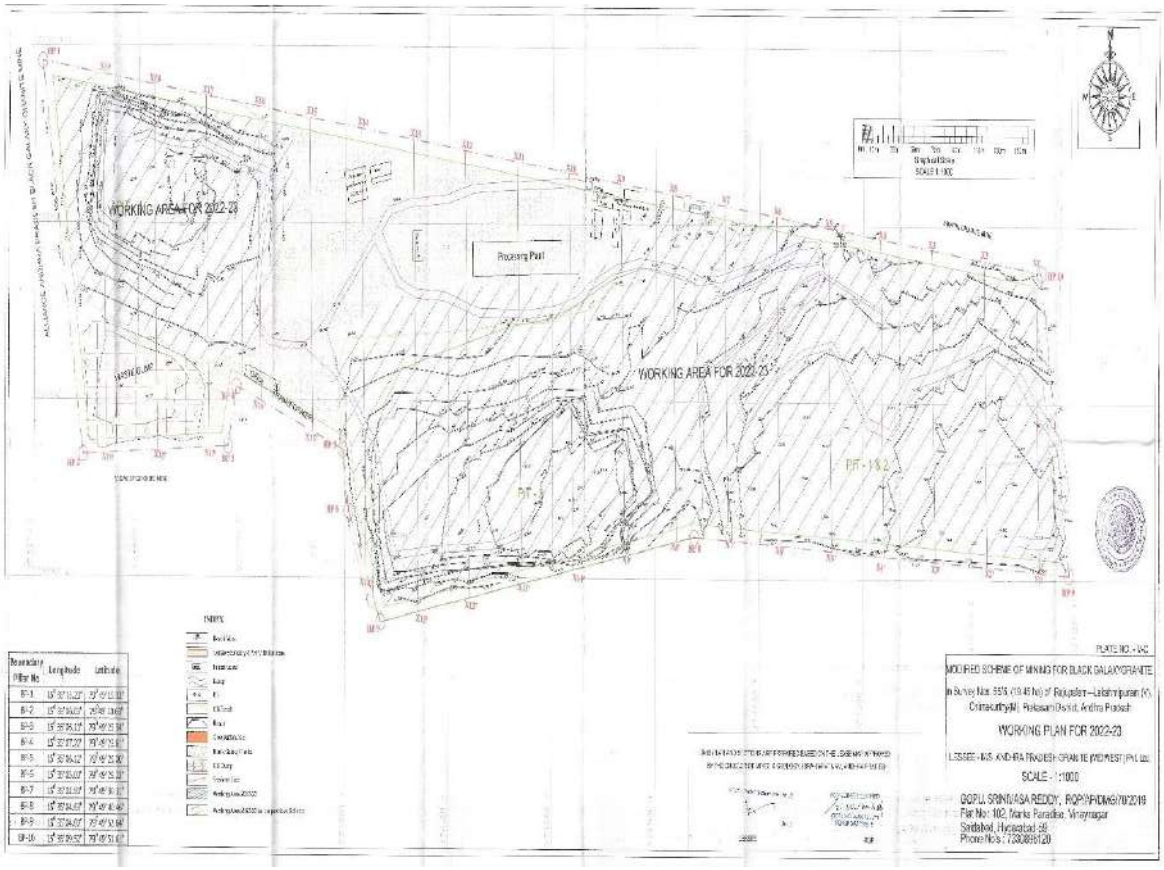
1st Year:



2nd Year:



3rd Year:



2.13 Anticipated life of mine

With the available reserves of 12,26,080 m3 and considering average annual production of 79,401 m3 the life of the mine is estimated as 15 Years.

2.14Top Soil &Waste Generation and Disposal

The granite occurs as surface deposit with soil cover and the weathering is limited to the inter-spaces of boulders and joint planes. Mining of black galaxy granite results in serious problem of generation huge quantity of waste rock. The year wise waste production details are given below.

Year	Total tentative Excavation (M3)	OB/ Waste rock (M3)	ROM		Mineral Rejects (M3)
			Ore @ 8 % (M3)	Tiles recovery 1.84% (M3)	
2020-21	9,67,600.0	82,550.0	77,408.0	17803.84	9,54,938.16
2021-22	10,00,350.0	0	80,028.0	18406.44	9,01,915.56
2022-23	10,09,600.0	0	80,768.0	18576.64	9,10,255.36
Total	**29,77,550.0**	**82,550.0**	**2,38,204.0**	**54,786.92**	**27,67,109.08**

Dump Handling: The lessee is having waste dump in Sy.No 938/1 to 7 of Chimakurthy Village & Mandal over 6.17 Acres, Lease was granted vide DMG proceedings No 25711/R3-2/2007 dt. 08.11.2007 and executed vide ADMG proceedings No 5483/Q/2007 dt. 20.11.2007 and The Lessee also having another waste dumps in Sy.No. 941/1 to 4, 944/1 & 944/4 to 7 of Chimakurthy Village & Mandal over 11.19 Acres and Sy.No 25/3B2 of R.L.Puram (V) over an extent of 11.00Acres, in Sy.No 938/1 to 7, over an extent of 6.17 acres of Chimakurthy (V) for dumping the waste The lease agreement over an extent of 1.98 acres Survey.No.25/3B2 of R.L.Puram (V) with Midwest Granite Pvt Ltd for dumping the waste.

The generated waste will be dumped in Sy. Nos. 25/3B2 of R.L. Puram Village, Chimakurthy Mandal & 938/1 to 7 of Chimakurthy Village & Mandal during the present scheme period.

2.15 Requirements for the Project

2.15.1 Site Services

Facilities like rest shed, store room and office room will be provided in 7.5 m barrier zone of the mining lease area.

2.15.2 Water Requirement

The total water requirement for the project is about 80 KLD. This requirement will be met from nearby villages by tankers.

2.15.3 Power Requirement

Power is available in the Mine site. Power is used for office and for wire saw machines, compressors and dewatering pumps. Mining equipments will be operated with both power and diesel motive power.

2.15.4 Man power requirements

Currently there are 448 people are working in the mine. For expansion activities, additional 40 people will be employed. Preference for local youth will be there in providing the employment.

Chapter - 3

Baseline Environmental Status

CHAPTER-3
BASELINE ENVIRONMENTAL STATUS

3.1Preamble

Collection of base line data is an integral aspect of the preparation of environmental impact assessment report. Baseline data reflects the present status of environment before the initiation of any activity of the Production Enhancement. The possible effects due to the proposed Production Enhancement mine site of M/s.Andhra Pradesh Granite Midwest Private Limited are estimated and superimposed on the compiled baseline data subsequently, to assess environmental impacts.

3.1.1 Study Area:

Data is collected in a circular area of 10 km radius surrounding the mine site.

3.1.2 Data Collection Period & Components:

Baseline environment assessment was conducted in the impact zone of 10 km surrounding the site area during the months of January 2019- March 2019. Studies were undertaken to generate baseline data of micrometeorology, air quality, water quality, noise levels, flora and fauna, land use, soil quality, Ecology biodiversity and socio-economic status of the community were collected in a circular area of 10 km radius surrounding the Mine site.

3.2 Methodology of Base line data collection

3.2.1 Scoping

The scope of the assessment for this study was determined by the Terms of Reference, granted by Ministry of Environment, Forests & climate change, by the State level Expert appraisal Committee and approved by SEIAA, Andhra Pradesh. An initial discussion with the project proponents and other stakeholders was conducted in the study area so that the focus would be on valued ecosystem components (VECs) they considered important.

3.2.2 Reconnaissance Surveys

The study team made initial site visit to ascertain basic information on various aspects like availability of infrastructure, physiological condition of the project site, surroundings of the site such as road connectivity, nearby population, nearby water bodies & neighboring industrial activity etc. The information collected has been used for identifying the valued eco components to be chosen for specific consideration in the study.

3.2.3 Assembly and Analysis of Data

The data from initial surveys was recorded. Various work sheets are used for collection of base line data by various functional area experts focusing on different components of EIA study. The Supplementary information was collected from Survey of India topo sheets, census handbooks, published flora and fauna data etc. Standard statistical techniques were used for analysis of the socio-economic data, etc. Qualitative analysis was done for more descriptive data.

3.2.4 Assessment of Alternatives

Alternatives of the project site are not relevant in this case as this project is production enhancement from existing mine

3.2.5 Assessment of Potential Impacts

Potential significant impacts were identified on the basis of analytical review of baseline data; review of land uses and environmental factors; analytical review of the socio-economic conditions within the PIA; biotic environment in study zone.

3.2.6 Integration of Environmental Impacts in the Design Process:

The design and decision – making process integrated environmental and social issues and prompted the early identification of appropriate actions. Such actions included, for example, optimizing the layout, use of existing infrastructure with both units, current air pollution control systems and improvements, measures for water conservation, measures for solid waste management etc.

As this is brown field project, existing environmental infrastructure is reviewed and gaps are identified for improvements in the production enhancement project

3.3 Land Environment

The area is located on Topo Sheet No.57M/14. At the junction of the quarry lease area consists of undulating with gently sloping to East and South. The mine activity is there from year 2007 onwards and current mine reached a depth 20-24 Meters from GL.

3.3.1 Physiography

The quarry lease area is located on plain with worked pits. The general ground level is gently sloping to south. The area is unfit for agriculture as it is sheet rock of granite.

3.3.2 Topography

The Mine lease area falls in the Survey of India, topographical Map No.57M/14 with Geographical Co-ordinates Latitude 15º35’09.70” to 15º35’03.66”N and Longitude 79º 49’51.60” to 79º 49’53.06”E.

The area located at a distance of about 0.75km due North of RL Puram village and 5.0 Km West of Chimakurthy on B.T. Road Ongole-Kurnool State Highway No.53. The quarry lease area can be approached by earthen road from this State Highway.

The mine site is approachable from Ongole – Kurnool state highway no. 53 which is about 0.34 KM from the mine site

The Google map showing the mine site area and its surroundings is as shown in figure 3.1

Figure-3.1: Google map showing the project area and its surroundings



3.4 Drainage

The area regionally forms part of the Paleru River Catchment System. The Musi (Paleru Basin River) river located 5.50 km west of the quarry site forms part of its stream (Figures 1 to 3). The Musi River originates in the Darsi -Donakonda hills and upland areas of Prakasam district, and flows 33 km southwardly to reach the study area, and further flows 23 km southeasterly and joins the Paleru River near Kamepalli (6 km south of Kondapi), which then flows 30 km eastwards to join the Bay of Bengal.

Major part of the pediplain in the Paleru catchment area is dissected by ephemeral streams and streamlets flowing in north and northeasterly directions. Some small and medium size lakes are situated along the topographic gradients, and most of which are connected by small

ephemeral nallahs (streamlets). Other than these many small ponds and lakes are situated around the study area. These lakes receive seasonal flows, otherwise mostly remain dry.

The AP Granite (Midwest) area lies within a local catchment area referred here as Pulikonda Watershed. The watershed is about 30 sq. km in areal extent, and forms part of the Musi river catchment.

Regional Map of the Area showing the Paleru River Catchment is given in **Figure 3.2**.

Figure 3.2: Regional Map of the Area showing the Paleru River Catchment

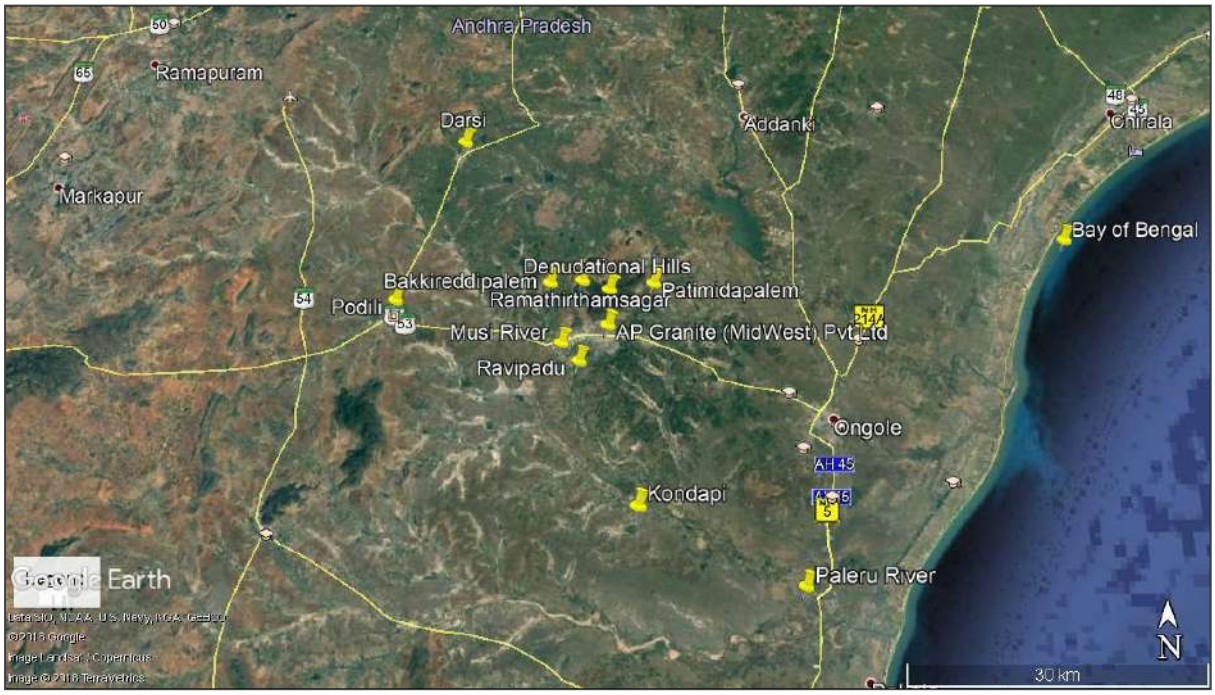
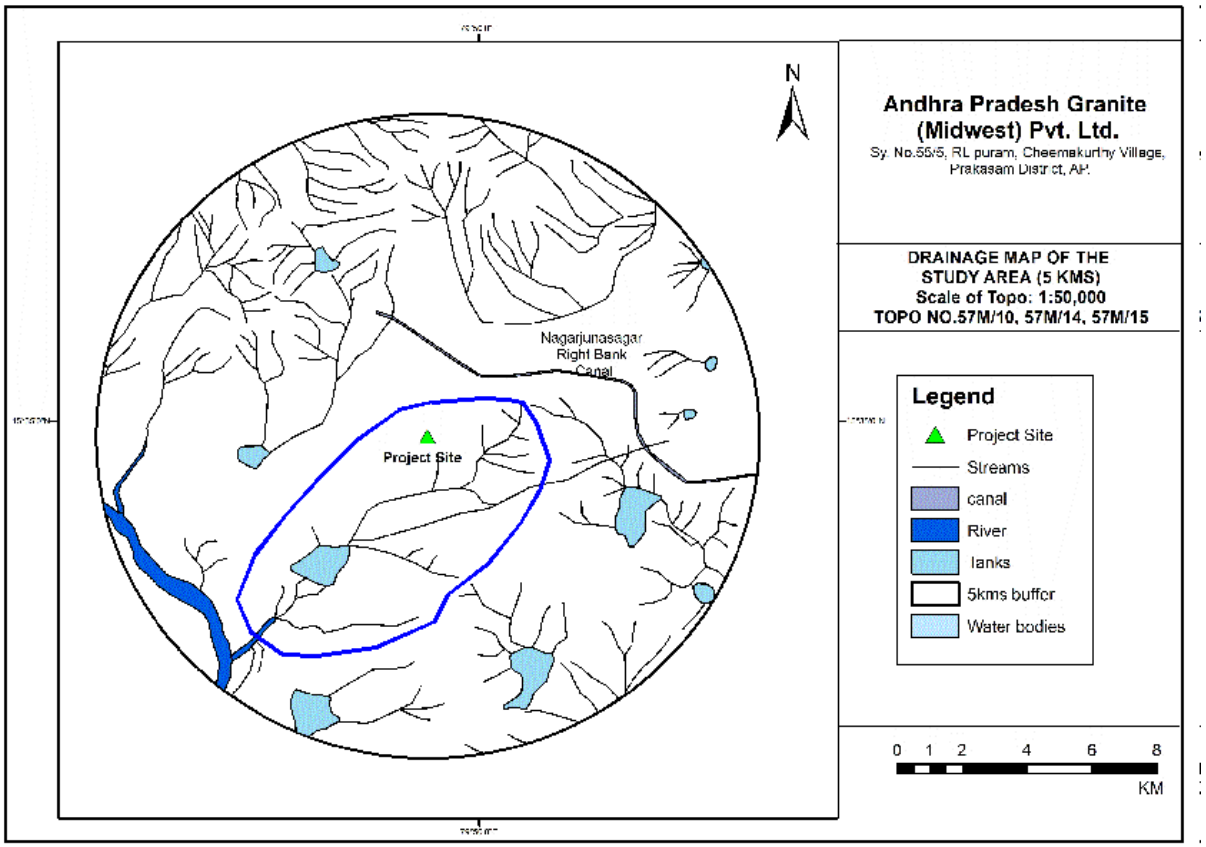


Figure 3.3 Drainage Distribution and Watershed Demarcation of the Study Area. (Watershed boundary is shown with Blue Outline, and Musi river is seen southwest of the watershed)



Out of the small and medium size tanks and water bodies situated around the quarry area, the notable one is the Ramathirtham Cheruvu, which is about 4 km (geographically) northwest direction of the site, and about 4 sq.km in areal extent (Figure 2). Other notable tanks and water bodies in the area are situated near Patimedapalem (Peddacheruvu), southeast of Bakkireddipalem, west of Gonugunta, west of Muvvavaripalem, south of Pulikonda, and north of Mailavarm. All these are less than 1.5 km in areal extent, and receive inflows from the surrounding hills and uplands during monsoon season, and usually get dried up before the onset of premonsoon (Feb/March). The distributary of Nagarjuna sagar Right Canal partly covers the areas, and takes route north of Chimakurthi and project site areas.

The overall area is regionally characterised by dendritic pattern of drainage, with a general slope towards main river and stream courses. The percolation tanks and water bodies in the area are either dry or gradually getting reduced over the years with the slowing down of inflows, and erratic rainfall.

3.5 Land use Pattern

Land use / Land cover Classification of the Study area

Based on the scale and the satellite resolution the Land Use/Land Cover classification can be made. The satellite data of IRS LISS IV was used while carrying out the analysis. Digital image processing was carried out to delineate various land use / land cover categories viz. build up area, crop areas and forests, waste lands such as land with or without scrub, water bodies and Transportation. The interpreted map was verified on ground at limited points and final land use / land cover map was prepared.

The Satellite image of the study area is given at **Figure 3.4** and land use and land cover map of the study area is given at **Figure3.5**

Figure 3.4 Satellite image of the study area

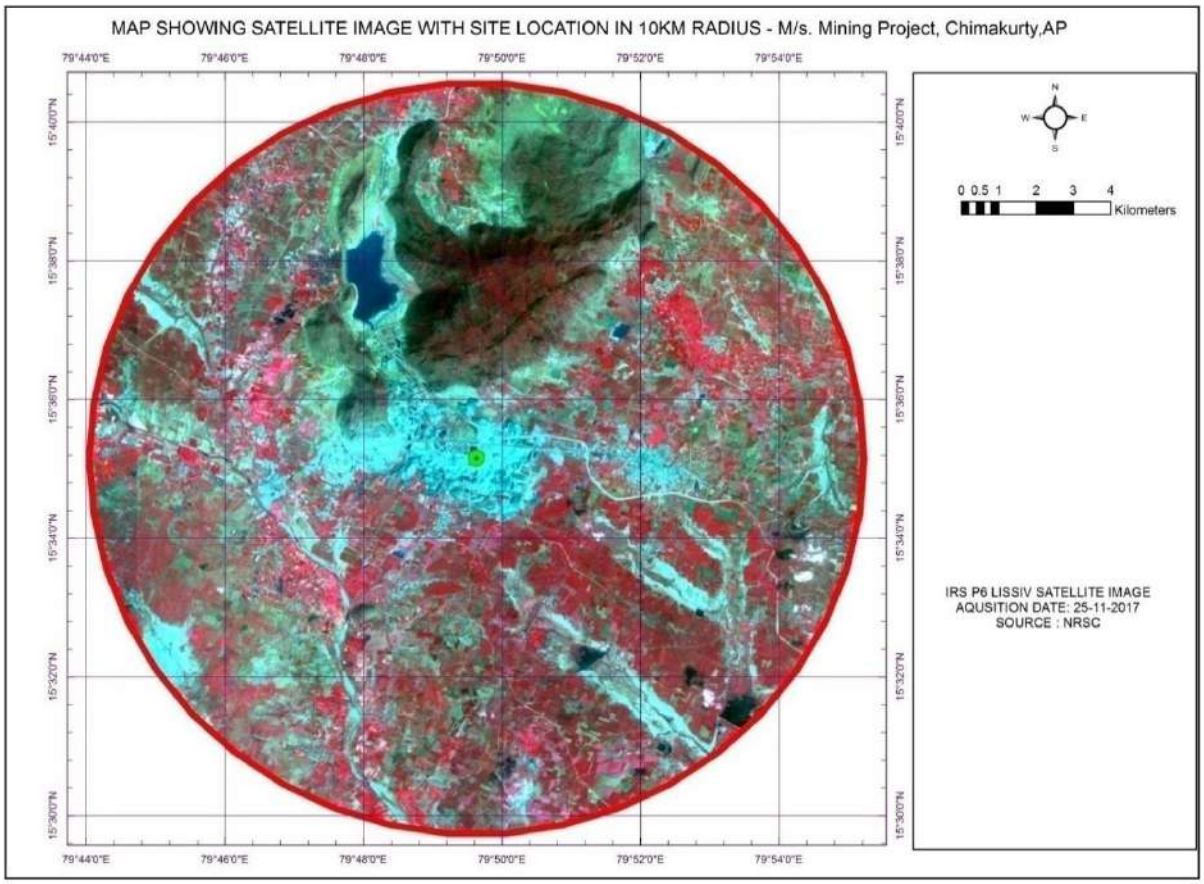
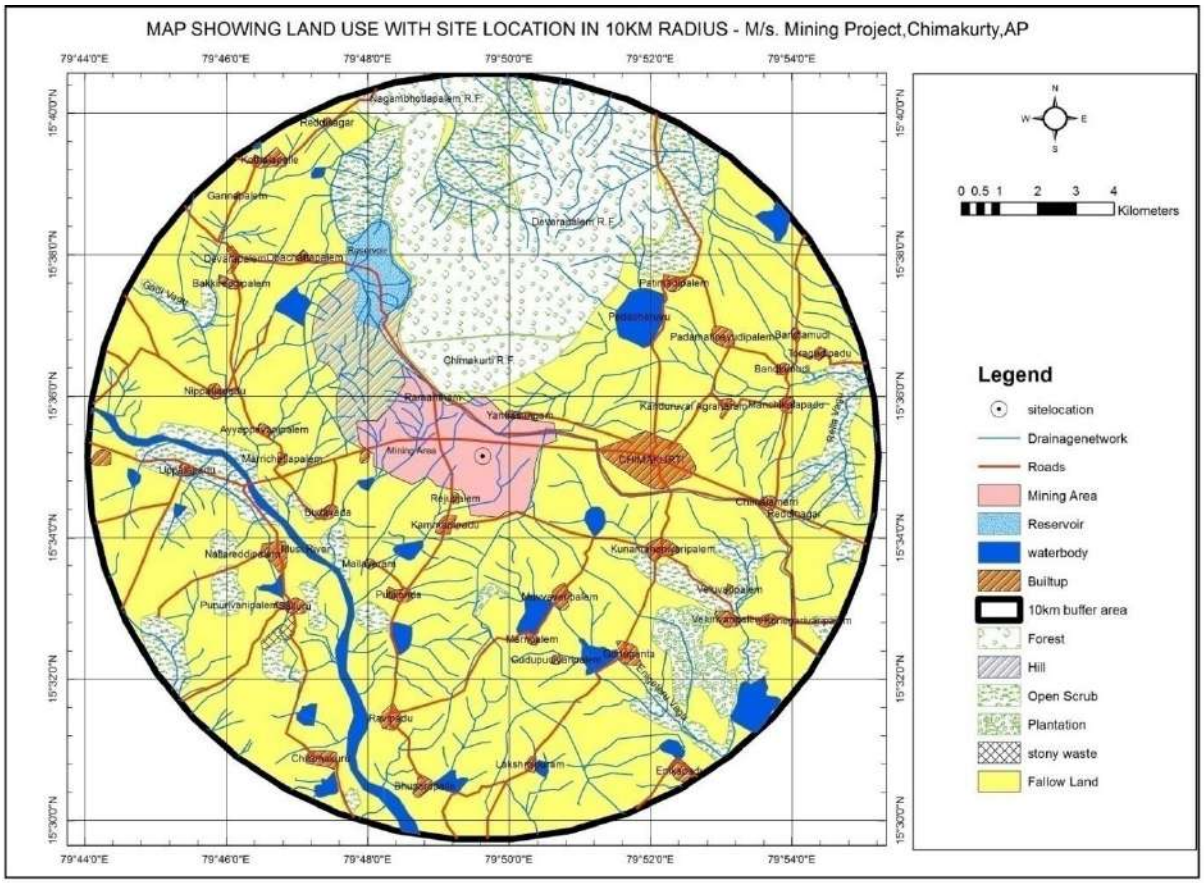


Figure 3.5 Land Use and Land Cover map of the study area

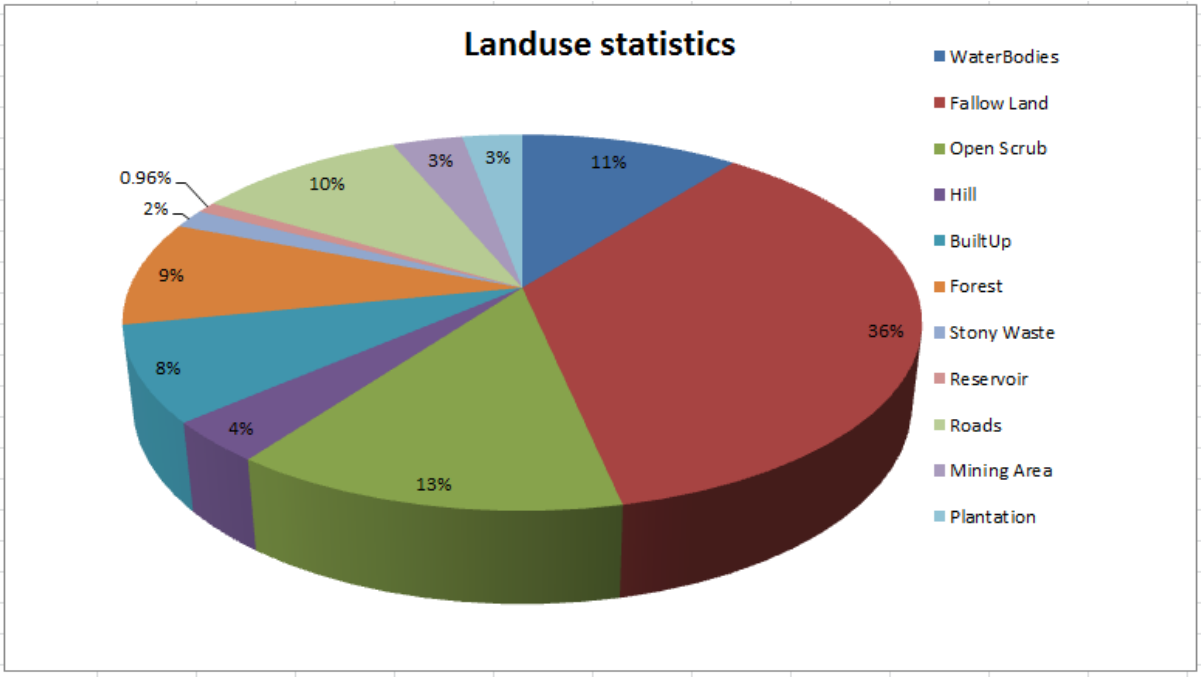


Below table provides land use for various requirements in the study area

Table 3.1: Land Use/Land Cover classes in Study area

S. No	LULC Class	Area (Ha)	Area (%)
1	Water Bodies	3268.31	11%
2	Fallow Land	11180.44	36%
3	Open Scrub	4173.13	13%
4	Hill	1100.12	4%
5	Built-Up	2534.13	8%
6	Forest	2800.12	9%
7	Stony Waste	501.44	2%
8	Reservoir	297.00	0.96%
9	Roads	3166.75	10%
10	Mining Area	1054.00	3%
11	Plantation	901.44	3%
Total		**30976.87**	**100%**

Figure 3.6: Pie Diagram Showing Land use & Land cover area



3.6 Geology

Prakasam district has come into prominence as a granite producing district in the State in recent years, thanks to the discovery of what is known as "Galaxy Granite" in the Chimakurthy area of the District; this unique type won international recognition. The main litho units in the area comprise of granites, gneisses, gabbro and schists of Archaean and Dharwar Super group. The Proterozoic rocks comprising the Srisailam Quartzites of Cuddapah Super Group occur as the structural hills near to Markapur – Vinukonda area, which is about 80 km northwest of the study area. Shallow fluvial alluvium is restricted along the Musi river and its stream courses. The locally available and worldwide popular Chimakurthy granite is different from other granitic rocks, because of its unique origin from a 'massive concentric type' intrusive magmatic igneous body.

The granite bearing areas in Prakasam district can be divided geologically into two parts, the southern part falling in the mica-schist belt and the northern part forming a part of the granite terrain. Black granite deposits around Chimakurthy basically are anorthosite gabbro pyroxenites which have intruded into biotite-hornblende schist. The rocks are of black colour, medium grained and take good polish. Some places near Chimakurthy and R.L.Puram, the anorthosite gabbro-pyroxenite has developed crystals of bronzite (a pyroxene which is golden yellow in colour). The golden yellow crystal gives the appearance of 'star studded galaxy sky',

hence the name 'Galaxy granite'. This variety is being quarried around Chimakurthy and R.L.Puram, whereas the black gabbroic-pyroxenite is being quarried near Budavada (AP Mines and Geology, 2012).

The gabbro rock occurring in the mine area is black coloured, medium grained, and equigranular with evenly distributed brown coloured bronzite specks. Petrographically the gabbro essentially consists of plagioclase and pyroxenes with accessory minerals like bronzite, ilmenite, and garnet. The specks of bronzite impart 'galaxy' appearance to the solid gabbro after right cutting and polishing.

Petrogenesis: The Chimakurthy Igneous complex in the area is an oval shaped igneous body, comparable to the Bushveld Igneous Complex of South Africa and Skeeargard Intrusive of Greenland. With the source and solidification of magma from deeper levels in the form of deep continuity of bronzite gabbro is supposed to be occurring few kilometres below the surface. The geological field investigations indicate that the pre-existing older migmatised biotite-hornblende schist and quartzo-feldspathic formation is intruded by the emplacement of huge igneous body formed by *insitu* fraction differentiation and crystallisation process. Geomorphological expression shows that the outer bronzite gabbro rim and central core of anorthosite and pyroxinite forms plains, where as the semi-circular, crecentric shaped clino-pyroxene gabbro occur as a very high ridge with steep flanks. The detailed geological evidences suggest that the lineation of the gabbro in this area is generally trending in $N60^0E$ direction, which continues across the boundary in view of the preferred orientation of the minerals (Mining Plan, 2013).

Structural Features: Structurally three sets of major joints can be delineated in gabbro-gneissic rocks occurring in the area, especially along $N40^0E - S40^0W$, $N30^0W - S30^0E$, and $N20^0W - S20^0E$. In addition there is common occurrence of vertical to sub-vertical, horizontal to sub-horizontal sheet joints of 3 to 4 in number (Mining Plan, 2013).

3.6.1 Climate and Rain Fall:

The climate of the area is moderate and sub-tropical. The summer season is from March to May. This is followed by monsoon period from June to December, the post monsoon from October to November and the winter season from December to February. The average annual rainfall of Prakasam district is 924 mm.

Table 3.2. Climate Data of Ongole, District Head Quarters, Prakasam District, AP.

	January	February	March	April	May	June	July	August	September	October	November	December
Avg. Temperature (°C)	24.1	25.8	27.8	30.3	33	32.9	30.2	30.2	29.7	28.1	25.8	24.2
Min. Temperature (°C)	19.6	20.9	23.3	26	27.9	28.4	26.4	26.3	25.8	24.6	21.8	19.9
Max. Temperature (°C)	28.7	30.7	32.4	34.6	38.2	37.4	34.1	34.1	33.7	31.7	29.8	28.6
Avg. Temperature (°F)	75.4	78.4	82.0	86.5	91.4	91.2	86.4	86.4	85.5	82.6	78.4	75.6
Min. Temperature (°F)	67.3	69.6	73.9	78.8	82.2	83.1	79.5	79.3	78.4	76.3	71.2	67.8
Max. Temperature (°F)	83.7	87.3	90.3	94.3	100.8	99.3	93.4	93.4	92.7	89.1	85.6	83.5
Precipitation / Rainfall (mm)	4	6	5	12	39	68	98	96	128	211	145	27

December is the coldest month with normal mean maximum temperature of about 24.2°c and mean minimum temperature of 19.9°c. Temperature begins to rise after February. May is the hottest month with mean daily maximum temperature of about 38.2°C and the mean daily minimum temperature of about 27.9°C. During May and early June the maximum temperature rises occasionally to 46°C and with the one set of southwest monsoon by about second week of June, temperature begins to drop rapidly.

The relative humidity is generally high throughout the year and is of the order of 80% in the morning and 75% in the evening. During summer season particularly in the month of May the relative humidity is 71% in the morning and 64% in the evening hours. The wind speed in the district is generally light to moderate throughout the year. The special weather phenomenon prevails in the district viz., during post monsoon season wide spread heavy rain and strong winds occur when depressions in the Bay of Bengal move north-westerly direction (CGWB, 2013).

The average annual rainfall of the district is 798.6 mm. The mean seasonal rainfall distribution is 384 mm in southwest monsoon (June-September), 331 mm in northeast monsoon (Oct-Dec), 17 mm rainfall in Winter (Jan-Feb) and 68 mm in summer (March – May).

3.7 Water Environment

The industrial development of any region is contingent on the availability of sufficient water resources, as most of the process industries would require water for process or cooling purposes. The potential for exploitation of ground water resources increases as development of new projects increases in industrial and agricultural areas. With the increasing industrialization

and urbanization the possibilities of contamination of surface water and ground water sources are rapidly increasing. The water resources in the area broadly fall into following categories:

1. Ground Water resources : Accumulation in deeper strata of ground.
2. Surface Water resources : Streams, ponds, etc.

3.7.1 Ground Water Conditions

General Features:

Geomorphic features, Climate and Rainfall, Topography and Land Slope, Drainage pattern, Soils and Agricultural activities and ultimately the nature, thickness, distribution and structure of different geological formations, independently or collectively or in association with each other play a distinct role in the occurrence, movement, quality and availability of Ground Water.

Geographical set-up and Geomorphic features control, in a large measure the amount of precipitation that contributes to run off and ground water recharge. Dry conditions in arid and semi-arid climatic conditions results in large scale evaporation losses, promotes oxidation processes and results in the deep ground water level.

High rainfall contributes higher amounts of infiltration and deep percolation to recharge ground water. Development of landforms mainly depends upon the nature of the rocks, their resistance to erosion and the geologic structure, climatic conditions and vegetative cover.

Morpho-Metric parameters such as drainage density and slope characteristics provide a basis for evaluation of run-off and ground water potentials of a drainage basin. Total drainage basin arial extent determines the total quantity of water available in a basin. A low network of drainage courses is indicative of the presence of highly resistant or highly permeable rocks on the surface. High drainage density characterizes hilly terrain and areas underlain by weak or impermeable rocks result in more run-off than infiltration. Low network drainage density corresponds to high infiltration rates.

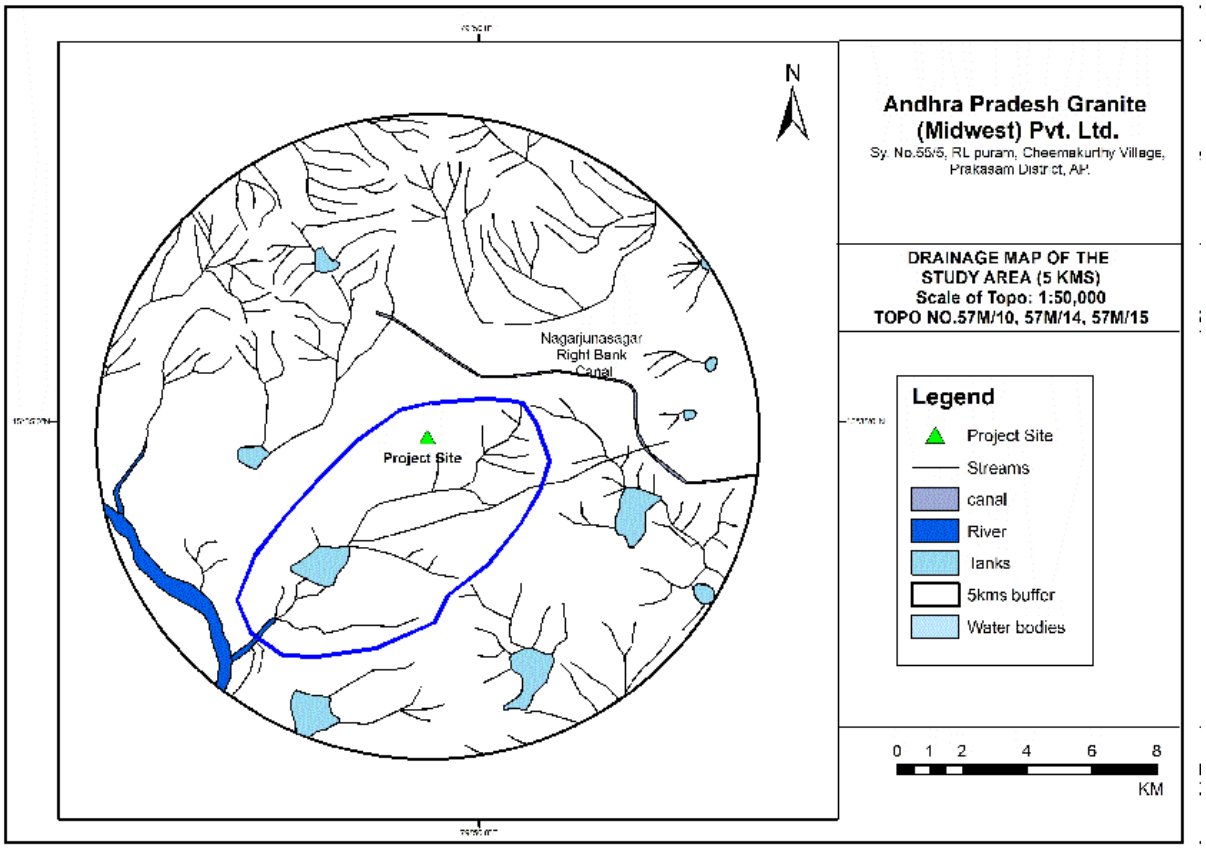
3.7.2 Water-shed and Neighbourhood Areas:

The area regionally forms part of the Paleru River Catchment System. The Musi river located 5.50 km west of the quarry site forms part of its stream. The Musi River originates in the Darsi -Donakonda hills and upland areas of Prakasam district, and flows 33 km southwardly to reach the study area, and further flows 23 km southeasterly and joins the Paleru River near Kamepalli (6 km south of Kondapi), which then flows 30 km eastwards to join the Bay of Bengal.

Major part of the pediplain in the Paleru catchment area is dissected by ephemeral streams and streamlets flowing in north and northeasterly directions. Some small and medium size lakes are situated along the topographic gradients, and most of which are connected by small ephemeral nallahs (streamlets). Other than these many small ponds and lakes are situated around the study area. These lakes receive seasonal flows, otherwise mostly remain dry.

The AP Granite (Midwest) area lies within a local catchment area referred here Pulikonda Watershed. The watershed is about 30 sq. km in areal extent, and forms part of the Musi river catchment.

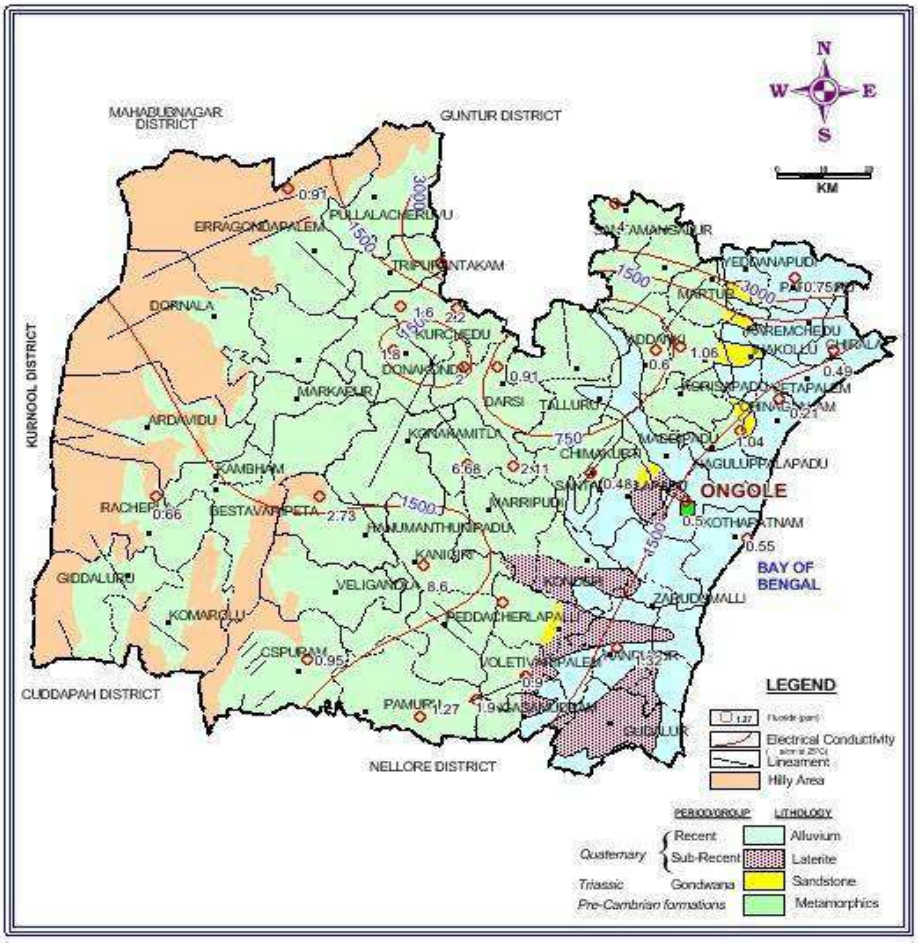
Figure-3.7: Topography and Drainage of the Project Site Neighborhood



3.7.3 Field study and discussion:

During the field study, depth of water levels and total depth of wells were measured using embossed steel measuring tape. Quality of ground water in terms of its Specific Electrical Conductance values was measured by using a sensitive pocket type Electrical Conductivity Meter. The hydro-geological data of wells inventoried and location of wells is shown in the Google Map Figure-3.8

Figure 3.8. Hydrogeology Map of Prakasam District
(Source: CGWB, 2013)



3.7.4 Ground Water Resource Estimation

In collaboration with the state departments, Central Ground Water Board (CGWB) carried out ground water resource estimation and categorization studies for the Chimakurthy Mandal of Prakasam District for the year 2008-09. The CGWB has categorized the area as 'safe' with the stage of ground water utilization of 28%, indicating little scope for further ground water development. The details are given in Table 4. These details are applicable to the study area also, as it forms part of this Mandal.

The study area mainly comprises of unconsolidated hard rock formations. The chief sources of surface irrigation are through distributaries of Nagarjuna sagar Right Canal and tanks. Ground water irrigation in the study area is not intensive, but in some areas is developed by means of dug-cum-bore wells and bore wells.

Table3.3: Ground Water Resource Estimations

S. No.	Item	Total (ham)
1.	Net Annual Ground Water Availability (1-2)	3302
2.	Existing Gross Ground Water Draft for all uses	919
3.	Provision for domestic and industrial requirement supply to the Yr 2025	187
4.	Net Ground Water Availability for future irrigation development	2266
5.	Stage of Ground Water Development %	28
6.	Category	Safe

3.8 Ground Water Quality

Quality of ground water can be expressed in terms of Specific Electrical Conductance (E.C.) which is a measure of dissolved salt content. The location of Bore wells and other sources examined are shown in the figure 3.8. The permissible limits of E. C. for drinking and irrigation purposes respectively are 1500 and 2000 μ/cm at 25 °C.

The samples were collected from 8 Bore Wells around Mine site in 10 km radius i.e., Muvvavaripalem, Mailavarm, RL puram, Mine pit, Pulikonda, Marrichettypalem, Ramateertham and chimakurthy and got them analyzed for associated parameters. The sampling location map of the area is given in Figure 3.9. Summarized Results of Chemical Analysis of Ground Water Samples is shown in the Table-3.4.

Figure3.9: Ground water sampling locations in Study area

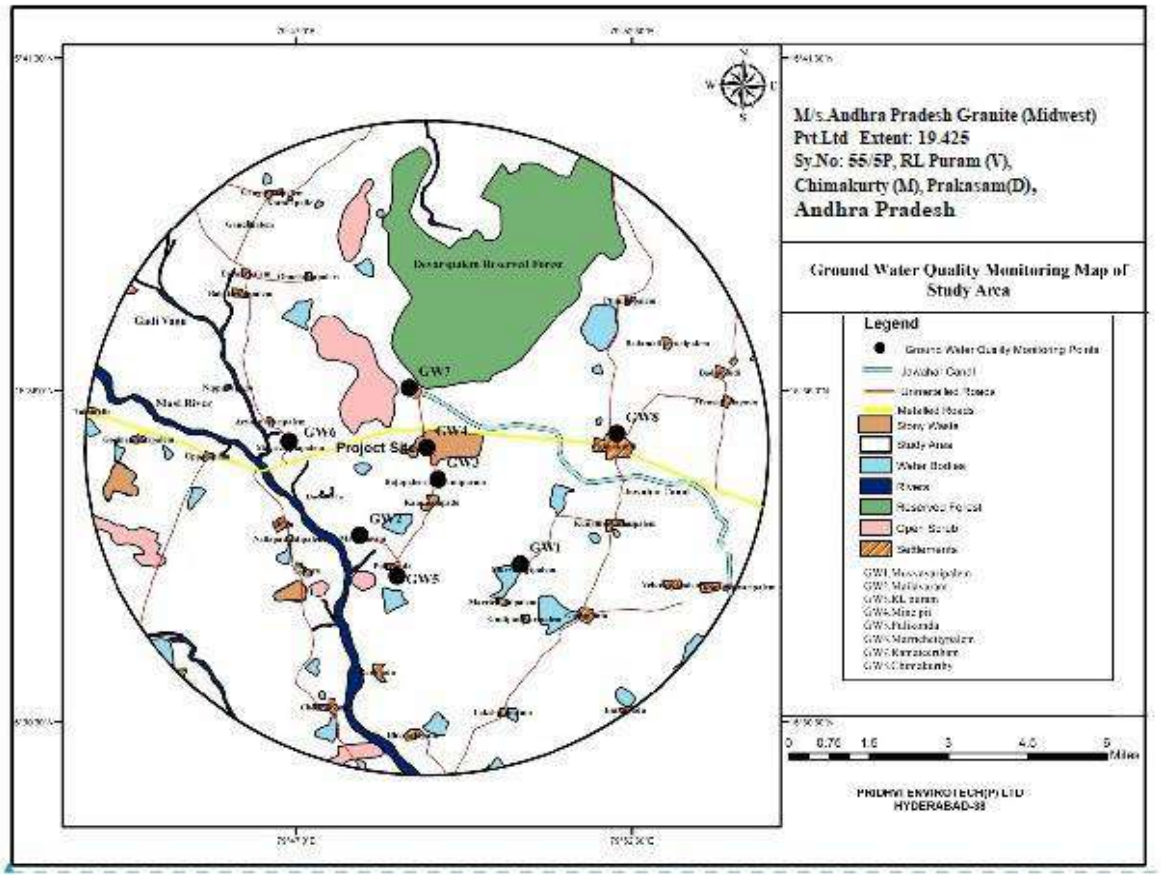


Table.3.4: Summarized Chemical analysis of Ground Water

S.No.	PARAMETER	Method APHA 22nd Edition	Unit	RESULTS				As per IS:10500 - 2012
				Muvvavaripalem Borewell	Mailavaram Borewell	RL Puram Borewell	Requirement Acceptable limit	Permissible limit in the absence of alternative source
1	pH	4500 H+B	––	7.19	7.83	7.45	6.50 - 8.50	NS
2	Color (Hazen units)	2120 B	Hazen	<1.0	<1.0	<1.0	5 max	15 max
3	Total dissolved solids	2540 C	mg/l	532	496	860	500 max	2000 max
4	Total alkalinity as $CaCo_3$	2320 B	mg/l	222.90	196.00	354.60	200 max	600 max
5	Total hardness as $CaCo_3$	2340 C	mg/l	336.00	296.00	404.00	200 max	600 max
6	Calcium as Ca	3500 Ca B	mg/l	72.00	67.00	86.40	75 max	200 max
7	Magnesium as Mg	3500-Mg B	mg/l	37.92	31.24	45.70	30 max	100 max
8	Sulphates as SO4	4500 SO_4D	mg/l	28.32	24.93	68.40	200 max	400 max
9	Chlorides as Cl	4500 Cl-C	mg/l	144.80	127.42	197.62	250 max	1000 max
10	Lead as Pb	3111B	mg/l	<0.01	<0.01	<0.01	0.01 max	No relaxation
11	Cadmium as Cd	3111B	mg/l	<0.01	<0.01	<0.01	1 max	No relaxation
12	Total Chromium as Cr	3111B	mg/l	<0.01	<0.01	<0.01	0.05 max	No relaxation
13	Copper as Cu	3111B	mg/l	0.02	0.05	0.12	0.05 max	1.5 max
14	Zinc as Zn	3111B	mg/l	0.21	0.24	0.29	5.0 max	15 max
15	Nickel as Ni	3111B	mg/l	<0.01	<0.01	0.03	0.02 max	No relaxation

16	Flourides as F	4500 F-D	mg/l	0.58	0.63	0.71	1.0 max	1.5 max
17	Aluminium as Al	3500 Al B	mg/l	<0.02	<0.02	<0.02	0.03 max	0.2 max
18	Boron as B	4500 B B	mg/l	<0.10	<0.10	<0.10	0.5 max	1 max
19	Manganese as Mn	3111 B	mg/l	<0.01	<0.01	<0.01	0.1 max	0.3 max
20	Iron as Fe	3500 Fe B	mg/l	0.16	0.19	0.23	0.3 max	No relaxation
21	Nitrates as NO_3	4500 NO_3B	mg/l	4.84	4.26	13.56	45 max	No relaxation
22	Sodium as Na	3500 Na B	mg/l	55.74	49.05	132.80	NS	NS
23	Potassium as K	3500 K B	mg/l	2.60	2.29	17.83	NS	NS
24	Odour	2150 C	––	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable
25	E.C (micromhos/cm)	2510 B	µmho/cm	921	810	1,431	NS	NS
26	Phosphorus as P	4500 P B	mg/l	<0.10	<0.10	<0.10	NS	NS
27	Carbonates as CaCo3	2320 B	mg/l	Nil	Nil	Nil	NS	NS
28	Bicarbonates as Hco3	2320 B	mg/l	222.90	239.12	432.61	NS	NS

Table3.4: Summarized Chemical analysis of Ground Water

S.No.	PARAMETER	Method APHA 22nd Edition	Unit	RESULTS				As per IS:10500 - 2012	
				Mine Pitwater Borewell	Pulikonda Borewell	Marrichettu Palem Borewell	Ramateertam Bore Well	Cheemakurthy Borewell	Permissible limit in the absence of alternative source
1	pH	4500 H+B	––	7.24	7.76	7.93	7.61	7.82	NS
2	Color (Hazen units)	2120 B	Hazen	<1.0	<1.0	<1.0	<1.0	<1.0	15 max
3	Total dissolved solids	2540 C	mg/l	468	504	802	858	1371	2000 max
4	Total alkalinity as CaCo3	2320 B	mg/l	247.52	260.55	376.10	399.00	519.15	600 max
5	Total hardness as CaCo3	2340 C	mg/l	300.00	316.00	352.00	373.00	490.00	600 max
6	Calcium as Ca	3500 Ca B	mg/l	68.80	67.20	72.00	76.30	96.00	200 max
7	Magnesium as Mg	3500-Mg B	mg/l	31.11	35.97	41.80	44.30	60.76	100 max
8	Sulphates as SO4	4500 SO4D	mg/l	55.58	58.50	86.60	91.80	94.51	400 max
9	Chlorides as Cl	4500 Cl-C	mg/l	70.54	74.25	131.10	139	347.45	1000 max
10	Lead as Pb	3111B	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	No relaxation
11	Cadmium as Cd	3111B	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	No relaxation
12	Total Chromium as Cr	3111B	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	No relaxation
13	Copper as Cu	3111B	mg/l	0.03	0.02	0.04	0.03	0.05	1.5 max
14	Zinc as Zn	3111B	mg/l	0.07	0.05	0.08	0.06	0.10	15 max
15	Nickel as Ni	3111B	mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	No relaxation

16	Flourides as F	4500 F-D	mg/l	0.71	0.62	0.80	0.83	0.91	1.5 max
17	Aluminium as Al	3500 Al B	mg/l	<0.01	<0.01	<0.01	<0.01	0.04	0.2 max
18	Boron as B	4500 B B	mg/l	<0.10	<0.10	<0.10	<0.10	<0.10	1 max
19	Manganese as Mn	3111 B	mg/l	<0.01	<0.01	<0.01	<0.01	0.05	0.3 max
20	Iron as Fe	3500 Fe B	mg/l	0.15	0.13	0.17	0.14	0.21	No relaxation
21	Nitrates as NO3	4500 NO3B	mg/l	5.51	5.80	8.60	9.12	18.48	No relaxation
22	Sodium as Na	3500 Na B	mg/l	48.36	50.90	136.12	144.29	216.90	NS
23	Potassium as K	3500 K B	mg/l	3.23	3.40	7.90	8.38	126.00	NS
24	Odour	2150 C	––	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable
25	E.C (micromhos/cm)	2510 B	µmho/cm	819	862	1,316	1,395	2,245	NS
26	Phosphorus as P	4500 P B	mg/l	<0.10	<0.10	<0.10	<0.10	0.46	NS
27	Carbonates as CaCo3	2320 B	mg/l	Nil	Nil	Nil	Nil	Nil	NS
28	Bicarbonates as Hco3	2320 B	mg/l	301.98	317.80	458.84	486.78	519.15	NS

The results indicate that ground water is of average natural type with pH ranging from 7.19 to 7.93. Against desirable limit of 250 mg/l and permissible limit of 1000 m/l of Chloride, ground water samples had a minimum of 70.5 mg/l and maximum of 347.4 mg/l indicating that ground water is within the limits of good taste, its palatability is not affected. Total Hardness of ground water ranged between 296 mg/l and 490 mg/l and is well within the limits of 300 – 600 mg/l (permissible and acceptable levels). Total Dissolved Solids ranged from 468 mg/l to 1371 mg/l. showing low salt content at all places well within the permissible ranges of 500 and 2000 mg/l respectively. The low values are attributable to the dilution through good recharge from upper terrain storm water flow. Ground water is found less mineralized with no harmful substances and all the chemical constituents well within the suitable limits for drinking and all other purposes.

3.9 Surface Water Quality

The AP Granite (Midwest) area lies within a local catchment area referred here Pulikonda Watershed. The watershed is about 30 sq. km in areal extent, and forms part of the Musi river catchment

Surface water samples were collected from 2 minor irrigation tanks i.e. Ramateertham reservoir and Jawahar canal (Nagarjuna Sagar Canal). The map showing the surface water bodies is as shown Figure 3.10.

Summarized results of chemical analysis of surface water samples are shown in the Table-3.5.

Figure3.10: Locations of surface water samples in the study area

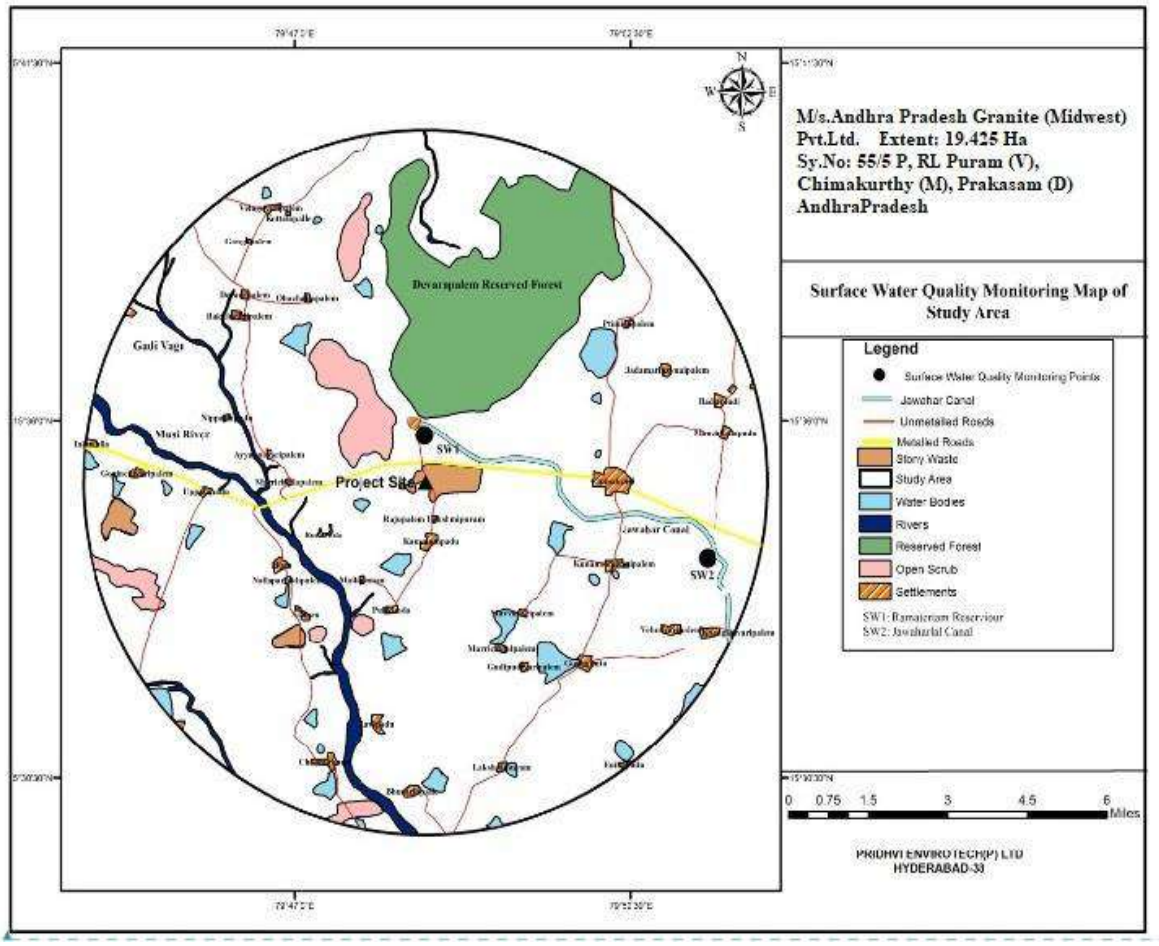


Table3.5: Summarized Chemical Analysis of Surface Water

S.No.	PARAMETER	Method APHA 22nd Edition	Unit	Ramateertham Reservoir	Jawaharlal Canal
1	pH	4500 H+B	––	8.26	8.39
2	Color (Hazen units)	2120 B	Hazen	<1.0	<1.0
3	Odour	2150 B	––	Agreeable	Agreeable
4	E.C (micromhos/cm)	2510 B	µmho/cm	431	468.00
5	Dissolved Oxygen	4500-O C	mg/l	5.30	4.90
6	Turbidity (NTU)	2130 B	NTU	<1.0	<1.0
7	Total Suspended Solids	2540 D	mg/l	<5.0	<5.0
8	Total dissolved solids	2540 C	mg/l	242	263
9	Chemical Oxygen Demand	5220 B	mg/l	8.0	6.0
10	Biochemical Oxygen Demand	IS 3025 (Part-44) 2009	mg/l	<1.0	<1.0
11	Total alkalinity as CaCo3	2320 B	mg/l	97.70	106.20
12	Total hardness as CaCo3	2340 C	mg/l	120	130
13	Calcium as Ca	3500 Ca B	mg/l	26	28
14	Magnesium as Mg	3500-Mg B	mg/l	13.37	14.59
15	Sulphates as SO4	4500 SO4D	mg/l	16.78	18.24
16	Chlorides as Cl	4500 Cl-C	mg/l	70.20	76.30
17	Lead as Pb	3111 B	mg/l	<0.01	<0.01
18	Cadmium as Cd	3111 B	mg/l	<0.01	<0.01

19	Total Chromium as Cr	3111 B	mg/l	<0.05	<0.05
20	Copper as Cu	3111 B	mg/l	<0.01	<0.01
21	Zinc as Zn	3111 B	mg/l	<0.01	<0.01
22	Nickel as Ni	3111 B	mg/l	<0.01	<0.01
23	Flourides as F	4500 F-D	mg/l	0.39	0.36
24	Aluminium as Al	3500 Al B	mg/l	<0.01	<0.01
25	Boron as B	4500 B B	mg/l	<0.10	<0.10
26	Manganese as Mn	3111 B	mg/l	<0.01	<0.01
27	Iron as Fe	3500 Fe B	mg/l	0.14	0.11
28	Nitrates as NO3	4500 NO3B	mg/l	1.50	1.62
29	Sodium as Na	3500 Na B	mg/l	41.80	45.43
30	Potassium as K	3500 K B	mg/l	3.78	4.10
31	Phosphorus as P	4500 P B	mg/l	<0.10	<0.10
32	Oil & Grease	5520 B	mg/l	<1.0	<1.0
33	Total Coliform	IS 1622	MPN/100 ml	Absent	Absent
34	Feacal Coliform	IS 1622	MPN/100 ml	Absent	Absent
35	Bicarbonates as Hco3	2320 B	mg/l	119.20	129.56

Surface water is found to be of poor quality in terms of odour (in two samples) and presence of coliform bacteria (In all samples). The results indicate the water with pH ranging from 8.26 to 8.39 and fall in the average natural type pH range of 6.5 to 8.5. chlorides of the samples ranges between 70.2mg/l and 76.3 mg/l. Total Hardness of ground water ranged between 120 mg/l and 130 mg/l and is well within the limits of 300 – 600 mg/l and Total Dissolved Solids ranged from 242mg/l to 263 mg/l. The dissolved oxygen is varying between 4.90 to 5.30 mg/lit. Overall the surface water quality in the region comes in the category B and C of IS 2296: 1982.

3.10 Soil Environment

The area is mostly covered by brownish red loamy soils. These soils are largely originated by the granitic parental rock. The thickness of soils usually varies from 0.50 to 0.80 m. along the river stream courses the alluvial soils are found. The area around the granite quarries is largely unfit for agriculture as it is strewed with dust, mined out granitic blocks, boulders and rock waste.

The Soil characteristics include both physical and chemical parameters. M/s. Pridhvi Enviro tech (P) Limited field team carried out soil survey to assess the soil characteristics of the study area. The soil sampling was done at representative locations and these locations are shown in Figure 3.11.

Figure3.11: Soil Sampling Locations

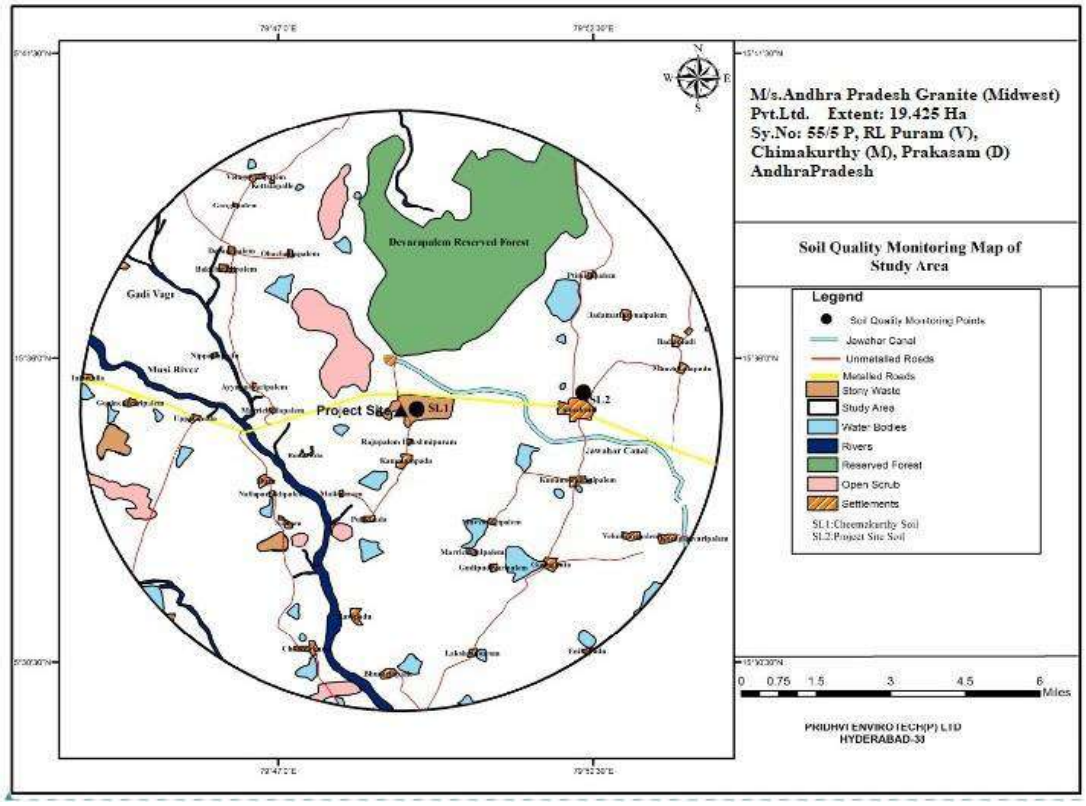


Table 3.6: Analytical data of Soil Samples

S.No.	Parameters	Unit	Chimakurthi Soil	Project Site Soil
1	pH (1:5 Soil water extract)		8.01	6.53
2	E.C (1:5 Soil water extract)	µmhos/cm	840	890
3	Bulk Density	g/cc	1.26	1.24
4	Moisture	%	5.90	4.39
5	Nitrates as N	kg/ha	90.50	86.20
6	Phosphorous as P	kg/ha	28.60	25.40
7	Potassium as K+	kg/ha	73.10	67.60
8	Sodium as Na	kg/ha	82.48	75.92
9	Calcium as Ca	mg/kg	437.44	349.53
10	Magnesium as Mg^{2+}	mg/kg	248.90	216.26
11	Sulphates as So4	mg/kg	699.40	839.28
12	Organic Carbon	%	0.82	0.94
13	Texture			
	a) Sand %	%	19.00	16.00
	b) Silt %	%	38.20	58.60
	c) Clay %	%	42.80	25.40
	Type of Soil		Clayee	Sity Loan
14	Copper as Cu	mg/kg	0.38	0.49
15	Zinc as Zn	mg/kg	0.96	1.02
16	Lead as pb	mg/kg	0.27	0.36
17	Cadmium as Cd	mg/kg	0.14	0.11
18	Chromium as Cr	mg/kg	0.05	0.03
19	Nickel as Ni	mg/kg	0.12	0.15
20	Ca/ Mg Ratio		1.75	1.62
21	Cation Exchange Capacity (CEC)	meq/kg	81.00	68.00
22	Base Saturation	%	4.60	5.30
23	Total Nitrogen as N	%	0.023	0.021

The analysis results of soil samples collected in the impact area are interpreted referring to the book; “Interpreting soil test results”. The reference tables are presented in Table 3.7.

The pH of soil ranges from Neutral to moderately alkaline. The cation exchange capacity of the soils is very high. The level of extractable potassium is low. The Ca/Mg ratio varies from balanced to low. The total Nitrogen as N is low in the area. The base saturation of the samples is predominantly very strongly leached. Bulk density of soil of impact varies from 1.24- 1.26 g/cc. Soil texture is predominantly Clayee and silty Loam.

Table 3.7 Soil Test Results – Reference Tables

General interpretation of pH measured		
pH	**Range**	**Classification**
	<4.5	Extremely Acidic
	4.51 -5.0	Very Strong Acidic
	5.1-5.5	Strong Acid
	5.6- 6.0	Moderately Acid
	6.1-6.5	Slightly acid
	6.6-7.3	Neutral
	7.4-7.8	Mildily Alkaline
	7.9 -8.4	Moderately Alkaline
	8.5-9.0	Strongly Alkaline
	>9.0	Very Strongly Alkaline

Source: Bruce and Rayment (1982).

Rating for Cation exchange Capacity	
	CEC (Cmol)+)/kg
Very low	<6 *
Low	6-12
Moderate	12-25
High	25-40
Very High	>40

Source: Metson (1961)
* Soils with CEC less than three are often low in fertility and susceptible to soil acidification.

Ca/mg Ratio	
	Description
<1	Ca Deficient
1-4	Ca (Low)
4-6	Balanced
6-10	Mg (Low)
>10	Mg deficient

Source: Eckert (1987)

Base Saturation as a criterion of leaching	
Range (%BS)	**Rating**
70-100	Very Weakly Leached
50-70	Weakly Leached
30-50	Moderately Leached
15-30	Strongly Leached
0-15	Very Strongly Leached

Source: Metson (1961)

Rating of Total Nitrogen	
Rating (% by W)	**Description**
<0.05	Very low
0.05-0.15	Low
0.15-0.25	Medium
0.25-0.50	High
>0.5	Very High

Source: Bruce and Rayment (1982)

Extractable Potassium (K)	
	K
low	<150 ppm* (< 0.4 meq/100 g soil)
medium	150–250 ppm (0.4–0.6 meq/100 g soil)
high	250–800 ppm (0.6–2.0 meq/100 g soil)
excessive	>800 ppm (>2.0 meq/100 g soil)

Source: Abbott (1989)

3.11 Air Environment

3.11.1 Meteorology

Methodology plays a vital role in effecting the dispersion of pollutants, once discharged into the atmosphere, their transport, dispersion and diffusion into the environment. The meteorological data is very useful for interpretation of the baseline information and for model study of air quality impacts also. Since meteorological data show wide fluctuations with time, meaningful interpretation can only be drawn from long term and reliable data. Such source of data is the India Meteorological Department (IMD) that maintains a network of meteorological stations at several important locations. The data recorded for nearest IMD station at Ongole is summarized for the period 1979 to 2000 and the same is represented Figure 3.12.

Also Micro Meteorological studies are conducted at the site during study period.

Figure 3.12: IMD data of meteorological station Ongole

जलवायवी सारणी
CLIMATOLOGICAL TABLE

BACK

स्टेशन : ओंगोल STATION : Ongole — अक्षांश LAT. 15° 34' N — देशांतर LONG. 80° 03' E — समुद्री तल माध्य से ऊँचाई HEIGHT ABOVE M.S.L. 12 मीटर METRES — पक्षणों पर आधारित BASED ON OBSERVATIONS 1961-1990

माह		स्टेशन का मतह दाब	वायु तापमान										आर्द्रता		मेघ की मात्रा		वर्षा						
			माध्य						चरम														
			शुष्क बल्ब	नम बल्ब	दैनिक अधिक तम	दैनिक न्यून तम	माह में उच्चतम	माह में निम्नतम	उच्चतम	दिनांक और वर्ष	निम्नतम	दिनांक और वर्ष	सापेक्ष आर्द्रता	वाष्प दाब	समस्त मेघ	निम्न मेघ	मासिक योग	वर्षा के दिनोंकी संख्या	वर्षसहित सबसे नम महीने का योग	वर्षसहित शुष्कतम महीने का योग	24 घंटोंकी सबसे भारी वर्षा	दिनांक और वर्ष	माध्य पवन गति
MONTH		STATION LEVEL PRESSURE	AIR TEMPERATURE MEAN						EXTREMES				HUMIDITY		CLOUD AMOUNTS		RAINFALL						
			DRY BULB	WET BULB	DAILY MAX	DAILY MIN	HIGHEST IN THE MONTH	LOWEST IN THE MONTH	HIGHEST	DATE AND YEAR	LOWEST	DATE AND YEAR	RELATIVE HUMIDITY	VAPOUR PRESSURE	ALL CLOUDS	LOW CLOUDS	MONTHLY TOTAL	NO. OF RAINY DAYS	TOTAL IN WETTEST MONTH WITH YEAR	TOTAL IN DRIEST MONTH WITH YEAR	HEAVIEST FALL IN 24 HOURS	DATE AND YEAR	MEAN WIND SPEED
		एच.पी.ए. hPa	डि. से °C	डि. से °C	डि. से °C	डि. से °C	डि. से °C	डि. से °C	डि. से °C		डि. से °C		प्रतिशत %	एच.पी.ए. hPa	आकाश के अष्टमांश Oktas of sky		मि.मि. mm		मि.मि. mm	मि.मि. mm	मि.मि. mm		कि.मी. प्र. घं. Kmph
जनवरी JAN	I	1014.3	22.2	20.5	29.8	20.0	32.2	18.0	36.5	22 1990	14.0	06 1962	84	22.9	2.3	1.1	11.6	0.6	88.4 1986	0.0	67.4	11 1981	4.7
	II	1010.8	26.6	22.2									67	23.3	2.1	0.6							
फरवरी FEB	I	1012.6	24.6	22.5	31.9	21.7	35.0	19.2	38.3	14 1954	14.4	22 1945	81	25.3	2.1	1.2	7.0	0.5	99.9 1959	0.0	47.0	21 1962	5.7
	II	1009.0	28.5	23.6									65	25.4	1.6	0.5							
मार्च MAR	I	1010.8	27.1	24.4	34.1	23.8	37.8	21.0	43.3	29 1953	17.5	03 1967	79	28.6	2.4	1.6	10.8	0.3	79.3 1987	0.0	55.0	11 1989	7.3
	II	1007.0	30.2	25.2									65	28.0	1.4	0.3							
अप्रैल APR	I	1007.9	29.9	26.5	36.3	26.2	40.6	23.6	44.4	26 1956	19.4	14 1945	75	31.9	2.8	1.7	11.1	0.5	89.4 1963	0.0	81.1	23 1963	9.0
	II	1003.9	31.9	27.4									69	32.7	2.4	0.7							
मई MAY	I	1004.2	32.5	27.1	39.7	28.2	44.6	24.4	46.7@	25 1980	16.1	22 1949	66	31.7	3.3	1.1	52.0	2.2	368.8 1969	0.0	196.0	18 1969	9.4
	II	1000.1	35.1	28.0									59	32.0	3.5	1.2							
जून JUN	I	1002.7	31.4	26.4	37.8	28.2	42.7	24.1	46.1	04 1953	22.4	19 1961	67	30.5	5.1	1.6	65.1	4.9	212.2 1961	5.3 1952	81.4	07 1961	11.3
	II	998.8	35.2	27.6									55	30.7	5.4	2.4							
जुलाई JUL	I	1003.1	29.6	25.5	35.4	26.8	39.0	23.7	41.8	02 1982	21.2	25 1957	72	29.5	5.6	1.9	113.6	6.4	301.9 1985	22.8 1977	176.3	20 1985	10.8
	II	999.6	33.3	26.6									59	29.4	5.8	2.6							
अगस्त AUG	I	1003.8	29.2	25.3	34.7	26.5	38.0	23.8	39.8	28 1968	21.7	27 1956	73	29.2	5.3	1.8	105.3	7.5	235.0 1957	2.8 1984	93.4	24 1971	10.3
	II	1000.2	32.7	26.4									60	29.4	5.6	2.5							
सितम्बर SEP	I	1005.8	29.0	25.5	34.7	26.0	38.2	23.4	41.0	15 1987	21.6@	29 1964	74	29.9	4.8	1.8	118.8	7.3	395.7 1949	17.3 1952	232.7	22 1949	6.9
	II	1002.2	31.6	26.3									65	29.9	5.2	2.4							
अक्तूबर OCT	I	1008.9	27.7	25.0	32.8	24.8	36.5	22.4	39.5	07 1965	18.3	24 1947	79	29.6	4.2	1.9	219.2	7.7	589.9 1969	19.7 1965	219.7	16 1944	5.1
	II	1005.7	29.5	25.5									72	29.4	4.5	1.8							
नवम्बर NOV	I	1012.0	25.1	22.7	30.6	22.5	33.4	19.6	35.3	03 1965	16.7	17 1955	80	25.9	3.5	1.6	177.8	5.9	740.4 1946	0.0	265.0	23 1990	5.3
	II	1008.9	27.6	23.8									72	26.5	3.7	1.4							
दिसम्बर DEC	I	1014.2	22.8	20.8	29.6	20.5	31.7	17.9	33.9	13 1951	15.1	27 1959	83	23.2	2.9	1.1	36.5	1.6	269.6 1967	0.0	115.8	13 1988	5.0
	II	1010.9	26.4	22.3									68	23.7	3.0	0.9							
वार्षिक योग या माध्य ANNUAL TOTAL OR MEAN	I	1008.4	27.6	24.3	34.0	24.6	44.9	17.3	46.7		14.0		76	28.2	3.7	1.5	928.8	45.4	1559.5	457.0	265.0		7.6
	II	1004.8	30.7	25.4									65	28.4	3.7	1.4			1969	1984			
वर्षों की सं NUMBER OF YEARS	I	30	30	30	30	29	30	29	46		46		30	30	30	29	30	30	47	47	47		30
	II	30	30	30									30	30	30	30							

623

जलवायवी सारणी
CLIMATOLOGICAL TABLE

स्टेशन : ओंगोल
STATION : Ongole

माह	मौसम परिघटना — के साथ दिनों की संख्या						पवन — पवन की गति के साथ दिनों की संख्या (कि. मी. प्र. घं.)				पवन — पवन की दिशा के दिनों की संख्या का प्रतिशत									मेघ — मेघ मात्रा (सघन मेघ) सहित दिनों की संख्या - अष्टमांश					मेघ — निम्न स्तरी मेघ मात्रा सहित दिनों की संख्या - अष्टमांश						दृश्यता — दृश्यता सहित दिनों की संख्या				
	उर्पण 0.3 मि.मि.या अधिक	ओले	गर्जन	कुहरा	धूल धरी आंधी	चंड वात	62 या अधिक	20-61	1-19	0	उ	उपू	पू	सपू	द	दप	प	उप	शांत	0	से-2	3-5	6-7	8	0	से-2	3-5	6-7	8	कुहरा 8	1 कि.मी. तक	1-4 कि.मी.	4-10 कि.मी.	10-20 कि.मी.	20 कि.मी. से अधिक
	WEATHER PHENOMENA						WIND													CLOUD											VISIBILITY				
	No. OF DAYS WITH						No. OF DAYS WITH WIND SPEED (Km. p. h.)				PERCENTAGE No. OF DAYS WIND FROM									No. OF DAYS WITH CLOUD AMOUNT (ALL CLOUDS) O K T A S					No. OF DAYS WITH LOW CLOUD AMOUNT O K T A S						No. OF DAYS WITH VISIBILITY				
MONTH	FPT 0.3 mm Or more	HAIL	THUN DER	FOG	DUST STORM	SQU ALL	62 Or more	20-61	1-19	0	N	NE	E	SE	S	SW	W	NW	CALM	0	T-2	3-5	6-7	8	0	T-2	3-5	6-7	8	FOG 8	UP TO 1 Km.	1-4 Kms.	4-10 Kms.	10-20 Kms.	OVER 20 Kms.
जनवरी I	0.9	0.0	0.0	2.5	0.0	0.0	0	0	21	10	26	21	3	2	2	6	2	6	32	12	7	7	4	1	19	7	4	1	0	0	0.6	3.8	9.1	9.7	7.8
JAN II							0	0	28	3	1	16	39	33	1	0	0	0	10	13	7	7	3	1	21	7	2	1	0	0	0.0	0.0	0.9	20.5	9.6
फ़रवरी I	0.8	0.0	0.2	1.2	0.0	0.0	0	0	18	10	10	8	4	9	10	16	3	4	36	11	6	7	3	1	16	7	4	1	0	0	0.1	1.6	7.5	11.9	6.9
FEB II							0	0	27	1	0	6	22	64	3	0	0	0	5	14	7	5	2	0	21	6	1	0	0	0	0.0	0.0	0.3	20.0	7.7
मार्च I	0.6	0.0	0.5	0.3	0.0	0.0	0	0	24	7	2	4	3	30	16	18	2	1	24	11	7	9	4	0	13	9	7	2	0	0	0.0	0.4	5.2	18.0	7.4
MAR II							0	0	30	1	0	2	13	78	4	0	0	0	3	17	7	5	2	0	26	4	1	0	0	0	0.0	0.0	0.3	20.8	9.9
अप्रैल I	0.8	0.0	1.0	0.2	0.0	0.0	0	0	26	4	1	2	2	43	20	16	1	2	13	8	7	10	5	0	13	8	7	2	0	0	0.0	0.0	1.4	20.7	7.9
APR II							0	1	29	0	0	2	5	86	4	1	0	0	2	11	6	8	4	1	22	6	1	1	0	0	0.0	0.1	0.4	20.2	9.3
मई I	2.8	0.0	2.7	0.0	0.0	0.0	0	1	26	4	2	3	2	25	11	21	13	10	13	6	7	10	6	2	18	7	4	1	1	0	0.0	0.2	1.4	21.2	8.2
MAY II							0	2	28	1	2	3	6	60	5	5	7	8	4	6	6	11	6	2	16	9	4	1	1	0	0.0	0.1	1.0	20.5	9.4
जून I	7.7	0.0	2.2	0.0	0.0	0.0	0	2	26	2	1	1	0	5	3	36	33	14	7	2	4	9	8	7	16	8	3	1	2	0	0.0	0.1	2.4	19.9	7.6
JUN II							0	2	26	2	2	2	3	17	6	29	23	11	7	2	3	7	12	6	7	11	8	2	2	0	0.0	0.2	2.0	19.8	8.0
जुलाई I	10.8	0.0	2.0	0.0	0.0	0.0	0	2	27	2	1	0	0	4	4	43	31	10	7	2	3	6	11	9	13	11	3	1	3	0	0.0	0.2	2.3	22.0	6.5
JUL II							0	1	28	2	2	1	2	11	5	32	26	14	7	2	2	7	11	9	6	13	8	2	2	0	0.0	0.1	2.5	21.2	7.2
अगस्त I	11.1	0.0	1.8	0.0	0.0	0.0	0	2	27	2	0	0	1	5	3	42	28	14	7	2	4	7	10	8	12	11	4	2	2	0	0.0	0.2	2.9	20.8	7.1
AUG II							0	1	27	3	2	1	3	10	4	31	23	17	9	1	3	7	13	7	5	13	9	2	2	0	0.0	0.3	1.8	21.3	7.6
सितम्बर I	10.7	0.0	4.1	0.0	0.0	0.0	0	0	24	6	4	2	1	5	4	33	19	13	19	2	5	9	9	5	11	10	6	1	2	0	0.0	0.1	2.9	19.3	7.7
SEP II							0	0	26	4	3	6	7	21	4	17	15	14	13	1	4	9	11	5	5	13	9	1	2	0	0.0	0.3	2.2	19.4	8.1
अक्तूबर I	10.3	0.0	4.5	0.4	0.0	0.0	0	0	21	10	17	17	2	3	2	12	6	8	33	4	6	9	7	5	11	10	7	1	2	0	0.0	0.7	5.0	18.2	7.1
OCT II							0	0	25	6	4	20	24	19	2	4	3	4	20	3	5	10	8	5	10	13	5	1	2	0	0.1	0.4	3.1	19.8	7.6
नवम्बर I	7.1	0.0	1.1	0.8	0.0	0.0	0	0	24	6	26	38	2	2	1	2	1	8	20	7	7	6	6	4	16	7	5	1	1	0	0.0	0.9	6.2	16.5	6.4
NOV II							0	0	25	5	7	34	26	12	1	1	0	2	17	6	6	7	7	4	15	9	4	1	1	0	0.0	0.4	3.1	19.7	6.8
दिसम्बर I	2.3	0.0	0.2	0.5	0.0	0.0	0	0	24	7	32	34	2	0	0	1	0	7	24	8	7	9	5	2	18	7	4	1	1	0	0.3	1.2	8.4	14.2	6.9
DEC II							0	0	27	4	3	31	40	12	0	0	0	2	12	7	7	10	5	2	18	9	2	1	1	0	0.0	0.0	1.6	21.5	7.9
वार्षिक योग या माध्य I	65.9	0.0	20.3	5.9	0.0	0.0	0	7	288	70	10	11	2	11	6	21	12	8	20	75	70	98	78	44	176	102	58	15	14	0	1.0	9.4	54.7	212.4	87.5
ANNUAL TOTAL OR MEAN II							0	7	326	32	2	10	16	35	3	10	8	6	9	83	63	93	84	42	172	113	54	13	13	0	0.1	1.9	19.2	244.7	99.1
वर्षों की संख्या I	30						30				30									29					29						29				
NUMBER OF YEARS II							30				30									30					30						30				

624

3.11.2 Meteorological Station at Industry Site

The micro meteorological data in the area is collected simultaneously with the ambient air quality monitoring. The station was installed in such a way that there are no obstructions for free flow of wind. Wind speed, wind direction, humidity &temperature are recorded on hourly basis in the study period. Salient features of micro meteorological data collected are as follows:

Wind Direction and Speed

The hourly wind speed and wind direction observations are computed during the seasons of study period and the same are presented in Table 3.8and the wind rose diagrams are presented in Figure 3.13. The following observations can be made from the collected data;

- The predominant wind direction is SE.
- Calm period is observed to be 31.99% during the time of monitoring.
- Mostly the wind speeds are observed to be in the range of 0 – 32 m/s.

The maximum and minimum temperature and relative humidity and total seasonal rainfall are summarized in below table. The salient features are discussed in brief as follows.

Temperature:

(a) Maximum: 43.50C, (b) Minimum: 15.0 0C and (c) Average: 29.3^{o}C during study period from micro meteorological station

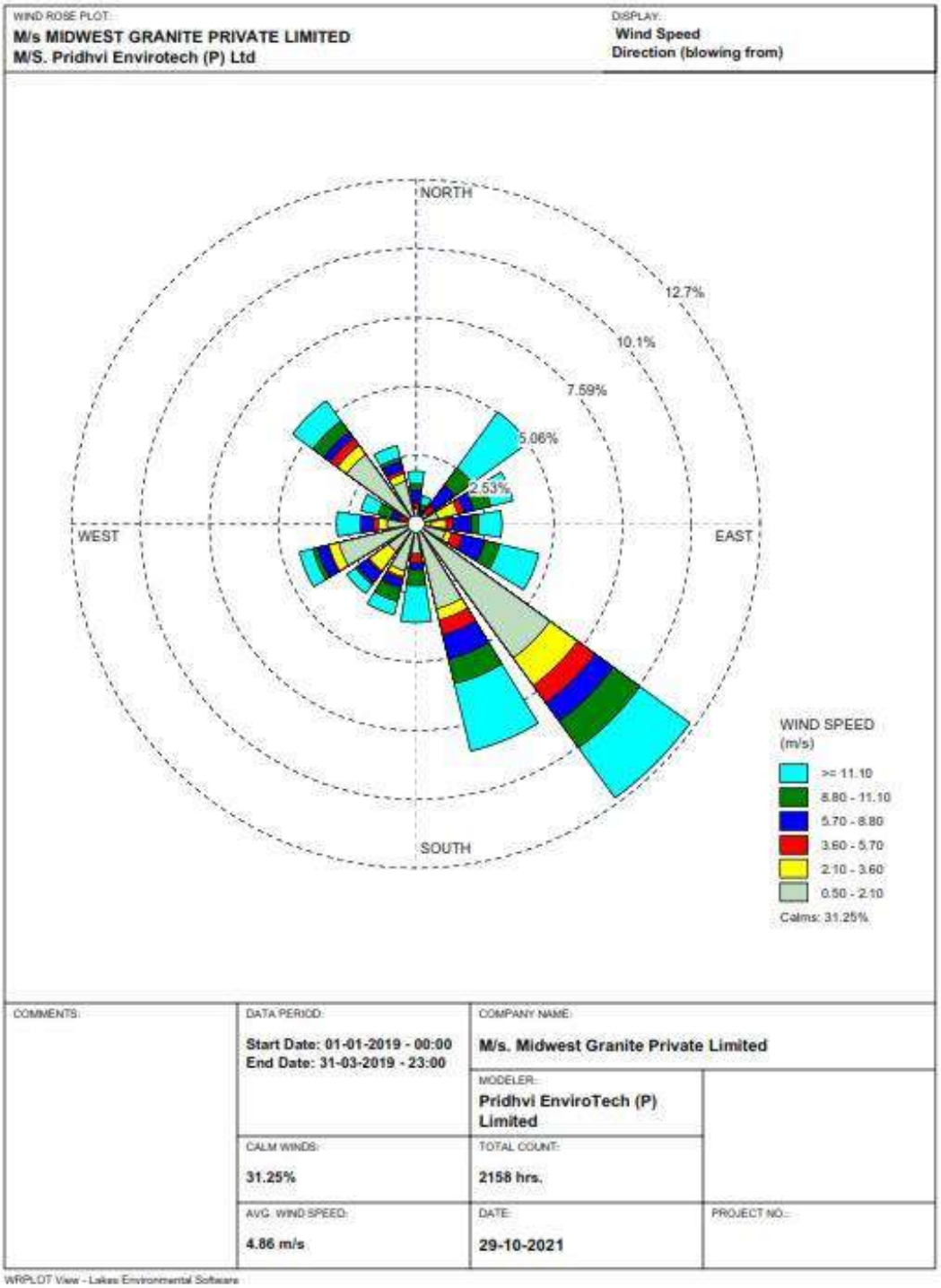
Humidity:

The daily relative humidity values are observed to range between 7.1 – 97.1%. During study period from micro meteorological station

Table 3.8: Frequency Distribution of Wind Speeds and Wind Directions

S.no	Directions / Wind Classes (m/s)	0.5 - 2.1	2.1 - 3.6	3.6 - 5.7	5.7 - 8.8	8.8 - 11.1	>= 11.1	Total (%)
1	N	0.54	0.05	0.18	0.45	0.23	0.45	1.90
2	NNE	0.05	0.18	0.18	0.14	0.18	0.32	1.04
3	NE	0.32	0.14	0.32	0.91	0.82	2.54	5.03
4	ENE	0.82	0.73	0.23	0.45	0.63	0.82	3.67
5	E	0.50	0.54	0.23	0.73	0.27	0.86	3.13
6	ESE	0.95	0.23	0.45	0.82	0.54	1.59	4.58
7	SE	5.76	1.32	0.86	0.91	1.18	2.40	12.42
8	SSE	2.99	0.41	0.63	0.95	0.91	2.72	8.61
9	S	1.00	0.09	0.27	0.23	0.63	1.45	3.67
10	SSW	1.59	0.23	0.09	0.32	0.50	0.54	3.26
11	SW	1.09	0.86	0.05	0.45	0.09	0.45	2.99
12	WSW	2.81	0.36	0.05	0.36	0.18	0.59	4.35
13	W	1.04	0.32	0.18	0.45	0.00	0.86	2.85
14	WNW	0.27	0.09	0.27	0.27	0.50	0.73	2.13
15	NW	2.95	0.45	0.32	0.27	0.50	0.91	5.39
16	NNW	1.59	0.27	0.14	0.32	0.14	0.45	2.90
	Sub-Total	24.24	6.25	4.44	8.02	7.29	17.67	67.92
	Calms							31.99
	Missing/Incomplete							0.09
	Total							100

Figure 3.13 Wind rose Diagram for the study January-2019 to March 2019



3.11.3 Ambient Air Quality

Air pollution means the presence of one or more contaminants or combinations thereof in such quantities and of such duration as are or may tend to be injurious to human, plant or animal life or property. Air pollutants include smoke, vapors, soot, fumes, gases, mist, odors, particulate matter, radioactive material or noxious chemicals. With increasing industrial activity a range of different pollutants are released into the atmosphere that are dispersed and can have a significant impact on neighborhood air environment. Thus, collection of base line data of air environment occupies a predominant role in the impact assessment statement. The ambient air quality status across the study zone forms basis for prediction of, the impacts due to the proposed project.

The data required to assess air quality impacts in and around neighborhood is achieved by designing such a network, which encompasses micro meteorological conditions, quantity and quality of emissions, locations, duration, resources/monitoring technology and operational criteria. The optimal scheme for air quality monitoring should consider all the above factors.

3.11.4 Scope of Field Study

The scope of baseline status of the ambient air quality can be accessed through a well-designed ambient air quality stations network. An intensive ambient air quality monitoring of the study area consisting of 10 km. radius from the site was carried out during the period from January 2019-March 2019.As AP Midwest Pvt Ltd are going for Production Enhancement activity. The ambient air quality was monitored at 8locations spread over entire study area. Figure 3.16 presents the locations of ambient air quality-monitoring stations. At each sampling station monitoring was carried out for 24 hours in a day for 2 days a week, and for three months. The major air pollutants monitored on 24 hourly basis are, PM10, PM2.5, Carbon Monoxide, Sulfur dioxide and Oxides of Nitrogen. Sampling and analysis of the above variables is according to the guidelines of Central Pollution Control Board.

3.11.5 Description of Sampling Locations

The location of ambient air quality stations is contingent on the meteorological status of the area. Hence, the micro meteorological data was collected before initiating the ambient air quality monitoring. Table 3.9 presents the ambient air quality locations and their distances and directions from the plant site.

Table 3.9: Locations of Ambient Air Quality Monitoring Stations

S. No	Location Name	Direction	Distance	Wind Frequency	Latitude	Longitude
1	Mine Site	--	--	--	N 15^0 35 10.60”	E 79^0 49’37.33”
2	RL Puram	SW	1.5	Down	N 15^0 34 32.87”	E 79^0 49’12.52”
3	Puli konda	SW	4.3	Down	N 15^0 33 09.46”	E 79^0 48’27.97”
4	Chimakurthy	East	3.5	Cross	N 15^0 35 10.82”	E 79^0 51’52.03”
5	Ramathirtham	NW	2.0	Down	N 15^0 35 27.07”	E 79^0 49’00.49”
6	Marri chetlapalem	NW	4.5	Down	N 15^0 34 52.31”	E 79^0 46’44.98”
7	Ayyapavaripalem	NW	5.7	Down	N 15^0 35 30.28”	E 79^0 46’39.37”
8	Muvvavaripalem	SE	3.7	UP	N 15^0 33 11.24”	E 79^0 50’39.31”

3.11.6 Ambient Air Quality Status

The existing baseline levels with respect to Particulate Matter (Size Less than 10µm) or PM10 µg/m3, Particulate Matter (Size Less than 2.5µm) or PM2.5, Sulphur dioxide, oxides of nitrogen, VOC and CO at 8 locations and ambient air quality status are presented in Table 3.10

Table 3.10: Ambient Air Quality Status

(24 hourly)(Unit;µg/m3)(VOC &CO-ppm)

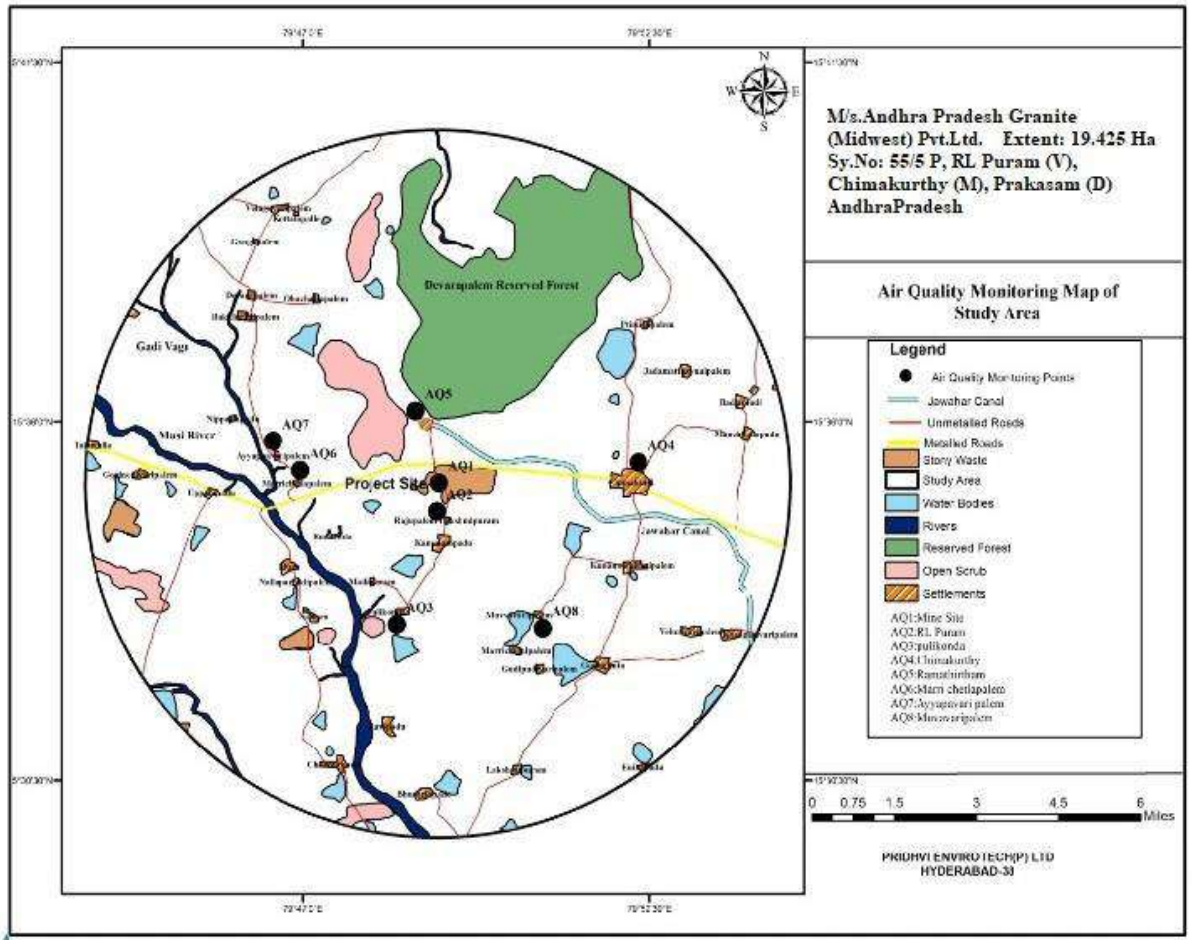
Pollutant	Maximum	Minimum	Mean	98%
1) Location: Mine Site				
PM_{10}	92.5	83.2	88.2	92.0
$PM_{2.5}$	48.8	28.5	36.0	47.1
SO_2	18.2	13.3	15.5	17.7
NOx	28.5	15.2	19.0	25.2
VOC	3.6	2.0	2.7	3.4
CO	2.3	1.2	1.8	2.3
2) Location: RL Puram				
PM_{10}	85.3	60.3	69.7	80.65
$PM_{2.5}$	40.5	20.1	24.9	35.44
SO_2	15.4	11.2	13.3	15.3
NOx	20.2	13.4	16.6	19.4
VOC	3.1	1.2	2.2	3.0
CO	2.3	1.2	1.7	2.3
3) Location: Puli Konda				
PM_{10}	72.3	63.2	67.7	71.5
$PM_{2.5}$	30.2	20.2	25.8	30.2
SO_2	16.4	10.2	12.2	15.9
NOx	19.4	14.3	16.4	19.0
VOC	3.1	1.4	2.3	3.1
CO	2.2	1.2	1.8	2.2

4) Location: Chimakurthy				
PM10	85.3	62.3	68.3	73.5
PM2.5	40.5	20.1	24.8	28.5
SO2	15.1	9.8	12.4	14.7
NOx	18.5	14.2	16.4	18.5
VOC	2.8	1.2	2.0	2.7
CO	1.8	1.2	1.5	1.8
5) Location: Ramathirtham				
PM10	69.3	55.7	64.0	68.8
PM2.5	25.4	19.2	22.7	25.3
SO2	16.4	11.5	13.6	16.1
NOx	18.2	13.0	15.1	17.4
VOC	2.8	1.0	2.0	2.8
CO	1.8	1.2	1.5	1.8
6) Location: Marri chetlapalem				
PM10	69.3	56.3	63.4	68.8
PM2.5	28.3	19.4	23.9	27.8
SO2	15.3	10.0	13.0	15.3
NOx	18.3	13.9	15.7	17.9
VOC	2.5	1.0	1.7	2.5
CO	2.0	1.2	1.5	2.0
7) Location: Ayyapavaripalem				
PM10	64.3	54.6	59.6	63.9
PM2.5	25.3	16.7	20.3	25.3
SO2	15.9	11.2	13.4	15.8
NOx	17.3	12.3	15.0	17.0
VOC	2.9	1.4	2.1	2.9
CO	2.0	1.1	1.4	2.0
8) Location: Muvvavaripalem				
PM10	62.4	53.2	58.8	62.4
PM2.5	24.1	16.8	20.1	23.3
SO2	15.3	10.0	12.8	15.3
NOx	17.3	4.7	14.6	17.0
VOC	2.8	1.5	2.3	2.7
CO	1.9	1.2	1.6	1.9

Three months Ambient Air Quality data is given at **Annexure - IV**

The Location Map of Ambient Air Quality Monitoring Stations in the study area is as shown in figure 3.14.

Figure 3.14: Location Map of Ambient Air Quality Monitoring Stations



3.11.7 Noise Environment

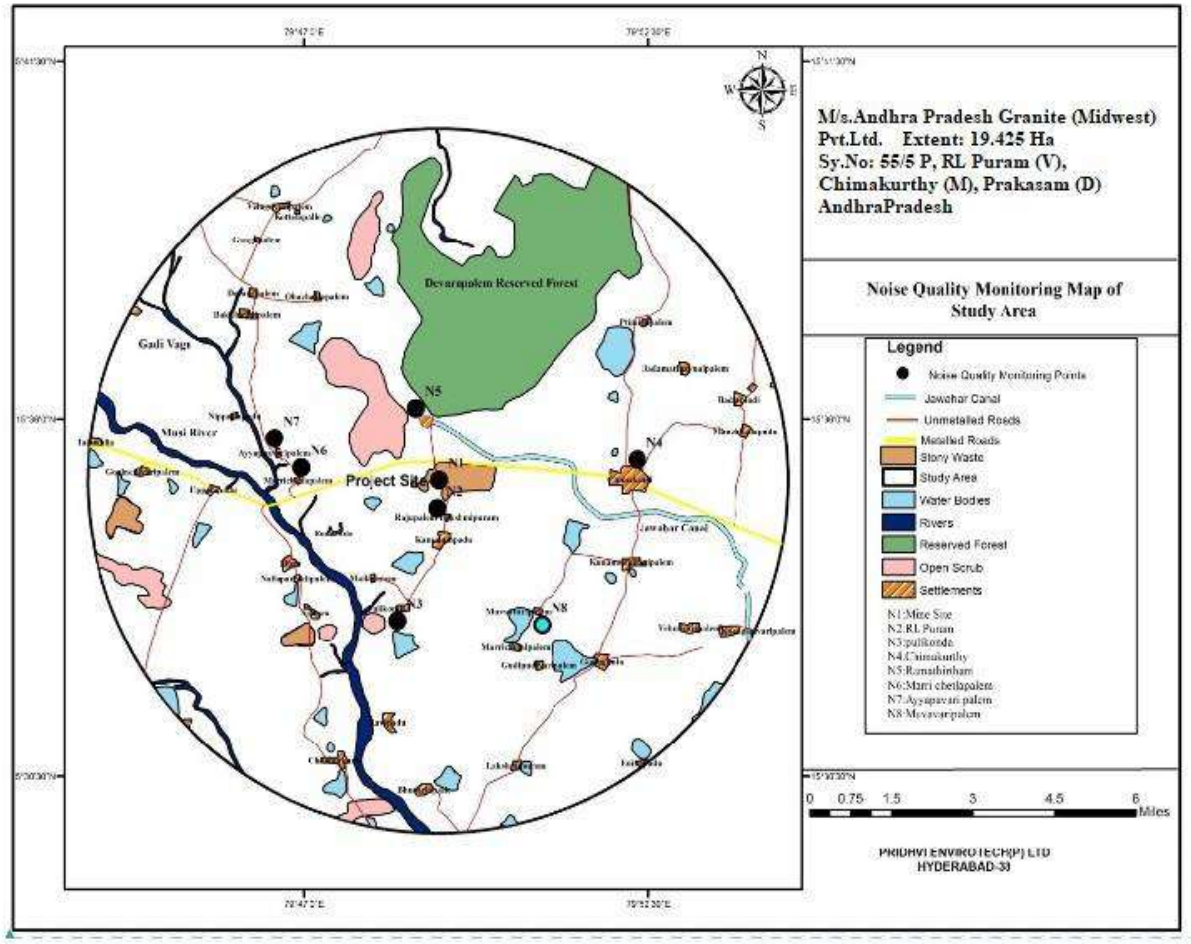
Noise is an unwanted sound without musical quality.Artificial noise and its impact on environment, grown apace with advancing human civilization. Noise pollution is equally hazardous to environment as air, water and other forms of pollution. Various noise measurement units have been introduced to describe, in a single number, the response of an average human to a complex sound made up of various frequencies at different loudness levels. The most common scale is, weighted decibel dB (A), measured as the relative intensity level of one sound with respect to another sound (reference sound).

The noise monitoring locations and their directions with respect to the project site are given in Table 3.11. And Location Map for Noise Monitoring stations is presented in Fig 3.14.

Table No.3.11: Noise Monitoring Locations

S. No	Location Name	Direction	Distance	Wind Frequency	Latitude	Longitude
1	Mine Site	--	--	--	N 15^0 35 10.60”	E 79^0 49’37.33”
2	RL Puram	SW	1.5	Down	N 15^0 34 32.87”	E 79^0 49’12.52”
3	Puli konda	SW	4.3	Down	N 15^0 33 09.46”	E 79^0 48’27.97”
4	Chimakurthy	East	3.5	Cross	N 15^0 35 10.82”	E 79^0 51’52.03”
5	Ramathirtham	NW	2.0	Down	N 15^0 35 27.07”	E 79^0 49’00.49”
6	Marri chetlapalem	NW	4.5	Down	N 15^0 34 52.31”	E 79^0 46’44.98”
7	Ayyapavaripalem	NW	5.7	Down	N 15^0 35 30.28”	E 79^0 46’39.37”
8	Muvavaripalem	SE	3.7	UP	N 15^0 33 11.24”	E 79^0 50’39.31”

Figure 3.15 Location Map for Noise Monitoring stations



The impact of noise depends on its characteristics (instantaneous, intermittent or continuous in nature), time of day (day or night) and location of noise source. Table 3.13 shows the effects of different noise levels on human beings. The environmental impact of noise can have several effects varying from noise induced hearing loss to annoying depending on noise levels.

The assessment of noise pollution on neighborhood environment due to the proposed industry was carried out keeping in view, all the considerations mentioned above. The existing status of noise levels is measured at 8 locations at various villages including the site within the study area. Figure 3.14 shows noise level measured locations. The measured noise values are shown in Table 3.13. Noise levels are high at the urban traffic junctions compared to the industrial and village areas.

Table 3.12: Effects on Human Beings at Different Noise Levels

Source	Noise Level dB(A)	Effects
Large Rocket Engine (Near By)	180	Threshold of Pains
Hydraulic Press (1m)	130	
Jet take off (60 m)	120	Maximum vocal effort possible
Automobile Horn (1m)	120	
Construction Noise (3m)	110	
Shout, Punch, Press, Circular Saw	100	Very annoying
Heavy Truck (15m), Farm Machinery	90	Prolonged exposure Endangers Lathes, Sports Car, Noisy Machines hearing loss
Automobile (15m)	80	Annoying
Loud Conversations	60	
Living Room in Home	50	Quiet
Power Station (15m)	50	
Bed Room in Home	40	
Tick of Wall clock (1m)	30	
Whisper	20	
Rattling of Leaves by Breeze	10	Barely audible

Table 3.13: Equivalent Noise levels in the Study Area

Time	N1	N2	N3	N2	N5	N6	N7	N8
6:00:00 AM	59.3	50.9	42.9	52.2	37.1	40.9	43.8	44.5
7:00:00 AM	61.4	49.1	44.9	50.5	38.8	42.8	46.3	47.0
8:00:00 AM	64.3	44.2	49.9	45.4	41.4	45.7	51.1	51.8
9:00:00 AM	65.4	52.7	53.1	54.3	44.1	52.8	50.0	50.8
10:00:00 AM	68.7	50.9	55.1	52.4	46.9	51.7	53.1	53.9
11:00:00 AM	65.3	56.0	54.2	57.6	46.5	51.3	53.2	54.0
12:00:00 PM	70.4	56.8	53.1	58.6	48.7	53.7	53.1	53.9
1:00:00 PM	65.5	52.7	52.3	54.6	41.5	49.2	51.1	51.8
2:00:00 PM	63.4	49.7	49.9	51.2	46.1	50.8	50.0	50.8
3:00:00 PM	69.3	48.9	47.9	50.4	36.9	49.8	52.0	52.7
4:00:00 PM	64.5	59.2	54.0	61.0	43.7	51.7	48.9	49.7
5:00:00 PM	69.3	50.7	54.2	52.3	44.5	50.8	45.9	46.6
6:00:00 PM	71.2	50.4	53.4	52.2	39.4	46.7	50.9	51.6
7:00:00 PM	67.3	51.0	51.6	52.6	36.8	43.6	49.9	50.7
8:00:00 PM	64.3	48.9	49.0	50.8	35.0	42.5	45.9	46.6
9:00:00 PM	63.3	44.2	46.4	45.8	33.4	41.1	40.8	41.4
10:00:00 PM	57.3	42.9	43.8	44.4	33.4	41.6	39.8	40.4
11:00:00 PM	56.7	45.2	43.3	46.9	33.1	41.2	42.0	42.7
12:00:00 AM	59.2	43.4	41.8	45.0	35.9	44.7	41.1	41.7
1:00:00 AM	58.3	41.5	43.2	42.9	35.0	43.6	42.1	42.8
2:00:00 AM	60.3	43.0	41.4	44.4	37.3	43.1	41.1	41.7
3:00:00 AM	62.3	43.8	43.9	45.1	38.3	42.2	43.2	43.9
4:00:00 AM	61.4	50.3	42.0	51.9	37.4	40.8	41.4	42.0
5:00:00 AM	62.3	52.9	43.8	55.9	34.9	42.7	40.0	40.6
Min	56.7	41.5	41.4	42.9	33.1	40.8	39.8	40.4
Max	71.2	59.2	55.1	61.0	48.7	53.7	53.2	54.0
Ld	66.9	52.8	51.9	54.5	43.4	49.6	50.2	51.0
Ln	60.2	47.4	43.0	49.6	36.0	42.7	41.5	42.1

3.11.8 Observations of Noise Levels

The noise data reveals that the values were well within the standards. In the study area maximum value 71.2 dB (A) was observed at Mine site and minimum value 33.1 dB (A) was observed at Ramathirtham village. The higher noise levels in the Mine area is due to the Mining operation and allied activities. All remaining stations are under rural (Residential) Category and all the noise levels are well within the CPCB prescribed standards as shown in Table 3.14.

Table 3.14: Ambient Noise Standards

Area Code	Category of Area/Zone	Limits in dB(A) Leq*	
		Day Time	Night Time
A	Industrial Area	75	70
B	Commercial Area	65	55
C	Residential Area	55	45
D	Silence Zone	50	40

3.12 Traffic Studies

The major road access to the site is kutcha road which is connecting to the Ongole –Kurnool state highway which is 0.3 Kms from the mine site. Traffic study is conducted on the ongole – Kurnool state highway near the mine site on 10th and 11th September, 2021 and data is presented in the table. The total volume of traffic on both directions on this double-lane road is 2505 PCUs/Day. As per the IRC code 64-1990 Guidelines for capacity of the double lane road in plane area is 15000 PCUs/day.Table-3.15 shows traffic data of Ongole –Kurnool state highway.

Table 15: Traffic Data of Ongole –Kurnool state highway

Time (Hours)	Heavy Vehicles (Buses, Single and Multi-axle trucks)		Light Vehicles (LCV's)		Two Wheels		Cars		Autos		Other slow moving traffic(Tractor trolleys)	
	Towards		Towards		Towards		Towards		Towards		Towards	Towards
	Ongole	Podili	Ongole	Podili	Ongole	Podili	Ongole	Podili	Ongole	Podili	Ongole	Podili
1to2am	3	6	1	2	1	1	1	5	1	1	0	0
2to3am	4	9	0	0	3	2	2	1	2	2	0	0
3to4am	3	8	1	1	4	2	2	4	2	1	0	0
4to5am	5	7	1	2	5	4	3	3	3	3	0	0
5to6am	5	5	3	4	4	3	4	4	3	3	0	0
6to7am	6	7	2	5	8	4	4	8	5	6	0	0
7to8am	2	6	4	6	5	5	3	7	3	9	0	0
8to9am	6	15	5	4	5	10	2	7	10	6	0	0
9to10am	15	13	2	3	9	15	10	6	9	7	0	0
10to11am	20	6	7	8	10	10	8	9	8	8	1	1
11to12pm	10	15	4	3	12	9	5	12	6	4	1	0
12to1pm	15	10	10	5	20	6	9	15	8	3	0	1
1to2pm	22	9	5	3	2	8	6	20	12	5	1	0
2to3pm	6	11	6	1	3	3	9	9	9	8	0	2

3to4pm	12	10	5	5	20	10	9	15	9	9	1	1
4to5pm	12	5	6	3	15	12	6	9	8	10	1	1
5to6pm	30	15	10	6	20	15	2	11	8	3	1	1
6to7pm	15	9	6	9	19	20	4	16	10	5	0	0
7to8pm	12	9	8	2	10	20	7	15	7	4	0	0
8to9pm	10	6	4	2	25	3	3	9	5	3	0	0
9to10pm	12	7	8	1	6	3	5	3	1	3	0	0
10to11pm	11	4	5	0	5	3	4	4	2	1	0	0
11to12am	15	5	6	1	8	6	3	1	3	2	0	0
12to1am	10	8	5	5	9	2	7	2	1	0	0	0
TOTAL Vehicles	466		195		404		313		241		13	
Factor for converting PCUs *	3		1.5		0.5		1		1		4.5	
Equivalent PCUs/day	1398		292.5		202		313		241		58.5	
PCU/Hour	58.25		12		8		13		10		2	
Total PCU/Hour	76											

***As per the IRC CODE 64-1990 Guidelines for capacity of double lane roads in Plain areas**

The estimated traffic during the mine road is 248 PCU/Day

Estimated Traffic Data – in the mine access road due to mining activity

Time (24 hours)	Heavy Vehicles (Buses, Single and Multi-axle trucks)	Two Wheels	Cars	Other slow moving traffic(Tractor trolleys)
Total	30	43	23	15
Factor for converting PCUs *	3	0.5	1	4.5
Equivalent PCUs/day	90	21.5	23	67.5
Total PCU/Day	**248**			

***As per the IRC CODE 64-1990 Guidelines for capacity rural roads in Plain areas**

3.13 Socio Economic Environment

Industrial development reflects in social development, i.e., growth in infrastructure facilities, growth in employment rates, increased demands for housing, and other amenities etc., which will have a bearing on the socio economic status. The Study area is rural in nature. The study area is dependent mostly on dry crop agriculture. The area is irrigated by ground water sources and few tanks. In this context it is necessary to study the present socioeconomic status with respect to the proposed development in the coming future.

3.13.1 Objectives of socio economic study

The primary objectives of the socio-economic impact assessment are:

i. Understanding the baseline socio-economic environment obtaining in the impact zones.
ii. Identifying the key stakeholders who are likely to be impacted by the proposed project expansion.
iii. Predicting the positive and negative impacts of the project on the socio-economic environment in the area.
iv. Suggesting mitigation measures to minimize the negative impacts.

3.13.2 Scope

The scope of socio-economic impact assessment extends to:

a) Assessing the baseline socio-economic environment prevailing in the impact area focusing the core and buffer zones.
b) Identifying key economic sectors and major sources of livelihood in the study area.

c) Understanding social structures and lifestyles of people who are likely to be affected the most.

d) Assessing physical and social infrastructure facilities accessible to the project affected people.

e) Identifying heritage precincts and archaeological monuments, if any in the area which may be affected by the proposed project expansion.

f) Predicting the likely socio-economic impacts as a consequence of the project activity.

g) Suggesting adverse impact mitigation measures in line with the felt needs, aspirations and expectations of the project affected people.

h) Preparing an appropriate Socio-economic Environment Management Plan.

3.13.3 Methodology

The socio-economic impact assessment of the proposed expansion project expansion of Chimakurthy Mining Cluster relied on a judicious mix of secondary and primary data collected from diverse sources. The SEIA was carried out in three different stages – Desk Research, Field Survey and Data Processing, Analysis and Report Preparation. The methodology adopted at each stage is explained below.

Desk Research

A fairly comprehensive desk research to understand the socio-economic setting in and around the project site was the first initiative towards carrying out the SEIA of the project. Accordingly, published and unpublished information available on the subject was referred, reviewed and critical information gaps identified by the SEIA team. The major documents and information sources extensively referred to are:

S.No.	Publication/Document	Source
1	Handbook of Statistics – Prakasam,	Planning department Govt of Andhra Pradesh
2	Census – 2011 Provisional Population Tables for Prakasam, District in Andhra Pradesh	Registrar General of India
3	Prakasam District Industrial Profile	District Industries Centre, Ongole
4	Brief Industrial Profile of Prakasam	MSME Development Institute
5	Brief Educational Institutional Details (Schools are Colleges)	District DEO Office and MEO office
6	Brief Health Status	Commissioner of Health & Family Welfare, Gollapudi, Vijayawada

Besides, unpublished information of the Departments of Industries, Education, Health, Irrigation, Agriculture and Social Welfare was also referred to gain deeper insights into the socio-economic setting of Prakasam District in general and the project area in particular.

The desk research has enabled the study team to effectively leverage the macro level socio-economic information available and identify information gaps at the village and household levels. It was during this stage, the key stakeholders were identified and study instruments – schedules and checklists – prepared, tested and finalized. Similarly, the sampling frame and sample size were also designed and finalized. The sampling frame for the study consisted of villages, households and District and Mandal level officials as also local opinion leaders.

3.13.4 Project Site

M/s. Andhra Pradesh Granite (Midwest) Pvt. Ltd., was granted Quarry Lease for mining of Black Galaxy Granite over an extent of 19.425 Hectares in Survey No. 55/5 of Rajupalem Lakshmi Puram village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh State. The site is a part of Rajupalem Lakshmipuram and is surrounded all-around by many quarries.

Chimakurthy Mandal, is situated in the North West direction of Prakasam district Head quarter, is bounded by the mandals of Santhanuthalapadu(South East), Kondapi (south), Marripudi (South West), Podili (west), Darsi, Tallur (North West) and Addanki (North East).

The project site is easily accessible from Kurnool to Ongole road at a distance of about 0.34 km, North Direction of the project site.

3.13.5 Geographical Coverage

The geographical coverage for SEIA extends over the area falling within the 10 km radius from Chimakurthy and comprises as many as 50 villages which are administratively distributed across 1 Districts, 5Mandals as detailed in Table3.16 - below.

SEIA of M/s. Andhra Pradesh Granite (Midwest) Private Limited, Survey No. 55/5 of Rajupalem Lakshmi Puram Village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh State.

VITAL STATISTICS OF SURVEY VILLAGESAS PER CENSUS – 2011

S. No.	Category/ Area Name	Mandal	Locational Direction from Project Site	Distance from Project Site (Km Aerial)	Total House Holders	Total Population	Males	Females	SC	ST	Average Literacy (%)
1	Kambhampadu	Chimakurthi	SW	1.55 Km	879	3437	1766	1671	1366	96	49.81
Sub Total -I					**879**	**3437**	**1766**	**1671**	**1366**	**96**	**49.81**
2	Rajupalem Lakshmipuram	Chimakurthi	W	2.40 Km	Under Rajupalem Lakshmipuram Revenue Population Data is Same						
3	Chimakurthy	Chimakurthi	E	2.50 Km	7577	30279	15213	15066	4058	1640	62.96
4	Muvvavaripalem	Chimakurthi	SE	3.70 Km	Under Chimakurthy Revenue Population Data is Same						

5	Mayalavaram	Chimakurthi	SW	3.70 Km	161	678	352	326	274	49	45.43
6	Pulikonda	Chimakurthi	S	3.90 Km	422	1622	827	795	754	14	57.40
7	KunanneniVariPalem	Chimakurthi	SE	4.40 Km	Under Chimakurthy Revenue Population Data is Same						
8	Budavada	Chimakurthi	SW	4.45 Km	827	3442	1684	1758	1034	115	43.93
9	Marrichetlapalem	Chimakurthi	SW	4.50 Km	Under Pulikonda Revenue Population Data is Same						
10	Ayyapa Raju Palem	Marripudi	W	4.80 Km	317	1196	594	602	261	22	40.30
Sub Total -II					**9304**	**37217**	**18670**	**18547**	**6381**	**1840**	**59.9**
11	Eluru	Podili	SW	5.05 Km	682	2669	1398	1271	638	9	37.47
12	Nallapareddipalem	Podili	SW	5.20 Km	Under Eluru Revenue Population Data is Same						
13	Gudipaduvaripalem	Chimakurthi	SE	5.20 Km	Under Gonugunta Revenue Population Data is Same						
14	T Salluru	Podili	SW	5.40 Km	113	540	278	262	228	0	30.00
15	Harijanawada	Marripudi	W	5.50 Km	Under Ayyapa Raju Palem Revenue Population Data is Same						
16	Gonugunta	Chimakurthi	SE	5.70 Km	1679	6841	3439	3402	2448	180	54.14
17	Kandurivari Agraharam	Chimakurthi	NE	5.75 Km	161	689	340	349	363	23	42.38
18	China Ravipadu	Chimakurthi	SW	5.80 Km	263	1080	537	543	282	0	44.44
19	Punurvaripalem	Podili	SW	5.90 Km	Under Eluru Revenue Population Data is Same						
20	Patimedapalem	Chimakurthi	NE	6.10 Km	Under P Naidu Palem Revenue Population Data is Same						
21	P Naidu Palem/Padamatinaidupalem	Chimakurthi	NE	6.20 Km	856	3933	1997	1936	2098	25	41.57
22	Nippatlapadu	Chimakurthi	NW	6.30 Km	645	2798	1440	1358	616	152	35.99
23	Obachettapalem	Chimakurthi	NW	6.30 Km	Under Nippatlapadu Revenue Population Data is Same						
24	Raipadu	Chimakurthi	S	6.65 Km	Under China Ravipadu Revenue Population Data is Same						
25	Uppalapadu	Podili	SW	6.70 Km	732	2991	1553	1438	934	161	46.10
26	Elurivaripalem	Chimakurthi	SE	6.80 Km	Under Chimakurthy Revenue Population Data is Same						
27	Chimalamari	Chimakurthi	E	7.10 Km	Under Manchikalapadu Revenue Population Data is Same						
28	Bakkireddypalem	Chimakurthi	NW	7.10 Km	Under Nippatlapadu Revenue Population Data is Same						
29	Manchikalapadu	Chimakurthi	NE	7.35 Km	543	2165	1110	1055	1158	2	52.66
30	Bamdlamudi	Chimakurthi	NE	7.40 Km	383	1504	773	731	583	0	43.42
31	Devarapalem	Chimakurthi	NW	7.40 Km	820	3408	1730	1678	865	164	42.22
32	Reddy Nagar	Chimakurthi	E	7.50 Km	Under Manchikalapadu Revenue Population Data is Same						
33	Bodapalem	Santhanuthala Padu	SE	7.80 Km	170	566	285	281	0	32	61.13
34	Lakshmipuram	Santhanuthala Padu	SE	7.80 Km	210	809	408	401	298	32	49.20
35	TorragudiPadu	Chimakurthi	NE	8.05 Km	329	1311	664	647	594	6	46.00
36	Busarapalli	Chimakurthi	S	8.30 Km	216	859	434	425	339	2	49.59
37	GunduvariLakshmipuram	Chimakurthi	SE	8.30 Km	198	801	410	391	105	13	55.56
38	Chilamkur	Marripudi	S	8.50 Km	373	1592	795	797	544	1	42.21
39	Gangapalem	Tallur	NW	8.65 Km	1812	7028	3590	3438	2055	72	41.63
40	Goginenivaripalem	Podili	SW	8.70 Km	Under Uppalapadu Revenue Population Data is Same						
41	Enikapadu	Santhanuthala Padu	SE	8.90 Km	513	2119	1085	1034	964	57	54.46
42	VeluguvariPalem	Tallur	NW	9.0 Km							
43	AyyalePalem	Tallur	NW	9.20 Km	Under BoddikuraPadu Revenue Population Data is Same						
44	Pandurangapuram	Chimakurthi	NE	9.70 Km	781	3436	1736	1700	1187	61	47.79

45	Yarragudipadu	Chimakurthi	NE	9.80 Km	193	898	478	420	405	74	50.22
46	Gudipadu	Santhanuthala Padu	SE	9.80 Km	544	2148	1062	1086	752	14	49.39
47	Ramachandrapuram	Chimakurthi	SE	9.90 Km	307	1271	658	613	312	76	47.13
48	Kakarla	Marripudi	SW	9.95 Km	717	3112	1581	1531	1256	43	44.60
49	Somavarappadu	Tallur	NW	9.95 Km	345	1388	689	699	355	237	45.97
50	NagambotlaPalem	Tallur	N	10.0 Km	2162	8972	4641	4331	1794	123	49.93
Sub Total -III					**15747**	**64928**	**33111**	**31817**	**21173**	**1559**	**46.38**
Grand Total					**25930**	**105582**	**53547**	**52035**	**28920**	**3495**	**51.27**

3.13.6 Impact Zones

For a more systematic impact assessment, the study area has been divided into three impact zones: core, buffer and transition.

It is evident from the data contained in Table -3.16 that of the total 50 villages/towns comprising the overall project impact zone, only 1 village fall in the core impact zone (up to 2 km in aerial distance from the project site); 10 fall in the buffer zone (>2-5 km) and the remaining 39 fall in the transition zone (beyond 5 km and up to 10 km).

Thus only 3.4% of the impact area villages accounting for 3.3% of the total population inhabiting the area fall in the core impact zone while 35.9% of villages housing 35.2% of the population fall in the buffer zone. A whopping 60.7% of the villages housing 61.5% of the study area population fall in the transition zone.

The SEIA study lays more emphasis on covering more villages in the core and buffer zones vis-à-vis the transition zone.

3.13.7 Sampled households:

A stratified random sampling technique was followed to select the sample villages and households. Accordingly, the sample villages were picked up at random from the three impact zones considered – core, buffer and minimally affected. The number of households to be contacted in each sample village was determined on the basis of the size of population of the respective village. In the absence of household level information, the respondent households were selected randomly during the course of visit to the respective village. However, while selecting the respondent households, emphasis was on contacting households, who are economically poor, susceptible to shifts in livelihood patterns and belonged to vulnerable social communities. Accordingly, the sample villages selected for the survey together with the number of households across each village are presented in Table

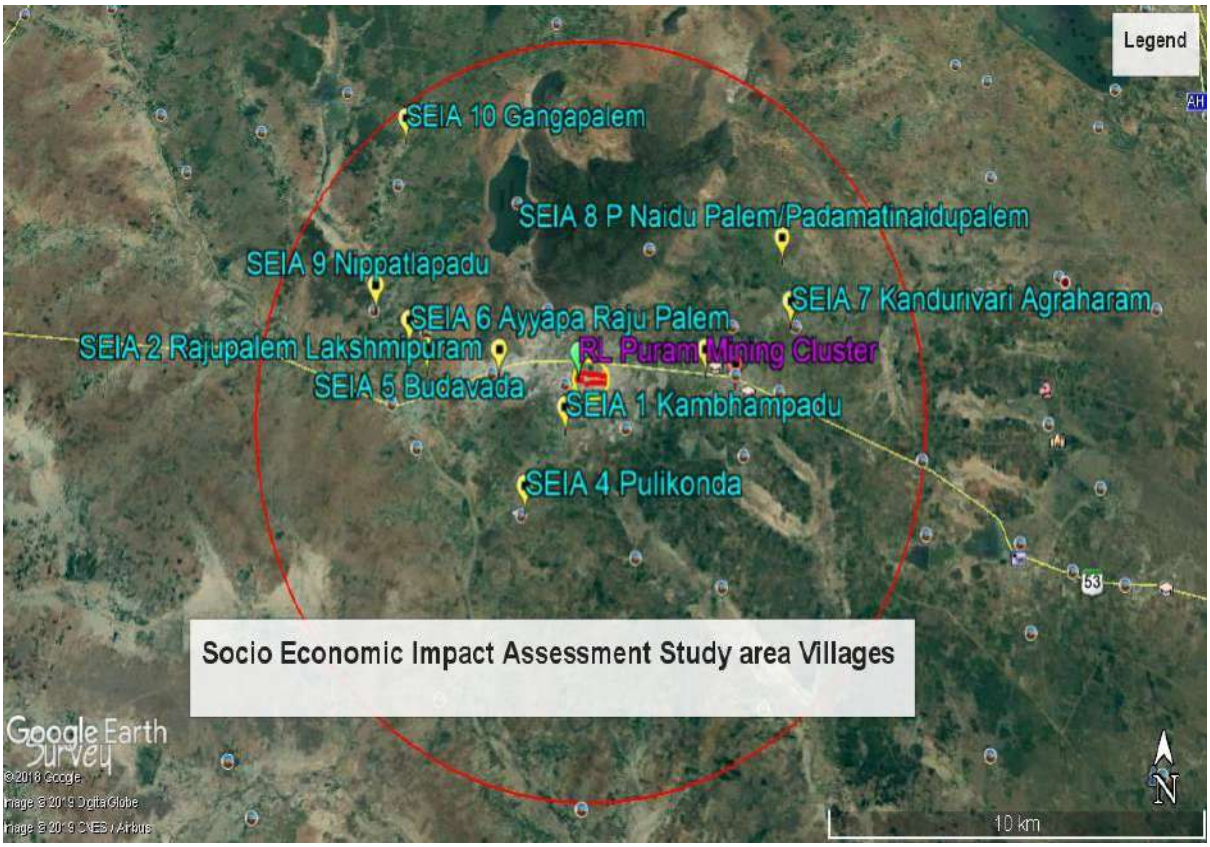
Table: 3.17: Sample Villages Selected for Socio-Economic Impact Assessment

Sample Villages Selected for Socio-Economic Impact Assessment							
S. No	**Town / Village Name**	**Mandal Name**	**Location from Project Site**		**Population(2011 Census)**	**House Holds (No.)**	
			Directions	**Distance**		**Total**	**Samples**
1	Kambhampadu	Chimakurthy	SW	1.55 Km	Under Rajupalem Lakshmipuram Revenue Population Data is Same		11
2	Rajupalem Lakshmipuram	Chimakurthy	W	2.40 Km	3437	879	12
3	Chimakurthy	Chimakurtyi	E	2.50 Km	30279	7577	14
4	Pulikonda	Chimakurthy	S	3.90 Km	1622	422	9
5	Budavada	Chimakurthy	SW	4.45 Km	3442	827	10
6	Ayyapa Raju Palem	Marripudi	W	4.80 Km	1196	317	12
7	Kandurivari Agraharam	Chimakurthy	NE	5.75 Km	689	161	11
8	P Naidu Palem/Padamatinaidupalem	Chimakurthy	NE	6.20 Km	3933	856	12
9	Nippatlapadu	Chimakurthy	NW	6.30 Km	2798	645	11
10	Gangapalem	Tallur	NW	8.65 Km	7028	1812	10
Total					**54424**	**13496**	**112**

The sample villages account for 51.54% of the population and 52.04% of the number of households of the total 50 villages

It needs to be mentioned in this context that an element of purposiveness was induced into selection of sample villages with a view to focusing more on villages in the core and buffer zones of impact. Such a selection of the sample villages was well justified as the entire area beyond the core and buffer impact zones is likely to have only minimal impact of the project and the same cannot be distinguished.

Figure: 3.16 Google Map of SEIA Sample Villages



3.13.8 Field Surveys

On the conclusion of the desk research, a four member study team was constituted. The team members have had considerable exposure and experience in conducting large size socio-economic research studies especially in Andhra Pradesh and Telangana. They are well versed with the socio-economic setting in the project area and understood the local dialect.

Planned and in-depth field surveys for Four days, between 20th to 23thJanuary, 2019 constituted the most important element of the methodology. Field surveys helped collect fairly reliable primary data with respect to the major livelihood sources, family incomes and expenditure, education and health status, basic amenities available, lifestyles, standard of living etc., of residents and small businesses in the project impact zone. They also helped in eliciting information from the natives about the negative environmental impacts of industrial units already existing in the area and the measures initiated by them to mitigate the impacts.

Field surveys were carried out in all the 10 sample villages contacting a total of 112 households. The potential respondents in the sample households were approached personally by the Field Investigators who explained the purpose of the visit and solicited their participation by sharing the relevant information unabashedly. The Field Investigators also clarified the doubts and

apprehensions expressed by the respondents. Once the respondents were willing and ready to participate, household level socio-economic information was collected with the help of a structured questionnaire. A number of questions were open ended to facilitate capturing perceptions of the respondents objectively.

In addition to household surveys, Participatory Rapid Assessment (PRA) tools comprising Transect Walks, Focus Group Discussions, and Opinion Leader Interviews were used for collecting village level qualitative information.

Accordingly, the study team carried out:

- Transect walks around all the sample villages along with a few natives.
- Carried out 3 Focus Group Discussions, comprising 10 members per group representing women, youth, household heads and opinion leaders to elicit information pertaining to their awareness, expectations and apprehensions about the proposed expansion projects.
- Detailed discussions with Sarpanch, MPDO and Chairperson-ZPTC, District Planning Officer, District Agriculture Officer, District Industries Centre Mandal Medical Officer and Lead Bank Manager to understand the major livelihood sources – existing and alternative – in the area.

3.13.9 Data Analysis

The data collected during the field survey and desk research phases was processed, tabulated and analyzed with the help of basic quantitative and qualitative analytical tools. The emerging results were validated through in-house brainstorming and discussions with district level officials.

The socio-economic impact of the proposed expansion project was assessed in terms of its effects on:

- Livelihoods and incomes
- Life styles and quality of life
- Community infrastructure – physical and social – facilities available.

3.13.10 Household Survey Results

3.13.10.1Demographics

The 112 sample households have a total population of 587 persons implying an average 5.2 members per household.51.3% of the household population are males and 48.7% females. 53.1% are adults; 29.6% children and 17.3% aged.

On the 50.0% of the households are BCs followed by SCs (15.0%), STs (8.1%), OCs (20.3%), and Minorities (6.6%).

On an average, 60.9% of the household population is formally educated – 56.0% of males and 44.0% of females.

The education profile points to that 33.5% have completed primary education followed by secondary education (49.2%), Graduation (11.6%), Post-graduation (4.3%) and vocational (1.4%).

3.13.10.2 Occupational Distribution

As per Census, 2011 the study area has a total workforce of 51,980 persons – 91.8% Main Workers and 8.2% Marginal Workers. Male workforce (60.3%) predominates.

The occupational distribution predominantly agriculture. Cultivators (19.3%) Agricultural Laborers (46.5%) and Household Industry (1.5%)

3.13.11 Livelihoods

3.13.11.1 Agriculture

The Project 10km Radios area more than village's people depend on s, agricultural activity mining activity and related services. Some members gone to work purpose in nearest cities. Subsistence farming is a traditional source of livelihood in the study area. As per Census, 2011 there are 0.09 lakh cultivators and 0.22 lakh agricultural labourers, who depend on the agricultural and allied activities for livelihood.

Agriculture in the area is primarily monsoon dependent and by and large is limited to a single crop. This area main cultivation of Cotton, Paddy, Maize and vegetables are the common food crops cultivated. Occasionally paddy cultivation is noticed in areas where sustainable irrigation facilities are available. Cultivation of a few horticultural crops.

As agricultural operations are highly seasonal and subject frequent droughts, employment is provided to rural poor under Mahatma Gandhi National Rural Employment Guarantee Act (NREGA) up to 150 days.

The younger generations do not evince much interest in pursuing agriculture and allied activities for a livelihood; they look for alternative employment opportunities based on skills.

3.13.11.2 Industry

Industrial growth in the area is relatively limited and is confined largely to micro and small enterprises. Majority of these enterprises are engaged in agriculture and other agro processing activities. All most all the enterprises are survivalist micro enterprises which are essentially as self-employment ventures. As such, less than 15% of the population depends on the industry sector for livelihood.

3.13.11.3Mineral-Based Industries

There are base in the surrounding area is predominated about 1080 Granite cutting and polishing units in and around Chimakurthi, Ballikurava, Martur, Gundlapalli, Santhanuthalapadu, Ongole, Budawada, Veluru where in Black Galaxy Granite and Colour Granite are being cut and polished. Apart from the above, there are 112 stone crusher units, 15 Silica Sand processing units, 10 Slate Cutting and polishing units, and 04 Quartz stock points are also available in Prakasam District. There is further scope for establishment of Granite cutting and polishing units basing on the granite deposits available in the District. majority of the enterprises comprise growth-oriented and survivalist micro and macro enterprises exist as self-employment ventures.

The younger population in the area looks more towards industry and services sectors for sustainable livelihoods. Roughly around 15% of them are keen to explore self-employment through establishment of micro enterprises in manufacturing and services segments. But they lack the required skills and need proper capacity building support.

3.13.11.4 Lifestyles

No reliable information/data are available, either at the district or mandal levels, about the lifestyles of people. As such, it will be very difficult to assess the lifestyles in the area with reasonable accuracy. However, based on an impressionistic assessment during the field surveys, it is felt that a majority of the population in the area residence in own accommodation. People have access to basic amenities such as protected drinking water, electric power, education and healthcare. A majority of the people in the area also have access to a mobile phone and a cable TV connection.

Generally, participation and interaction among people in the area is more common at work and during festivals, functions and cultural events. Participation in marriages and mourning is reported to be more common among homogenous social groups.

The recreation facilities in the area are reflected in watching TV, playing, cinema going and partying.

3.13.11.5 Assets Owned

The Surveyed villages various assets owned of household population owning these assets are Land, House, Tv, Fridge, Toilet, Mobile Phone, Cooking Gas, Two wheeler, Sewing Machine and other transport facilities available and good facilities available.

3.13.11.6 Sources of Livelihood

The Project 10km Radios area more than village's people depend on agricultural activity. Surveyed households reported Source of livelihood varying, Daily Wage Labour for 32.1%, Agriculture are the primary sources of livelihood for 33.2%, Pension for 7.4%, Self-employment and Private sector employment for 22.1% and Government service for 5.2%.

3.13.11.7 Family Incomes

Surveyed households reported family incomes varying predominantly between 85.0% of the family incomes >Rs. 10,000/- p.m, while 15% had incomes ranging between Rs 5001/- and Rs. 10000/- p.m.

3.13.11.8 Family Expenditures

57.6% of the households reported family expenditures ranging from >Rs. 10,000, 43.4 % Rs. 5001-10000 p.m.

3.13.11.9 Educational Facilities

The Government of India with an objective to provide universal education to all of its citizens has enacted the Right of Children to Free and Compulsory Education Act, 2009 to provide free and Compulsory Education to all the children in the age group of 6 to 14 years, in the country. In order to achieve this objective, need exists for ensuring access to education, gender parity in school enrolment, overall enrolment and retention and the quality of education being imparted and collected the data of Andhra Pradesh government Educational Institutions, Enrolment and Teachers Employed at a Glance for the year, 2016-17 detailed table in below

SCHOOLS & ENROLLMENT			
S.NO	**CATEGORY**	**TOTAL NO. OF SCHOOLS**	**TOTAL ENROLLMENT**
1	Primary	2857	188571
2	Upper Primary	630	76773
3	High Schools(Secondary)	800	200000
4	Higher Secondary	24	9298
Total		**4311**	**474642**
Source: report by DEO Prakasam 2017			

The Project 10km Radius area more than members people studying in Govt & Private educational Institutes and collecting only 10sample villages educational Intuitional information in DEO & MEO office these are detailed table in below:

Table-3.18						
Category of Educational Intuitional Details in Mandal wise						
Category	Chimakurthy		Marripudi		Tallur	
	Govt	Private	Govt	Private	Govt	Private
Anganiwadi	21	0	1	0	1	0
Primary and Upper Primary Schools	14	3	1	0	1	0
Z.P.H.Schools	2	5	0	0	0	0
Jr. Collges	2	2	0	0	0	0
Senior Colleges	0	4	0	0	0	0
Engg. Colleges	0	1	0	0	0	0
DIET/Bed/ Pharma Colleges	0	1	0	0	0	0
Model Schools, KGBV Schools & Vocational Colleges	1	0	0	0	0	0
Total	**40**	**16**	**2**	**0**	**2**	**0**
Source : MEO Chimakurthi, Tallur, Marripudi						

3.13.12 Health Status

3.13.12.1 Health and Family Welfare

Director of Public Health and Family Welfare monitors the Public Health Services, prevention and control of communicable dieses and implementation of National and State Health programmes. The DPH&FW is responsible for prevention, control and management of endemic and epidemic diseases, health promotion, management of all primary health institutions in rural and urban areas of the state and implementation of Births and Deaths ACT.

The status of health Facilities of Praksam District is detailed Table 3.19 in below:

Table3.19: The status of health Facilities of Praksam District

S.No	Health Facility available in Prakasam district	Prakasam
1	Number of Sub Centers	**536**
2	Number of PHCs	**78**
3	CHC (30 beds)	**8**
4	CHC (50 beds)	**3**
5	Area Hospital (100 beds)	**3**
6	District Hospital (200-300 beds)	**1**
7	District Hospital (300-400 beds)	**-**
Source: Sekhar, P., S. et al., 2008. Facility Survey of Public Health Institutions – 2008		

The information was collected state Sample Registration system. The right to health is the economic, social and cultural right to the highest attainable standard of health. Estimated Birth, Death, Infant Mortality, Total Fertility and Under 5 Mortality rates from 2013 to 2015 of Prakasam District detailed in Table 3.18 below:

Table 3.18

Sl. No	District	Live Births		Maternal Deaths		(MMR) Reported (per 1000 lakh live births)	
		2013-14	2014-15	2013-14	2014-15	2013-14	2014-15
1	2	3	4	5	6	7	8
	Prakasam	57,270	64,073	28	34	49	53
Sl. No	District	Live Births		Infant Deaths		(IMR) Reported (per 1000 lakh live births)	
		2013-14	2014-15	2013-14	2014-15	2013-14	2014-15
1	2	3	4	5	6	7	8
1	Prakasam	57,270	64,073	512	**439**	**9**	**7**
Source: Commissioner of Health & Family Welfare, Gollapudi, Vijayawada							

3.13.12.2 Health Status of Surveyed Villages

There are 1 Community Health Center, 4 Sub Center 3 Private clinics, 2Private hospitals and 12 RMP's in the sampled villages.7.9% of the household population reported incidence of lifestyle and other diseases. The decease incidents reported 58.4% of aged, 41.6% of active adults. The following diseases are prevalent among the sampled population in the area below in table 3.19

Table 3.19: Disease patterns observed in sampled villages

S.No	Diseases	%percentage
1	Hypertension	20.3
2	Asthma	8.0
3	T.B	1.3
4	Diabetes	23.9
5	Others (Fever, Diarrhoea, Skin allergies, backaches, arthritis etc.)	46.5

3.13.13 Physical Infrastructure

The entire study area has access to well-developed and reliable physical infrastructure - road, rail, power and telecommunication network. Both public and private modes of road transport are common.

3.13.14 Social Infrastructure

Fairly well developed and reliable infrastructure exists in the area for basic education, healthcare, housing, banking, recreation and marketing of goods and services.

3.13.15 Archaeological Monuments

No archaeological monuments and heritage precincts are observed to be present in the project impact area.

3.14 Aspirations and Apprehensions

As revealed during focus group discussions, nearly 60% of the respondents are aware of the proposed project. While they are convinced with the positive impacts of the project, they are also wary of the negative impacts on the surrounding environment. The perceptions of the respondents as regards the establishment of mining projects in the area could be summarized as follows:

- 82.2% of the households feel it would generate alternative employment opportunities for some of the locals as unskilled and skilled workers.
- 83.2% of the respondents feel it would help enhance income levels for some through the development of secondary and tertiary sector small business opportunities.
- 75.1% of the respondents feel it would help strengthen physical infrastructure, especially roads in the area.
- 76.2% of the respondents feel it would help improve the vocational skill sets of the people.
- 42.4% of the respondents feel it would adversely affect the livelihoods of the small farmers in the vicinity.
- 44.3% feel that it may impair land productivity due to dust pollution.
- 47.1% feel dust pollution may also lead to respiratory disorders of the natives.

3.15 FLORA & FAUNA

Biological environment includes the study of the relationships of organisms or groups of organisms to their environment. Ecology in essence is the study of the abiotic (non-living) and biotic (living) components, interaction of community and the environment and exchange of material (energy and nutrient) between living and non-living parts. A community includes all organisms in the given area interacting with the abiotic environment. The main aim of Conservation of Biodiversity is to ensure "No Net Loss".

The following approach has been chosen by the IAIA to help achieve 'no net loss' of biodiversity:

- Avoidance of irreversible loss of biodiversity.
- Seeking alternative solutions to minimize biodiversity losses.
- Use of mitigation to restore biodiversity resources.
- Compensation for unavoidable loss by providing substitutes of at least similar biodiversity value.
- Looking for opportunities for enhancement.

This approach can be called "positive planning for biodiversity." It helps achieve no net loss by ensuring the safety and survival of rare or endangered or endemic or threatened (REET) species. This approach has been adopted by the proposed project in the study under report.

3.15.1 Scope of work for this study is in line with the ToR that includes

Scope of work is to identify of ecologically sensitive receptors based on literature survey and field investigations, quantification of impacts on flora and fauna in core and buffer zones and to suggest appropriate mitigation measures with conservation and management plan. Biological assessment of the site was done to identify whether there are any rare, endangered, endemic, threatened (REET) species of flora or fauna in the project site or core area as well its buffer zone (upto 10 km radius). The study also designed to suggest suitable mitigation measures and conservation plan if necessary for REET species if any.

The proposed project falls in

- 6D – Deccan Peninsula Deccan Plateau as per the Biogeography Classification of India.
- Hot Semi-arid type as per the India's Köppen climate classification.

The vegetation of the study area falls under

- 5A: Southern tropical dry deciduous forests C3: Southern dry mixed deciduous forest;
- 6A: Southern tropical thorn forests DS1: Southern thorn scrub, 2S1: Secondary dry deciduous forest

by revised classification of Indian forest types (Champion and Seth, 1968). These types of forests are seen throughout the Eastern Ghats and few parts of Western Ghats of the country. The existing Mine lease area and Dump sites does not encounter with any kind of forest types like Reserve Forest, Protected forest or un- classed Forest (declared Protected under " The Indian Forest Act, 1927") and "Forest (Conservation) Act, 1980 with Amendments Made in 1988". (Source: Forest Department). No wetland notified under "The Ramsar Convention – 1971" or listed under "the National wetland Conservation Programme – 2009" is reported within 10 km from project boundary.

Crops raised in the buffer zone: Out of the total croplands, more than 70% was under Paddy, Maize, Chilies, Red gram, Black gram, Green gram and Groundnut. Remaining was a current fallow without any crops but ready for cultivation.

3.15.2 Methodology:

The ToR specific detailed survey has been made during Dec, 2018 to understand the existing situation at ground level and to suggest the mitigation measures based on impacts quantified. The core zone means the existing Mine lease area and buffer zone refers to area covered within 10 km radius of Mine lease area. The vegetation present in the existing mine lease area (MLA), dumpsite (DS) and Green belt (GB) were estimated through quadrate sampling method or belt transect method. The number and type of trees present were calculated through standard statistical methods. There are two Reserve forests (Chimakurthy RF and Devarapalem RF) are present at 2.3 km from mine lease area. Hence the tree density of the RFs also estimated by quadrate method. The proposed area for greenbelt and type of species to be planted will be also estimated and suggest the budgetary terms.

3.15.3 Equipment / Instruments deployed

- Digital Camera (NIKON 42 X zoom)
- GPS (Accurate readings available in Mobile and inbuilt camera)
- Binoculars (OLYMPUS 10 X 50 DPSI)
- Field observation book, Field guides, Pen, Measuring tape etc
- PAST –statistical software for Biodiversity.
- MS-Excel for Phyto-sociological calculations and graphs.

3.15.4 Floral analysis:

The vegetation structure of the region was randomly checked at selected habitats and sampling was done near reserve forests. Most of the buffer zone area is under cultivation and hence sampling near pond side and roadside are restricted to dense covered regions. To avoid the biased result, statistical analysis done near two reserve forests. The remaining buffer zone

survey was carried out for validating the secondary data collected at regional level. The status of each identified species was given in terms of four categories such as "Dominant", "Common", "Sporadic", and "Rare". This status is mainly based on Density, Frequency and Abundance estimated during the field visit for Trees, Shrubs and Herbs.

Phyto-sociological aspects of the study were carried out by perambulating and sampling through quadrat sampling method. Sample plots were selected in such a way to get maximum representation of different types of vegetation and plots were laid out near reserve forest blocks. Belt transect method (100m X 10m) was followed during the study. In this method, transect (a straight line) of 100 m is marked with rope and tape. The trees are sampled 5 m on either side of the central line of transect, while sampling, recorded the Girth at Breast Height (GBH) for trees with greater than 30 cm GBH. The shrubs, Herbs, Climbers, Grass species and Hydrophytes are documented with their status availability in all sampling points along with tree species. Unidentified species in the field are photographed and given a code for further investigation. Only photographs were taken during the field survey and no damage is created to flora and fauna during the sampling. None of the specimens are collected as voucher specimens for herbarium. The plants were identified using floras by Gamble (1915-36), Saxena and Brahmam (1994-96), Nair and Henry (1983), Henry et al. (1987), Henry et al. (1989) and also by using updated checklist from www.theplantlist.org.

Standard phyto-sociological methods of Mishra (1968) for density, frequency, abundance calculated during the present study. Relative values of these were calculated by following Philips (1959). Important Value Index (IVI) was calculated by adding up these three values of relative frequency, relative density and relative dominance (Curtis, 1959). In case of shrubs, herbs and saplings abundance status was given as per direct field observation. Diversity indices such as Shannon – Wiener Indices of Diversity index, Evenness, Dominance, A/F ratio are also calculated through standard methods and by using software tools such as PAST.

3.15.5 Faunal analysis:

A random survey for mammals were conducted by in all major habitats and recorded the species through direct and indirect evidences. Species were identified using "A pictorial guide to the Mammals of the India" by Vivek menon (2014), Prater (1997).

For Birds, random based observation followed with point count method was applied near water bodies and reserve forest s. Birds seen or heard 50 m radius from has been recorded during survey period. Bird surveys were carried out in dawn and desk of the field visit days. A special note on migratory status of birds were also recorded through secondary data. Birds sighted at

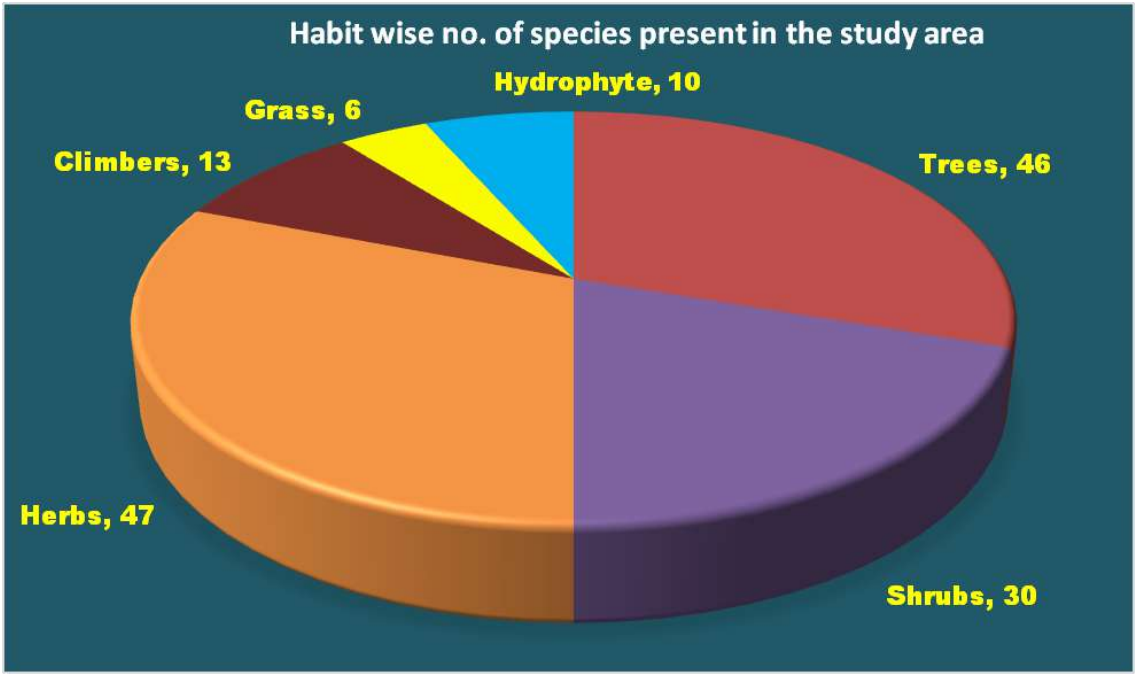
the study area were identified using "A field guide to the birds of the Indian Sub-Continent" by Ranjit Manakadan et.al (2011).

Reptiles were identified by direct or indirect evidences and literature gathered from the working plans of the forest department and other publications. The directly observed species are identified using the field book on Indian Reptiles and Amphibians by J.C. Daniel (2002). Snakes of India by Whitaker (2016).

Amphibians are surveyed both at aquatic and terrestrial systems searching under the logs and stones, digging through litter and soil, searching short bushes and tree hollows and under fallen barks. The books referred are Amphibians of Peninsular India by Ranjit Daniel (2004).

Invertebrates such as Butterflies, Dragonflies and spiders sighted during the survey period were identified by species-specific field guides. Bugs, Beetles and other insect data were gathered from publications and forest working plan data.

Vegetation structure and composition: During the present study, around 152 floral species are recorded from primary and secondary sources. The overall study area consists of agricultural fallow / Barren / uncultivable / wasteland / Reserve forests. Some part of it has commercial mango plantation and few parts are currently used as grazing land. The list of plant species recorded during field survey and literature from the study area are given in Annexure V.The Wastelands fully covered by Prosopis juliflora and the rest of the area is covered by rocky uplands with scattered succulents and thorny shrubs.Thus entire buffer area is either a cropland or current fallow land or a non-forest wasteland. There are good number is social forestry plantations of Eucalyptus and Casuarina within the buffer zone.



3.15.6 Mine Lease Area:

The existing mine lease area is adjoining to other mines under operation. Hence, there is no particular space for developing any vegetation near the boundary of the mining area. No vegetation near the approach road to Mining site. None of the community resources present in and around 500 m radial distance from the mine site boundary.

The vegetation raised near the office premises are recorded. The major tree species raised within the office premises are:

Botanical Name	**Common name**	**Family**	**Habit**
Albizia lebbeck (L.) Benth.	Dirisanam	Leguminosae	Tree
*Azadirachtaindica*A.Juss.	Vepa	Meliaceae	Tree
Dalbergia sissoo DC.	Sissoo	Leguminosae	Tree
Delonix regia (Hook.) Raf.	Chittikesaram	Leguminosae	Tree
Eucalyptus globulus Labill.	Neelagirithylam	Myrtaceae	Tree
Ficusbenghalensis L.	Marri	Moraceae	Tree
Leucaena leucocephala(Lam.) de	Jabarichettu	Leguminosae	Tree
*Nerium odoratum*Lam.	Erraganneru	Apocynaceae	Tree
Peltophorumpterocarpum(DC.)	Kondachintha	Leguminosae	Tree
Senna siamea Lam.	Seema thangedu	Leguminosae	Tree
*Phyllanthus reticulatus*Poir.	Nallapuli	Phyllanthaceae	Shrub
*Cissusquadrangularis*L.	Nalleru	Vitaceae	Climber
Pergulariadaemia (Forssk.)	Dustapu-Teega	Apocynaceae	Climber

3.15.7 Dump

Few sporadically naturally grown trees such as Eucalyptus and Senna auriculataare observed along with Prosophis which is a common weed throughout the region. No possibility of growing grass on dump slopes to stabilise them and to avoid the dust as there is no soil binding capacity.

3.15.8 Reserve forest vegetation:

Chimakurthy RF contain good number of trees such as Buchananialanzan, Azadirachtaindica, Balanitesaegyptiaca, Butea monosperma, Phoenix sylvestris, Acacia nilotica, Dalbergia sissoo etcand shrubs such as Senna auriculata,Calotropis procera, Carallumaumbellata, Euphorbia antiquorum, Euphorbia tortillis, Dodonaea viscoseJatropha gossypifolia, Carissaspinarum, Prosopis juliflora, Acacia nilotica, Acacia leucophloea, and Lantana camara and other all common herbs and climbers.

3.15.9 Quantitative analysis

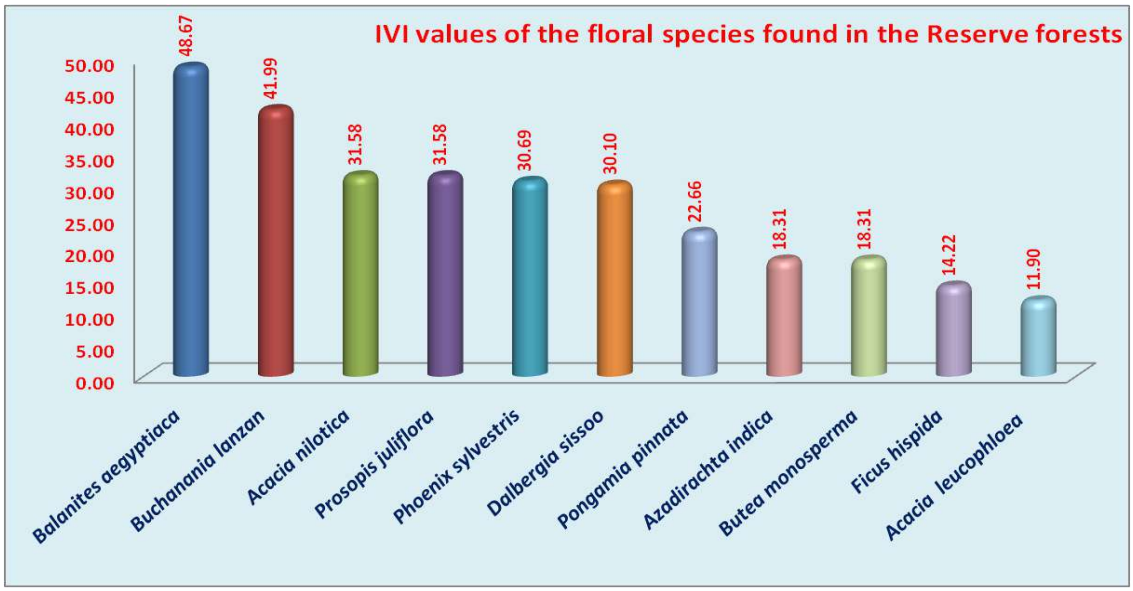
The study area is mainly focused through secondary data validation from primary observations. Checklist is prepared and marked the species noticed during rapid assessment. 10 sampling

points are through Area-Species graph. And the quantitative analysis done based on the methodology adopted for various species.

Table3.20: Phytosociological data of study area

S.No	Scientific Name	Rel Density	Rel Frequency	Rel Abundance	IVI
1	*Azadirachtaindica*	4.69	7.69	5.93	18.31
2	*Balanitesaegyptiaca*	20.31	12.31	16.05	48.67
3	*Acacia leucophloea*	2.34	4.62	4.94	11.90
4	*Buchananialanzan*	16.41	10.77	14.81	41.99
5	*Dalbergia sissoo*	10.16	10.77	9.17	30.10
6	*Ficushispida*	3.13	6.15	4.94	14.22
7	*Phoenix sylvestris*	10.16	7.69	12.84	30.69
8	*Butea monosperma*	4.69	7.69	5.93	18.31
9	*Acacia nilotica*	10.94	10.77	9.88	31.58
10	*Pongamiapinnata*	6.25	10.77	5.64	22.66
11	*Prosopis juliflora*	10.94	10.77	9.88	31.58
		100	100	100	300.00

Figure3.17: Graph showing Important Value Index of dominant tree species within study area



3.15.10 Vegetation Analysis:

Table3.21: Biodiversity indices values of the sampling sites in project site

a	A/F value	0.039
b	Shannon H	2.855
	Simpson 1-D	0.940
c	Dominance D	0.060
	Evenness e^H/S	0.946

a. Distribution pattern (A/F ratio): The ratio between abundance and frequency was used to interpret the distribution pattern of species (Whitford, 1949). Abundance to frequency ratio (A/F) has been calculated to assess the distribution pattern of species and depending upon the ratios, distribution may be regular (<0.025), random (0.025-0.05) and contagious (>0.05). In natural conditions, contagious distribution is most common type of distribution due to significant variation in environmental conditions (Odum, 1971). Distribution pattern of species in the study area is identified as **random distribution** as the value of A/F ratio is **0.039.** This distribution of species is random because of several ecosystems randomly distributed in the study area.

b. The Shannon indices value of study area is 2.855indicates Good diversity (Normal diversity in ecological studies is 1.5 to 3.5 range (Kerkhoff, 2010). As there are two RFs within the buffer zone of the study area, diversity is more near undisturbed habitats.

c. Population size and Dominance of the species is 6% and Evenness is around 95% (Indicates the species are evenly distributed in the study area). This might be due to contiguous patches of natural species such asBalanitesaegyptiaca, Butea monosperma, Phoenix sylvestris, Acacia nilotica, Dalbergia sissoo, Prosopis juliflora(near the villages),Peltophorumpterocarpum, Delonix regia and Subabul (near road side).

d. Frequency indicates A<B<C>D>E as per the Raunkiaer's law of frequency classification indicates that species diversity is distributed maximum for 40 to 60%

3.15.11 Terrestrial Fauna

The presence or absence of an animal or plant in a certain region depends on its ecological and geographical setting. Wild animals can exist in a region only if prevailing set of conditions are congenial for their survival and perpetuation. Diversity of the species and their abundance is largely dependent on the availability of required habitats.

3.15.12 Terrestrial fauna of the Core area and the Buffer zone:

As the animals, especially vertebrates move from place to place in search of food, shelter, mate or other biological needs, separate lists for core and buffer areas are not feasible however, a separate list of fauna pertaining to core and buffer zone are listed separately. Though there are two reserved forest blocks near dump and in the buffer zone. As such there are no chances of occurrence of any rare or endangered or endemic or threatened (REET) species within the core or buffer area. There are no Sanctuaries, National Parks, Tiger Reserve or Biosphere Reserve or Elephant Corridor or other protected areas within 10 km radius from core area. It is evident from the available records, reports and circumstantial evidence that the entire study area including the core and buffer areas were free from any endangered animals. There were no

resident birds other than common bird species such as Paddy egrets, Green bee eaters, Indian rollers, Parakeets, common babblers, Weaver birds, Mynas, Black dragons, Crows, Sparrows. The list of Mammalian species directly sighted or recorded during field survey and from literature reviewed from the study area are given in Annexure V The list of bird species recorded during field survey and literature from the study area are given in Annexure V. The list of reptilian species recorded during field survey and literature from the study area are given in Annexure V. The list of Amphibian species recorded during field survey and literature from the study area are given in Annexure V. The list of Fish species recorded during field survey and literature from the study area are given in Annexure V. The list of Butterfly species recorded during field survey and literature from the study area are given in Annexure V. It is apparent from the list that none of the species either spotted or reported is included in Schedule I of the Wildlife Protection Act. Similarly, none of them comes under the REET category. Some of the birds listed were rare locally but they do not fall under any of the REET categories.

3.15.13 Flora and fauna of Aquatic ecosystem

There is Ramathirtham Reservoir exist at 4.4 km from the Mine area. Lemna minor, Ipomoea aquatica, Neptunia oleracea, Typha domingensis are most common in the study area. There were no threatened category species of aquatic and semi aquatic plants in the study area. The floristic survey of the existing farming site witnesses scattered growth of grasses (mainly weeds), rooted hydrophytes, emerging hydrophytes, shrubs in undisturbed area where human movement is absent. Among herbaceous species, weeds are commonly reported such as Congress grass (Parthenium hysterophorus), Lantana (Lantana Camara), and Datura (Datura stromonium). Rooted hydrophytes are reported along the banks, where water level is less than foot or water has recedes after flooding. The commonly reported hydrophytes are Ipomoea aquatica, Argeratumconyzoides, Typha angustataetc.

Chapter - 4

Anticipated Environmental Impacts & Mitigative Measures

CHAPTER -4
ANTICIPATED ENVIRONMENTAL IMPACTS & MITIGATIVE MEASURES

4.1 Identification of Environmental Impacts

Any developmental activity in its make will bring about some impacts associated with its origin, which can be broadly classified as reversible, irreversible, long and short-term impacts. In this chapter, an endeavor has been made to identify various environmental impacts associated with the mining processes and other related activities wherein, there may be a chance of pollution.

The major phases in the mining which are likely to impact the environment are:

1. **Exploration:** Including surveys, field studies, drilling and exploratory excavations; some land disturbances and waste already occur at this stage
2. **Project Development:** Includes roads and buildings, access trenches, over burden stripping and placing, preparation of disposal areas, construction of service infrastructure, power lines, water supplies and sewerage.
3. **Associated transport & Storage of minerals** may be a handling risk and can result in localized site contamination
4. **Mine Closure:** Rehabilitation is best done progressively rather than at the end of life of the mine; while mine closure is intended to mitigate environmental impacts, it is important that it does not itself create secondary effects through excessive fertilizer use, spread of weeds, silting and incompatible landscape features

4.2.2 Impacts on Air Quality

The primary air pollutant of concern at mining sites is particulate matter. Following are various activities will be carried out in the mine site due to which Dust emission will be generated.

- Removal of top soil will generate dust emissions.
- Excavation can be source of fugitive dust. Loading of Rough Stone, Road Metal & Manufactured Sand generate dust.
- Machinery and vehicles travel on access and haul roads are major sources of dust emission.

- Dust will be generated due to the vehicular movement. Also material fall from transport vehicles will also contribute to dust emissions.

Predicted Emissions:

Emissions from the production of Rough Stone, Road Metal & Manufactured Sand consist primarily of particulate matter (PM) and particulate matter less than 10 micrometres (PM10) in aerodynamic diameter, which are emitted by many operations at Rough Stone, Road Metal & Manufactured Sand mines, such as excavation and transportation operations. The main air pollutants respirable dust arises from excavation and transportation operation.

The tippers will move with a speed of 20 to 30 kmph on the unpaved roads constructed for gravel transportation. The width of all unpaved mine roads within the mine site is around 10 m.

The details of emissions expected from vehicular movement during mining activity are given in **Table 4.1**. The line source emission rates for the proposed Rough Stone, Road Metal & Manufactured Sand mine are derived from the following equation:

$$E_{vt} = 5.9 \left(\frac{S_p}{12}\right) \left(\frac{V_v}{30}\right) \left(\frac{M_v}{3}\right)^{0.7} \left(\frac{W_v}{4}\right)^{0.5} \left(\frac{365-D_p}{365}\right)$$

Where,

E_{vt}	-	Emission factor (lb / vehicle / mile travelled)
Sp	-	Silt content of road surface material
Vv	-	Mean vehicle velocity (miles / hour)
Mv	-	Mean vehicle mass (tons)
Wv	-	Mean number of wheels
Dp	-	Number of days per year with at least 0.01 inch of precipitation

Simulation Model for Prediction using Industrial Source Complex ISC AERMOD View

The mathematical model used for predictions on air quality impact in the present study is ISC-AERMOD View-9.9.0. It is the next generation air dispersion model, which incorporates planetary boundary layer concepts.

The AERMOD is actually a modelling system with three separate components:

AERMOD (AERMIC Dispersion Model), AERMAP (AERMOD Terrain Pre-processor) and AERMET (AERMOD Meteorological Pre-processor).

Special features of AERMOD include its ability to treat the vertical in homogeneity of the planetary boundary layer special treatment of surface releases, irregularly shaped area sources, a plume model for the convective boundary layer, limitation of vertical mixing in the stable boundary layer, and fixing the reflecting surface at the stack base.

The AERMET is the meteorological pre-processor for the AERMOD. Input data can come from hourly cloud cover observations, surface meteorological observations and twice-a-day upper air soundings. Output includes surface meteorological observations and parameters and vertical profiles of several atmospheric parameters.

The AERMAP is a terrain pre-processor designed to simplify and standardize the input of terrain data for the AERMOD. Input data include receptor terrain elevation data. Output includes, for each receptor, location and height scale, which are elevations used for the computation of airflow around hills.

4.2.2.1 Micro Meteorological Data

Meteorological data was collected during Study Period (January 2019 – March 2019) a cumulative wind rose diagram of all the 3 months were compiled using mathematical model WRPLOT. The seasonal wind rose diagram has been presented in Fig: 4.1 which covers all the three months meteorological data.

Table 4.1: Frequency Distribution of Wind Speeds and Wind Directions

S.no	Directions / Wind Classes (m/s)	0.5 - 2.1	2.1 - 3.6	3.6 - 5.7	5.7 - 8.8	8.8 - 11.1	>= 11.1	Total (%)
1	N	0.54	0.05	0.18	0.45	0.23	0.45	1.90
2	NNE	0.05	0.18	0.18	0.14	0.18	0.32	1.04
3	NE	0.32	0.14	0.32	0.91	0.82	2.54	5.03
4	ENE	0.82	0.73	0.23	0.45	0.63	0.82	3.67
5	E	0.50	0.54	0.23	0.73	0.27	0.86	3.13
6	ESE	0.95	0.23	0.45	0.82	0.54	1.59	4.58
7	SE	5.76	1.32	0.86	0.91	1.18	2.40	12.42
8	SSE	2.99	0.41	0.63	0.95	0.91	2.72	8.61
9	S	1.00	0.09	0.27	0.23	0.63	1.45	3.67
10	SSW	1.59	0.23	0.09	0.32	0.50	0.54	3.26
11	SW	1.09	0.86	0.05	0.45	0.09	0.45	2.99
12	WSW	2.81	0.36	0.05	0.36	0.18	0.59	4.35
13	W	1.04	0.32	0.18	0.45	0.00	0.86	2.85
14	WNW	0.27	0.09	0.27	0.27	0.50	0.73	2.13
15	NW	2.95	0.45	0.32	0.27	0.50	0.91	5.39
16	NNW	1.59	0.27	0.14	0.32	0.14	0.45	2.90
	Sub-Total	24.24	6.25	4.44	8.02	7.29	17.67	67.92
	Calms							31.99
	Missing/Incomplete							0.09
	Total							100

Figure 4.1: Wind rose Diagram for the study period January 2019 to March 2019

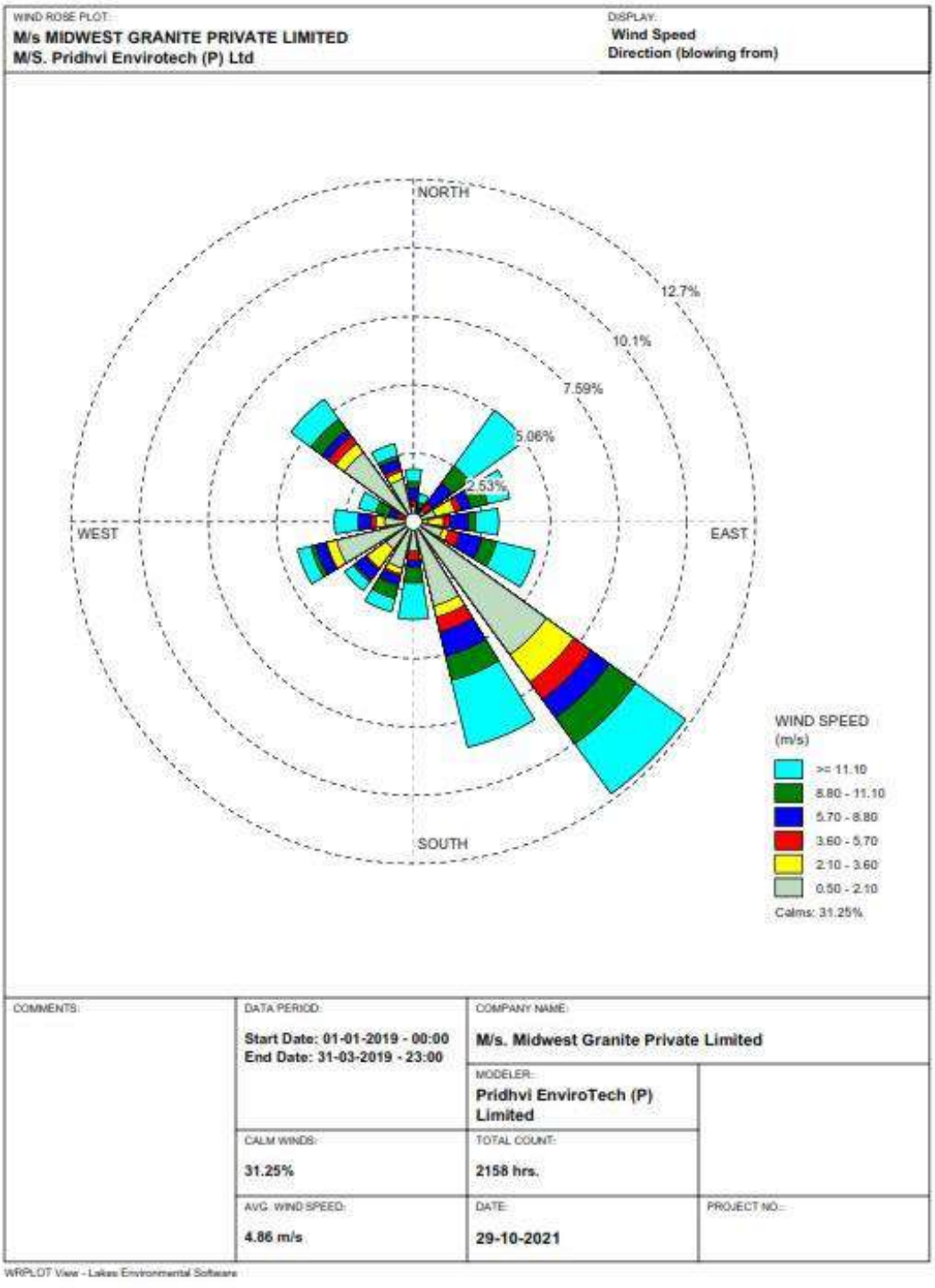


TABLE 4.2 - EMISSION DETAILS

(Basis –AP-42: Emission Estimation Technique Manual for Mining)

A. POINT SOURCE EMISSION

Total drills/day = 56

Drill dust emission = 0.59 kg/hole = 33.04 gm/sec

(I) Area Emissions – Excavation of Top Soil

Quantity, TPA	1,48,590
Operational Hours Per Year	2400
Activity Rate, t/hr.	61.9125
Emission of dust, g/t.	*0.136
Emission of dust, g /hr.	8.4201
Area of influence, m^2	625
Uncontrolled emission rate g/s/m^2	0.00000374
Controlled emission rate g/s/m^2	0.00000374

(II) Area Emissions – Total Material handling

Quantity, TPA		25,24,000
Operational Hours Per Year		2400
Activity Rate, t/hr.		1051.67
Emission of dust, g/t.	PM_{10}	*0.136
Emission of dust, g /hr.	PM_{10}	143.027
Area of influence, m^2		625
Uncontrolled emission rate g/s/m^2	PM_{10}	0.229
Controlled emission rate g/s/m^2	PM_{10}	0.0229

(III) Line Source – Transport of Granite from Pit to Stock yard

Quantity, TPA	248362.5
Operational Hours Per Year	2400
Capacity of each Dumper (T)	35
Total No. of Tippers/ year	7096
Lead length/trip, Km	1.28
Total VKT/Year	9083
Emission Kg/VKT	+0.133
Total emission Kg/Year	1208.0352
Uncontrolled emission rate g/s/m	0.0002185
Controlled emission rate g/s/m	0.00002185

(IV) Line Source – Transport of Waste from pit to Waste dump

Quantity, TPA	22,75,637.5
Operational Hours Per Year	2400
Capacity of each Tipper (T)	25
Total No.of Tippers/ year	91025.5
Lead length/trip, Km	1.024
Total VKT/Year	93210.112
Emission Kg/VKT	+0.133
Total emission Kg/Year	12396.945
Uncontrolled emission rate g/s/m	0.0028

Controlled emission rate g/s/m	0.00028

(V) Line Source – Transport of Overburden from pit to Waste dump

Quantity, TPA	1,48,590
Operational Hours Per Year	2400
Capacity of each Tipper (T)	16
Total No. of Tippers/ year	9287
Lead length/trip, Km	1.024
Total VKT/Year	9509.76
Emission Kg/VKT	+0.133
Total emission Kg/Year	1264.8
Uncontrolled emission rate g/s/m	0.000286
Controlled emission rate g/s/m	0.0000286

Note : *Emission factor computed based on wind speed of 2 m/s, and moisture content of 10 %.

+ Emission factor computed based on silt content of 10 % and moisture content of 10 %

4.3.3 Air Quality Predictions

Simulation Model for Prediction using Industrial Source Complex ISC AERMOD View

The pollutants released into the atmosphere will disperse in the down wind direction and finally reach the ground at farther distance from the source. The concentration of ground level concentrations mainly depends upon the strength of the emission source and micrometeorology of the study area.

In order to estimate the ground level concentrations due to the emission from the proposed project, EPA approved Industrial Source Complex ISC AERMOD View Model has been employed.

The mathematical model used for predictions on air quality impact in the present study is ISC-AERMOD View-6.8.6. It is the next generation air dispersion model, which incorporates planetary boundary layer concepts.

The AERMOD is actually a modeling system with three separate components:

AERMOD (AERMIC Dispersion Model), AERMAP (AERMOD Terrain Preprocessor) and AERMET (AERMOD Meteorological Preprocessor).

Special features of AERMOD include its ability to treat the vertical in homogeneity of the planetary boundary layer special treatment of surface releases, irregularly shaped area sources, a plume model for the convective boundary layer, limitation of vertical mixing in the stable boundary layer, and fixing the reflecting surface at the stack base.

The AERMET is the meteorological preprocessor for the AERMOD. Input data can come from hourly cloud cover observations, surface meteorological observations and twice-a-day upper air soundings. Output includes surface meteorological observations and parameters and vertical profiles of several atmospheric parameters.

The AERMAP is a terrain preprocessor designed to simplify and standardize the input of terrain data for the AERMOD. Input data include receptor terrain elevation data. Output includes, for each receptor, location and height scale, which are elevations used for the computation of airflow around hills.

4.3.3.1 Post Project Scenario

Predicted maximum ground level concentrations considering micro meteorological data of December 2013 to February 2014 are superimposed on the maximum baseline concentrations obtained during the study period to estimate the post project scenario, which would prevail at the post operational phase. The overall scenario with predicted concentrations over the maximum baseline concentrations is shown in the following table along with isopleths Figures 4.3.

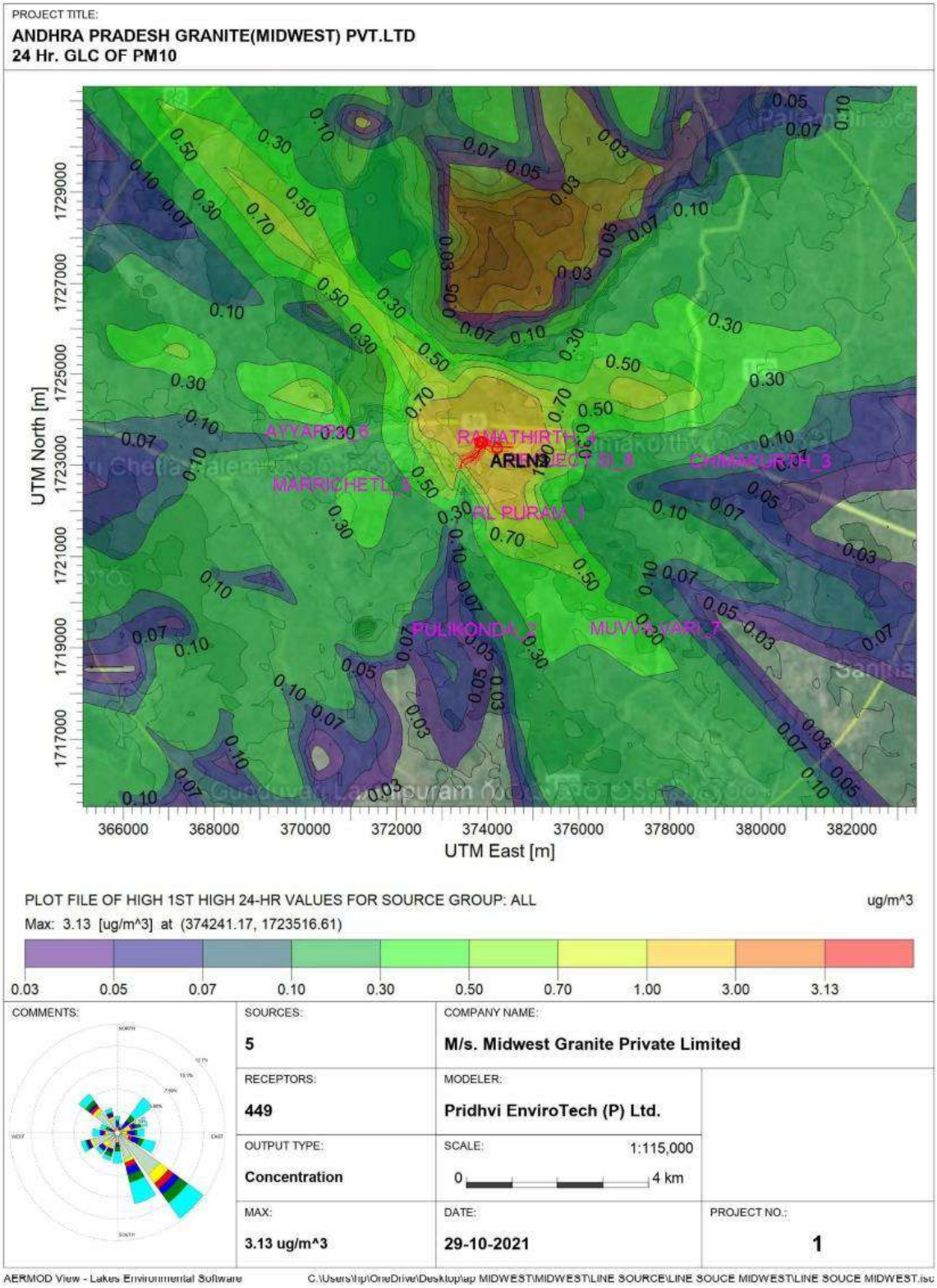
4.3.3.2 Study Period

Meteorological data was collected during Study Period (January - March 2019) a cumulative wind rose diagram of all the 3 months were compiled using mathematical model WRPLOT. The seasonal wind rose diagram has been presented in Fig: 4.2, which covers all the three months meteorological data.

TABLE 4.3 EMISSION DETAILS

ID	Location	Baseline	Predicted	Cumulative
		PM10	PM10	PM10
AAQ1	Mine Site	92.5	3.13	95.63
AAQ2	RL Puram	85.3	0.14	85.44
AAQ3	Puli konda	72.3	0.11	72.41
AAQ4	Chimakurthy	85.3	0.14	85.44
AAQ5	Ramathirtham	69.3	2.50	71.8
AAQ6	Marri chetlapalem	69.3	0.15	69.45
AAQ7	Ayyapavari palem	64.3	0.14	64.44
AAQ8	Muvavaripalem	62.4	0.41	62.81

Figure: 4.3: Predicted 24 hourly of PM10 in µg/m^3



4.4 Noise Environment

4.4.1Prediction of Impacts Due To Quarry Activity

If the sound source is directive and is located very near to hard flat surface, then the sound pressure level at a distance "r" is calculated by,

Lp= Lw + DI - 20 Log(r) - Ae - 8

Where,

Lp = Sound pressure level at a receiver located in a particular direction

Lw = Sound power level of the source

DI = Directivity index of the source in that direction

r = Distance of receiver from the source (meters)

Ae = Excessive attenuation caused by the environmental conditions

For hemispherical wave divergence in a homogeneous loss free atmosphere (Ae = 0), and over loss free ground, free of barrier the sound pressure drops off by 6 dB for each doubling of distance.

The noise generated due to mining operations and movement of equipments and vehicles affects the acoustical environment surrounding the mining area. Cumulative effect at observer is calculated by the equation

Lp = 10 Log [10Lp1/10 + 10Lp2/10 + 10Lp3/10 + -------]

4.4.2 Noise Due to vehicular Traffic

The traffic density in the adjoining villages surrounding the proposed mine site is low. The speed of medium and light weight vehicles will be around 40 to 50 kms/hr and for heavy vehicles it will be in the range of 35 to 40 kms/hr.

To estimate the impact of vehicular activity in the study area, total numbers of vehicles passing per hour on road were measured. Equivalent noise levels due to traffic, at required locations was estimated using integrating sound level meter based on the following Federal Highway Administration (FHWA) noise equation:

$$Log(h)i = Loe + 10Log(Ni/SiT) + 10Log(15/d)\ 1+B + Ds - 13$$

Where

Loe : Reference mean energy level for i^{th} vehicle type (from published data)

Ni : Number of class i^{th} vehicle passing during time T

Si : Average speed for i^{th} vehicle class in km/hr

T : Duration for which Leq is desired corresponding to Ni

D : Perpendicular distance in meters from the centre line of the traffic lane to the location of the observer

B : Factor relating to the absorption characteristics of the ground cover between roadways and observer

Ds : Shielding factor such as provided by a noise barrier

The above equation is the basis on which integrating sound level meter records and integrates the noise levels generated due to light, medium and heavy vehicles.

The preceding equation is used three times, once for automobiles, once for medium trucks and once for heavy trucks to obtain three values as Leq(A) for Autos, Leq(M) for medium trucks and Leq(H) for heavy trucks. The final Leq can be calculated by logarithmically adding the three Leq values as follows,

$$Leq\ (Total) = 10\ Log\ [10\ Leq\ (A) / 10 + 10\ Leq\ (M) / 10 + 10\ Leq\ (H) / 10]$$

The total impact on observer, standing at a distanced kms from the centre line of the traffic lane is given by the above equation.

Due to wave divergence and attenuation, the noise level decreases as distance from source increases. Hence the effect of traffic at 1km distance will be around 45-50 dB(A). The expected Noise from mining machineries are given in Table: 4.3.

Table: 4.3 -Expected Noise from Mining Machinery

S. No.	Type	dB(A)
1.	Excavators	75-80
2.	Trippers/ Dumpers	65-80
3.	Water Sprinklers	60-70
4.	Maintenance Van	70-75
5.	Truck	55-70
6.	Jeeps	55-60
7.	Ambulance Van	55-60

4.5 Mine Wastewater

Domestic wastewater will be sent to septic tank followed by soak pit.

4.5.1 Mine Pit Water

The pit water is expected to be good quality except the pollutant like high-suspended solids. Mine pit water will be pumped into garland drains with sedimentation pits. The silt free water will be let into the nearby seasonal stream.

4.6 Land Environment

In the long term, the quarry pit is proposed to be secured and biologically reclaimed by converting it into an aquatic reservoir. The boundaries of the pit will be fully secured and provision will be made for the people to sit and relax. The water body will help in storage of water, increase the water level in the vicinity and will also support the fauna in order to avoid breeding of mosquitoes. Suitable species of fish will be introduced into the pond to feed on the larva of mosquitoes. Water lilies and other aquatic plants will be planted to improve the aesthetic and visual impact and landscape of the area. The overall effect is that the mined out area will render a pleasant view after closer of teh mine benefiting the local communities and beautifying the landscape apart from helping recharging the ground water apart from meeting the local requirements of water.

4.7 Vibration Levels

Impact of blasting operations involved in the mining activity with respect to vibration and noise level will be envisaged.

4.7.1Disasters

The potential disasters in the mining operation are sliding of mine slope, water inundation and water inrush into the mine pit, fire hazard at mine and stock yard. Detail slope stability study, on-site and off-site disaster management plan will be prepared before operation of the mine.

4.7.2 Mitigation Measures

Following environmental mitigation measures are recommended to minimize the impacts caused by project activities:

4.8 Air pollution mitigation measure includes following:

- Regular sprinkling of water on hauling roads.

- Water sprinkling on dumps.
- Greenbelt would be provided around the lease area.
- Regular maintenance of the Diesel operated HEMMs to reduce emission of noxious gases.
- Covering of the trucks carrying granite on public road.

4.8.1 Water pollution mitigation measure includes following:

- Provision of Garland drains around the mine pit to check inflow of surface run-off into mine-pit.
- Mine water will be pumped out into the garland drains and silt free water will be let into the nearby seasonal stream.
- Catch drains would be provided around the external dumps & ore stock area.
- Surface run-off from external dumps & ore stocks area will be collected & diverted to garland drains with sedimentation ponds for treatment.
- Septic tank would be provided to treat wastewater generated from domestic front.

4.9 Impact on Traffic & Road Network:

The major road access to the site is kutcha road which is connecting to the Ongole –Kurnool state highway which is 0.3 Kms from the mine site. Traffic study is conducted on the ongole – Kurnool state highway near the mine site on 10th and 11th September, 2021 and data is presented in the table. The total volume of traffic on both directions on this double-lane road is 2505 PCUs/Day. As per the IRC code 64-1990 Guidelines for capacity of the double lane road in plane area is 15000 PCUs/day

Mitigation measures:

The connecting road to mine will not cross village. Hence traffic on connecting road will not impact village much. It is suggested to regulate traffic on connecting road in nonpeak hours i.e., between 11 AM to 5 PM or after 10 PM so that no inconvenience is caused to village. It is suggested to contribute for road strengthening and repairs if any at regular interval

4.10 Land Reclamation

As discussed earlier land is to be degraded on account of following activities:

- Quarrying operation that will create pit.
- Dumping of overburden

Land reclamation scheme has been developed that covers the following:

- Backfilling of degraded pit with OB removal in course of mining. Once the backfill is over, it will be graded and leveled. Saplings would be planted on the prepared surface.
- For plantation, pits would be dig which will be filled with mixture of top soil & fertilizers. Such plantations would be done with onset of monsoon.

Similarly, when the formation of external dump is over, it will be physically reclaimed. This will include:

- Grading & leveling of flat top surface.
- Plantation of sapling on the prepared surface.
- Installation of drainage structure around the dump.
- Construction of toe wall at the foot of the dump.

4.11 Predicted Impacts on Socio-Economic Environment

The socio-economic impacts of the Mining project of APGMPL could be assessed in terms of demand-supply aspects for goods and services in the area, pressure on natural resources and infrastructure, growth of industry, sustainability of livelihoods and employment

Some indirect impacts can also be understood in terms of public health and safety, preserving local culture and community aesthetics.

It is however important to appreciate that given the size of the project, its socio-economic impacts will be confined largely to the core and buffer zones and not beyond.

Based on a close understanding and assessment, the socio-economic impacts of APGMPL mine are predicted as follows:

S. No.	Impact Area	Predicted Impacts	
		Positive	**Negative**
1	Natural Resource Sustainability	▪ No acquisition of land and change of land use pattern	▪ Increased pressure on exploitation of groundwater resources in the area. ▪ Possible contamination of groundwater if adequate safeguards are not taken.
2	Human Settlement	▪ No displacement of habitations will occur.	▪ Nil

S. No.	Impact Area	Predicted Impacts	
		Positive	**Negative**
3	Livelihoods	▪ No loss of existing livelihoods in the area is expected to occur. ▪ Additional nonfarm livelihood opportunities are expected directly and as spinoffs.	▪ Nil
4	Employment Generation	▪ Employment to about an additional 40 skilled and unskilled workers will be provided during project execution phase. ▪	▪ Nil
5	Incomes and Revenues	▪ An average Rs.15,000 per capita money income generation to additional workforce engaged. ▪ Improved tax revenues of RL Puram Village Panchayat	▪ Nil
6	Demographics	▪ The population levels of the neighboring areas are not likely to change in any significant manner. ▪ The lifestyles are expected to improve in tune with the rise in incomes and improvement in infrastructure facilities. ▪ The skill sets of the local residents are expected to improve in keeping with the emerging employment opportunities.	▪ Health of people residing in the core impact zone is likely to be impacted adversely as a consequence of air pollution.

4.12 Biological Environment

The project proposes to increase the production Capacity of Black Galaxy Granite from 54,643 m^3/Annum to 79,401 M3/annum..

4.12.1 Prediction of Impacts on local flora and fauna during operational phase:

Proposed activity does not involve in removal of natural vegetation during operational phase of existing mining activity at site as well as at dump area. Thus, there will be no major adverse impacts are envisaged. There are no migratory corridors, nesting and Breeding sites within the core zone. No need to take any mitigation measures specific to these parameters. From the list

of floral and faunal species, it is very clear that there are no Rare or Endangered or Endemic or Threatened (REET) species present in core or buffer zones.

4.12.2 Impact on Terrestrial and Aquatic Ecology

Prediction of impacts is based on the direct and indirect; short-term as well as long-term; reversible and irreversible impacts that are most likely to occur owing to the proposed industrial activity during operation. The following ecological factors that are considered most significant as far as the impact on flora and fauna are concerned:

The overall impact of the project on ecological aspects is:

Magnitude: less

Extent: The extent of the impact is on-site as the impacts will be limited to the boundaries of the Site.

Duration: The duration would be short-term as the natural vegetation of the site would not be affected during the construction phase.

Intensity or magnitude: The intensity is very less during construction and operation phases.

Reversible or Irreversible: Nil

Ecological factors and their impacts due to proposed activity:

Species diversity	No reduction
Habitat loss or fragmentation	Nil
Effect on any additional risk or threat to the rare or endangered or endemic or threatened (REET) species	Nil
Any impairment of ecological functions such as (i) disruption of food chains, (ii) decline in species population and or (iii) alterations in predator-prey relationships	Nil
Is it possible to attain the global objectives of “no net loss” of biodiversity	No

From the above table, it is very clear that proposed activity shall not impact on the biodiversity of the region. Hence, it is recommended to proposed project in the proposed locality by following certain conservation measures. However, some positive impacts of the proposed unit

are enriching the habitat and species with greenbelt activities in core and buffer region will enhance the beauty of the surrounding region.

As there are no REET species in the study area, the proposed project will not pose any threat to local flora and fauna. No direct or indirect damage is expected to the flora and fauna of the buffer zone. Hence, the anticipated environmental impacts on the flora and fauna of the study area are negligible and easily reversible if any. It will not create any kind of environmental stress to the local flora and fauna.

4.12.3 Impacts Identified:

Major impacts on ecology due to additional traffic near buffer zone, which will be created due to the movement of mined granite blocks. The approximate increase of trucks/tippers will be 8 dumpers/day during operations. Due to this, dust and vehicular emissions on the natural and agriculture plants get affected. The crop yield will be reduced and growth rate of trees in the buffer zone at villages side plants and near reserve forest blocks will get affected.

The direct impacts: Dust and noise pollution are direct impacts due to mining activity. The noise and vibrations are affecting the reptilian fauna present in the reserve forests. The bird nesting and breeding grounds will be shifted to adjoining stabilized habitats. Higher mammals changes their habitats. The dust particles will be directly affecting the aquatic ecosystem near Ramatheertham Reservoir. The impact will be there on local fishes and amphibians of the region.

Indirect Impacts: Indirect impacts on small mammals and other fauna on the roads. Road kills will occur during vehicular movement.

Duration of the Impacts: Long term Negative impacts on the ecosystem because of mining process, and transportation will occur on ecological environment as secondary impacts due to environmental contamination caused mainly by gaseous & particulate emissions released. Such environmental contamination caused by emissions may lead to heavy toxic load on ecological components (biotic & abiotic) which may have very adverse effects on flora, fauna and ecological habitat depending up on the toxicity type & level of the contaminant.

Such particulate matter will get settled on leaves of flora of the region specially surround agricultural ecosystem which may in turn cause long term negative impacts on flora causing hindrance in photosynthesis and adverse effects on growth & quality due to toxicity. There are few aquatic bodies nearer to the existing unit. The cumulative impacts of all the surrounding

mining sites and proposed project activities causes deterioration of the quality of these water bodies which may intern causes death to many aquatic flora and fauna.

The above said impacts due to the toxicity of contaminants released may also occur in the distant fauna due to the interaction & migration of the fauna between micro habitats of the region. The eventual toxic effects of the said impacts may increase as a result of bio-magnification in food chain and food web. The heat radiated from site may result in considerable adverse impacts on ecology. These impacts are controllable by planning & implementation of efficient & adequate mitigation measures.

Minor Impacts

Minor negative impact in terms on fauna may also occur during transportation route in case of major road accident. Negative impacts in term of contamination of soil due to mixing of heavy load of pollutants (Dust and particulate matter), which may alter many soil quality parameters.

Positive impacts on Ecological aspects:

Aesthetics:

Due the present plantation activities within the buffer zone of the site, the aesthetic value of the region will be enhanced besides reduction of effects of pollutants. The beauty of the region will be enhanced. More native plants species that are suitable to the region taken into criteria. Involving local villagers in plantation activities and educating the people on importance of biodiversity leads to protection and conservation of flora and fauna in long term.

Carbon sink: Within the study area i.e in the buffer zone, the mass avenue plantation activities, rehabilitating degraded lands contribute to mitigating climate change. These actions increase the rate and quantity of carbon sequestration in biomass. Introduction of trees on non-forest or degraded forestlands, industrial plantations, Village plantations, restoration of natural forest, watershed protection, orchards and perennial cultures, agro-forestry activities enhance the ecological and economic values.

4.13 Conclusion

Based on the EIA study it is observed that there will be a marginal increase in the dust pollution, which will be controlled by sprinkling of water and transportation in properly controlled movement. There will be minimal impact on ambient environment & ecology due to mining activities, moreover the mining operations will lead to direct and indirect employment generation and improves the quality of life of the people in the region.

Hence, it can be summarized that the Black Galaxy Granite Mine of APGMPL will have a positive impact on the socio-economics of the area and lead to overall sustainable development of the region, therefore environmental clearance may be granted.

Chapter - 5

Analysis of Alternatives (Technology & Site)

CHAPTER 5
ANALYSIS OF ALTERNATIVES (TECHNOLOGY & SITE)

5.1 Analysis of Alternative Technology

Andhra Pradesh Granite Midwest Private Limited is adopting semi mechanized open - cast method of mining. Granite is a siliceous rock. During dimensional quarrying of granite blocks drilling cannot be avoided completely though primary sawing is executed using diamond wire saw. The process of drilling which is percussive-cum-rotary, invariably releases fine dust which is injurious when inhaled by the workmen.

The proponents developed an improvised dust collection system in association with Ripamonti, Italy with enhanced number of filters and incorporating superior quality filters. Adoption of this dust collection system greatly improves the efficacy of dust collection mechanism and minimise the release of dust in the work area.

5.2 Analysis of Alternative Sites

Alternative site selection does not arise as this is existing mine going for expansion

Chapter - 6

Environmental Monitoring Programme

CHAPTER-6

ENVIRONMENTAL MONITORING PROGRAMME

Andhra Pradesh Granite (Midwest) Private Limited will ensure the implementation of the EMP measures within the mine area and Carry out efficient monitoring.

6.1 Environmental Monitoring

Monitoring of various environmental parameters will be carried out on a regular basis. Midwest Granite Private Limited will monitor Ambient Air, Water and Soil quality at following locations:

Particulars	Monitoring Frequency	Standards	Duration of Sampling	Important Monitoring Parameters
Ambient Air Quality Monitoring				
At 3 Locations: Project Site RL Puram Chimakurthy	Quarterly	Air (Prevention and control of Pollution) Rules, CPCB,1994	24 Hours	PM10, PM2.5, SO2, NOX
Ground Water Quality Monitoring				
At 3 Locations: Project Site, RL Puram Chimakurthy	Quarterly	IS:10500		pH, TDS, TH, TSS, Metals & Chlorides
Surface Water Quality Monitoring				
At 2 Locations: At quarry pit, Ramathirtham	Quarterly	IS:10500		pH, TDS, Chlorides, Hardness
Noise Quality Monitoring				
Noise Levels at 3 Locations: Project Site, RL Puram Chimakurthy	Quarterly	Noise standards by CPCB		Equivalent Noise Levels in dB(A)
Soil Quality Monitoring				
Soil Sampling at 2 Locations: Project Site, Chimakurthy	Yearly			EC, pH, CEC, Moisture, Texture, Bulk Density etc.,

Monitoring will be carried out as per the CPCB guidelines using Fine dust samplers, and Respirable Dust samplers.

6.2 Environmental Management Cell

Andhra Pradesh Granite (Midwest) Private Limited hired services of experienced Environmental Professionals periodically for consulting and implementing the Environmental Management measures from time to time to comply with environmental norms stipulated for the project.

6.3 Environment Policy

Environmental Policy has been formulated for implementation. Mines Manager will be responsible for maintaining ecological balance by adequate environmental protection. He will monitor the environmental performance and guide the employees for maintaining good environmental Practices and to adopt newer, safe and cleaner technologies.

6.4 Responsibility of Monitoring and Reporting System

Records will be maintained for the analysis of ambient air quality, noise levels and water quality. These records are not only required for the perusal of the statutory authorities. The management will maintain the records as EPA regulations and apply for the annual consents for air and water. The mine shall obtain the consent for establishment (CFE) and consent for operation (CFO) from State Pollution Control Board as required under section 25/26 of the Water act, 1974 and under section 21/22 of Air Act, 1981, before enhancement the production from the Pollution Control Board as directed by the Hon'ble Supreme Court of India. The CFO will be renewed every 5 year by the management in consultation with APPCB. The mine submitting environmental statement every year. The management ensures that it will comply with all the directions and regulations issued by the Ministry of environment, forests and climate change, New Delhi, State and Central Pollution Control Boards. The Consent for Establishment, Consent for Operation displayed in a conspicuous location for the information of the inspecting authorities of different departments and all stake holders.

6.5 Environmental Monitoring Budget

The Budget for environmental monitoring for the parameters mentioned below:

Component	Item	Unit	Unit Cost (INR)	Total Cost / Year (INR)
Ambient Air Quality	3 Locations Specified in monitoring plan	Quarterly	10500	126000
Surface & Ground water Quality	3Locations Specified in monitoring plan	Quarterly	2000	24000
Noise	3 Locations Specified in monitoring plan	Quarterly	1200	14400
Soil	2 Locations Specified in monitoring plan	Yearly	1800	3600
Total				**1,68,000**

Chapter - 7

Additional Studies

CHAPTER – 7

ADDITIONAL STUDIES

ADDITIONAL STUDIES

7.1 Public Consultation

The Draft EIA Report is prepared as per the EIA notification,2006 and being submitted to APPCB Regional office, Ongole for conducting public hearing for the enhancement of Black Galaxy Granite mine of Andhra Pradesh Granite (Midwest) Private Limited located in Sy.No.55/5, of RL Puram Village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh. After completion of public hearing, inputs received during public hearing process and details will be submitted in final EIA report.

7.2 Risk Assessment of Black Galaxy Granite Mining

Risk assessment is a process whereby risks are analysed, assessed and risk management priorities are evaluated. It is defined as the characterization of the potential adverse effect to human health & environment due to environmental hazards.

Objectives of risk assessment are:

- Identifying hazardous activities
- Assessment of risk level and severity in different operations
- Identification of control measures
- Setting monitoring process
- Reduce the impact of mishaps of all kinds
- Reduce the inherent potential for major accidents.

Methodology of Risk assessment: -

- Collection of information & identification of hazard
- Classify their severity and probability of occurrence.
- Identification of exposed risks
- Assess the risk and risk rating based on
- Probability

- Exposure
- Consequence
- Prioritization of the risks
- Implementation of control measures
- Monitoring risk assessment
- Evaluation and correction

Factors of risks involved due to human induced activities in connection with mining operations are 1) removal of O.B and side burden 2) drilling 3) blasting 4) excavation of ore and 5) transportation of ore 6) Safeguarding of explosives.

Other risk factors due to natural activities are 1) water inundation 2) natural calamities such as earth quakes

For the various risks, likely to arise, as above, detailed analysis of causes and control measures is given in **below Table:**

Table 7.1: Identified Risks & Control measures

S. No	Factors	Causes of risks	Control measures
1	Removal of O.B, ore & its dump	a) Sliding of benches due to its unconsolidated nature. b) Vibration due to movement of vehicles in the benches.	- The maximum bench slope and ultimate pit slope of 70° and 40° will be maintained. - There is no overburden in this case as mine is operational since 2007 - Mine waste is transported to Dump site which is away from the mine and hence no possibility of slip of dump into mine - Adequate slopes are maintained for dump waste to avoid any slippage

2	Drilling	a) Due to high pressure of compressed air hoses may burst b) Jack hammer rod may break due to improper maintenance of the rod	- Periodical preventive maintenance and replacement of worn out accessories in the compressor and drill equipment. - As per manufacturers recommendation rod to be replaced and bits will be changed
3	Blasting	a) Fly rock, ground vibration and noise etc., b) Improper charging of explosives	- Spacing will be kept optimum on trial basis. - Delay detonators will be used. - Maximum charge per delay will be restricted and by optimum blast hole pattern, vibrations will be controlled within the permissible limit and blast can be conducted safely. - Blast warning system is in place and people presence during blasting will be restricted
4	Excavation of waste & granite blocks	a) Hauling and loading equipment are used in mine. Derick lifts are used for loading granite blocks b) Failure of Derick lift ropes c) Falling of Heavy Blocks during loading using loaders	- Operator shall not operate the machine when person & vehicles are in such proximity. - Shall not swing the bucket over the cab and operator leaves the machine after ensuring the bucket is on ground. - Shall not allow any unauthorized person to operate the machines by effective supervision - Derick lift ropes are tested periodically and replaced when required - Helmets, Safety shoes, ear plugs, Dust masks shall be used during loading process

5	Storage of explosives	a) Accidental detonation of explosives b) Theft of explosives by unscrupulous elements and miss use	- A separate magazine storage area approved by Mine Safety department is maintained with all necessary safe guards - Security arrangement for 24 hours is maintained to avoid any theft and mis-use of explosives
6	Transportation of Waste and Blocks	a. Operating the vehicle "nose to tail" b. Overloading of material c. While reversal & overtaking of vehicle d. Operator of truck leaving his cabin when it is loaded.	- It will be ensured that all these causes will be nullified by giving training to the operators. - No overloading is permitted - Audio visual reverse horn will be provided for vehicles - No un-authorised persons will be allowed during loading operations
7	Fire due to electricity and Oil	a) Due to the short circuit of cables & other electrical parts b) Due to the leakage of inflammable liquid like diesel, oil etc.,	- Commutator & electrical parts shall be cleaned frequently with the help of dry air blower. - Checking will be done every day for proper earthing of electrical equipment - All fastening parts and places will be tightened. - Vehicles will be maintained every week to detect leaks
8	Water inundation	a) Inrush of storm water due to heavy rain b) Unusual seepage of water c) Sudden collapse of peripheral bund due to torrential pour	- Ground water levels yet to reach in the mine - Mining operations will be suspended during heavy rain period - No over burden is stored at the site
9	Natural calamities	Unexpected happenings	- The mine management is capable to deal with the situation

7.3 Disaster Management:

The planning for mitigating the emergency involves recognizing the accident or possible, assessing the consequences of such possible accidents and deciding on emergency planning / procedures in advance both on site & off site that would be necessary to be implemented in the event of emergency without any delay and confusion.

Proper bench design in mines and observing safety measures for transportation, storage, handling of explosives & fuels, etc, good maintenance of vehicles, roads, fire prevention measures shall go in preventing disaster/accidents.

Mining shall be carried out as per MMR 1961 & Rules and regulations applicable the project in charge is having a mobile communication for quick Intimation of information if need arises.

As the quarry operation is open cast, possible risk is collapse of mine pit when people are working. For this proper bench slope will be maintained as per the mining plan. Mining operations will not be conducted during rainy days to avoid any collapse of the overburden into mine pit. All precautions would be taken to control fly rock during the blasting operations.

The quarry is located at about 25KM from Ongole town having good transport and communication systems. The quarry area is provided with good communication facilities like telephone & mobile phone.

In case of any minor/major accidents within the mine, the same will be communicated to the local Revenue Officials, Police, Fire Department, DGMS etc for action.

A vehicle is kept all the time at the quarry site for conveyance. The workmen in the project are periodically trained at group VT center for proper safety training as per the DGMS circulars.

First aid kits are kept at mine site for providing first aid. Management staff at mine are trained to provide immediate first aid

At quarry level first aid facility is provided and person(s) will be shifted to nearest major hospital. A key person will be assigned duties and responsibilities for incident response and emergency response at the quarry.

At mine, level first aid will provided and person(s) shall be shifted to nearby hospital at Chimakurthy or Ongole, which are having all basic amenities. There is qualified medical officer in the Government Hospital at Chimakurthy, who is capable of handling normal accidents. In case of high risk accidents injured will be shifted by department vehicle to well-equipped Hospitals at Ongole town, which is 25 Km from quarry site, as per the requirement.

Quarry Manager is the person responsible of the disaster management and he needs to be contacted in case of any disaster.

7.4 Social Impact Assessment:

Total 50 Villages are falling in 10.0 km radius of project area. In the study area, the population is 1,05,582 which males are 56.71% and females are 49.28%. The literate population in the study area is 51.27% of the total population of the study area.

The working population in the project area are 51,980 which is 49.23% of total population out of which males are constituted to 60.3% and females account to 39.7%.

Sampled households:

A stratified random sampling technique was followed to select the sample villages and households. Accordingly, the sample villages were picked up at random from the three impact zones considered – core, buffer and minimally affected. The number of households to be contacted in each sample village was determined on the basis of the size of population of the respective village. In the absence of household level information, the respondent households were selected randomly during the course of visit to the respective village.

However, while selecting the respondent households, emphasis was on contacting households, who are economically poor, susceptible to shifts in livelihood patterns and belonged to vulnerable social communities. Accordingly, the sample villages selected for the survey together with the number of households across each village are presented in Table.

Table –7.2 Sample Villages Selected for Socio-Economic Impact Assessment

S. No	Town / Village Name	Mandal Name	Location from Project Site		Population(2011 Census)	House Holds (No.)	
			Directions	Distance		Total	Samples
1	Kambhampadu	Chimakurthy	SW	1.55 Km	Under Rajupalem Lakshmipuram Revenue Population Data is Same		11
2	Rajupalem Lakshmipuram	Chimakurthy	W	2.40 Km	3437	879	12
3	Chimakurthy	Chimakurthy	E	2.50 Km	30279	7577	14
4	Pulikonda	Chimakurthy	S	3.90 Km	1622	422	9
5	Budavada	Chimakurthy	SW	4.45 Km	3442	827	10
6	Ayyapa Raju Palem	Marripudi	W	4.80 Km	1196	317	12
7	Kandurivari Agraharam	Chimakurthy	NE	5.75 Km	689	161	11

8	P Naidu Palem/Padamatinaidupalem	Chimakurthy	NE	6.20 Km	3933	856	12
9	Nippatlapadu	Chimakurthy	NW	6.30 Km	2798	645	11
10	Gangapalem	Tallur	NW	8.65 Km	7028	1812	10
Total					**54424**	**13496**	**112**

The sample villages account for 51.54% of the population and 52.04% of the number of households of the total 50 villages

It needs to be mentioned in this context that an element of purposiveness was induced into selection of sample villages with a view to focusing more on villages in the core and buffer zones of impact. Such a selection of the sample villages was well justified as the entire area beyond the core and buffer impact zones is likely to have only minimal impact of the project and the same cannot be distinguished.

Based on a close understanding and assessment, the socio-economic impacts of APGMPL mine are predicted as follows:

S. No.	Impact Area	Predicted Impacts	
		Positive	**Negative**
1	Natural Resource Sustainability	▪ No acquisition of land and change of land use pattern	▪ Increased pressure on exploitation of groundwater resources in the area. ▪ Possible contamination of groundwater if adequate safeguards are not taken.
2	Human Settlement	▪ No displacement of habitations will occur.	▪ Nil
3	Livelihoods	▪ No loss of existing livelihoods in the area is expected to occur. ▪ Additional nonfarm livelihood opportunities are expected directly and as spinoffs.	▪ Nil
4	Employment Generation	▪ Employment to about an additional 40 skilled and unskilled workers will be provided during project execution phase.	▪ Nil

S. No.	Impact Area	Predicted Impacts	
		Positive	**Negative**
5	Incomes and Revenues	▪ An average Rs.15,000 per capita money income generation to additional workforce engaged. ▪ Improved tax revenues of RL Puram Village Panchayat	▪ Nil
6	Demographics	▪ The population levels of the neighboring areas are not likely to change in any significant manner. ▪ The lifestyles are expected to improve in tune with the rise in incomes and improvement in infrastructure facilities. ▪ The skill sets of the local residents are expected to improve in keeping with the emerging employment opportunities.	▪ Health of people residing in the core impact zone is likely to be impacted adversely as a consequence of air pollution.

Chapter - 8

Project Benefits

CHAPTER – 8

PROJECT BENEFITS

8.1 Employment Potential

APGMPL has employed 448 persons for carrying out the mining operations. In addition there will be indirect employment to many more people in the form of contractual jobs, business opportunities, service facilities etc., This will enhance the economic status.

Apart from the jobs, APGMPL will contribute to medical and educational facilities to the employees.

8.2 Social Welfare Measures

APGMPL has spent Rs.5,85,74,743/- towards Infrastructure Development Fee to APMDC which in turn spends the amount for development works in the district.

8.3 Corporate Environmental Responsibility

As a part of Corporate Environmental Responsibility APGMPL, During the period of Pandemic Project proponent spend Rs.21,31,920/- towards providing Covid vaccination for entire employees, some of their family members, vendors and local villagers.

APGMPL proposed to spend additional funds towards Corporate Environmental Responsibility as per Office Memorandum F.No. 22-65/2017. 1A.III dated 1st May, 2018, an amount of Rs. 30.5 lakhs will be earmarked and spent in the nearby villages based on feedback from the villages

8.4 Other Intangible benefits:

The proposed expansion activities will help to increase trade in the area in terms of maintenance activities of equipment, spares and transport activity. This will in turn result in economic benefits to local people. The increased mine production results in royalty to Government and GST and other taxes which will in turn help the local panchayats

Chapter - 9

Environmental Cost Benefit Analysis

CHAPETR – 9

ENVIRONMENTAL COST BENEFIT ANALYSIS

Not Applicable as it is not recommended at the scoping stage.

Chapter - 10

Environmental Management Plan

CHAPTER – 10

ENVIRONMENAL MANAGEMENT PLAN

10.1 Environnemental Management Plan

The activities at the QL area shall have both positive and negative impacts. In order to reduce the negative impacts owing to activities in the QL area under consideration, measures have been integrated in the process at the planning stage itself. Environmental management approach includes actions to protect environment by using various means at different stages of the project. The measures result in reduction of the impacts at the source itself. Some of the measures result in the change in technology / process thereby reducing and or eliminating some of the adverse impacts. In a few inevitable cases, the remedial solutions are provided to counter / minimize the negative impacts.

10.2 Air Pollution Control Measures

Air pollution from mining activities in the present case is not anticipated to significantly extend beyond the immediate vicinity of the Quarry Lease area. Following measures shall be taken to mitigate the effect of mining operations.

The dust rising due to mining activity will be controlled by covering dust extractors will also be employed.

- Wet drilling implemented and dust free drilling machine developed and adopted.
- Dust suppression on roads will be done with sprinkling of water.
- Regular maintenance of vehicles, machinery and DG set shall be carried out in order to minimize the emissions.
- Development of green belt to minimize dust emission.

10.3 Noise Pollution Control Measures

The major sources of noise in mining are from the operations of drilling and blasting, trucks, material handling machinery, loading equipment etc.

Mitigation measures that are proposed to be adopted for control of noise levels are:

- DG Set will be installed with Acoustic Enclosures and Silencers.

- The machinery will be maintained properly to reduce the noise.
- Proper maintenance of equipment is taken care of by the proponent at the Quarry Lease Area.
- It is proposed to use low strength explosives and in optimum quantity to minimize the effects so that the vibration generated will be feeble within 8 Hz.
- Limiting the speed of vehicles entering and leaving the quarry to 25 kmph.

10.3.1 Measures to minimise ground vibrations and noise

Controlled Blasting will be done with optimum quantity explosives. Compressor is kept far from working spots to minimize ground vibrations.

Reduction of noise to acceptable occupational exposure levels are as follows:

- Ensuring that large equipment (e.g. excavators, dumpers, dozers, drillers and other automated equipment that requires an operator) are equipped with a soundproof cabin.
- Workers working in noisy areas are provided with ear plugs.
- Exposure to hand-arm vibration from hand and power tools or whole-body vibration from surfaces on which the worker stands or sits should be adequately controlled through the selection and maintenance of equipment which meets occupational vibration exposure standards.
- The operators will be provided with hand gloves or cotton cloth with which they grip the machine handle. Thus the vibration transmission is reduced.

10.4 Water Pollution Control Measures

Water is used for dust suppression, Green belt development and domestic purpose only. Domestic waste water be generated from the toilets. Effect of water regime at the surrounding mines is not observed.

However, following measures shall be taken:

- Garland drains around pit and waste dump were constructed.
- Retention wall at the foot of the dump is made.
- Check dam is constructed against drainage channels to arrest the silt.
- Growing vegetation on the slopes of the dumps is made.
- The spillages from fuelling of machinery shall be avoided and in case of any spillage the same shall be handled properly. Likewise the spent lubricating oils etc. shall be safely collected and properly disposed of.

- Surface runoff will be collected in the mine pit and will be used for wire saw cutting and dust suppression operations.
- Domestic wastewater will be sent to septic tank followed by soak pit.
- No effect on ground water table as work pits have not touched the ground water table.

10.4.1 Water Conservation Measures

In the pre monsoon period the area experiences non-availability of water. Under Environmental Social Responsibility it is suggested to undertake campaign on water conservation plans by involving local environmentally conscious citizens and construct Rain Water Harvesting structures in the private farmer's agricultural lands. Locally best suited structures in the area are Farm Ponds, Nala Bunds and Continuous Contour Trenches and Bunds.

Farm Pond: The area is underlain by thick soil cover of 2 to 3 meters. It is possible excavate and create dug out pits on the elevated region in the individual farm plots and divert the storm water to store in the pond. Dug out material can be used for developing bunds around the farm pond as already being practiced in the area.

Nala Bund / Stream Checks: The area is gently sloping terrain having a number of linear drainage courses. In the upper run-off zones of the drainage courses, land morphology is changed; grading is done for cultivating the lands. As a result the storm water flow paths have disappeared and runoff water escapes from the region as a sheet wash along with silt from higher elevation to lower regions. It is suggested to identify such original / natural flow paths and to construct Nala Bunds / Stream Checks with local stones and soil excavated in the area across the nala courses.

Continuous Contour Trenches and Bunds: The hills in the study area are having slope in all directions. All along the foot hills Continuous Contour Trenches and Bunds will enhance ground water recharge to increase yield of irrigation wells.

10.5 Socio-Economic Environment

The commencement of mining activity in this area improved the socio-economic status of the local people by creation of employment and paying taxes to gram Panchayat.

APGMPL would be required to initiate the following measures to minimize the possible negative impacts, as a consequence of setting up his proposed Granite mining project at RL Puram village in Prakasam District, on the surrounding socio-economic environment:

- Install appropriate dust control equipment to check air pollution and developing Green belt on road surrounding areas

- Organize monthly health camps in the area to check the occurrence of any respiratory and other disorders.
- Organizing vocational and soft skills development programmes for the youth.
- Conduct Entrepreneurship Development Camps to nurture entrepreneurial talents among the local youth.

10.6 Ecological Management

All the plantation activities of the expansion project if any will be commenced along with construction activity, so that it can grow and help in the construction and operation stage of the expansion project.

10.6.1 Greenbelt :

Green belt developed in a separate area of 2.74acres in sy.no 954/1 of chimakurthy village near dump site and developed 1100 trees in this area.

Table 10.1: List of existing plantation

S.no	Botanical name	Common Name	No of trees planted
1	*Syzygium cumini (L.) Skeels*	Neredu	300
2	*Pongamia pinnata* (L.) Pierre	Kanuga	150
3	*Delonix regia (Hook.) Raf.*	*Chittikesaram*	150
4	*Buchanania lanzan* Spreng.	Sara	200
5	*Azadirachta indica A.Juss.*	Neem	300
	Total		**1100**

Proposed Greenbelt species:

Development of a greenbelt reduces the intensity to pollution as well as improves the aesthetics. Hence species selection is very important. Local species which are also useful to local communities will be given most priority during the plantation.

Guidelines for Greenbelt Development

Design and development of greenbelt should be in adherence to site-specific requirements and prevalent climatic conditions.

- Lessee shall ensure healthy & dens greenbelt throughout the life.
- Lessee shall follow CPCB guidelines for development & maintenance of greenbelt area.
- Lessee shall ensure regular irrigation & fertilization of greenbelt area as required timely.

- Lessee shall ensure re-plantation in greenbelt area depending upon the survival rate of planted vegetation to maintain greenbelt.
- Lessee shall plant trees with density of about 1500 trees per hectare or developing the greenbelt area.
- Indigenous species with fast growth are only selected or form the base of selection as Green Belt can come in view as fast as possible.
- The plantation shall also include fruit bearing trees/ species which shall be uniformly distributed which shall act as dwelling place for Varity of birds and other fauna and form a breeding ground for them. The tree products should have acceptable characteristics to suit local customs and traditions flowering Herbs & shrubs species

As per the present project activities, it is recommended to raise the following plant species. These species can be also available near the forest department nurseries.

Among the various native plant species evaluated during the research studies *Pongamia pinnata, Mangifera indica* and *Azadiracta indica* are identified as keystone species which is coming under the excellent category.

Table 10.2: List of Trees recommended for plantation under Greenbelt

S.no	Botanical Name	Common name	Number
1	Azadirachta indica A.Juss.	Neem	200
2	Pongamia pinnata (L.) Pierre	Kanuga	200
3	Mangifera indica L.	Mamidi	250
4	Syzygium cumini (L.) Skeels	Neredu	250
Total			**900**

Table 10.3: Proposed financial Budget for the Green belt development (Rs in Lakhs)

S.no	Component	1st year	2nd year	3rd year	4th year	5th year	Total
1	Plant seeds/ saplings	1.5	1.5	-	-	-	3.0
2	Gardeners Cost	2.0	2.0	2.0	2.0	2.0	10.0
3	Fertilizers and Miscellaneous cost	1.0	1.0	1.0	1.0	1.0	5.0
Total		**4.5**	**4.5**	**3.0**	**3.0**	**3.0**	**18.0**

10.6.2 Mitigation Measures proposed

- Even though the mining process is within the standard levels prescribed by CPCB, sensitive floral and aquatic ecosystems will get long-term impacts. Hence, all necessary control & measures for prevention of release of toxic contaminants from mining site as well as for prevention of pollution due to emissions of the project shall be implemented as suggested in other relevant sections of this chapter as well as mentioned in EMP.
- Environmental education activities to operation and transportation staff are also control the negative impacts of the operational activities.
- Safety measures & action plan for prevention of spreading of dust and particulate matter shall be ensured.
- Proper maintenance of roads in the buffer zone and at the entrance of the core zone reduces the dust emissions and major accidents on fauna.
- Creation & Maintenance of dense greenbelt near the villages and buffer zone reduces the local temperature and dust pollution.

10.7 Occupational Safety & Health Management

Under Occupational Safety and Health Management, the project proponent will take all the necessary steps to provide better working environment. Occupational health and safety issues occur during all phases of the mine cycle. These are classified according to the following categories:

- General workplace health and safety
- Use of explosives
- Noise and vibration
- Hazards during mining (Fires, explosions)

10.7.1 General Workplace Health and Safety

Mining exploration and development activities should manage occupational health and safety hazards as part of a comprehensive health and safety management plan incorporating the following aspects:

- Preparation of emergency response plans specifically applicable to exploration and production activities (considering the often geographically isolated nature of mining

sites) and including the provision and maintenance of necessary emergency response and rescue equipment.

- First aid trained employees to respond to emergencies.
- Implementation of specific personnel training on worksite health and safety management including a communication program with a clear message about corporate management's commitment to health and safety.
- Training of employees on the recognition and prevention of occupational hazards, protection against the elements; thermal stress; acclimatization; disease exposure.
- Illumination systems should be adequate and safe for the planned working conditions in travel paths, mine working areas, and within and around surface facilities and dumpsites of mines.
- Management will provide workers and visitors with the necessary personal protective equipment (PPE), and provide instruction and monitoring in their appropriate maintenance and use. Applicable PPE include, at a minimum, safety helmets and footwear, in addition to ear, eye, and hand protection devices.
- Occupational health assessments should be conducted for employees on annual basis, based on exposure to risk.

10.8 Use of explosives

Blasting activities that may result in safety impacts are typically related to accidental explosion and poor coordination and communication of blasting activities. The proponent is taking safety measures by using soundless stone expansive mortar to break the mother rock to avoid accidents, noise impact on personnel and environment.

Use handling and transporting explosives in accordance with local and/ or national explosives Safety regulations;

- Assigning certified blasters or explosives experts to conduct blasts
- Actively managing blasting activities in terms of loading, priming, and firing explosives, drilling near explosives, misfired shots and disposal
- Adoption of consistent blasting schedules, minimizing blast- time changes
- Specific warning devices (e.g. horn signals, flashing lights) and procedures should be implemented before each blasting activity to alert all workers and third parties in the surrounding areas (e.g. the resident population).

- Blasting-permit procedures should be implemented for all personnel involved with explosives (handling, transport, storage, charging, blasting, and destruction of unused or surplus explosives).
- Blasting sites should be checked post-blast by qualified personnel for malfunctions and unexploded blasting agents, prior to resumption of work.
- Qualified security personnel should be used to control transport, storage, and use of explosives on site.

For any blasting activity, the same shall be outsourced to professional agencies. The agency shall be responsible for taking care of all the precautions and safety measures. No explosives shall be stored or kept at the QL area. The blasting material shall be carried to the QL area in specified quantity as and when required. However, the overall responsibility shall lie with the lessee.

10.8.1 Machine and Equipment Safety

To prevent and control hazards related to mining equipment use, measures for the enhancement of visibility should be applied throughout the mine. Specific visibility management practices may include the following-

- Use of contrast colouring on equipment / machinery, including the provision of reflective markings to enhance visibility.
- Use of moving equipment / machinery equipped with improved operator sight lines.
- Issuing workers high visibility clothing
- Use of reflective markings on structures, traffic junctions, and other areas with a potential for accidents

10.9 Environment Management Cell

10.9.1 Environmental Policy and management reporting system

Midwest has established itself as a very strong player in the domain of Exploration, Mining, Processing and export of natural Stone products. Midwest is among the top 3 natural stone players in India.

10.9.2 Environmental Policy

Environmental policy framed considering conservation and protection of Environment

Environmental Policy Statement of the Company:

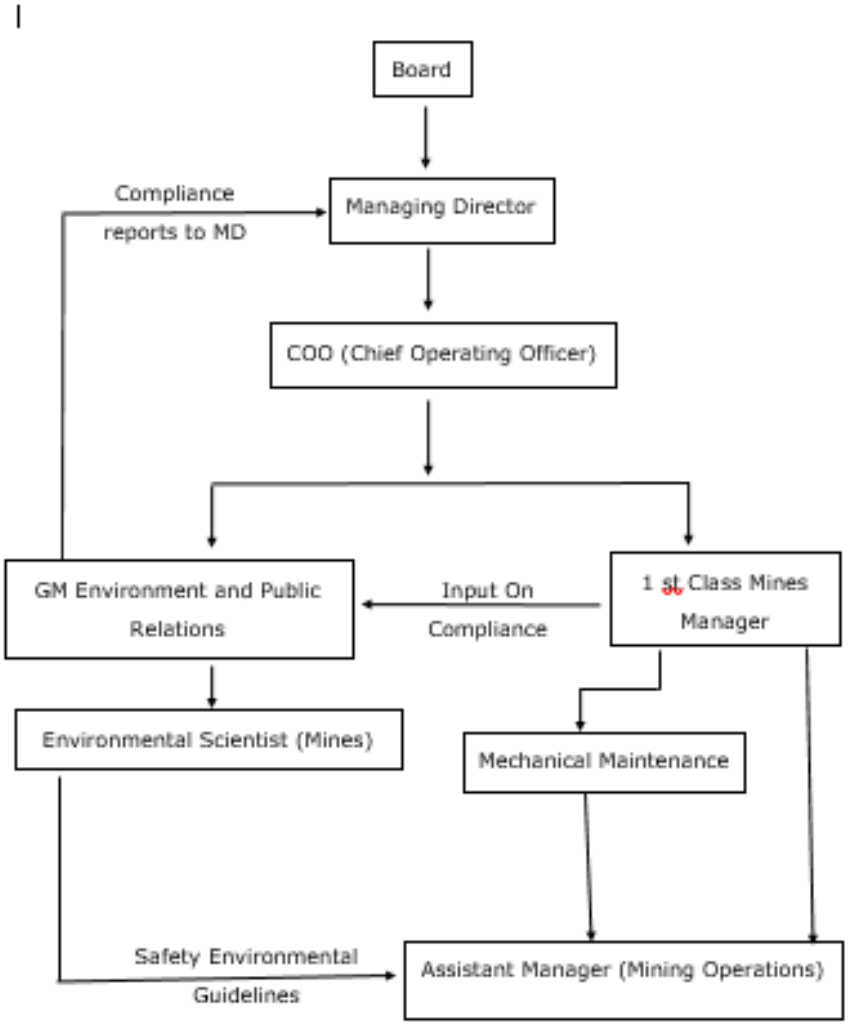
Midwest Granite Private Limited is committed to promote sustainable mining through mitigation of pollution & conservation of natural Resources in all its group companies and process units

It commits to the basic sustainable principles of Reduce & Re-use of waste and dispose all wastes with adequate environmental Safe guards without harming the nature. It continually strive to improve the environmental standards of its operations and process through objectively verifiable targets and environmental management Programs

It is committed to include stake holders in the process and comply all legal and other requirements to which it subscribes

This environmental policy is available to all stake holders, public and employees through appropriate communication processes

Figure 10.1: AP Midwest Granites Pvt Limited Group Companies Organo Gram



10.9.3 Reporting system to the Board of Directors

Non compliances /Violations on any matter which is material in nature will be apprised to Board at the Board Meeting. all Non compliances/Violations on Environmental norms if any, will be reported to this Committee

10.9.4 Reporting System

Reporting system provides the necessary feedback for project management to ensure quality of the works and that the program is on schedule. The rationale for a reporting system is based on accountability to ensure the measures proposed as part of the Environment Management Plan get implemented in the project.

The reporting system will operate by mines manager and implementation system reporting to the Supervision, who in turn shall report to the top management of Andhra Pradesh Granite

Midwest Private Limited Every month, implementation of EMP shall be reviewed by top management during implementation stage and thereafter during operational stage.

10.9.5 Records maintenance and reporting to statutory bodies

Records shall be maintained for the analysis of ambient air quality data, micro-meteorological data and noise levels and ground water analysis. These records are not only requires for the perusal of the pollution control board authorities but also to derive at the efficiencies of the pollution control equipment as the objective of the project proponent is not only compliance with statutory regulations, but also a serious commitment towards clean environment.

The industry shall regularly maintain the records as per the hazardous waste regulations & shall apply for the renewal of consents for air and water and renewal of authorization for the Hazardous waste.

10.10 Reclamation

The proposal for reclamation of land affected by mining activity will be done at the end of mining lease. No reclamation is proposed during the three years period because, the mining will be continued to further depths and reclamation is possible only after completion of mining up to the proposed depths. In future, once the economically viable depth is reached in a part of the area, back filling shall be adopted. Alternately, the excavated pit shall be left for water storage

Based on the mining Plan the life of mine up to 15 years. Quarrying operation will create pit.

Land reclamation scheme will be developed that covers the following:

- Creation of water reservoirs with guard fence will be done at the end of mining
- Green belt will be developed progressively on closed mine area which will improve overall environmental conditions.

10.11 Environmental Management Budget

Based on the proposed expansion activity, it is proposed invest a capital amount of Rs. 56.0 lakhs and recurring expenditure of Rs 39.0 lakhs per annum

Table 10.4: Budget for EMP

S. no.	Component	Amount (Rs in Lakhs) Capital	Recurring expenditure per annum Lakhs of Rs.
1.	Dust Suppression measures	14.0	10.0
2.	Greenbelt Development	18.0	4.0
3.	Dump Management such as construction of stone wall, garland drains etc.	24.0	7.0
4	Safety gadgets	-	10.0
5	Health Surveillance of employees	-	5.0
6	Environment Monitoring	-	3.0
	Total	**56.0**	**39.0**

10.12 Environnemental Social Responsbilities

As responsible corporate, M/s AP Granite Midwest Private Limited proposed to support the local villages for implementing environmental protection measures such as maintenance of Roads, greenery development.

It is proposed to spend Rs 30.0 lakhs in the next 5 years period for maintenance of Road, Development of green belt on public road, providing water harvesting structures, conducting health camp in villages etc. However, these activities will be taken based on the inputs received during public hearing process and details will be submitted in final EIA report.

Chapter - 11

Summary and Conclusion

CHAPTER – 11

SUMMARY & CONCLUSION

11.1 Introduction

M/s. Andhra Pradesh Granite (Midwest) Private Limited (APGMPL) has been granted a Black Galaxy Granite mining lease over an area of 19.425 ha. in Sy. No. 55/5 of Rajupalem- Lakshmi Puram village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh.

11.2 Project Description

The Black Galaxy Granite mine application is made for the proposed production capacity of 79,401 m3/Annum in 19.425 Ha., based on the approved mine plan.

11.3 Description of the Environment

Table 11.1: Salient features of the Project

S. No	Details	Description
1	Location	Sy.No,55/5 Rajupalem-Lakshmipuram village, Chimakurthy Mandal, Prakasam District, Andhra Pradesh
2	State	Andhra Pradesh
3	Temperature	➢ Mean Daily Maximum: 43.5°C ➢ Mean Daily Minimum: 15.0°C
4	Wind Speed	➢ The predominant wind direction is South East ➢ Maximum Mean wind speed is 11.10 m/s ➢ Minimum Mean wind speed is 0.5 m/s
5	Rainfall	➢ Annual rainfall: 798.6 mm
6	Present Land use	It is a barren land
7	Seismicity	Zone-III (medium risk zone)
8	Nearest Road Connectivity	Kurnool to Ongole road – 0.34 Km (N)
9	Nearest Rail Connectivity	Ongole-27 Km
10	Nearest Seaport	Krishna Patnam Port -150 kms
11	Nearest Airport	Vijayawada International Airport- 148 Kms (ESE)
12	Nearest Town/ Village	Podili-23 Km
13	Reserve Forests	Chimakuthy RF-2.3 Km (N) Devarapalem RF – 6.0Km (N)

14	Surface water Bodies	Musi River-3.1 Km E (Tributary of Paleru) Ramathirtham Reservoir-4.6 (NW) Jawahar canal (Right canal of Nagarjuna Sagar)- 5.6 Km (W) Gaddi Vagu-8.1 km (NW)
15	Nearest Reserve Forest	Chimakuthy RF-2.3 Km (N) Devarapalem RF – 6.0Km (N)
16	Any ecologically sensitive areas	There is No ecological sensitive areas within 10 Km radius

The area regionally forms part of the Paleru River Catchment System. The Musi river located 5.50 km west of the quarry site forms part of its stream. The Musi River originates in the DarsDonakonda hills and upland areas of Prakasam district, and flows 33 km southwardly to reach the study area, and further flows 23 km southeasterly and joins the Paleru River near Kamepalli (6 km south of Kondapi), which then flows 30 km eastwards to join the Bay of Bengal.

11.3.1 Land Use Details

The total mining lease area of 19.425 ha. About15.0 ha will be used for mining during the life of the mine. Details of the land use pattern of the mining lease area given in table

Table 11.2: Land Use Pattern

S No	Purpose	Area in Ha.
1	Area under Quarrying	15.00
2	Hauling Road	1.00
3	Finished Granite storage area	2.50
4	Other Infrastructure	0.925
Total		**19.425**

11.4 Project Cost

Total cost of the project is Rs. 35.51 crores and the proponent proposed to incur about Rs. 56.0 Lakhs towards capital cost and Rs. 39.0 Lakhs as recurring cost for implementing Environment Management Plan.

11.5 Anticipated Environmental Impacts & Mitigation Measures

Project Influence Area (PIA)/Study Area: An area covering 10 km radius from project site has been earmarked as study area for baseline environmental surveys.

Study Period: The baseline environmental surveys were carried out during Post Monsoon season (January 2019-March 2019).

11.5.1 Environment

- As site is barren land. No Settlements are there within the site.
- Project site falls between Latitude 15°35'03.66"N to 15°35'09.70"N and Longitude 79°49'51.60"E to 79°49'53.06"E.
- Proposed project site falls under Seismic Zone III (Low Risk Zone area) as per IS 1893 (Part I):2002.
- As per the Project site specific meteorological data, the predominant wind direction is South East.

11.5.1.1 Air Environment

The primary air pollutant of concern at mining sites is particulate matter. As it is a open cast semi mechanized mining, various activities are involved during mining operations, which are removal of top soil, excavation, loading & Vehicle movement due to which dust emissions will be generated.

Maximum concentrations of all the parameters are well with in the National Ambient Air Quality Standards(CPCB,NAAQS,2009):

- PM_{10} ranged between 53.2 µg/m3 to 92.5 µg/m3.NAAQ stipulated standard for PM_{10} for 24 hr. average is 100 µg/m3.
- $PM_{2.5}$ values varied from 16.7 µg/m3 to 48.8 µg/m3. NAAQ stipulated standard for $PM_{2.5}$ for 24 hr. average is 60 µg/m3.
- SO_2 levels varied from 9.8 µg/m3 to 18.2 µg/m3. NAAQ stipulated standard for SO_2 for 24 hr. average is 80 µg/m3.
- NO_X ranged between 15.3 µg/m3 to 28.5 µg/m3. NAAQ stipulated standard for NO_X for 24 hr. average is 80 µg/m3.
- CO is ranging from 1.1 mg/ m^3 to 2.3 mg/ m^3, NAAQ stipulated standard for CO for one hour is 4.0 mg/ m^3.

Air Pollution Control Measures

The mine ambient air quality levels in the mine area are well within the limits. The main activities for air pollution are unloading, transportation and heavy duty mobile equipment.

The environmental control measures to control the fugitive dust will release for the proposed production are given below:

- Regular water sprinkling on haul roads with water tankers.
- Regular grading of haul roads and service roads to clear accumulation of loose material.
- The vehicles will be kept in well-maintained condition so that emissions will be minimized.

11.5.1.2 Noise Environment

Noise will be produced due to mining operations & vehicular movement.

Day equivalent noise levels ($L_{(day)}$) ranged between 43.4 dB(A) to 66.9 dB (A); Night equivalent noise levels (Ln) ranged between 36.0 dB(A) to 60.2 dB (A) in the study area.

Noise Pollution Control Measures

Mitigation measures that are proposed to be adopted for control of noise levels are:

- DG Sets will be installed with Acoustic Enclosures and Silencers.
- The machinery will be maintained properly to reduce the noise.
- Proper maintenance of equipment is taken care of by the proponent at the Quarry Lease Area.
- Limiting the speed of vehicles entering and leaving the quarry to 20 kmph.
- Development of thick greenery in buffer area which will control the noise levels.

11.5.1.3 Water Environment

- Water is used for dust suppression, Green belt development and domestic purpose only. Domestic waste water of 8.0 KLD likely to be generated from the toilets.
- There are no streams/canals passing through the site. Silting of water courses and tanks due to rain water flow from the mine site.
- The general ground water level in the area is 20 meters.

Control Measures for Water Environment:

- Only wastewater generated from mine operations is Domestic wastewater which is about 8.0 KLD and which will be sent to septic tank followed by soak pit.
- In order to avoid siltation of water tank, it is suggested to provide garland canal on the site of water courses in SW corner of the site with silt trap
- It is suggested obtain ground water permission whenever the ground water encountered in mine site and it is suggested to follow the guidelines given by the central ground water board for mining material
- It is proposed to develop water harvesting structure from where water is drawn.

11.5.2 Traffic & Road Network:

The total estimated traffic due to mining activity is 248 PCU/Day.

Mitigation measures:

The connecting road to mine will not cross village. Hence traffic on connecting road will not impact village much. It is suggested to regulate traffic on connecting road in nonpeak hours i.e., between 11 AM to 5 PM or after 10 PM so that no inconvenience is caused to village. It is suggested to contribute for road strengthening and repairs if any at regular interval

11.5.3 Afforestation

As mine area is hard rock. In order to meet the green belt norms prescribed by the MoEF& CC, a separate area of 2.74acres in sy.no 954/1 of chimakurthy village developed for green belt. Now developed 1184 trees in this area and proposed to develop additional 900 plants.

11.6 Socio Economic Environment

- Total 50 Villages are falling in 10.0 km radius of project area.
- In the study area, the population is 1,05,582 which males are 50.72% and females are 49.28%. The average literacy rate in the study area is 51.27%.
- As per Census, 2011 the study area has a total workforce of 51,980 persons – 91.8% Main Workers and 8.2% Marginal Workers. Male workforce (60.3%) predominates.

The mine area does not cover any habitation. Hence the mining activities do not involve any displacement of human settlement. No public buildings, places, monuments etc. exist within

the lease area or in the vicinity. The mining operations will not disturb/relocate any village or need resettlement. Hence no R&R plan is involved.

11.7 Occupational Health and Safety Measures

M/s. Andhra Pradesh Granite (Midwest) Private Limited (APGMPL) will provide essential medicines at the site. The medicines and other test facilities will be provided at free of cost. The first aid box will be made available at the mine for immediate treatment.

11.8 Environmental Management Plan

- Mining operations will be done in compliance with the EC conditions.
- As per the requirement, after obtaining EC, CFE and CFO will be obtained and all conditions as laid by APPCB will be implemented.
- EMP as detailed in this report will be implemented.
- Plantation programme will be implemented to the green cover in the area.

11.9 Social Impact Assessment

- This assessment is specifically addressing the potential impacts which will be generated as a result of proposed development.
- Greenbelt will be developed on road surrounding areas.
- Organize health camps in the area to check the occurrence of any respiratory and other disorders.
- Regular water sprinkling on haul roads.
- improve the socio-economic status of the local people by creation of employment and paying taxes to gram Panchayath.
- During operational phase generate direct employment of 488 no’s from nearby villages.

11.10 Project Benefits

- M/s. Andhra Pradesh Granite (Midwest) Private Limited (APGMPL) will provide employment of 488 people in mine operations.
- Royalty payment to the Government of Andhra Pradesh and District Mineral fund for the development of infrastructure in the nearby villages.
- ESR implementation to improve living standard for the nearby villagers.

11.11 Environmental Social Responsibility (ESR)

As responsible corporate, M/s. Andhra Pradesh Granite (Midwest) Private Limited proposed to support the local villages for implementing environmental protection measures such as maintenance of Roads, greenery development.

It is proposed to spend Rs. 30.0 lakhs in the next 5 years period for maintenance of Road, Development of green belt on public road, Provide water harvesting structures, Conducting health camp in village etc.

11.12 Budget for Environmental Management Plan

It is proposed invest a capital amount of Rs. 56.0 lakhs and recurring expenditure of Rs 39.01 lakhs per annum.

11.13 Conclusion

M/s. Andhra Pradesh Granite (Midwest) Private Limited strongly believes in the concept of eco-friendly mining. Various socio economic development activities proposed will bring about overall socio economic development in the area.

The environmental impact study identified various mining activities that may result in change in baseline status, and accordingly the proposed mitigation measures mentioned in mine plan and the environment management plan were observed to be sufficient to ensure negligible impact on the surrounding areas. The residual impact due to mining is mainly loss of habitat which has to be compensated, loss of biodiversity which can be offset, and worked out mine pit which has to be reclaimed. The management assures to adopt extensive plantation in surrounding areas to compensate habitat and biodiversity loss, thereby reducing the impact to the maximum extent, while taking responsibility for implementing an effective mine closure plan. The environmental impact study identified various mining activities that may result in change in baseline status, and accordingly the proposed mitigation measures mentioned in the environment management plan were observed to be sufficient to ensure negligible impact on the surrounding areas.

Chapter - 12

Disclosure of Consultant

CHAPTER-12

DISCLOSURE OF CONSULTANT

Name of the project	**:**	**M/s Andhra Pradesh Granite (Midwest) Private Limited**
Environmental coordinator	**:**	**Mrs. T. Srilatha**
EIA Consultant Organization	**:**	**Pridhvi Envirotech (P) Limited**
Contact Information	**:**	# 184/C, 4th Floor, Lawn house, Vengalrao Nagar, Hyderabad-500038
Phone	**:**	040-40179770, 9849023662
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Website	**:**	www.pridhvienviro.com
Status of Accreditation with QCI/NABET	**:**	S.No140, List of Accredited EIA Consultant Organizations (Alphabetically) / Rev.18, 5th January 2022, published by QCI/NABET
Laboratory Engaged in EIA Project	**:**	Lawn Enviro Associates
Status of Laboratory	**:**	MoEF Accreditation S.O.857(E), dated 26th February 2018

Company Profile

Pridhvi Envirotech (P) Limited was established by professionals in the field of environment with nearly 30 years of experience with clear vision to serve customers in the field of environment.

Accredited by National Accreditation Board for Education and Training (NABET) under Quality Council of India as consultant for preparation of Environmental Impact Assessments for new and existing industries, Pridhvi Envirotech (P) Limited provides the following services:

- Provide complete and one stop consultancy to get Environmental clearances from MOEF and CFE and CFO from state Pollution Control Boards
- Provide complete assistance to industry by conducting legal gap analysis and advise actions to be taken
- Conduct EIA and EMP reports
- Conduct Due diligence audits for mergers and take overs

- Periodical monitoring of effluents/emissions/Hazardous Wastes from MOEF approved laboratory
- Provide assistance to industries to help them in case of legal compliances
- Provide consultancy services in the area of implementation of ISO 14001 systems and OHSAS 18001
- Provide consultancy services in design and execution of effluent treatment plants & Sewage treatment plants
- Provide consultancy services in hazardous waste handling and disposal services
- Waste mapping and audits and provide solutions for waste reduction
- Provide training to personnel on environment and safety management systems, legal requirements and other host of environmental, occupational Health and Safety issues

EIA Team members involved in the preparation of the EIA

Approved FAEs		
S.No	**Name of the FAE**	**Area of Expertise**
1	T. Srilatha	Environmental Coordinator
2	P.V.Raju	AQ/AP/SC/SHW
3	Dr.G.Pallavi	WP
4	Dr. N. Harinath Reddy	GEO/HG
5	T. Srilatha	NV/LU
6	I. Durga Prasad	SE
8	G. Rama Swami Reddy	EB
9	K. Raji Reddy	RH
Team Members		
10	Ch. Prasanth	AQ
11	L. Goverdhan Reddy	SE
12	G. Rama swami Reddy	SHW

Declaration by the head of the Accredited Consultant Organization

Declaration by the Head of the accredited consultant organization/ authorized person

I, **P. Venkata Raju**, here by, confirm that the above mentioned experts prepared the EIA **M/s Andhra Pradesh Granite (Midwest) Private Limited,** Pridhvi Envirotech (P) Limited is accredited to the sector 1 (a) as per the MoEF&CC and S.no 1 as per NABET certificate. All the conditions in the TOR is reviewed and addressed in the EIA report prepared. I also confirm that EIA Coordinator (EC) has gone through the report, and the consultant organization shall be fully accountable for any misleading information. It is certified that no unethical practices, plagiarism involved in carrying out the work and external data / text has not been used without proper acknowledgement while preparing this EIA report.

Signature:

Name: **P. Venkata Raju**

Designation: **Managing Director**

Name of the EIA consultant organization: **M/s. Pridhvi Envirotech Private Limited**
NABET Certificate No. & Issue Date: **NABET/EIA/1922/RA0132 date 20th June 2019**

Annexure – VII

Declaration by Experts contributing to the EIA of

M/s Andhra Pradesh Granite (Midwest) Private Limited

Sy.no 55/5 of Rajupalem- Lakshmipuram Villages, Chimakurthy Mandal, Prakasham District, Andhrapradesh

I, hereby, certify that I was a part of the EIA team in the following capacity that developed the above EIA.

	EIA Coordinator
Name:	Mrs. T.Srilatha
Signature:	T. Sri Latha
Date :	27/10/2021
	Team Member
Name:	**G.Rama Swami Reddy**
Signature:	GRSReddy
Date :	27.10.2021

Period of involvement: January 2019-September 2021
Study period: January 2019 – March 2019

Contact information: M/s. Pridhvi Enviro Tech Private Limited,
4th Floor, 184/C, Lawn house, Vengal Rao nagar,
Hyderabad – 500038; Ph.No 040-40179770
Email: pvr@pridhvienviro.in

Functional area experts:

S. No.	Functional areas	Name of the expert/s	Involvement (period and task**)	Signature and date
1	AP	P.Venkata.Raju	Period : January 2019-March 2019 Identification of air pollution sources and their impacts and mitigation measures	P.Venkata.R 27/10/21
2	WP	Dr. G. Pallavi	Period : January 2019-March 2019 Estimation of water and waste water generation and proposed treatment facilities	27/10/21
3	SHW	P.Venkata .Raju	Period : January 2019-March 2019 Identified hazardous and non-hazardous wastes from proposed project. Provided solid waste management.	P.Venkata.R 27/10/21
		G.Rama Swami Reddy (TM)		GRSReddy 27.10.2021

4	SE	I. Durga Prasad	Period : January 2019-March 2019 Conducted baseline socio economic survey. Identified Socio Economic impacts by the proposed project on nearby settlements.	
		L. Goverdhan Reddy (TM)		27/10/21
5	EB	G. Rama Swami Reddy	Period : January 2019-March 2019 Collected primary and Secondary data. Analysed the data. Identified impacts and suggested mitigation methods. Prepared green belt development plan	27/10/21
6	HG	N. Harinath Reddy	Period : January 2019-March 2019 Reviewed the surface hydrological data. Reviewed storm water drainage plan including management of storm water. Identified ground water potential and also identified watershed areas for surface water.	
7	GEO	N. Harinath Reddy	Period : January 2019-March 2019 Analysis of geology and geomorphology. Review of maps	
8	AQ	P. Venkata Raju	Period : January 2019-March 2019 Review of secondary data on meteorology. Air Quality modelling using AERMOD. Representation of GLCS using isopleths for different Scenarios.	27/10/21
9	NV	T.Srilatha	Period : January 2019-March 2019 Reviewed the baseline noise monitoring data. Identified probable noise and vibration impacts of the proposed sectors. Representation of same using relevant models. Identified impacts of noise from proposed industries and Suggested mitigation and control measures.	27/10/2021
10	LU	T.Srilatha	Period : January 2019-March 2019 Reviewed the analysis of generated data for land use and review of maps. Identified probable impacts on Land Use by the proposed project. Generated data related to land use pattern. Prepared GIS based maps like land use map. Contribution to EIA documentation.	27/10/2021

		M.Divya (TM)		27/10/21
11	SC	P. Venkata Raju	Period : January 2019-March 2019 Identification of impacts and suggested mitigation measures	P. Venkatesh 27/10/21
12	RH	K. Raji Reddy	Period : January 2019-March 2019 Identified hazardous processes and risks in proposed sectors. Prepared on- site & off-site Emergency preparedness plan and Disaster Management Plan.	
13	The Base line analysis of the all parameters EIA of "M/s Andhra Pradesh Granite (Midwest) Private Limited" is done by M/s Lawn Enviro Associates",184/C, lawn House, Vengalrao nagar, Hyderabad-500038. Which is accredited by the NABL & MoEF Accreditation no S.O 857(E) Dated: 26.02. 2018.			

*One TM against each FAE may be shown

**Please attach additional sheet if required

Declaration by the Head of the accredited consultant organization/ authorized person

I, **P. Venkata Raju**, here by, confirm that the above mentioned experts prepared the EIA **M/s Andhra Pradesh Granite (Midwest) Private Limited,** Pridhvi Envirotech (P) Limited is accredited to the sector 1 (a) as per the MoEF&CC and S.no 1 as per NABET certificate. All the conditions in the TOR is reviewed and addressed in the EIA report prepared. I also confirm that EIA Coordinator (EC) has gone through the report, and the consultant organization shall be fully accountable for any misleading information. It is certified that no unethical practices, plagiarism involved in carrying out the work and external data / text has not been used without proper acknowledgement while preparing this EIA report.

Signature:

Name: **P. Venkata Raju**

Designation: **Managing Director**

Name of the EIA consultant organization: **M/s. Pridhvi Envirotech Private Limited**
NABET Certificate No. & Issue Date: **NABET/EIA/1922/RA0132 date 20th June 2019**

Annexure –I

Copy of Current Environmental Clearance

सत्यमेव जयते	**State Level Environment Impact Assessment Authority (SEIAA)** **Andhra Pradesh** **Ministry of Environment, Forests & Climate Change** **Government of India** D.No.33-26-14 D/2, Near Sunrise Hospital, Pushpa Hotel Centre, Chalamavari Street, Kasturibaipet, Vijayawada-520010

REGD POST WITH ACK.DUE

Order No. SEIAA/AP/PKM/MIN/MIO/08/2018/672 [021] **Dt:13.09.2019**

Sub: SEIAA, A.P - 19.425 Ha. Black Galaxy Granite Mine of M/s.Andhra Pradesh Granite (Midwest) Private Limited, Sy.No.55/5, Rajupalem Lakshmipuram (V), Chinnakurthy (M), Prakasam District, Andhra Pradesh - Environmental Clearance – Expansion - Issued - Reg.

I. This has reference to your application submitted through online on 09.07.2019 (Proposal No. SIA/AP/MIN/**27076**/2019), seeking Environmental Clearance for expansion of **19.425 Ha Black Galaxy Granite Mine at Sy No: 55/5, Rajupalem Lakshmipuram (V), Chinnakurthy (M), Prakasam District, Andhra Pradesh** in favour of **M/s. Andhra Pradesh Granite (Midwest) Private Limited.** It was reported that the nearest human habitation viz., Rajupalem - Laxmipuram (V) exists at a distance of about 0.7 km from the mine lease area. It was noted that the capital investment of the project is Rs.3051.97 Lakhs and capacity of the project is as follows:

Mining of Black Galaxy Granite - 1,026 m^3/annum to 54,643 m^3/annum in 19.425 Ha.

II. The location of the mine is as follows:

Sl.No	Latitude	Longitude
1.	15°35'09.70"N	79°49'51.60"E
2.	15°35'13.10"N	79°49'19.20"E
3.	15°35'06.00"N	79°49'20.60"E
4.	15°35'09.70"N	79°49'51.60"E
5.	15°35'06.18"N	79°49'25.21"E
6.	15°35'07.20"N	79°49'25.50"E
7.	15°35'06.10"N	79°49'28.90"E
8.	15°35'03.12"N	79°49'30.28"E
9.	15°35'04.90"N	79°49'29.10"E
10.	15°35'04.50"N	79°49'40.20"E
11.	15°35'03.66"N	79°49'53.06"E

III. It is an open cast semi-mechanized mine. The total mine lease area is 19.425 Ha. The present proposal comes under category 'B2'. The proposed project falls under Item No.1(a) of the schedule of the EIA Notification 2006-(i) Mining of Minerals (<100 Ha of mining lease area in respect of non-coal mine lease).

The proposal has been examined and processed in accordance with EIA Notification, 2006 and its amendments thereof. The State Level Expert Appraisal Committee (SEAC) examined the application, in its meeting held on **20.07.2019**. The representatives of the project proponent and their M/s Pridvi Enviro tech have attended the meeting. The Committee noted that SEIAA issued TOR& Specific TOR on 24.01.2019. The public hearing was conducted on 01.07.2019. The project proponent submitted the final EIA report along with Public Hearing minutes through online on 09.07.2019. The Committee recommended for issue of Environmental clearance for expansion to this existing Black Galaxy Granite mining project for the expansion of production quantities: Black Galaxy Granite from– **1026 m^3 /annum to 54,643 m^3 /annum**, duly stipulating a condition that the project proponent shall maintain the setback distance 7.5 meters buffer zone all around the mine lease area for greenbelt development and other conditions are to be fulfilled. The project proponent shall allocate sufficient funds for implementation of CSR activities as committed by the representative along with the EMP. The avenue plantation (tall plants), of at least 1.5m height, for 1 km length of the approach road on either side of the road is to be developed and maintained. The recommendation of the committee is without prejudice to the final outcome of the judgement in *W.P.No.44348/2017 of Hon'ble High Court of A.P., and any* order passed by the Hon'ble Courts. The State Level Environment Impact Assessment Authority (SEIAA), in its meeting held on **13.08.2019** examined the proposal and Agreed with recommendation of the SEAC, A.P., for issue of Environmental Clearance subject to final outcome of the judgment of the Hon'ble High Court of A.P., in W.P.No.44348 of 2017. The SEIAA, A.P **hereby accords Environmental Clearance to the project** as mentioned at Para No. I under the provisions of the EIA Notification 2006 and its subsequent amendments issued under Environment (Protection) Act, 1986 subject to implementation of the following cluster, specific and general conditions:

A. Specific Conditions:

i. The project proponent shall maintain the setback distance 7.5 meters buffer zone all around the mine lease area for greenbelt development and other conditions are to be fulfilled.
ii. The project proponent shall allocate sufficient funds for implementation of CSR activities as committed by the representative along with the EMP
iii. The avenue plantation (tall plants), of at least 1.5m height, for 1 km length of the approach road on either side of the road is to be developed and maintained

1) **Air Pollution:-**

i. Wet drilling & wire saw cutting method shall be adopted to control dust emissions. Delay detonators and shock tube initiation system for controlled sequential blasting shall be used so as to reduce vibration and dust.

ii. Greenbelt shall be developed along the boundary of mining lease area and also in back filled and reclaimed areas with tall growing native species in consultation with the local DFO/Agriculture Department. The proponent of mine shall carry mining operations in such a manner so as to cause least damage to the flora of the mining area and nearby areas. He shall take immediate measures for planting in the same area or any other area selected by authorities not less than twice the number of trees going to be felled by mining operations. He shall also take measures for restoration of other flora /fauna if damaged by mining operations.

iii. Effective safe guard measures such as regular water sprinkling shall be carried out in critical areas prone to air pollution and having high levels of SPM and RPM such as haul road, loading and unloading point and transfer points. It shall be ensured that the ambient air quality parameters conform to the norms prescribed by the Central Pollution Control Board in this regard.

iv. The proponent of mine shall carry air quality monitoring in the core zone as well as buffer zone for RSPM (PM10) and Noise levels. Location of monitoring stations should be decided based on the metallurgical data topographical features and environmentally and ecologically sensitive targets and frequency of monitoring should be undertaken in consultation with Andhra Pradesh Pollution Control Board. The data so recorded should be regularly submitted to the Ministry including its Regional office located at Chennai and the Andhra Pradesh Pollution Control Board/Central Pollution Control Board once in six months.

v. The proponent shall construct graded roads connecting the mining area to the nearby roads to avoid dust nuisance due to vehicular movements.

vi. The proponent shall take precautions against noise arising out of mining operations and shall be abated or controlled at the source so as to keep it within the permissible limits notified under Environmental (Protection) Act, 1986 / Noise Pollution(Regulations & Control) Rules, 2010 by implementing the following noise control measures.

- Proper and regular maintenance of vehicles and other equipment
- Limiting time exposure of workers to excessive noise.
- The workers employed shall be provided with protection equipment and earmuffs etc.
- Speed of trucks entering or leaving the mine is to be limited to moderate speed of 25 kmph to prevent undue noise from empty trucks.

vii. Whenever any damage to public buildings or monuments is apprehended due to their proximity to the mining lease area, scientific investigations shall be carried out by the holder of mining lease so as to keep the ground vibrations caused by mining operations within safe limit.

2) Water Pollution:-

i. The source of water is Bore well. Total water requirement is 60 KLD which is used for dust suppression; wire saw cutting; development of green belt and for domestic purpose.

ii. Garland drain and Siltation ponds of appropriate size should be constructed for the working pit to arrest flow of silt and sediment. The water so collected should be utilized for watering the mine area, roads, green belt development etc. The drains should be regularly desilted, particularly after monsoon, and maintained properly.

iii. The proponent of the mine shall take all possible precautions to prevent or reduce the discharge of toxic and objectionable liquid effluents from mine, workshop, tailing ponds into surface bodies, ground water aquifier and useable lands to a minimum. The effluents shall be suitably treated, if required, to conform to the general standards notified under Environmental (Protection) Act, 1986.

iv. Monitoring of ground water level and quality should be carried out quarterly by the project proponent in and around the project area in consultation with State Ground Water Department/Central Ground Water Authority and data thus collected shall be submitted regularly to the MOEF&CC and its Regional Office Chennai, CGWA, and the Regional Director, Central Ground Water Board, Hyderabad. If at any stage, it is observed that the ground water table is getting depleted due to the mining activity, necessary correction measures shall be carried out.

v. Suitable conservation measures to augment groundwater resources in the area shall be planned and implemented in consultation with Regional Director, CGWB, Southern Region, Hyderabad. Suitable measures should be taken for rainwater harvesting.

vi. Permission from the competent authority should be obtained for drawl of ground water, if any, required for this project.

4) Solid Waste :-

i. **Topsoil:** Wherever top soil exists and is to be excavated for mining operations, it shall be removed and stacked separately and top soil so removed shall be utilized for restoration or rehabilitation of the land, which is no longer required for mine operations or for stabilizing or landscaping the external dumps. Whenever the top soil cannot be utilized concurrently, it shall be stored separately for future use.

ii. **Overburden:** The proponent of mine shall take steps so that the overburden, waste rock, rejects and fines generated during mining operations shall be stored in separate dumps preferably on impervious grounds. The waste rock, overburden etc. shall be concurrently backfilled into the mine excavations so as to restore the land to its original use as far as possible. In the case of non feasibility of back filling, the waste dump shall be suitable terraced and stabilize through the vegetation. The proponent shall maintain proper angle of repose to ensure stability to the dump.

iii. The proponent of the mine shall construct required number of retaining walls to provide stability to the dumps. Dimensions of the retaining walls shall be based on the rainfall data.

iv. The proponent of mines shall construct required number of garland drains to arrest mineral particles being carried away as runoff during rainy seasons around the dump yards. Dimensions of the garland rains shall be based on rainfall data.

v. Waste oils, used oils generated from the EM machines, mining operations, if any, shall be disposed as per the Hazardous Wastes (Management, Handling, and transboundary movement) Rules, 2008 and its amendments thereof to the recyclers authorized by APPCB.

vi. The proponent of the mine shall undertake phased restoration, reclamation and rehabilitation of the lands affected by the mining operations and shall complete this work before the conclusion of such operations and the abandonment of the mine.

B. General Conditions:

i. This order is valid for a period of 20 years or the expiry date of mine lease period issued by the Government of A.P., whichever is earlier.

ii. "Consent for Establishment" & "Consent for Operation" shall be obtained from Andhra Pradesh Pollution Control Board under Air and Water Act and effectively comply with all the conditions stipulated thereof.

iii. No change in mining technology and scope of working should be made without prior approval of the SEIAA, A.P. No further expansion or modifications in the mine shall be carried out without prior approval of the SEIAA, AP/ MOEF&CC, GoI, New Delhi, as applicable.

iv. Personnel working in dusty areas shall be provided with protective respiratory devices and they should wear, and they should also be provided with adequate training and information on safety and health aspects.

v. The project proponent shall ensure that no natural watercourse and/or water resources shall be obstructed due to any mining operations. Necessary safeguard measures to protect the first order streams, if any, originating from the mine lease shall be taken.

vi. Occupational health surveillance program of the workers should be undertaken periodically to observe any contractions due to exposure to dust and take corrective measures, if needed.

vii. A separate environmental management cell with suitable qualified personnel should be set-up under the control of a Senior Executive, who will report directly to the Head of the Organization.

viii. The funds earmarked for environmental protection measures **(Capital cost is Rs.52.37 Lakhs and Recurring cost is Rs.35.5 Lakhs/annum)** should be kept in separate account and should not be diverted for other purpose. Year wise expenditure should be reported to the Ministry and its Regional Office located at Chennai.

ix. The Regional Office of MOEF&CC located at Chennai / The SEIAA, Andhra Pradesh through the Regional Offices of Andhra Pradesh Pollution Control Board, shall monitor compliance of the stipulated conditions. The project authorities should extend full cooperation to the officer (s) of the Regional Office by furnishing the requisite data/information/monitoring reports.

x. The project proponent shall submit six monthly reports on the status of compliance of the stipulated environmental clearance conditions including results of monitored data (both in hard copies as well as by e-mail) to the Ministry of Environment & Forests, its Regional Office, Chennai, SEIAA, A.P., Zonal Office of Central Pollution Control Board, Bangalore, District Collector and A.P. Pollution Control Board. The proponent shall upload the status of compliance of the environmental clearance conditions including results of monitored data on their websites and shall update the same periodically

xi. The project proponent shall submit the copies of the environmental clearance to the Heads of local bodies, Panchayats and Municipal Bodies in addition to the relevant offices of the Government who in turn has to display the same for 30 days from the date of receipt.

xii. The environmental statement for each financial year ending 31st march in Form-V as mandated is to be submitted by the project proponent to the A.P. Pollution Control Board as prescribed under the Environment (Protection) Rules, 1986, as amended subsequently, shall also be put on the website of the company along with the status of compliance of environmental clearance conditions and shall also be sent to the Regional office of the Ministry of Environment and Forests, Chennai by e-mail.

xiii. The project authorities should advertise at least in two local newspapers widely circulated, one of which shall be in the vernacular language of the locality concerned, within 7 days of the issue of the clearance letter informing that the project has been accorded environmental clearance and a copy of the clearance letter is available with the State Pollution Control Board and SEIAA, A.P.

xiv. The proponent shall obtain all other mandatory clearances from respective departments.

xv. Any appeal against this Environmental Clearance shall lie with the National Green Tribunal, if preferred, within a period of 30 days as prescribed under Section 16 of the National Green Tribunal Act, 2010.

xvi. Concealing the factual data or submission of false / fabricated data and failure to comply with any of the conditions mentioned above may result in withdrawal of this clearance and attract action under the provisions of Environment (Protection) Act, 1986.

xvii. The SEIAA may revoke or suspend the order, if implementation of any of the above conditions is not satisfactory. The SEIAA reserves the right to alter/modify the above conditions or stipulate any further condition in the interest of environment protection.

xviii. The above conditions will be enforced inter-alia, under the provisions of the Water (Prevention & Control of Pollution) Act, 1974, the Air (Prevention & Control of Pollution) Act, 1981, the Environment (Protection) Act, 1986 and the Public Liability Insurance Act, 1991 along with their amendments and rules.

Sd/-	**Sd/-**	**Sd/-**
MEMBER SECRETARY,	**MEMBER,**	**CHAIRMAN,**
SEIAA, A.P.	**SEIAA, A.P.**	**SEIAA, A.P.**

To

M/s. AP Granites (Midwest) Private Limited,
No. 8-2-684/3/25 &26, Road No.12,
Banjara Hills, Hyderabad,
Telangana -500034,
Ph. 9912237333

//T.C.F.B.O//

MEMBER SECRETARY
SEIAA, A.P.

Annexure –II

Copy of combined CFE and CFO

ANDHRA PRADESH POLLUTION CONTROL BOARD
D.No.33-26-14 D/2, Near Sunrise Hospital, Pushpa Hotel Centre,
Chalamalavari Street, Kasturibaipet, Vijayawada – 520 010.

RED CATEGORY

COMBINED CONSENT FOR ESTABLISHMENT, OPERATION & HW AUTHORIZATION ORDER (AFTER EXPANSION)

Consent Order No: APPCB/VJA/NLR/188/HO/CFO/2017- **Date: 21.10.2019**

CONSENT is hereby granted for Operation under section 25/26 of the Water (Prevention & Control of Pollution) Act, 1974 and under section 21 of Air (Prevention & Control of Pollution) Act 1981 and amendments thereof and Authorisation under Rule 6 of the Hazardous and Other Wastes (Management & Transboundary Movement) Rules, 2016 & Amendments thereof and the rules and orders made there under (hereinafter referred to as 'the Acts', `the Rules') to:

M/s. Andhra Pradesh Granite (Mid West) Pvt. Ltd.,
Sy. No. 55/5P, Rajupalem – Lakshmipuram (V),
Chimakurthy (M), Prakasam District – 523 226, A.P
E-mail: mlnsharma@midwestgranite.com

The Consent Order No: APPCB/VJA/NLR/188/HO/CFO/2017-469, Dt:02.06.2017 issued to the industry stands cancelled.

The Board, after careful scrutiny of the application, verification report of Regional Officer and recommendations of the CFO Committee, hereby issues **COMBINED CONSENT FOR ESTABLISHMENT, OPERATION & HWA ORDER** to the activity duly considering the CFE Expansion application and EC expansion order.

(Hereinafter referred to as 'the Applicant') authorizing to operate the industrial plant to discharge the effluents from the outlets and the quantity of emissions per hour from the chimneys as detailed below:

(i) Outlets for discharge of effluents:

Outlet No.	Outlet Description	Max Daily Discharge After Expansion (KLD)	Point of Disposal
1	Domestic	16.0	Septic tank followed by soak pit

ii) Emissions from chimneys:

Chimney No.	Description of Chimney	Quantity of Emissions at peak flow (m^3/hr) After Expansion
1	Common stack attached to 200 KVA & 125 KVA D.G. sets (Existing)	---
2	Common stack attached to 500 KVA & 125 KVA D.G. sets (Additional)	---

iii) Hazardous Waste Authorisation (Form – II) [See Rule 6 (2)]:

M/s. Andhra Pradesh Granite (Mid West) Pvt. Ltd. (19.425 Ha), Prakasam District is hereby granted an authorization to operate a facility for collection, reception, storage, treatment, transport and disposal of Hazardous Wastes namely:

- **Hazardous Wastes With Disposal Option:**

S. No	Name of Hazardous Waste	Stream	Quantity After Expansion	Method of disposal
1.	Used oil	5.1 of Schedule-I	1000 LPA	Disposed to authorized reprocessing agencies.

This consent order is valid for mining the following quantities only:

S. No	Product	Quantity After Expansion
1	**Mining of Black Galaxy Granite in an extent of 19.425 Ha.**	**54,643 m^3/annum**

This order is subject to the provisions of `the Acts' and the Rules' and orders made thereunder and further subject to the terms and conditions incorporated in the schedule A, B & C enclosed to this order.

This Combined Order of Consent for Establishment, Operation & HWA (After Expansion) is valid up to 31st July, 2022.

Bandla Siva Sankar Prasad

Digitally signed by Bandla Siva Sankar Prasad
DN: c=IN, o=APPCB, ou=EFS and T Head Office, cn=Bandla Siva Sankar Prasad, postalCode=520010, 2.5.4.20=c040c01bf7f6f398976702e0cd b9b47945 4f18c98b21c03bc47f7349d65 55b35, st=Andhra Pradesh
Date: 2019.10.22 12:50:29 +05'30'

CHAIRMAN

To

M/s. Andhra Pradesh Granite (Mid West) Pvt. Ltd.,
Sy. No. 55/5P, Rajupalem – Lakshmipuram (V),
Chimakurthy (M), Prakasam District – 523 226, A.P
E-mail: mlnsharma@midwestgranite.com

Copy to:

The JCEE, Zonal Office, **Vijayawada** for information and necessary action.
The EE, Regional Office, **Ongole** for information and necessary action.

SCHEDULE-A

1. Any up-set condition in any industrial plant / activity of the industry, which result in, increased effluent / emission discharge and/ or violation of standards stipulated in this order shall be informed to this Board, under intimation to the Collector and District Magistrate and take immediate action to bring down the discharge / emission below the limits.
2. The industry shall carryout analysis of waste water discharges or emissions through chimneys for the parameters mentioned in this order on quarterly basis and submit to the Board.
3. The industry shall put up two sign boards (6x4 ft. each) at publicly visible places at the main gate indicating the products, effluent discharge standards, air emission standards, hazardous waste quantities and validity of CFO and exhibit the CFO order at a prominent place in the factory premises.
4. Notwithstanding anything contained in this consent order, the Board hereby reserves the right and powers to review / revoke any and/or all the conditions imposed herein above and to make such variations as deemed fit for the purpose of the Acts by the Board.
5. The industry shall file the water cess returns in Form-I as required under section (5) of Water (Prevention and Control of Pollution) Cess Act, 1977 on or before the 5th of every calendar month, showing the quantity of water consumed in the previous month along with water meter readings. The industry shall remit water cess as per the assessment orders as and when issued by Board.
6. The applicant shall submit Environment statement in Form V before 30th September every year as per Rule No.14 of E (P) Rules, 1986 & amendments thereof.
7. The applicant shall make applications through Online for renewal of Consent (under Water and Air Acts) and Authorization under HWM Rules at least 120 days before the date of expiry of this order, along with prescribed fee under Water and Air Acts and detailed compliance of CFO conditions for obtaining Consent & HW Authorization of the Board. The industry should immediately submit the revised application for consent to this Board in the event of any change in the raw material used, processes employed, quantity of trade effluents & quantity of emissions. Any change in the management shall be informed to the Board. The person authorized shall not let out the premises / lend / sell / transfer their industrial premises without obtaining prior permission of the State Pollution Control Board.

8. Any person aggrieved by an order made by the State Board under Section 25, Section 26, Section 27 of Water Act, 1974 or Section 21 of Air Act, 1981 may within thirty days from the date on which the order is communicated to him, prefer an appeal as per Andhra Pradesh Water Rules, 1976 and Air Rules 1982, to Appellate authority constituted under Section 28 of the Water (Prevention and Control of Pollution) Act, 1974 and Section 31 of the Air (Prevention and Control of Pollution) Act, 1981.

SCHEDULE-B

Water Pollution:

1. The following is the permitted water consumption:

S. No	Purpose	Quantity After Expansion (KLD)
1.	Dust suppression & Wire saw cutting	36.0
2.	Development of Greenbelt	20.0
3.	Domestic	4.0
	Total	**60.0**

Separate meters with necessary pipe-line shall be maintained for assessing the quantity of water used for each of the purposes mentioned above for Cess assessment purpose.

2. The mining unit shall not cause ground water pollution in and around the mining area.

Air Pollution:

3. The emissions shall not contain constituents in excess of the prescribed limits mentioned below:

Chimney No.	Parameter	Emission Standards
1 & 2	Particulate matter	115 mg/Nm^3

4. The industry shall comply with ambient air quality standards of PM_{10} (Particulate Matter size less than 10μm) - 100 μg/m^3; $PM_{2.5}$ (Particulate Matter size less than 2.5 μm) - 60 μg/m^3; SO_2 - 80 μg/m^3; NO_x - 80 μg/m^3, outside the factory premises at the periphery of the industry.

   Standards for other parameters as mentioned in the National Ambient Air Quality Standards CPCB Notification No.B-29016/20/90/PCI-I, dated 18.11.2009.

   Noise Levels: Day time (6 AM to 10 PM) - 75 dB (A)
   Night time (10 PM to 6 AM) - 70 dB (A)

5. The industry shall comply with emission limits for DG sets of capacity upto 800 KW as per the Notification G.S.R.520 (E), dated 01.07.2003 under the Environment (Protection) Amendment Rules, 2003 and G.S.R.448(E), dated 12.07.2004 under the Environment (Protection) Second Amendment Rules, 2004. In case of DG sets of capacity more than 800 KW shall comply with emission limits as per the Notification G.S.R.489 (E), dated 09.07.2002 at serial no.96, under the Environment (Protection) Act, 1986.

6. The industry shall comply with the following for controlling fugitive emissions.
   - Machineries shall be properly maintained to prevent undesirable noise. Attention shall be paid towards rigorous maintenance of the silencers of diesel engines
   - Drills should be water-jacked. Local exhaust ventilation systems should be installed at dust generation points and the dust is led to a dust collection system
   - Surface drilling and blasting operations shall not be carried out at night.
   - The drilling shall be done with sharp drilling bits to reduce the generation of noise during drilling
   - Safety equipment shall be given to the mine operators.
   - Periodical monitoring of noise level of mining machines and at some locations in the mining site shall be done with the help of noise level meter.

7. The industry shall provide water sprinklers for wetting the roads and at dust generating sources to control fugitive dust emissions.
8. The industry shall maintain the following records and the same shall be made available to the inspecting officers of the Board:
    a) Daily production details
    b) Quantity of effluents generated, treated, recycled / reused and disposed
    c) Log Books for pollution control systems.
    d) Characteristics of ambient air, effluents and emissions
    e) Hazardous / non hazardous Solid waste generated and disposed.
    f) Inspection book.
    g) Manifest copies of hazardous waste.
9. The industry shall dispose solid waste (NON HAZARDOUS) as follows :

S. No	Name of the Solid Waste	Quantity After Expansion	Disposal
1.	Over burden & mine rejects	1296760 m^3/annum	Shall be stored in designated area as per the approved mine plan.

General :

10. **The industry shall handover the Original Consent order dated 02.06.2017 to the RO: Ongole after receipt of this CFO order.**
11. **The industry shall not carry out the mining activity till industry obtains Revocation of Stop Production Order issued on 18.05.2019.**
12. The industry shall maintain the setback distance 7.5 meters buffer zone all around the mine lease area for greenbelt development and other conditions are to be fulfilled.
13. The industry shall allocate sufficient funds for implementation of CSR activities as committed by the representative along with the EMP.
14. The avenue plantation (tall plants), of at least 1.5m height, for 1 km length of the approach road on either side of the road is to be developed and maintained.
15. The industry shall comply with the conditions stipulated in the EC Order Dt.13.09.2019.
16. The industry shall construct graded roads connecting the mining area to the nearby roads to avoid dust nuisance due to vehicular movements.
17. Garland drain and Siltation ponds of appropriate size should be constructed for the working pit to arrest flow of silt and sediment. The water so collected should be utilized for watering the mine area, roads, green belt development etc. The drains should be regularly desilted, particularly after monsoon, and maintained properly.
18. The industry shall construct required no. of garland drains to arrest mineral particles being carried away as runoff during rainy seasons around the dump yards. Dimensions of the garland rains shall be based on rainfall data.
19. Soil binding and nitrogen fixing plants shall plant in the mining lease area. Biological reclamation should be done in two phases, the first phase shall be paint appropriate quick growing grass and shrubs and the second phase shall grow slower growing native shrubs and trees.
20. Ground water table level shall be monitored every season. Any lowering of ground water table in comparison to the previous seasons should be reported to Board immediately. Discarded pits should be allowed to fill with water.
21. Vehicles should be well maintained and engine idling should be minimized. Vehicle cabs shall be made dust-proof and to avoid oil spillages within the mining area.
22. The mine rejects shall be stacked in earmarked area as per approved mine plan in a pyramid form with garland drain around the stacking area and mining shall be carried out as per approved mine plan.

23. Suitable tree species should be planted on either side of the haul roads.

24. Greenbelt shall be developed along the boundary of the site and shall develop compensative afforestation at anywhere outside the mining area.

25. A separate environmental management cell with suitable qualified personnel should be set-up under the control of a Senior Executive, who will report directly to the Head of the Organization.

26. The funds earmarked for environmental protection measures (Capital cost is Rs.52.37 Lakhs and Recurring cost is Rs.35.5 Lakhs/annum) should be kept in separate account and should not be diverted for other purpose. Year wise expenditure should be reported to the Ministry and its Regional Office located at Chennai marking a copy to this office.

27. The industry shall comply with standards and directions issued by CPCB/MoEF & CC as and when notifications are issued.

28. The industry shall submit six monthly reports on the status of compliance of the stipulated environmental clearance conditions including results of monitored data (both in hard copies as well as by e-mail) to the Ministry of Environment & Forests, its Regional Office, Chennai, SEIAA, A.P., Zonal Office of Central Pollution Control Board, Bangalore, District Collector and A.P. Pollution Control Board for every 6 months as on 01st January and 01st July of every year and the same shall be uploaded to the concerned websites.

SCHEDULE – C

[See rule 6 (2)]

[CONDITIONS OF AUTHORISATION FOR OCCUPIER OR OPERATOR HANDLING HAZARDOUS WASTES]

1. The authorised person shall comply with the provisions of the Environment (Protection) Act, 1986, and the rules made there under.

2. The authorisation or its renewal shall be produced for inspection at the request of an officer authorised by the State Pollution Control Board.

3. The person authorised shall not rent, lend, sell, transfer or otherwise transport the Hazardous and other wastes except what is permitted through this authorisation.

4. Any unauthorised change in personnel, equipment or working conditions as mentioned in the application by the person authorised shall constitute a breach of his authorisation.

5. The person authorised shall implement Emergency Response Procedure (ERP) for which this authorisation is being granted considering all site specific possible scenarios such as spillages, leakages, fire etc. and their possible impacts and also carry out mock drill in this regard at regular interval of time;

6. The person authorised shall comply with the provisions outlined in the Central Pollution Control Board guidelines on “Implementing Liabilities for Environmental Damages due to Handling and Disposal of Hazardous Waste and Penalty”.

7. It is the duty of the authorised person to take prior permission of the State Pollution Control Board to close down the facility.

8. An application for the renewal of an authorisation shall be made as laid down under these Rules.

9. Any other conditions for compliance as per the Guidelines issued by the Ministry of Environment, Forest and Climate Change or Central Pollution Control Board from time to time.

10. Annual return shall be filed by June 30th for the period ensuring 31st March of the year.

11. The authorised person shall not store hazardous waste for more than 90 days as per the Hazardous & Other Wastes (Management and Transboundary Movement) Rules, 2016.

12. The authorised person shall store Used / Waste Oil and Used Lead Acid Batteries in a secured way in their premises till its disposal to the manufacturers / dealers on buyback basis.

13. The authorised person shall maintain 7 copy manifest system for transportation of waste generated and a copy shall be submitted to concerned Regional Office of APPCB. The driver who transports Hazardous Waste should be well acquainted about the procedure to be followed in case of an emergency during transit. The transporter should carry a Transport Emergency (TREM) Card.

14. The authorised person shall maintain proper records for Hazardous & other wastes stated in Authorization in FORM-3 i.e., quantity of Incinerable waste, land disposal waste, recyclable waste etc., and file annual returns in Form- 4 as per Rule 6 (5) of the Hazardous & Other Wastes (Management & Transboundary Movement) Rules, 2016 and amendments thereof by June 30th for the period ensuring 31st March of the year.

15. The authorised person shall submit the condition wise compliance report of the conditions stipulated in Schedule A, B & C of this Order on half yearly basis to Board Office and concerned Regional Office.

Bandla Siva Sankar Prasad

Digitally signed by Bandla Siva Sankar Prasad
DN: c=IN, o=APPCB, ou=EFS and T Head Office, cn=Bandla Siva Sankar Prasad, postalCode=520010, 2.5.4.20=c040c01bf76f3198976702e0cdb9fb479454ff18c98b21c03bc47f7349d6555b35, st=Andhra Pradesh
Date: 2019.10.22 12:51:08 +05'30'

CHAIRMAN

To

M/s. Andhra Pradesh Granite (Mid West) Pvt. Ltd.,
Sy. No. 55/5P, Rajupalem – Lakshmipuram (V),
Chimakurthy (M), Prakasam District – 523 226, A.P
E-mail: mlnsharma@midwestgranite.com

Annexure-III

Copy of certified compliance report

I/6958/2022

भारत सरकार / Government of India
पर्यावरण, वन एवं जलवायु परिवर्तन मंत्रालय
Ministry of Environment, Forest and Climate Change
एकीकृत क्षेत्रीय कार्यालय / Integrated Regional Office
ग्रीन हाउस, गोपालारेड्डी रोड, विजयवाड़ा, आंध्र प्रदेश – 520010
Green House, Gopalareddy Road,
Vijayawada, Andhra Pradesh-520010.
Tel: +91 866 – 2419787, +91 866 – 2419788, E-mail: iro.vijayawada-mefcc@gov.in, suresh.pasupuleti@gov.in

प्रमाणित अनुपालन रिपोर्ट / Certified Compliance Report

SN	Item	Details
1.	परियोजना का प्रकार **Type of Project**	Non-coal Mining (Galaxy Granite Mine)
2.	परियोजना का नाम **Name of Project**	19.425 Ha. Black Galaxy Granite Mine of M/s. Andhra Pradesh Granite (Midwest) Pvt. Ltd., Sy. No. 55/5P, RL Puram (V), Chimakurthy (M), Prakasam District.
3.	परियोजना प्राधिकरण का पता **Address of project authorities**	The Director, M/s. Midwest Granite Pvt. Ltd., D. No. 8-2-684/3/25 & 26, Road No. 12, Banjara Hills, Hyderabad – 500034.
4.	पर्यावरणीय स्वीकृति पत्र सं0 एवं तिथी **Env. Clearance Letter No. & Date**	SEIAA/AP/PKM/MIN/VIO/ 08/2018/672-1021 dated 13.09.2019 SEIAA/AP/PKM-119/2013-1593 dated 10.06.2013
5.	क्षेत्रीय कार्यालयपत्र सं0 **Regional Office File No.**	IRO/VIJ/EPA/MISC/111-01/2021
6.	स्थल दौरातिथी **Date of Site Visit**	30.08.2021
7.	परियोजना की स्थिति **Status of Project**	Operational

8. परियोजना की वर्तमान स्थिति / Present status of the Project:

The project was monitored by undersigned on 30.08.2021. During monitoring, Shri O. Madhusudhan Reddy, GM (Env), Shri K. Damodar Reddy, (Agent cum Manager) and other Officials were present. As per the discussions held with Project Authorities (PAs), it has been observed that the mine is galaxy granite stone mine with the total mine lease area of 19.425 ha. It has been observed that the PAs have obtained combined Consent for Establishment (CFE), Consent for Operation (CFO) and Hazardous Waste Authorization (HWA) vide Consent Order No. APPCB/VJA/NLR/188/HO/CFO/2017 dated 21.10.2019 for mining of black galaxy granite of 54,643 m^3/annum. The CFO is valid till 31.01.2022.

It has been observed that the PAs have obtained permission to extend opencast working at common boundary from Directorate General of Mines

Safety (DGMS), GOI vide letter No. Perm-111(3)/11 dated 20.3.2011 and Perm-111(3)/12 dated 30.5.2012. The details of common boundary between other mine leases are as follows:

1. M/s Golden Granite Mine, Survey No. 55/6 of R.L. of 19.43 ha.
2. Sri Surya Granite Mine, Survey Nos. 976/5, 977/9(P0, 10(P), 11, 978/1,2,3,4,5,6 and 979/1,2,8,9,10 over an extent of 14.38 acre.
3. M/s Midwest Granite Mine, Sy. No. 55/5, 59/P & 103/P over an extent of 10.761 ha
4. M./s Alliance AP Black Galaxy Granites Pvt. Ltd, Sy. No. 52, 55/3C, 55/4A over an extent of 54.06 acre.

The photographs of the mine area are as follows:





The satellite image of the project location by google maps is as follows:



It has been observed that the Project Authorities (PAs) have complied or are in process of complying the conditions stipulated in EC. The detailed observations are as follows:

9. Stipulated Conditions:

EC: SEIAA/AP/PKM/MIN/VIO/08/2018/672-1021 dated 13.09.2019

A: Specific Conditions:

i. The project proponent shall maintain the setback distance 7.5 meters buffer zone all around the mine lease area for greenbelt development and other conditions are to be fulfilled.

Status: Being complied

It has been observed that the mine lease area is rocky and highly impossible to develop green belt in and around the mine lease area. It has been observed that the PAs are in process of developing green belt at Sy.No. 954/1 of Chimakurthy village. As per the information provided, it has been observed that the PAs have developed green belt in an area of 2.74 acres with 800 saplings till date.

Photographs of green belt development status is as follows:



ii. The project proponent shall allocate sufficient funds for implementation of CSR activities as committed by the representative along with the EMP.

Status: Being complied

It has been observed that the PAs have allocated sufficient funds for implementation of CSR activities as committed by the representative along with the EMP.

iii. The avenue plantation (tall plants), of at least 1.5 m height, for 1 km length of the approach road on either side of the road is to be developed and maintained.

Status: Being complied

It has been observed that the PAs have developed avenue plantation (tall plants), of at least 1.5 m height, for 1 km length of the approach road on either side of the road and is being maintained.

1) Air Pollution: -

i. Wet drilling method is to be adopted to control dust emissions. Delay detonators and shock tube initiation system for blasting shall be used so as to reduce vibration and dust.

Status: Being complied

During the day of monitoring, it has been stated that the wet drilling method is being adopted to control dust emissions. Delay detonators and shock tube initiation system for blasting is being used to reduce vibration and dust. Photograph of wet drilling during the day of monitoring is as follows:



ii. Greenbelt shall be developed along the boundary of mining lease

area and also in back filled and reclaimed areas with tall growing native species in consultation with the local DFO/Agriculture Department. The proponent of mine shall carry mining are and nearby areas. He shall take immediate measures for planting in the same area or any other area selected by authorities not less than twice the number of trees going to be felled by mining operations. He shall also take measures for restoration of other flora/funna if damaged by mining operations.

Status: Being complied

It has been observed that the mine lease area is rocky and highly impossible to develop green belt in and around the mine lease area. It has been observed that the PAs are in process of developing green belt at Sy.No. 954/1 of Chimakurthy village. As per the information provided, it has been observed that the PAs have developed green belt in an area of 2.74 acres with 800 saplings till date.

Photographs of green belt development status is as follows:



iii. **Effective safe guard measures such as regular water sprinkling shall be carried out in critical areas prone to air pollution and having high levels of SPM and RPM such as haul road, loading and unloading point and transfer points. It shall be ensured that**

the ambient air quality parameters confirm to the norms prescribed by the Central Pollution Control Board in this regard.

Status: Being complied

It has been observed that the PAs are in process of taking effective safe guard measures of regular water sprinkling on regular basis in critical areas prone to air pollution and having high levels of SPM and RPM such as haul road, loading and unloading point and transfer points. As per the monitoring reports submitted, it has been observed that the AAQ levels are within the prescribed limits.

iv. **The proponent of mine shall carry air quality monitoring in the core zone as well as buffer zone for RSPM (PM10) and Noise levels. Location of monitoring stations should be decided based on the metallurgical data topographical features and environmentally and ecologically sensitive targets and frequency of monitoring should be undertaken in consultation with Andhra Pradesh Pollution Control Board. The data so recorded should be regularly submitted to the Ministry including its Regional Office located at Chennai and the Andhra Pradesh Pollution Control Board/Central Pollution Control Board once in six months.**

Status: Being complied

It has been observed that the PAs are conducting ambient air quality and noise levels monitoring in core as well as buffer zone. As per the monitoring reports submitted, the parameters are within the prescribed limits.

v. **The proponent shall construct graded roads connecting the mining area to the nearby roads to avoid dust nuisance due to vehicular movements.**

Status: Being complied

It has been observed that the PAs have constructed graded roads which connecting the mining area to the nearby road to avoid dust nuisance due to vehicular movements.

vi. **The proponent shall take precautions against noise arising out of mining operations and shall be abated or controlled at the source so as to keep it within the permissible limits notified under Environmental (Protection) Act, 1986 / Noise Pollution (Regulations & Control) Rules, 2010 by implementing the following noise control measures:**

- **Proper and regular maintenance of vehicles and other**

equipment

- **Limiting time exposure of workers to excessive noise.**
- **The workers employed shall be provided with protection equipment and earmuffs. Etc.**
- **Speed of trucks entering or leaving the mine is to be limited to moderate speed of 25 kmph to prevent undue noise from empty trucks.**

Status: Being complied

It has been observed that the PAs are in process of implementing above measures to reduce noise pollution during transportation of mineral.

vii. Whenever any damage to public buildings or monuments in apprehended due to their proximity to the mining lease area, scientific investigations shall be carried out by the holder of mining lease so as to keep the ground vibrations caused by mining operations within safe limit.

Status: Being complied

It has been observed that the condition is being complied with.

2) Water Pollution: -

i. The source of water is Bore well. Total water requirement is 60 KLD which is used for dust suppression; wire saw cutting; development of green belt and for domestic purpose.

Status: Being complied

As per the discussions held, it has been observed that the source of water is mine pit water and by water tankers from nearby villages. The total water requirement is under 60 KLD and being used for dust suppression, wire saw cutting, development of green belt and for domestic purpose.

ii. Garland drain and siltation ponds of appropriate size should be constructed for the working pit to arrest flow of silt and sediment. The water so collected should be utilized for watering the mine area, roads, green belt development etc. The drains should be regularly desilted, particularly after monsoon, and maintained properly.

Status: Partially complied

It has been observed that the PAs have constructed garland drains and siltation pond of appropriate size for the OB/waste dumps.

However, garland drains and siltation pond are not constructed for the working pit to arrest flow of silt and sediment.

It is required to construct garland drain and siltation pond of appropriate size for the working pit to arrest flow of silt and sediment. The water so collected should be utilized for watering the mine area, roads, green belt development etc. The drains should be regularly desilted, particularly after monsoon, and maintained properly.

iii. **The proponent of the mine shall take all possible precautions to prevent or reduce the discharge of toxic and objectionable liquid effluents from mine, workshop, tailing ponds into surface bodies, ground water aquifer and useable lands to a minimum. The effluents shall be suitably treated, if required, to conform to the general standards notified under Environmental (Protection) Act, 1986.**

Status: Being complied

It has been observed that the PAs are in process of taking all possible precautions to prevent or reduce the discharge of toxic and objectionable liquid effluents from mine, workshop, tailing ponds into surface bodies, ground water aquifer and useable lands to a minimum.

iv. **Monitoring of ground water level and quality should be carried out quarterly by the project proponent in and around project area in consultation with State Ground Water Department/Central Ground Water Authority and data thus collected shall be submitted regularly to the MOEF&CC and its Regional office, Chennai, CGWA, and the Regional Director, Central Ground Water Board, Hyderabad. If at any stage, it is observed that the ground water table is getting depleted due to the mining activity, necessary correction measures shall be carried out.**

Status: Partially complied

As per the discussions held, it has been stated that ground water level and quality is being monitored on regular basis. However, as per records, the monitoring reports are not being submitted along with six monthly compliance reports.

It is required to submit the quarterly monitored reports of ground water level and quality in and around project area along with six monthly compliance reports on regular basis to Ministry's Integrated Regional Office, Vijayawada.

v. **Suitable conservation measures to augment groundwater resources in the area shall be planned and implemented in consultation with Regional Director. CGWB, Southern Region, Hyderabad. Suitable measures should be taken for rainwater harvesting.**

Status: Being complied

It has been observed that the PAs are using mine pit as conservation measures to augment groundwater resources in the area.

vi. **Permission from the competent authority should be obtained for drawl of ground water, if any, required for this project.**

Status: Being complied

As stated above, it has been observed that the PAs are not withdrawing ground water. The water source of the project is from mine pit which is being used for wire saw processing, green belt development and water sprinkling. For drinking and domestic purpose, the source of water is by water tankers from nearby villages.

3) Solid Waste: -

i. **Topsoil: Wherever top soil exists and is to be excavated for mining operations, it shall be removed and stacked separately and top soil so removed shall be utilized for restoration or rehabilitation of the land, which is no longer required for mine operations or for stabilizing or landscaping the external dumps. Whenever the top soil cannot be utilized concurrently, it shall be stored separately for future use.**

Status: Being complied

It has been observed that currently top soil is not being generated. It has been stated that the generated top soil has been used for plantation purpose.

ii. **Overburden: The proponent of mine shall take steps so that the overburden, waste rock, rejects and fines generated during mining operations shall be stored in separate dumps preferably on impervious grounds. The waste rock, overburden etc. shall be concurrently backfilled into the mine excavations so as to restore the land to its original use as far as possible. In the case of non feasibility of back filling, the waste dump shall be suitable terraced and stabilize through the vegetation. The proponent shall maintain proper angel of response to ensure stability to the dump.**

Status: Being complied

It has been observed that the PAs have two dump sites viz., one is at Sy. No. 938/1 to 7 of Chimakurthy village over an extent of 6.17 acres and another dumpsite at Sy.No. 941/1 to 4 , 944/1, 944/4 to 7 over extent of 11.19 acre in Chimakurty village and Mandal. Dumpsite exhausted and second dump site is under operational. It has been stated that the PAs have acquired patta land at Sy No. 25/3B2 of R.L. Puram Village, Chimakurthy Mandal, Prakasam District over an extent of 11.00 acres for dumping as third dump site. It has been observed that the PAs are in process of stabilizing dump site.

iii. **The proponent of mine shall construct required number of retaining walls to provide stability to the dumps. Dimensions of the retaining wall shall be based on the rainfall data.**

Status: Being complied

It has been observed that the PAs have constructed required number of retaining wall to provide stability to the dumps.

iv. **The proponent of mines shall construct required number of garland drains to arrest mineral particles being carried away as runoff during rainy seasons around the dump yards. Dimensions of the garland drains shall be based on rainfall data.**

Status: Being complied

It has been observed that the PAs have constructed garland drains and settling ponds around the dump yards. The photographs are as follows:



v. **Waste oils. used oils generated from the EM machines, mining operations, if any, shall be disposed as per the Hazardous Wastes**

(Management, Handling, and transboundary movement) Rules. 2008 and its amendments thereof to the recyclers authorized by APPCB.

Status: Being complied

It has been observed that the waste oils. used oils generated from the EM machines, mining operations, if any, is being disposed as per the Hazardous Wastes (Management, Handling, and transboundary movement) Rules. 2008 and its amendments thereof to the recyclers authorized by APPCB.

vi. **The proponent of the mine shall undertake phased restoration, reclamation and rehabilitation of the lands affected by the mining operations and shall complete this work before conclusion of such operations and abandonment of the mine.**

Status: Being complied

It has been observed that no restoration, reclamation and rehabilitation of the lands affected by the mining operation were not undertaken yet. As per the discussions held, it has been stated that the deposit is symptomatic in persistence at depth and there is no habitation in the lease area. Therefore, no reclamation and rehabilitation proposal were made and the excavated pit shall be left for water storage.

B. Specific Conditions:

i. **This order is valid for a period of 20 years or the expiry date of mine lease period issued by the Government of A.P., whichever is earlier.**

Status: Being complied

It has been observed that the condition is being complied with.

ii. **"Consent for Establishment" & "Consent for Operation" shall be obtained from Andhra Pradesh Pollution Control Board under Air and Water Act and effectively comply with all the conditions stipulated thereof.**

Status: Being complied

It has been observed that the PAs have obtained combined Consent for Establishment (CFE), Consent for Operation (CFO) and Hazardous Waste Authorization (HWA) vide Consent Order No.

APPCB/VJA/NLR/188/HO/CFO/2017 dated 21.10.2019 for mining of black galaxy granite of 54,643 m^3/annum. The CFO is valid till 31.01.2022. It has been observed that the PAs are in process of complying with all the conditions stipulated thereof.

iii. **No change in mining technology and scope of working should be made without prior approval of the SEIAA, A.P. No further expansion or modifications in the mine shall be carried out without prior approval of the SEIAA, AP/ MOEF&CC. Gol. New Delhi, as applicable.**

Status: Being complied

It has been observed that there is no change in mining technology and scope of working made without prior approval of the SEIAA, A.P. No further expansion or modifications in the mine shall be carried out without prior approval of the SEIAA, AP/ MOEF&CC. Gol. New Delhi. as applicable. It has been observed that there is no change in the calendar plan including excavation, quantum of granite, and waste. As per the information provided, the production details for last five years are as follows:

Year	Quantity (m^3)	Waste Rock (m^3)	OB (m^3)
Dec, 2019-20	15196.225	97618.769	2970
2020-21	54408.740	350793.260	15992
2021-22 (till July)	20930.405	131163.595	5217

iv. **Personnel working in dusty areas should wear protective respiratory devices and they should also be provided with adequate training and information on safety and health aspects.**

Status: Being complied

It has been stated that the persons working in the dusty areas are provided with the suitable PPEs and giving suitable training on safety and health aspects and all the corrective actions is being taken.

v. **The project proponent shall ensure that no natural watercourse and/or water resources shall be obstructed due to any mining operations. Necessary safeguard measures to protect the first order streams, if any, originating from the mine lease shall be taken.**

Status: Being complied

It has been observed that no natural watercourse and/or water

resources during mining operations are being obstructed by mining operations. Measures are being taken to protect the first order streams, if originating in mine lease.

vi. **Occupational health surveillance program of the workers should be undertaken periodically to observe any contractions due to exposure to dust and take corrective measures. if needed.**

Status: Being complied

It has been observed that PAs are conducting occupational health surveillance program of the workers periodically and records are being maintained.

vii. **A separate environmental management cell with suitable qualified personnel should be set-up under the control of a Senior Executive, who will report directly to the Head of the Organization.**

Status: Partially complied

It has been observed that the PAs have a separate environment management cell with suitable qualified personal under the control of a Senior Executive, who will report directly to the head of the organization. However, it has been observed that the environmental management cell does not have a suitable qualified personal.

It is required to establish a separate Environment Management Cell with suitable qualified personal under the control of a Senior Executive, who will report directly to the head of the organization.

viii. **The funds earmarked for environmental protection measures (Capital cost is Rs 52.37 Lakhs and Recurring cost is 35.5 Lakhs/annum) should be kept in separate account and should not be diverted for other purpose. Year wise expenditure should be reported to the Ministry and its Regional Office located at Bangalore.**

Status: Partially complied

It has been stated that the funds earmarked for environmental protection measures has kept in separate account and not be diverted for other purpose. However, year wise expenditure is not being reported to Regional Office.

It is required to submit the detailed report regarding item wise expenditure incurred under environmental protection measures for last five years.

It is also required to submit the detailed report regarding item wise expenditure incurred under environmental protection measures along with six monthly compliance reports to Ministry's Integrated Regional Office, Vijayawada on regular basis.

ix. **The Regional Office of MOEF&CC located at Chennai / The SEIAA, Andhra Pradesh through the Regional Offices of Andhra Pradesh Pollution Control Board, shall monitor compliance of the stipulated conditions. The project authorities should extend full cooperation to the officer (s) of the Regional office by furnishing the requisite data/information/monitoring reports.**

Status: Being complied

The condition is being complied with.

x. **The project proponent shall submit six monthly reports on the status of compliance of the stipulated environmental clearance conditions including results of monitored data (both in hard copies as well as by e-mail) to the Ministry of Environment & Forests, its Regional Office, Chennai, SEIAA, A.P., Zonal Office of Central Pollution Control Board, Bangalore, District Collector and A.P. Pollution Control Board. The proponent shall upload the status of compliance of the environmental conditions including results of monitored data on their websites and shall update the same periodically.**

Status: Partially complied

As per the records, it has been observed that the PAs are regularly submitting six monthly compliance reports along with all monitored data to Ministry's Regional Office. However, the same is not being uploaded on company's website.

It is required to upload the status of compliance of the environmental conditions including results of monitored data on the website of company and shall update the same periodically. It requires immediate action.

xi. **The project proponent shall submit the copies of the environmental clearance to the Heads of local bodies, Panchayats and Municipal Bodies in addition to the relevant offices of the Government who in turn has to display the same for 30 days from the date of receipt.**

Status: Being complied

The condition is being complied with.

xii. The environmental statement for each financial year ending 31st March in Form-V as mandated is to be submitted by the project proponent to the A.P. Pollution Control Board as prescribed under the Environment (Protection) Rules, 1986, as amended subsequently, shall be put on the website of the company along with the status of the compliance of environmental clearance conditions and shall also be sent to the Regional office of the Ministry of Environment and Forests, Chennai by e-mail.

Status: Partially complied

As per the records, it has been observed that the PAs submitting environmental statement for each financial year ending 31st March in Form-V. However, the same is not uploading website of the company.

It is required to upload the copy of Environmental Statement (Form-V) on the website of company and shall update the same periodically. It requires immediate action.

xiii. The project authorities should advertise at least in two local newspapers widely circulated. one of which shall be in the vernacular language of the locality concerned, within 7 days of the issue of the clearance letter informing that the project has been accorded environmental clearance and a copy of the clearance letter is available with the State Pollution Control Board and SEIAA, A.P.

Status: Complied

It has been observed that the PAs have advertised details of EC in two local newspapers viz., "Sakshi" (Telugu) dated 27.09.2019 and "The New Indian Express" (English) dated 27.09.2019.

xiv. The proponent shall obtain all other mandatory clearances from respective departments.

Status: Being complied

It has been observed that the PAs have obtained statutory clearances from respective departments.

xv. Any appeal against this Environmental Clearance shall lie with the National Green Tribunal, if preferred, within a period of 30 days as prescribed under Section 16 of the National Green Tribunal Act. 2010.

Status: Being complied

The condition is being complied with.

xvi. Concealing the factual data or failure to comply with any of the conditions mentioned above may result in withdrawal of this clearance and attract action under the provisions of Environment (Protection) Act, 1986.

Status: Being complied

The condition is being complied with.

xvii. The SEIAA may revoke or suspend the order, if implementation of any of the above conditions is not satisfactory. The SEIAA reserves the right to alter/modify the above conditions or stipulate any further condition in the interest of environment protection.

Status: Being complied

The condition is being complied with.

xviii. The above conditions will be enforced inter-alia, under the provisions of the Water (Prevention & Control of Pollution) Act. 1974, the Air (Prevention & Control of Pollution) Act, 1981, the Environment (Protection) Act., 1986 and the Public Liability Insurance Act, 1991 along with their amendments and rules.

Status: Being complied

The condition is being complied with.

10. Observations:

The PAs have complied or are in process of complying the conditions stipulated by the Ministry. In this context, information/action plans have been sought on following points.

1. It is required to construct garland drain and siltation pond of appropriate size for the working pit to arrest flow of silt and sediment. The water so collected should be utilized for watering the mine area, roads, green belt development etc. The drains should be regularly desilted, particularly after monsoon, and maintained properly.
2. It is required to submit the quarterly monitored reports of ground water level and quality in and around project area along with six monthly compliance reports on regular basis to Ministry's Integrated Regional Office, Vijayawada.

3. It is required to establish a separate Environment Management Cell with suitable qualified personal under the control of a Senior Executive, who will report directly to the head of the organization.
4. It is also required to upload the status of compliance of the environmental conditions including results of monitored data on the website of company and shall update the same periodically. It requires immediate action.
5. It is required to upload environmental statement for each financial year ending 31st March in Form-V on website of the company. It requires immediate action.

11. Recommendations:

This project can be put up in the following tick-marked category of compliance status.

- Compliance status could not be judged as the project was shut down / not in operational during the site visit.
- No non-compliances detected. No any further action is required.

√ ***Minor non-compliances detected (not of immediate danger to health & safety of the people). Letter issued to project authorities for taking corrective measures.***

- Serious non-compliances detected. Recommended to issue a Show-Cause Notice to Project Authorities.

(डॉ। सुरेश बाबु पसुपुलेटी)
(Dr. Suresh Babu Pasupuleti)
संयुक्त निदेशक (एस) / वैज्ञानिक 'डी'
Joint Director (S) / Scientist 'D'

Copy to: Project File

Digitally Signed by Suresh Babu Pasupuleti
Date: 06-01-2022 15:42:27
Reason: Approved

Annexure-IV

Ambient Air Quality data

Project site						
	PM10	Pm2.5	SO2	Nox	VOC	CO
2.1.2019	91.3	48.8	14.8	16.4	2.8	2.1
3.1.2019	88.3	45.2	15.2	17.5	3.2	2.2
9.1.2019	84.3	39.2	16.6	18.5	2.6	1.8
10.1.2019	89.3	45.2	13.3	19.5	3	2.2
16.1.2019	90.3	42.3	16.4	17.5	2.4	2.3
17.1.2019	91.4	38.5	15.2	16.6	2	2.3
23.1.2019	85.3	37.2	17.2	20.1	2.2	2
24.1.2019	84.3	40.3	18.2	28.5	2.9	1.5
1.2.2019	85.3	39.4	14.2	17.2	2.9	1.6
2.2.2019	90.3	32.5	15.2	20.1	3.2	1.3
6.2.2019	86.7	30.2	17.2	18.2	3.2	1.2
7.2.2019	87.3	30.5	14.8	21.3	2.8	1.6
13.2.2019	90.3	29.5	14.9	20.8	2.5	1.2
14.2.2019	91.5	28.5	14.3	15.2	2.2	1.6
20.2.2019	86.5	31.3	16.5	21.3	2.4	2.1
21.2.2019	90.4	34.2	17.2	17.2	2.9	1.6
1.3.2019	83.2	29.4	15.3	19.2	2.5	2.1
2.3.2019	91.4	29.3	16.2	20.9	2.5	1.6
8.3.2019	90.3	35.7	15.2	17.2	3.6	1.8
9.3.2019	86.4	31.5	13.45	21.2	2.2	2.1
15.3.2019	92.5	34.2	16.6	20.1	2.6	1.7
16.3.2019	86.5	40.2	15.3	17.2	3.1	1.8
22.3.2019	90.4	35.2	14.2	18.7	2.2	1.6
23.3.2019	83.2	36.2	14.9	16.5	2.5	1.9
Max	92.5	48.8	18.2	28.5	3.6	2.3
Min	83.2	28.5	13.3	15.2	2	1.2
mean	88.19583	36.02083	15.51458	19.0375	2.683333	1.8
0.98	92.04	47.144	17.74	25.188	3.416	2.3

RL Puram						
	PM10	Pm2.5	so2	Nox	VOC	CO
2.1.2019	75.4	30.5	13.5	17.3	2.4	2.2
3.1.2019	63.2	25.7	12.3	15.3	2	1.7
9.1.2019	62.3	24.3	13.5	18.2	1.9	1.7
10.1.2019	72.3	26.7	15.4	17.3	2.3	1.9
16.1.2019	70.1	22.3	12.3	18.2	2.6	2
17.1.2019	68.3	23.2	12.4	15.4	2.9	1.8
23.1.2019	71.3	28.4	12.5	18.3	2.2	2.3
24.1.2019	85.4	40.5	14.6	15.3	2.3	1.6
1.2.2019	65.3	29.5	14.2	16.2	2.4	1.2
2.2.2019	70.2	20.1	13.2	14.3	2.9	1.2
6.2.2019	73.4	23.2	14.2	17.4	2.6	1.8
7.2.2019	66.5	26.2	12.3	16.3	1.2	1.2
13.2.2019	68.3	21.2	13.2	18.4	2.6	1.7
14.2.2019	67.3	23.2	13.2	15.4	1.2	1.6
20.2.2019	71.2	23.5	14.2	16.3	2.2	1.6
21.2.2019	63.4	20.8	11.5	14.2	2.5	1.7
1.3.2019	74.3	20.4	11.2	13.4	2.3	2
2.3.2019	75.2	25.2	12.1	15	2.2	1.5
8.3.2019	70.2	24.5	13.1	17.3	2.3	1.5
9.3.2019	69.3	24.2	14.2	20.2	3.1	1.5
15.3.2019	67.2	27.4	11.2	17.3	2.2	1.7
16.3.2019	70.2	26.2	14.2	17.4	2.5	1.7
22.3.2019	69.3	25.2	15.2	16.4	1.6	1.5
23.3.2019	72.3	23.5	14.3	18.4	1.5	1.6
Max	85.3	40.5	15.4	20.2	3.1	2.3
min	62.3	20.1	11.2	13.4	1.2	1.2
mean	68.3	24.8	13.25	16.63333	2.245833	1.675
0.98	73.5	28.5	15.308	19.372	3.008	2.254

Rajupalem						
	Pm10	Pm2.5	So2	Nox	VOC	CO
2.1.2019	68.6	28.5	13.5	16.5	2.2	1.8
3.1.2019	70.4	25.2	15.3	18.4	2.6	2
9.1.2019	66.3	24.2	13.2	16.4	1.6	1.8
10.1.2019	67.3	22.3	14.3	16.3	2.1	2.2
16.1.2019	70.2	25.2	16.4	15.3	1.8	2.1
17.1.2019	65.3	30.1	14.2	14.3	2.2	1.5
23.1.2019	69	26.2	15.2	17.4	2.6	1.7
24.1.2019	67.3	29.2	12.3	16.4	1.4	1.2
1.2.2019	70.6	25.5	11.4	18.4	1.7	2.1
2.2.2019	69.3	24.5	10.9	15.3	2.2	2.2
6.2.2019	69.3	26.2	10.4	14.7	3	2.1
7.2.2019	67.6	25.2	12.2	15.8	2.4	1.8
13.2.2019	65.7	27.2	11.2	15.8	2.7	1.2
14.2.2019	67.3	23.1	10.3	15.4	2.3	2.2
20.2.2019	68.3	26.2	10.3	15.3	1.9	1.8
21.2.2019	70.3	30.2	12.1	15.9	2.4	2
1.3.2019	72.3	27.3	10.7	19.4	2.2	1.7
2.3.2019	68.3	24.2	11.4	17.4	3.1	1.2
8.3.2019	65.3	26.3	10.6	16.5	2.4	1.6
9.3.2019	65.2	27.2	12.1	18.5	2.3	1.6
15.3.2019	63.4	24.2	11.2	16.5	2.6	1.7
16.3.2019	63.2	25.2	10.2	15.3	2.7	2
22.3.2019	68.3	26.2	12.1	15.3	1.8	2.1
23.3.2019	66.5	20.2	10.2	16.3	2.4	1.6
Max	72.3	30.2	16.4	19.4	3.1	2.2
min	63.2	20.2	10.2	14.3	1.4	1.2
mean	67.72083	25.825	12.15417	16.36667	2.275	1.8
0.98	71.518	30.154	15.894	18.986	3.054	2.2

Chimakurthy						
	Pm10	Pm2.5	So2	Nox	VOC	CO
2.1.2019	71.3	24.5	13.1	15.4	2.4	1.4
3.1.2019	70.7	28.5	11.4	17.4	2.3	1.3
9.1.2019	67.3	27.5	13.1	15.6	1.7	1.5
10.1.2019	70.3	25.2	10.2	16.9	1.5	1.2
16.1.2019	69.3	28.2	9.8	17.4	2.2	1.7
17.1.2019	67.3	28.1	10.2	15.6	1.5	1.4
23.1.2019	70.3	27.3	13.2	18.5	1.6	1.5
24.1.2019	65.3	23.2	10.5	16.3	1.2	1.3
1.2.2019	69.3	27.3	11.2	18.4	2.5	1.3
2.2.2019	70.3	28.4	10.6	15.3	2.3	1.8
6.2.2019	68.3	21.2	11.2	17.4	2.4	1.5
7.2.2019	66.3	22.4	13.2	14.6	2.5	1.7
13.2.2019	67.3	27.4	12.5	15.5	2.2	1.2
14.2.2019	66.4	25.4	13.2	17.5	2.5	1.4
20.2.2019	70.3	25.4	12.9	14.2	2.8	1.7
21.2.2019	65.4	22.5	14.2	15.3	1.8	1.5
1.3.2019	67.4	25.2	13.2	16.3	1.7	1.4
2.3.2019	68.3	23.4	13.2	18.4	1.3	1.4
8.3.2019	70.4	21.2	12.43	15.7	2.3	1.6
9.3.2019	75.3	20.1	15.1	17.4	1.5	1.5
15.3.2019	68.4	23.3	13.2	15.4	2	1.3
16.3.2019	62.3	24.1	10.9	15.6	2.1	1.5
22.3.2019	67.4	21.2	14.2	16.2	1.5	1.2
23.3.2019	65.3	25.2	13.21	17.4	1.4	1.5
Max	75.3	28.5	15.1	18.5	2.8	1.8
min	62.3	20.1	9.8	14.2	1.2	1.2
mean	68.34167	24.84167	12.41043	16.40417	1.966667	1.45
0.98	73.46	28.454	14.704	18.454	2.662	1.754

Ramathirtham						
	PM10	PM2.5	SO2	Nox	VOC	CO
2.1.2019	65.3	24.3	12.6	14.6	1.5	1.4
3.1.2019	63.4	20.2	11.5	14.2	1.4	1.8
9.1.2019	67.4	22.5	12.5	15.3	1.2	1.7
10.1.2019	59.3	24.3	13.1	14.3	1	1.5
16.1.2019	60.3	22.3	14.3	16.3	1.2	1.3
17.1.2019	55.7	21.5	12.5	18.2	2.3	1.6
23.1.2019	60.2	19.2	13.5	13.4	1.4	1.2
24.1.2019	58.3	21.2	15.7	14.4	2.2	1.6
1.2.2019	62.5	24.3	12.1	16.2	1.8	1.7
2.2.2019	63.5	22.1	12.4	13	2.2	1.5
6.2.2019	64.3	25.4	15.2	14.9	2.2	1.6
7.2.2019	68.3	21.5	16.4	16.4	1.9	1.3
13.2.2019	67.4	23.5	13.4	15.4	2.3	1.8
14.2.2019	69.3	20.3	13.2	14.2	2.7	1.2
20.2.2019	65.3	21.2	14.3	15.3	1.9	1.5
21.2.2019	63.2	25.2	12.5	16.3	2.3	1.5
1.3.2019	65.3	21.2	15.3	15.3	1.7	1.7
2.3.2019	67.4	23.2	13.2	14.3	2.2	1.2
8.3.2019	64.3	25.2	13.4	14.6	2.3	1.7
9.3.2019	67.4	20.3	13.2	15.3	2.4	1.6
15.3.2019	64.3	23.3	15.2	15.3	2.3	1.6
16.3.2019	65.3	23.2	11.6	13.5	2	1.4
22.3.2019	67.3	25.2	14.2	16.2	1.8	1.5
23.3.2019	60.3	23.5	15.2	15.3	2.8	1.3
Max	69.3	25.4	16.4	18.2	2.8	1.8
min	55.7	19.2	11.5	13	1	1.2
mean	63.97083	22.67083	13.60417	15.09167	1.958333	1.508696
0.98	68.84	25.308	16.078	17.372	2.754	1.8

Marrichetla palem						
	Pm10	PM2.5	SO2	Nox	VOC	CO
2.1.2019	63.4	25.2	13.5	15.3	1.2	1.7
3.1.2019	59.3	23.4	14.6	14.7	1.2	1.5
9.1.2019	65.4	22.3	12.5	18.3	1.5	1.3
10.1.2019	60.3	24.2	15.3	15.4	1	1.6
16.1.2019	65.3	26.2	14.2	17.5	1.4	1.2
17.1.2019	64	23.4	13.5	14.3	1.3	1.7
23.1.2019	68.3	22.1	15.3	16.3	1.5	1.5
24.1.2019	63.2	24.2	13.4	13.9	2.1	1.3
1.2.2019	69.3	24.2	12.3	15.3	2.3	1.9
2.2.2019	65.3	20.4	14.2	15.4	1.8	1.3
6.2.2019	63.5	22.2	12.8	17.4	1.7	1.3
7.2.2019	67.3	26.23	11.4	15.3	2	1.6
13.2.2019	59.3	26.3	15.3	15.2	1.9	2
14.2.2019	64.3	28.3	11.4	16.5	2.5	1.6
20.2.2019	65.3	23.2	12.8	14.2	2.3	1.3
21.2.2019	63.2	24.2	12.4	15.7	1.7	1.7
1.3.2019	59.3	26.2	13.2	16.4	1.6	1.3
2.3.2019	62.2	27.3	10.5	15.7	1.34	1.2
8.3.2019	64.3	24.2	13.1	15.4	2.3	1.5
9.3.2019	56.3	24.2	12.4	15.4	1.8	1.3
15.3.2019	62.3	21.2	13.6	14.3	2.4	1.3
16.3.2019	65.4	19.4	10	16.4	1.5	1.5
22.3.2019	62.5	20.3	13.2	16.3	1.8	1.3
23.3.2019	62.6	24.2	12.2	15.3	1.6	1.4
Max	69.3	28.3	15.3	18.3	2.5	2
min	56.3	19.4	10	13.9	1	1.2
mean	63.4	23.87625	13.04583	15.6625	1.739167	1.470833
0.98	68.84	27.84	15.3	17.932	2.454	1.954

Ayyapa Varipalam						
	Pm10	Pm2.5	SO2	Nox	VOC	CO
2.1.2019	58.3	24.4	13.5	14.9	1.4	1.3
3.1.2019	64.3	22.3	15	13.2	1.8	1.2
9.1.2019	63.5	21.3	13.3	15.3	2.2	1.4
10.1.2019	54.6	25.3	15.2	16.7	1.7	1.2
16.1.2019	59.5	25.3	13.4	14.8	1.9	1.4
17.1.2019	58.3	21.3	11.4	15.3	1.4	1.2
23.1.2019	60.3	22.2	12.4	14.3	1.8	1.6
24.1.2019	56.3	19.2	13.2	17.3	2.2	1.2
1.2.2019	63.5	23.2	12	12.4	2.4	1.5
2.2.2019	56.3	17.3	12.3	16.5	2.2	1.5
6.2.2019	63.5	20.1	11.4	13.4	1.9	1.1
7.2.2019	61.4	17.3	14.6	16.2	1.7	1.3
13.2.2019	62.5	16.7	12.3	15.3	2	1.2
14.2.2019	60.3	18.3	15.6	12.3	2.3	1.5
20.2.2019	59.3	17.2	14.6	15.3	1.9	1.4
21.2.2019	60.4	18.2	15.6	13.8	2.9	1.5
1.3.2019	59.3	19.3	11.2	14.3	1.9	1.4
2.3.2019	57.3	20.3	15.9	15.3	2.2	2
8.3.2019	60.1	18.2	12.2	14.7	2.8	1.1
9.3.2019	58.3	18.2	13.5	16.3	2.7	1.2
15.3.2019	57.3	20.3	11.2	15.3	1.9	1.5
16.3.2019	60.3	23.1	12.4	15.6	2.4	1.5
22.3.2019	55.3	18.2	13.2	16.3	1.6	1.9
23.3.2019	60.3	19	15.3	14.2	2.3	1.4
Max	64.3	25.3	15.9	17.3	2.9	2
min	54.6	16.7	11.2	12.3	1.4	1.1
mean	59.60417	20.25833	13.3625	14.95833	2.0625	1.391304
0.98	63.932	25.3	15.762	17.024	2.854	1.956

Muvvavari palem						
	Pm10	Pm2.5	SO2	Nox	VOC	CO
2.1.2019	60.3	21.4	15.3	14.9	2.2	1.5
3.1.2019	59.3	20.7	13.2	15.3	2	1.2
9.1.2019	62.4	19.2	12.2	16.6	2.3	1.5
10.1.2019	59.3	18.2	13.5	15.7	2.4	1.3
16.1.2019	57.6	17.2	15.3	13.6	1.5	1.7
17.1.2019	60.3	20.3	12.4	12.4	2.4	1.3
23.1.2019	61.4	22.1	11.2	15.3	2	1.6
24.1.2019	58.3	21.3	14.3	15.4	2.2	1.5
1.2.2019	60.5	19	15.3	17.3	2.3	1.5
2.2.2019	58.3	21.2	15.3	16.4	2.5	1.7
6.2.2019	60.2	18.3	11.3	14.9	2.4	1.3
7.2.2019	58.2	21.4	12.1	15.3	2.3	1.7
13.2.2019	60.2	24.1	13.4	14.3	2.6	1.4
14.2.2019	58.2	18.3	11.2	15.3	2.8	1.6
20.2.2019	59.2	16.8	10	14.2	2.5	1.7
21.2.2019	60.3	20.3	12.3	15.4	2	1.9
1.3.2019	62.4	19.2	11.4	15.2	2.5	1.7
2.3.2019	59.3	19.7	12.4	14.3	1.6	1.9
8.3.2019	55.3	21.2	13.5	15.9	2.2	1.6
9.3.2019	56.2	18.3	12.4	15.5	2.2	1.3
15.3.2019	58.2	20.3	13.3	14.6	2.5	1.5
16.3.2019	54.2	22.1	12.2	13.4	2.2	1.3
22.3.2019	53.2	22.3	11.8	4.7	2.5	1.9
23.3.2019	58.2	18.7	11.4	15.6	2.3	1.6
Max	62.4	24.1	15.3	17.3	2.8	1.9
min	53.2	16.8	10	4.7	1.5	1.2
mean	58.79167	20.06667	12.77917	14.64444	2.266667	1.55
0.98	62.4	23.272	15.3	16.978	2.708	1.9

Annexure-V

The list of Flora and Fauna of the study area

List of Local vegetation observed during the field visit:

S.No.	***Botanical Name***	**Common name**	**Family**	**Habit**
1	*Acacia auriculiformis* Benth.	Australia thumma	Leguminosae	Tree
2	*Acacia chundra*	Khadiramu	Mimosaceae	Tree
3	*Acacia leucophloea* (Roxb.)	Tella thumma	Leguminosae	Tree
4	*Aegle marmelos* (L.) Corrêa	Maredu	Rutaceae	Tree
5	*Albizia lebbeck* (L.) Benth.	Dirisanam	Leguminosae	Tree
6	*Azadirachta indica* A.Juss.	Vepa	Meliaceae	Tree
7	*Balanitesa egyptiaca* (L.) Delile	Gara	Zygophyllaceae	Tree
8	*Bauhinia racemosa* Lam.	Bidi leaf tree	Leguminosae	Tree
9	*Borassus flabellifer* L.	Thadi chettu	Arecaceae	Tree
10	*Buchanania lanzan* Spreng.	Sara	Anacaradaceae	Tree
11	*Butea monosperma* (Lam.) Taub.	Modhuga	Leguminosae	Tree
12	*Cascabela thevetia* (L.) Lippold	Pacha ganneru	Apocynaceae	Tree
13	*Cassia fistula* L.	Rela	Leguminosae	Tree
14	*Chukrasia tabularis* A.Juss.	Konda vepa	Meliaceae	Tree
15	*Dalbergia sissoo* DC.	Sisu	Leguminosae	Tree
16	*Delonix regia* (Hook.) Raf.	Chittikesaram	Leguminosae	Tree
17	*Dendrocalamus strictus* (Roxb.) Nees	Sanna vedru	Graminae	Tree
18	*Eucalyptus globulus* Labill.	Neelagiri thylam	Myrtaceae	Tree
19	*Euphorbia tirucalli L.*	Tirukalli	Euphorbiaceae	Tree
20	*Ficus benghalensis L.*	Marri	Moraceae	Tree
21	*Ficus hispida L.f.*	Medipandu	Moraceae	Tree
22	*Ficus racemosa L.*	Medi	Moraceae	Tree
23	*Ficus religiosa L.*	Ravi	Moraceae	Tree
24	*Leucaena leucocephala* (Lam.) de Wit	Jabarichettu	Leguminosae	Tree
25	*Limonia acidissima L.*	Velaga	Rutaceae	Tree
26	*Mangifera indica L.*	Mamidi	Anacardiaceae	Tree
27	*Morinda tinctoria Roxb.*	Maddichettu	Rubiaceae	Tree
28	*Nerium odoratum* Lam.	Erra ganneru	Apocynaceae	Tree
29	*Peltophorum pterocarpum* (DC.) K.Heyne	Kondachintha	Leguminosae	Tree
30	*Phoenix sylvestris* (L.) Roxb.	Eethachettu	Arecaceae	Tree
31	*Phyllanthus emblica* L.	Usiri	Phyllanthaceae	Tree
32	*Pithecellobium dulce* (Roxb.) Benth.	Sima chinta	Leguminosae	Tree
33	*Polyalthia longifolia* (Sonn.) Thwaites	Naramamidi	Annonaceae	Tree
34	*Pongamia pinnata* (L.) Pierre	Adivi ganuga	Leguminosae	Tree
35	*Prosopis chilensis* (Molina) Stuntz	Mulla thumma	Leguminosae	Tree
36	*Prosopis juliflora* (Sw.) DC.	English tumma	Mimosaceae	Tree
37	*Samanea saman* (Jacq.) Merr.	Nidraganeeru	Sapindaceae	Tree
38	*Senna siamea* Lam.	Seema thangedu	Leguminosae	Tree
39	*Senna siamea* Lam.	Seema thangedu	Leguminosae	Tree
40	*Syzygium cumini* (L.) Skeels	Neredu	Myrtaceae	Tree
41	*Tamarindus indica L.*	Chintha	Leguminosae	Tree
42	*Tecoma stans* (L.) Juss. ex Kunth	Pasupu ganneru	Bignoniaceae	Tree
43	*Terminalia bellirica* (Gaertn.) Roxb.	Tani	Combretaceae	Tree
44	*Terminalia catappa L.*	Badham	Combretaceae	Tree
45	*Thespecia populnea* (L.) Sol	Ganga Raavi	Malvaceae	Tree
46	*Vitex negundo L.*	Vaavili	Verbenaceae	Tree
47	*Abutilon indicum* (L.) Sweet	Thutturubenda	Malvaceae	Shrub
48	*Agave americana* L.	Gitta nara	Asparagaceae	Shrub

49	*Caesalpinia bonduc* (L.) Roxb.	Gacha podha	Leguminosae	Shrub
50	*Caesalpinia pulcherrima* (L.) Sw.	Rathna gandhi	Leguminosae	Shrub
51	*Calotropis gigantea* (L.) Dryand.	Tella jilledu	Apocynaceae	Shrub
52	*Calotropis procera* (Aiton) Dryand.	Erra jilledu	Apocynaceae	Shrub
53	*Capparis zeylanica* L.	Aarudonda	Capparaceae	Shrub
54	*Carissa carandas* L.	Vakkaya	Apocynaceae	Shrub
55	*Catunaregam spinosa* (Thunb.) Tirveng.	Manga	Rubiaceae	Shrub
56	*Clerodendrum phlomidis* L.f.	Kond-takal	Lamiaceae	Shrub
57	*Datura stramonium* L.	Ummetha	Solanaceae	Shrub
58	*Dodonaea viscosa* Jacq.	Bandaru	Sapindaceae	Shrub
59	*Erythroxylum monogynum* Roxb.	Dedaraaku	Erythroxylaceae	Shrub
60	*Holarrhena antidysenterica* (Roth)	girimallika	Apocynaceae	Shrub
61	*Ipomoea carnea* Jacq.	Rubber mokka	Convolvulaceae	Shrub
62	*Ixora coccinea*	Ramabanam	Rubiaceae	Shrub
63	*Jatropha curcas L.*	Nepalam	Euphorbiaceae	Shrub
64	*Jatropha glandulifera* Roxb.	Yerranepalamu	Euphorbiaceae	Shrub
65	*Lantana camara L.*	Pulikampa	Verbenaceae	Shrub
66	*Morinda pubescens* Sm.	Maddi	Rubiaceae	Shrub
67	*Opuntia dillenii* (Ker Gawl.) Haw.	Naga jamudu	Cactaceae	Shrub
68	*Phoenix acaulis* Roxb.	Chitteetha	Palmae	Shrub
69	*Phyllanthus reticulatus* Poir.	Nallapuli	Phyllanthaceae	Shrub
70	*Randia dumetorum*		Rubiaceae	Shrub
71	*Senna auriculata* (L.) Roxb.	Tangedu	Leguminosae	Shrub
72	*Solanum pubescens* Willd.	Uchintha	Solanaceae	Shrub
73	*Xanthium strumarium* L.	Marula-Mathangi	Asteraceae	Shrub
74	*Ximenia americana L.*	Nekkera	Olacaceae	Shrub
75	*Ziziphus oenopolia* (L.) Mill.	Parimi	Rhamnaceae	Shrub
76	*Acalypha indica* L.	Haritha manjari	Euphorbiaceae	Herb
77	*Achyranthes aspera* L	Uttareni	Amaranthaceae	Herb
78	*Aerva lanata* (L.) Juss	Thelaga pindi	Amaranthaceae	Herb
79	*Aeschynomene aspera* L.	Neeti jeeluga	Leguminosae	Herb
80	*Ageratum conyzoides* (L.) L.	Vasavi	Asteraceae	Herb
81	*Aloe vera* (L.) Burm.f.	Kithanara	Tiliaceae	Herb
82	*Alternanthera sessilis* (L.) R.Br. ex DC.	Ponagantiaku	Amaranthaceae	Herb
83	*Amaranthus spinosus* L.	Mulla thotakoora	Amaranthaceae	Herb
84	*Amaranthus viridis* L.	Chilakathotakoora	Amaranthaceae	Herb
85	*Argemone mexicana* L.	Brahmadandi	Papaveraceae	Herb
86	*Barleria prionitis* L.	Pachagorinta	Acanthaceae	Herb
87	*Blumea mollis* (D. Don) Merr.	Kukkapogaku	Asteraceae	Herb
88	*Boerhavia diffusa* L.	Atikimamidi	Nyctaginaceae	Herb
89	*Caralluma umbellata*	Kundeti kummulu	*Apocynaceae*	Herb
90	*Catharanthus roseus* (L.) G.Don	Billaganneru	Apocynaceae	Herb
91	*Celosia virgata* Jacq.	Guruga	Amaranthaceae	Herb
92	*Cleome viscosa* L.	Kukka vomintha	Cleomaceae	Herb
93	*Colocasia esculenta* (L.) Schott	Atuka tiga	Araceae	Herb
94	*Croton bonplandianus* Baill.	Vana mokka	Euphorbiaceae	Herb
95	*Euphorbia hirta* L.	Nanubalu	Euphorbiaceae	Herb
96	*Evolvulus alsinoides* (L.) L	Vishnukrantha	Convolvulaceae	Herb
97	*Gomphrena serrata L.*	Tella bendumalli	Amaranthaceae	Herb
98	*Hygrophila auriculata* (Schumach.) Heine	Mundla gobbi	Acanthaceae	Herb
99	*Justicia procumbens L.*	Water Willow	Acanthaceae	Herb

100	*Leucas aspera* (Willd.)	Tummi	Lamiaceae	Herb
101	*Mimosa pudica* L.	Atthi pathi	Leguminosae	Herb
102	*Ocimum canum* Sims	Kukka Tulasi	Lamiaceae	Herb
103	*Ocimum sanctum*	Tulasi	Lamiaceae	Herb
104	*Oldenlandia umbellata L.*	Chiru veru	Rubiaceae	Herb
105	*Oxalis corniculata L.*	Indian Sorrel	Oxalidaceae	Herb
106	*Parthenium hysterophorus L.*	Vayyaribhama	Asteraceae	Herb
107	*Phyllanthus niruri* L.	Nela Usiri	Euphorbiaceae	Herb
108	*Portulaca oleracea L.*	Pappu Kura	Portulacaceae	Herb
109	*Senna occidentalis* (L.) Link	Adavi Chennangi	Leguminosae	Herb
110	*Senna tora* (L.) Roxb.	Tagarisa	Leguminosae	Herb
111	*Sida acuta* Burm.f.	Medabirusaku	Malvaceae	Herb
112	*Sida cordifolia L.*	Chiru Benda	Malvaceae	Herb
113	*Sida spinosa L.*	Naga bala	Malvaceae	Herb
114	*Solanum surattense* Burm. f.	Nela vakudu	Solanaceae	Herb
115	*Sphaeranthus indicus* L.	Bodasaramu	Asteraceae	Herb
116	*Tephrosia purpurea* (L.) Pers.	Vempali	Leguminosae	Herb
117	*Tribulus terrestris* L.		Zygophyllaceae	Herb
118	*Tridax procumbens* (L.) L.	Gaddichamanthi	Asteraceae	Herb
119	*Urena lobata* L.	Pedda benda	Malvaceae	Herb
120	*Vanda tessellata* (Roxb.) Hook. ex G.Don	Kodikalla chettu	Orchidaceae	Herb
121	*Vernonia cinerea* (L.) Less.	Gariti Kamma	Compositae	Herb
122	*Waltheria indica* L.	Nalla Benda	Malvaceae	Herb
123	*Ziziphus mauritiana* Lam.	Reni	Rhamnaceae	Herb
124	*Aristida hystrix* L.f.	Gaddi	Poaceae	Grass
125	*Chloris virgata* Sw.	Gaddi	Poaceae	Grass
126	*Chrysopogon lancearius* (Hook.f.) Haines	Gaddi	Poaceae	Grass
127	*Cyperus castaneus* Willd.	Gaddi	Poaceae	Grass
128	*Cyperus rotundus* L.	Gaddi	Cyperaceae	Grass
129	*Fimbristylis cymosa* R.Br.	Pulupu gaddi	Cyperaceae	Grass
130	*Abrus precatorius* L.	Gurivinda	Leguminosae	Climber
131	*Asparagus racemosus* Willd.	Pilli Gaddalu	Asparagaceae	Climber
132	*Cissus quadrangularis* L.	Nalleru	Vitaceae	Climber
133	*Clitoria ternatea L.*	Sanku-Pushpamu	Leguminosae	Climber
134	*Cuscuta reflexa* Roxb.		Convolvulaceae	Climber
135	*Dioscorea pentaphylla* L.	Adavi gunusuthega	Dioscoreaceae	Climber
136	*Hemidesmus indicus* (L.) R. Br. ex Schult		Apocynaceae	Climber
137	*Ipomoea macrantha Roem. & Schult.*		Convolvulaceae	Climber
138	*Ipomoea nil* (L.) Roth.		Convolvulaceae	Climber
139	*Ipomoea obscura* (L.) Ker Gawl.		Convolvulaceae	Climber
140	*Merremia tridentata* (L.) Hallier f.	Elaka chevi	Convolvulaceae	Climber
141	*Mucuna pruriens* (L.) DC.	Dulagondi	Leguminosae	Climber
142	*Pergularia daemia* (Forssk.) Chiov.	Dustapu-Teega	Apocynaceae	Climber

Source: Forest department Working Plan (2004-14)

LIST OF FAUNA & THEIR CONSERVATION STATUS

Mammals: (* directly sighted animals)

Scientific Name	Common Name	Family	WPA Status	IUCN
Herpestes javanicus	Common Indian Mongoose*	Herpestidae	Sch II	LC
Pipistrellus coromandra	Indian Pipistrelle (bat)	Chiroptera		LC
Pteropus giganteus	Indian Flying fox bat	Chiroptera		LC
Rhinopoma microphyllum	Grater Mouse-Tailed Bat	Chiroptera		LC
Megaderma spasma	lesser false vampire bat	Chiroptera		LC
Pteropus giganteus	Indian flying fox	Chiroptera		LC
Lepus nigricollis	Black-naped Hare	Leporidae	Sch IV	LC
Bendicota bengalensis	Indian mole rat	Muridae		LC
Bendicota indica	Bandicoot rat	Muridae		LC
Mus booduga	Little Indian Field mouse	Muridae	Sch V	LC
Mus musculus	House Mouse*	Muridae	Sch V	LC
Rattus rattus	House rat*	Muridae		LC
Funambulus palmarum	Three striped palm squirrel*	Sciuridae		LC

Status assigned by the IUCN, where – CR – Critically Endangered; EN – Endangered; LC – Least Concern; NT – Near Threatened; VU – Vulnerable, DA – Data Deficient, NE – Not Evaluated

Source :

- Andhra Pradesh State Forest Department Working plan data
- **Vivek Menon (2014**), *Indian Mammals: A Field Guide.* Hachette Book Publishing India Pvt. Ltd., Gurgaon, India, pp 1-522; IUCN (2015).
- *The IUCN Red List of Threatened Species.* Version 2015-4; *Schedules I to VI: Indian Wildlife (Protection) Act, 1972.*

Aves:

All the listed birds are Least Concern under IUCN and Schedule - IV under Indian Wildlife Protection Act (1972) (* directly obsereved bird from the study area)

Scientific Name	**Common Name**	IUCN	IWLP
Apus affinis	House Swift	LC	Sch-IV
Cypsiurus balasiensis	Asian Palm Swift	LC	Sch-IV
Vanellus indicus	Red-wattled Lapwing*	LC	Sch-IV
Columba livia	Rock Pigeon	LC	Sch-IV
Streptopelia chinensis	Spotted Dove	LC	Sch-IV
Streptopelia decaocto	Eurasian Collared Dove*	LC	Sch-IV
Alcedo atthis	Common Kingfisher*	LC	Sch-IV

Ceryle rudis	Pied Kingfisher*	LC	Sch-IV
Halcyon symensis	White-breasted Kingfisher*	LC	Sch-IV
Coracias benghalensis	Indian Roller*	LC	Sch-IV
Centropus sinensis	Crow Pheasant	LC	Sch-IV
Cuculus micropterus	Indian Cuckoo	LC	Sch-IV
Eudynamys scolopacea	Asian Koel	LC	Sch-IV
Accipiter badius	Shikra	LC	Sch-IV
Haliastur indus	Brahminy Kite	LC	Sch-IV
Ictinaetus malayensis	Black Eagle	LC	Sch-IV
Milvus migrans	Black Kite	LC	Sch-IV
Amauromis phoenicurus	White breasted Waterhen	LC	Sch-IV
Fulica atra	Common Coot	LC	Sch-IV
Porphyrio porphyrio	Indian purple Moorhen	LC	Sch-IV
Corvus splendens	House Crow*	LC	Sch-V
Dicrurus adsimilis	Black Drongo*	LC	Sch-IV
Nectarinia asiatica	Purple Sunbird*	LC	Sch-IV
Passer domesticus	House Sparrow*	LC	Sch-IV
Ploceus philippininus	Baya Weaver*	LC	Sch-IV
Pycnonotus cafer	Red-vented Bulbul*	LC	Sch-IV
Acridotheres tristis	Common Myna*	LC	Sch-IV
Acrocephalus aedon	Thickbilled Warbler*	LC	Sch-IV
Acrocephalus agricola	Paddy field Warblers	LC	Sch-IV
Turdoides caudatus	Common Babbler	LC	Sch-IV
Turdoides striatus	Jungle Babbler*	LC	Sch-IV
Copsychus saularis	Oriental Magpie Robin*	LC	Sch-IV
Saxicoloides fulicata	Indian Robin*	LC	Sch-IV
Ardea alba	Large Egret*	LC	Sch-IV
Ardea cinerea	Grey heron*	LC	Sch-IV
Ardeola grayii	Pond Heron*	LC	Sch-IV

*Status assigned by the IUCN, where – CR – Critically Endangered; EN – Endangered; LC – Least Concern; NT – Near Threatened; VU – Vulnerable, DA – Data Deficient, NE – Not Evaluated

Sources:

- **Ali, S and Ripley, S.D. 1969.***Handbook of the Birds of India and Pakistan together with those of Nepal, Sikkim, Bhutan and Ceylon,3. Stone Curlews to Owls*. Oxford University Press, Bombay, 327pp.
- **BirdLife International 2012. In: IUCN 2012**. IUCN Red List of Threatened Species. Version 2012.2. <www.iucnredlist.org>. Downloaded on 11 March 2013.
- **Grimmett, R., Inskipp, C and T. Inskipp, 2001.** Pocket Guides to the Birds of the Indian Subcontinent. Cristopher Helm Publishers, Oxford University Press, 384pp.

List of Reptiles either spotted or reported from the study area. (* indicates Direct observations)

S.	Scientific Name	Common Name	IUCN	IWPA
1.	*Ahaetulla nasuta*	Green whip snake	LC	
2.	*Naja naja*	Indian Cobra	LC	II
3.	*Daboia siamensis*	Russel Viper	LC	II
4.	*Dendrelaphis tristis*	Tree Snake	LC	
5.	*Ptyas mucosa*	Common Rat snake	LC	II
6.	*Amphiesma stolata*	Buffstriped keelback	LC	
7.	*Trimeresurus gramineus*	Green pit viper	LC	IV
8.	*Typhlops hypomethes*	Common blind snake	LC	IV
9.	*Varanus bengalensis*	Common Indian	LC	II
10.	*Chamaeleo zeylanicus*	Chameleon*	LC	II
11.	*Mabuya carinata*	Common Skink*	LC	
12.	*Calotes rouxi*	Forest Calottes*	LC	
13.	*Calotes versicolor*	Common garden lizard*	LC	
14.	*Hemimidactylus brooki*	House gecko*	LC	
15.	*Hemidactylus forenatus*	Southern House Gecko	LC	

List of **Amphibians** either spotted or reported from the study area.

S. No.	Scientific Name	Common Name	IUCN	IWPA
1.	*Duttaphrynus melanostictus*	Asian common toad	LC	Sch-IV
2.	*Euphlyctis hexadactylus*	Indian green frog	LC	Sch-IV
3.	*Hoplobatrachus tigerinus*	Indian bullfrog	LC	Sch-IV
4.	*Polypedates maculatus*	Indian Tree Frog*	LC	Sch-IV

*Status assigned by the IUCN, where – CR – Critically Endangered; EN – Endangered; LC – Least Concern; NT – Near Threatened; VU – Vulnerable, DA – Data Deficient, NE – Not Evaluated

<u>Sources</u>*:* Indraneil Das (2002). *Snakes & other Reptiles of India.* New Holland Publishers (UK) Ltd pp. 1-144; Romulus Whitaker & Ashok Captain (2006). *Snakes of India; Dreko Books, Chennai, pp 1-146;* IUCN (2015); *The IUCN Red List of Threatened Species.*Version 2015-4; *Schedules I to VI: Indian Wildlife (Protection) Act, 1972.*

Annexure-VI

Copy of TOR and TOR Compliance

File No.SEIAA/AP/PKM/MIN/VIO/08/2018/672

Goverment of India

State Level Environment Impact Assessment Authority

Andhra Pradesh

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To,

M/s MIDWESTGRANITE PVT LTD

Midwest Granite Private Limited 8-2-684/25 and 26 Road no 12 Banjara Hills Hyderabad 500034,

Hyderabad-500034

Andhra Pradesh

Tel.No.-; Email:midwestgpl@yahoo.com

Sub. Terms of Reference to the Andhra Pradesh Granite (Midwest) Private Limited, Midwest Granite Private Limited 8-2-684/25 & 26, Road No 12, Banjara Hills, Hyderabad 500034

Dear Sir/Madam,

This has reference to the proposal submitted in the Ministry of Environment, Forest and Climate Change to prescribe the Terms of Reference (TOR) for undertaking detailed EIA study for the purpose of obtaining Environmental Clearance in accordance with the provisions of the EIA Notification, 2006. For this purpose, the proponent had submitted online information in the prescribed format (Form-1) along with a Pre-feasibility Report. The details of the proposal are given below:

1. Proposal No.:	SIA/AP/MIN/63337/2021
2. Name of the Proposal:	Andhra Pradesh Granite (Midwest) Private Limited
3. Category of the Proposal:	Non-Coal Mining
4. Project/Activity applied for:	1(a) Mining of minerals
5. Date of submission for TOR:	15 May 2021

Date : 01-07-2021

Sri. P.V.CHALAPATHI RAO

(MEMBER SECRETARY)

Office : **Special Secretary to Govt., EFS& T Dept, Govt.of A.P., Secretariat, Velagapudi.**

Phone No : Mobile : **9912237333**

Email id : **chalapathipasala@gmail.com**

Note : This is auto tor granted letter.

In this regard, under the provisions of the EIA Notification 2006 as amended, the Standard TOR for the purpose of preparing environment impact assessment report and environment management plan for obtaining prior environment clearance is prescribed with public consultation as follows:

STANDARD TERMS OF REFERENCE (TOR) FOR EIA/EMP REPORT FOR PROJECTS/ACTIVITIES REQUIRING ENVIRONMENT CLEARANCE

Terms of Reference (TOR) for preparation of Environmental Impact Assessment (EIA) and Environmental Management Plan (EMP) for "Mining of Minerals" as per the EIA Notification, 2006 has been devised to improve the quality of the reports and facilitate decision-making transparent and easy. TOR will help the project proponents to prepare report with relevant project specific data and easily interpretable information. TOR for mining of minerals is expected to cover all environmental related features.

Mining of minerals plays a positive role in the process of country's economic development. In addition to the contribution towards economic growth, mining can also be a major source of degradation of physical as well as social environment, unless it is properly managed. Environmental impacts can arise during all activities of the mining process. Minimizing the damage due to mining operations depends on sound environmental practices in a framework of balanced environmental legislation. The potential adverse effects of mining activities include air pollution, surface and groundwater pollution, noise and vibration, damage to local ecology, natural topography and drainage, depletion of water resources etc. All these environmental components are required to be considered while selecting a proper methodology of mining, mitigation measures to reduce pollution load, conservation of natural resources etc.

The projects of mining of minerals as stated in the schedule require prior environment clearance under the EIA notification, 2006. Category 'A' Projects are handled in the MoEF&CC and Category 'B' projects are being handled by the respective State Environment Impact Assessment Authorities (SEIAAs) notified by MoEF&CC and following the procedure prescribed under the EIA Notification, 2006. As per this Notification, as amended, the projects of mining of minor minerals with mining lease area equal to or greater than 50 hectare are to be handled at the level of the MoEF&CC for grant of EC. Such projects with mining lease area less than 50 hectare are to be handled by the respective State Environment Impact Assessment Authority (SEIAA).

1(a):STANDARD TERMS OF REFERENCE FOR CONDUCTING ENVIRONMENT IMPACT ASSESSMENT STUDY FOR NON-COAL MINING PROJECTS AND INFORMATION TO BE INCLUDED IN EIA/EMP REPORT

1) Year-wise production details since 1994 should be given, clearly stating the highest production achieved in any one year prior to 1994. It may also be categorically informed whether there had been any increase in production after the EIA Notification 1994 came into force, w.r.t. the highest production achieved prior to 1994.

2) A copy of the document in support of the fact that the Proponent is the rightful lessee of the mine should be given.

3) All documents including approved mine plan, EIA and Public Hearing should be compatible with one another in terms of the mine lease area, production levels, waste generation and its management, mining technology etc. and should be in the name of the lessee.

4) All corner coordinates of the mine lease area, superimposed on a High Resolution Imagery/ toposheet, topographic sheet, geomorphology and geology of the areashould be provided. Such an Imagery of

the proposed area should clearly show the land use and other ecological features of the study area (core and buffer zone).

5) Information should be provided in Survey of India Toposheet in 1:50,000 scale indicating geological map of the area, geomorphology of land forms of the area, existing minerals and mining history of the area, important water bodies, streams and rivers and soil characteristics.

6) Details about the land proposed for mining activities should be givenwith information as to whether mining conforms to the land use policy of the State; land diversion for mining should have approval from State land use board or the concerned authority.

7) It should be clearly stated whether the proponent Company has a well laid down Environment Policy approved by its Board of Directors? If so, it may be spelt out in the EIA Report with description of the prescribed operating process/procedures to bring into focus any infringement/deviation/ violation of the environmental or forest norms/ conditions? The hierarchical system or administrative order of the Company to deal with the environmental issues and for ensuring compliance with the EC conditions may also be given. The system of reporting of non-compliances / violations of environmental norms to the Board of Directors of the Company and/or shareholders or stakeholders at large,may also be detailed in the EIA Report.

8) Issues relating to Mine Safety, including subsidence study in case of underground mining and slope study in case of open cast mining, blasting study etc. should be detailed. The proposed safeguard measures in each case should also be provided.

9) The study rea will comprise of 10 km zone around the mine lease from lease periphery and the data contained in the EIA such as waste generation etc. should be for the life of the mine / lease period.

10) Land use of the study rea delineating forest area, agricultural land, grazing land, wildlife sanctuary, national park, migratory routes of fauna, water bodies, human settlements and other ecological features should be indicated. Land use plan of the mine lease area should be prepared to encompass preoperational, operational and post operational phases and submitted. Impact, if any, of change of land use should be given.

11) Details of the land for any Over Burden Dumps outside the mine lease, such as extent of land area, distance from mine lease, its land use, R&R issues, if any, should be given.

12) A Certificate from the Competent Authority in the State Forest Department should be provided, confirming the involvement of forest land, if any, in the project area. In the event of any contrary claim by the Project Proponent regarding the status of forests, the site may be inspected by the State Forest Department along with the Regional Office of the Ministry to ascertain the status of forests, based on which, the Certificate in this regard as mentioned above be issued. In all such cases, it would be desirable for representative of the State Forest Department to assist the Expert Appraisal Committees.

13) Status of forestry clearance for the broken up area and virgin forestland involved in the Project including deposition of net present value (NPV) and compensatory afforestation (CA) should be indicated. A copy of the forestry clearance should also be furnished.

14) Implementation status of recognition of forest rights under the Scheduled Tribes and other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 should be indicated.

15) The vegetation in the RF / PF areas in the study area, with necessary details, should be given.

16) A study shall be got done to ascertain the impact of the Mining Project on wildlife of the study area and details furnished. Impact of the project on the wildlife in the surrounding and any other protected area and accordingly, detailed mitigative measures required, should be worked out with cost implications and submitted.

17) Location of National Parks, Sanctuaries, Biosphere Reserves, Wildlife Corridors, Ramsar site Tiger/ Elephant Reserves/(existing as well as proposed), if any, within 10 km of the mine lease should be clearly indicated, supported by a location map duly authenticated by Chief Wildlife Warden. Necessary clearance, as may be applicable to such projects due to proximity of the ecologically sensitive areas as mentioned above, should be obtained from the Standing Committee of National Board of Wildlifeand copy furnished.

18) A detailed biological study of the study area [core zone and buffer zone (10 km radius of the periphery of the mine lease)] shall be carried out. Details of flora and fauna, endangered, endemic and RET Species duly authenticated, separately for core and buffer zone should be furnished based on such primary field survey, clearly indicating the Schedule of the fauna present. In case of any scheduled-I fauna found in the study area, the necessary plan alongwith budgetary provisions for their conservation should be prepared in consultation with State Forest and Wildlife Department and details furnished. Necessary allocation of funds for implementing the same should be made as part of the project cost.

19) Proximity to Areas declared as 'Critically Polluted' or the Project areas likely to come under the 'Aravali Range', (attracting court restrictions for mining operations), should also be indicated and where so required, clearance certifications from the prescribed Authorities, such as the SPCB or State Mining Dept. Should be secured and furnished to the effect that the proposed mining activities could be considered.

20) Similarly, for coastal Projects, A CRZ map duly authenticated by one of the authorized agencies demarcating LTL. HTL, CRZ area, location of the mine lease w.r.t CRZ, coastal features such as mangroves, if any, should be furnished. (Note: The Mining Projects falling under CRZ would also need to obtain approval of the concerned Coastal Zone Management Authority).

21) R&R Plan/compensation details for the Project Affected People (PAP) should be furnished. While preparing the R&R Plan, the relevant State/National Rehabilitation & Resettlement Policy should be kept in view. In respect of SCs /STs and other weaker sections of the society in the study area, a need based sample survey, family-wise, should be undertaken to assess their requirements, and action programmes prepared and submitted accordingly, integrating the sectoral programmes of line departments of the State Government. It may be clearly brought out whether the village(s) located in the mine lease area will be shifted or not. The issues relating to shifting of village(s) including their R&R and socio-economic aspects should be discussed in the Report.

22) One season (non-monsoon) [i.e. March-May (Summer Season); October-December (post monsoon season) ; December-February (winter season)]primary baseline data on ambient air quality as per CPCB Notification of 2009, water quality, noise level, soil and flora and fauna shall be collected and the AAQ and other data so compiled presented date-wise in the EIA and EMP Report. Site-specific meteorological data should also be collected. The location of the monitoring stations should be such as to represent whole of the study area and justified keeping in view the pre-dominant downwind direction and location of sensitive receptors. There should be at least one monitoring station within 500 m of the mine lease in the pre-dominant downwind direction. The mineralogical composition of PM10, particularly for free silica, should be given.

23) Air quality modeling should be carried out for prediction of impact of the project on the air quality of the area. It should also take into account the impact of movement of vehicles for transportation of mineral. The details of the model used and input parameters used for modeling should be provided. The air quality contours may be shown on a location map clearly indicating the location of the site, location of sensitive receptors, if any, and the habitation. The wind roses showing pre-dominant wind direction may also be indicated on the map.

24) The water requirement for the Project, its availability and source should be furnished. A detailed water balance should also be provided. Fresh water requirement for the Project should be indicated.

25) Necessary clearance from the Competent Authority for drawl of requisite quantity of water for the Project should be provided.

26) Description of water conservation measures proposed to be adopted in the Project should be given. Details of rainwater harvesting proposed in the Project, if any, should be provided.

27) Impact of the Project on the water quality, both surface and groundwater, should be assessed and necessary safeguard measures, if any required, should be provided.

28) Based on actual monitored data, it may clearly be shown whether working will intersect groundwater. Necessary data and documentation in this regard may be provided. In case the working will intersect groundwater table, a detailed Hydro Geological Study should be undertaken and Report furnished. The Report inter-alia, shall include details of the aquifers present and impact of mining activities on these aquifers. Necessary permission from Central Ground Water Authority for working below ground water and for pumping of ground water should also be obtained and copy furnished.

29) Details of any stream, seasonal or otherwise, passing through the lease area and modification / diversion proposed, if any, and the impact of the same on the hydrology should be brought out.

30) Information on site elevation, working depth, groundwater table etc. Should be provided both in AMSL and bgl. A schematic diagram may also be provided for the same.

31) A time bound Progressive Greenbelt Development Plan shall be prepared in a tabular form (indicating the linear and quantitative coverage, plant species and time frame) and submitted, keeping in mind, the same will have to be executed up front on commencement of the Project. Phase-wise plan of plantation and compensatory afforestation should be charted clearly indicating the area to be covered

under plantation and the species to be planted. The details of plantation already done should be given.The plant species selected for green belt should have greater ecological value and should be of good utility value to the local population with emphasis on local and native species and the species which are tolerant to pollution.

32) Impact on local transport infrastructure due to the Project should be indicated. Projected increase in truck traffic as a result of the Project in the present road network (including those outside the Project area) should be worked out, indicating whether it is capable of handling the incremental load. Arrangement for improving the infrastructure, if contemplated (including action to be taken by other agencies such as State Government) should be covered. Project Proponent shall conduct Impact of Transportation study as per Indian Road Congress Guidelines.

33) Details of the onsite shelter and facilities to be provided to the mine workers should be included in the EIA Report.

34) Conceptual post mining land use and Reclamation and Restoration of mined out areas (with plans and with adequate number of sections) should be given in the EIA report.

35) Occupational Health impacts of the Project should be anticipated and the proposed preventive measures spelt out in detail. Details of pre-placement medical examination and periodical medical examination schedules should be incorporated in the EMP. The project specific occupational health mitigation measures with required facilities proposed in the mining area may be detailed.

36) Public health implications of the Project and related activities for the population in the impact zone should be systematically evaluated and the proposed remedial measures should be detailed along with budgetary allocations.

37) Measures of socio economic significance and influence to the local community proposed to be provided by the Project Proponent should be indicated. As far as possible, quantitative dimensions may be given with time frames for implementation.

38) Detailed environmental management plan (EMP) to mitigate the environmental impacts which, should inter-alia include the impacts of change of land use, loss of agricultural and grazing land, if any, occupational health impacts besides other impacts specific to the proposed Project.

39) Public Hearing points raised and commitment of the Project Proponent on the same along with time bound Action Plan with budgetary provisions to implement the same should be provided and also incorporated in the final EIA/EMP Report of the Project.

40) Details of litigation pending against the project, if any, with direction /order passed by any Court of Law against the Project should be given.

41) The cost of the Project (capital cost and recurring cost) as well as the cost towards implementation of EMP should be clearly spelt out.

42) A Disaster management Plan shall be prepared and included in the EIA/EMP Report.

43) Benefits of the Project if the Project is implemented should be spelt out. The benefits of the Project shall clearly indicate environmental, social, economic, employment potential, etc.

44) Besides the above, the below mentioned general points are also to be followed:-

a) All documents to be properly referenced with index and continuous page numbering.

b) Where data are presented in the Report especially in Tables, the period in which the data were collected and the sources should be indicated.

c) Project Proponent shall enclose all the analysis/testing reports of water, air, soil, noise etc. using the MoEF&CC/NABL accredited laboratories. All the original analysis/testing reports should be available during appraisal of the Project.

d) Where the documents provided are in a language other than English, an English translation should be provided.

e) The Questionnaire for environmental appraisal of mining projects as devised earlier by the Ministry shall also be filled and submitted.

f) While preparing the EIA report, the instructions for the Proponents and instructions for the Consultants issued by MoEF vide O.M. No. J-11013/41/2006-IA.II(I) dated 4th August, 2009, which are available on the website of this Ministry, should be followed.

g) Changes, if any made in the basic scope and project parameters (as submitted in Form-I and the PFR for securing the TOR) should be brought to the attention of MoEF&CC with reasons for such changes and permission should be sought, as the TOR may also have to be altered. Post Public Hearing changes in structure and content of the draft EIA/EMP (other than modifications arising out of the P.H. process) will entail conducting the PH again with the revised documentation.

h) As per the circular no. J-11011/618/2010-IA.II(I) dated 30.5.2012, certified report of the status of compliance of the conditions stipulated in the environment clearance for the existing operations of the project, should be obtained from the Regional Office of Ministry of Environment, Forest and Climate Change, as may be applicable.

i) The EIA report should also include (i) surface plan of the area indicating contours of main topographic features, drainage and mining area, (ii) geological maps and sections and (iii) sections of the mine pit and external dumps, if any, clearly showing the land features of the adjoining area.

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Standard ToR Compliance report issued by the SEIAA, Andhra Pradesh

To

Andhra Pradesh Granite (Midwest) Private Limited,

Sy. No. 55/5 of Rajupalem- Lakshmi Puram village,
Chimakurthy Mandal, Prakasam District, Andhra Pradesh.

File No: SEIAA/AP/PKM/MIN/VIO/08/2018/672 **Dated 01.07.2021.**

S.No	Standard Terms of Reference Points	Status
1	Year-wise production details since 1994 should be given, clearly stating the highest production achieved in any one year prior to 1994. It may also be categorically informed whether there had been any increase in production after the EIA Notification 1994 came into force, w.r.t. the highest production achieved prior to 1994	Included in Chapter 2 of Draft EIA Report-Section no-2.11, Table No 2.4 Page no-2-16
2	A copy of the document in support of the fact that the Proponent is the rightful lessee of the mine should be given.	Included in Chapter 1 of Draft EIA Report-section no 1.2 page no.1-1.
3	All documents including approved mine plan, EIA and Public Hearing should be compatible with one another in terms of the mine lease area, production levels, waste generation and its management, mining technology etc. and should be in the name of the lessee	The modified mining plan approved by the Deputy Director of mines and Geology, Ongole for the period of 2020-2023 vide Lr.no.7141/MMS/BGG/OGL/2020 dated 22.01.2021.Draft EIA report is prepared for same extent, same method of mining, same production, Waste generation and its management. Mining technology, Production capacity is presented in Chapter -2.0 of Draft EIA report Section 2.10,Page No.2-14
4	All corner coordinates of the mine lease area, superimposed on a High Resolution Imagery/ topo sheet, topographic sheet, geomorphology and geology of the area should be provided. Such an Imagery of the proposed area should clearly show the land use and other ecological features of the study area (core and buffer zone).	Included in Chapter 2 of Draft EIA Report-Figure no-2.2, page no-2-5
5	Information should be provided in Survey of India Topo sheet in 1:50,000 scale indicating geological map of the area, geomorphology of landforms of	Included in Chapter 2 of Draft EIA Report-Figure no-2.3, page no-2-6

	the area, existing minerals and mining history of the area, important water bodies, streams and rivers and soil characteristics.	
6	Details about the land proposed for mining activities should be given with information as to whether mining conforms to the land use policy of the State; land diversion for mining should have approval from State land use board or the concerned authority.	Land use and Land cover of the area is presented in Chapter-3.0 of Draft EIA report section no 3.5.1, Page No 3-6
7	It should be clearly stated whether the proponent Company has a well laid down Environment Policy approved by its Board of Directors? If so, it may be spelt out in the EIA Report with description of the prescribed operating process / procedures to bring into focus any infringement/ deviation / violation of the environmental or forest norms / conditions? The hierarchical system or administrative order of the Company to deal with the environmental issues and for ensuring compliance with the EC conditions may also be given. The system of reporting of non-compliances / violations of environmental norms to the Board of Directors of the Company and/or shareholders or stakeholders at large, may also be detailed in the EIA Report	Environmental policy and Reporting structure given in Chapter 10, Draft EIA Report in 10.9, Page no:10-8
8	Issues relating to Mine Safety, including subsidence study in case of underground mining and slope study in case of open cast mining, blasting study etc. should be detailed. The proposed safeguard measures in each case should also be provided.	Occupational safety and Health measures are presented in chapter 10 of Draft EIA report section 10.7,Page No 10-6.
9	The study area will comprise of 10 km zone around the mine lease from lease periphery and the data contained in the EIA such as waste generation etc. should be for the life of the mine / lease period.	Included in Chapter 2 of Draft EIA Report
10	Land use of the study rea delineating forest area, agricultural land, grazing land, wildlife sanctuary, national park,	Included in Chapter 3 of Draft EIA Report-Section No-3.5, Page no-3-7.

	migratory routes of fauna, water bodies, human settlements and other ecological features should be indicated. Land use plan of the mine lease area should be prepared to encompass preoperational, operational and post operational phases and submitted. Impact, if any, of change of land use should be given.	
11	Details of the land for any Over Burden Dumps outside the mine lease, such as extent of land area, distance from mine lease, its land use, R&R issues, if any, should be given.	Dump site Details are Presented in Chapter 2 of Draft EIA Report 2.9 page no 2-13. Mine lease area is a government land, no forestland is involved, No R&R issue. Included in Chapter 2 of Draft EIA Report-Section no 2.8.1-Page no: 2-15.
12	A Certificate from the Competent Authority in the State Forest Department should be provided, confirming the involvement of forestland, if any, in the project area. In the event of any contrary claim by the Project Proponent regarding the status of forests, the site may be inspected by the State Forest Department along with the Regional Office of the Ministry to ascertain the status of forests, based on which, the Certificate in this regard as mentioned above be issued. In all such cases, it would be desirable for representative of the State Forest Department to assist the Expert Appraisal Committees	The mine lease area is a government land, no forestland is involved.
13	Status of forestry clearance for the broken up area and virgin forestland involved in the Project including deposition of net present value (NPV) and compensatory afforestation (CA) should be indicated. A copy of the forestry clearance should also be furnished.	The proposed mine lease area is a Govt., Land, no forest land is involved, and no permission is required from them.
14	Implementation status of recognition of forest rights under the Scheduled Tribes and other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 should be indicated.	The proposed mine lease area is a government land and the area does not belong to scheduled tribes and other traditional forest dwellers.
15	The vegetation in the RF / PF areas in the study area, with necessary details, should be given.	Presented in Chapter 3 of Draft EIA report-Section no-3.15,page no-3-56.

16	A study shall be got done to ascertain the impact of the Mining Project on wildlife of the study area and details furnished. Impact of the project on the wildlife in the surrounding and any other protected area and accordingly, detailed mitigative measures required, should be worked out with cost implications and submitted	There is no wildlife sanctuary within 10 km radius of the lease periphery and there is no wild animal movement in core and buffer zone. There is no Schedule-I flora and fauna present in the 10 km radius from the mine lease periphery. Presented in Chapter 4 of Draft EIA report-Section no-4.12 Page no-4-15.
17	Location of National Parks, Sanctuaries, Biosphere Reserves, Wildlife Corridors, Ramsar site Tiger/ Elephant Reserves/(existing as well as proposed), if any, within 10 km of the mine lease should be clearly indicated, supported by a location map duly authenticated by Chief Wildlife Warden. Necessary clearance, as may be applicable to such projects due to proximity of the ecologically sensitive areas as mentioned above, should be obtained from the Standing Committee of National Board of Wildlife and copy furnished.	Included in Chapter 1 of Draft EIA report table No.1.0,Reserve forests, Page no-1-3
18	A detailed biological study of the study area [core zone and buffer zone (10 km radius of the periphery of the mine lease)] shall be carried out. Details of flora and fauna, endangered, endemic and RET Species duly authenticated, separately for core and buffer zone should be furnished based on such primary field survey, clearly indicating the Schedule of the fauna present. In case of any scheduled I fauna found in the study area, the necessary plan along with budgetary provisions for their conservation should be prepared in consultation with State Forest and Wildlife Department and details furnished. Necessary allocation of funds for implementing the same should be made as part of the project cost.	Presented in Chapter 3 of Draft EIA report-Section no-3.15, page no-3-56.
19	Proximity to Areas declared as 'Critically Polluted' or the Project areas likely to come under the 'Aravali Range',	Not applicable. There is no critically polluted area within the 10km radius of the mine lease periphery.

	(attracting court restrictions for mining operations), should also be indicated and where so required, clearance certifications from the prescribed Authorities, such as the SPCB or State Mining Department should be secured and furnished to the effect that the proposed mining activities could be considered.	
20	Similarly, for coastal Projects, A CRZ map duly authenticated by one of the authorized agencies demarcating LTL. HTL, CRZ area, location of the mine lease w.r.t CRZ, coastal features such as mangroves, if any, should be furnished. (Note: The Mining Projects falling under CRZ would also need to obtain approval of the concerned Coastal Zone Management Authority).	Not applicable, the mine lease area is away from CRZ.
21	R&R Plan/compensation details for the Project Affected People (PAP) should be furnished. While preparing the R&R Plan, the relevant State/National Rehabilitation & Resettlement Policy should be kept in view. In respect of SCs /STs and other weaker sections of the society in the study area, a need based sample survey, family-wise, should be undertaken to assess their requirements, and action programmes prepared and submitted accordingly, integrating the sectoral programmes of line departments of the State Government. It may be clearly brought out whether the village(s) located in the mine lease area will be shifted or not. The issues relating to shifting of village(s) including their R&R and socio-economic aspects should be discussed in the Report.	The proposed mine lease area has no settlements and hence R&R of village does not arise. The mining lease area does not belong to any scheduled area and other weaker sections of the society. Detailed socio economic study Presented in Chapter 3 of Draft EIA report-Section 3.13, Page No.3-42
22	One season (non-monsoon) [i.e. March-May (Summer Season); October-December (post monsoon season) ; December-February (winter season)]primary baseline data on ambient air quality as per CPCB Notification of 2009, water quality,	The baseline study was carried out in the months of January 2019 – March 2019. Ambient Air Quality Status of the study area is presented in Chapter-3.0 of Draft EIA report. Water quality data, Noise level data, soil analysis and flora and fauna of the study area is presented in Chapter -3. in

	noise level, soil and flora and fauna shall be collected and the AAQ and other data so compiled presented date-wise in the EIA and EMP Report. Site-specific meteorological data should also be collected. The location of the monitoring stations should be such as to represent whole of the study area and justified keeping in view the pre-dominant downwind direction and location of sensitive receptors. There should be at least one monitoring station within 500 m of the mine lease in the pre-dominant downwind direction. The mineralogical composition of PM10, particularly for free silica, should be given.	Draft EIA report. The monitoring locations are covered all the directions in the study area.
23	Air quality modeling should be carried out for prediction of impact of the project on the air quality of the area. It should also take into account the impact of movement of vehicles for transportation of mineral. The details of the model used and input parameters used for modeling should be provided. The air quality contours may be shown on a location map clearly indicating the location of the site, location of sensitive receptors, if any, and the habitation. The wind roses showing pre-dominant wind direction may also be indicated on the map.	Included in Chapter 4 of Draft EIA Report-Section no 4.3.3, Page no: 4-7.
24	The water requirement for the Project, its availability and source should be furnished. A detailed water balance should also be provided. Fresh water requirement for the Project should be indicated.	Water requirement for the project is 80.0 KLD, which shall be drawn from nearby village. Included in Chapter 2 of Draft EIA Report-Section no 2.15.3-Page no: 2-19.
25	Necessary clearance from the Competent Authority for drawl of requisite quantity of water for the Project should be provided.	Not Applicable
26	Description of water conservation measures proposed to be adopted in the Project should be given. Details of rainwater harvesting proposed in the Project, if any, should be provided.	Included in chapter 10 of Draft EIA Report-Section no-10.4.1, Page no-10-3

27	Impact of the Project on the water quality, both surface and groundwater, should be assessed and necessary safeguard measures, if any required, should be provided.	Included in chapter 10 of Draft EIA Report-Section no-10.4, Page no-10-2
28	Based on actual monitored data, it may clearly be shown whether working will intersect groundwater. Necessary data and documentation in this regard may be provided. In case the working will intersect groundwater table, a detailed Hydro Geological Study should be undertaken and Report furnished. The Report inter-alia, shall include details of the aquifers present and impact of mining activities on these aquifers. Necessary permission from Central Ground Water Authority for working below ground water and for pumping of ground water should also be obtained and copy furnished.	Presented in Chapter 3 of Draft EIA report-Section no-3-7, page no-3-10.
29	Details of any stream, seasonal or otherwise, passing through the lease area and modification / diversion proposed, if any, and the impact of the same on the hydrology should be brought out.	No streams/rivers passing through the mine area. No modification/diversion required.
30	Information on site elevation, working depth, groundwater table etc. Should be provided both in AMSL and bgl. A schematic diagram may also be provided for the same.	Included in Chapter 3 of Draft EIA Report-Section no-3-7.4, page no-3-13.
31	A time bound Progressive Greenbelt Development Plan shall be prepared in a tabular form (indicating the linear and quantitative coverage, plant species and time frame) and submitted, keeping in mind, the same will have to be executed up front on commencement of the Project. Phase-wise plan of plantation and compensatory afforestation should be charted clearly indicating the area to be covered under plantation and the species to be planted. The details of plantation already done should be given. The plant species selected for green belt should have greater ecological value and	Included in Chapter 10 of Draft EIA Report-Section no-10.6., Page no-10-4

	should be of good utility value to the local population with emphasis on local and native species and the species which are tolerant to pollution.	
32	Impact on local transport infrastructure due to the Project should be indicated. Projected increase in truck traffic as a result of the Project in the present road network (including those outside the Project area) should be worked out, indicating whether it is capable of handling the incremental load. Arrangement for improving the infrastructure, if contemplated (including action to be taken by other agencies such as State Government) should be covered. Project Proponent shall conduct Impact of Transportation study as per Indian Road Congress Guidelines.	Included in Chapter 4 of Draft EIA Report-Section no: 4.9, page no: 4-13.
33	Details of the onsite shelter and facilities to be provided to the mineworkers should be included in the EIA Report.	Included in Chapter 2 of Draft EIA Report-Section no-2.15.1, Page no-2-19.
34	Conceptual post mining land use and Reclamation and Restoration of mined out areas (with plans and with adequate number of sections) should be given in the EIA report.	Conceptual Plan is presented in chapter-2.0 of Draft EIA report.
35	Occupational Health impacts of the Project should be anticipated and the proposed preventive measures spelt out in detail. Details of pre-placement medical examination and periodical medical examination schedules should be incorporated in the EMP. The project specific occupational health mitigation measures with required facilities proposed in the mining area may be detailed.	Presented in Chapter 10 of Draft EIA Report-Section no-10.7, Page no-10-6.
36	Public health implications of the Project and related activities for the population in the impact zone should be systematically evaluated and the proposed remedial measures should be detailed along with budgetary allocations.	The periodical medical examination conducted to the mineworkers. Healthy safety Included in Chapter 10 of Draft EIA Report Section 10.7.1 Page: 10-6

37	Measures of socio economic significance and influence to the local community proposed to be provided by the Project Proponent should be indicated. As far as possible, quantitative dimensions may be given with time frames for implementation.	Included in Chapter 10 of Draft EIA report-Section no 10.12, Page no-10-12.
38	Detailed environmental management plan (EMP) to mitigate the environmental impacts which, should inter-alia include the impacts of change of land use, loss of agricultural and grazing land, if any, occupational health impacts besides other impacts specific to the proposed Project.	The environmental management plan is presented in Chapter 10 of Draft EIA report Table No 10.4 Page No 10-12.
39	Public Hearing points raised and commitment of the Project Proponent on the same along with time bound Action Plan with budgetary provisions to implement the same should be provided and also incorporated in the final EIA/EMP Report of the Project.	Will be included in the final Draft EIA report after completion of Public hearing.
40	Details of litigation pending against the project, if any, with direction /order passed by any Court of Law against the Project should be given.	Not applicable. No court case is pending against the project.
41	The cost of the Project (capital cost and recurring cost) as well as the cost towards implementation of EMP should be clearly spelt out.	Cost of the project Included in Chapter 1of draft EIA Report, Section no: 1.5, Page no-1-8. EMP Budget Included in Chapter 10 of Draft EIA report Table No 10.4 , Page no-10-12
42	A Disaster management Plan shall be prepared and included in the EIA/EMP Report.	Included in Chapter 7, Section no-7.3, Page no-7.5 of Draft EIA Report
43	Benefits of the Project if the Project is implemented should be spelt out. The benefits of the Project shall clearly indicate environmental, social, economic, employment potential, etc.	Included in Chapter 8 of Draft EIA Report
44	Besides the above, the below mentioned general points are also to be followed: **a)** Executive Summary of the EIA/EMP Report	Enclosed as annexures in the Draft EIA Report

	b) All documents to be properly referenced with index and continuous page numbering. **c)** Where data are presented in the Report especially in Tables, the period in which the data were collected and the sources should be indicated. **d)** Project Proponent shall enclose all the analysis/testing reports of water, air, soil, noise etc. using the MoEF&CC/NABL accredited laboratories. All the original analysis/testing reports should be available during appraisal of the Project. **e)** Where the documents provided are in a language other than English, an English translation should be provided. **f)** The Questionnaire for environmental appraisal of mining projects as devised earlier by the Ministry shall also be filled and submitted. **g)** While preparing the EIA report, the instructions for the Proponents and instructions for the Consultants issued by MoEF&CC vide O.M. No. J-11013/41/2006-IA.II (I) dated 4th August 2009, which are available on the website of this Ministry, should be followed. **h)** Changes, if any made in the basic scope and project parameters (as submitted in Form-I and the PFR for securing the TOR) should be brought to the attention of MoEF&CC with reasons for such changes and permission should be sought, as the TOR may also have to be altered. Post Public Hearing changes in structure and content of the draft EIA/EMP (other than modifications arising out of the P.H. process) will entail conducting the PH again with the revised documentation. **i)** As per the circular no. J-11011/618/2010-IA.II (I) dated 30.5.2012, certified report of the status of compliance of the conditions	

	stipulated in the environment clearance for the existing operations of the project, should be obtained from the Regional Office of Ministry of Environment, Forest and Climate Change, as may be applicable. **j)** The EIA report should also include (i) surface plan of the area indicating contours of main topographic features, drainage and mining area, (ii) Geological maps and sections and (iii) sections of the mine pit and external dumps, if any, clearly showing the land features of the adjoining area.	